

The low-temperature Sherrington–Kirkpatrick free-energy limiting law

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Abstract

For every fixed inverse temperature $\beta > 1$, we prove that the zero-field Gaussian Ising Sherrington–Kirkpatrick free energy, centered by its expectation and divided by its standard deviation, converges in distribution to a nondegenerate law as the system size tends to infinity through all integers. We also prove that its variance divided by $n^{1/3}$ converges to a finite positive constant.

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1 Introduction

At a fixed temperature below the transition, how large are the sample-to-sample fluctuations of the Sherrington–Kirkpatrick free energy, and do they approach a law after centering and rescaling? We prove a limiting law for every fixed inverse temperature $\beta > 1$ in the zero-field Gaussian Ising model. The result includes a positive variance prefactor at scale $n^{1/3}$ and a characterization of the law by finite-system moments.

Let $(g_{ij})_{1 \leq i < j \leq n}$ be independent standard Gaussian variables. For $\sigma \in \{-1, 1\}^n$ and $\beta > 1$, put

$$H_n(\sigma) = \frac{1}{\sqrt{n}} \sum_{i < j} g_{ij} \sigma_i \sigma_j, \quad Z_n(\beta) = \sum_{\sigma \in \{-1, 1\}^n} e^{\beta H_n(\sigma)}, \quad F_n(\beta) = \log Z_n(\beta). \quad (1.1)$$

All expectations concern the Gaussian disorder unless another probability law is explicitly specified. There is no external field. Write

$$s_n(\beta)^2 = \text{Var } F_n(\beta), \quad X_n(\beta) = \frac{F_n(\beta) - \mathbb{E} F_n(\beta)}{s_n(\beta)}.$$

The variance is positive for every $n \geq 2$: with all couplings except g_{12} set to zero, the free energy is $n \log 2 + \log \cosh(\beta g_{12}/\sqrt{n})$, a nonconstant continuous function. Full support of the Gaussian law excludes almost-sure constancy. Standard Gaussian moment bounds give finite variance.

1.1 The fluctuation problem and its history

The model (1.1) was introduced by Sherrington and Kirkpatrick [27]. Their symmetric replica calculation gave an unphysical negative entropy at zero temperature. Parisi’s hierarchy [21] replaced a single overlap parameter by successive levels, encoded in the infinite-level limit by a function. This development led to a variational formula for the thermodynamic free energy. Gaussian interpolation established the thermodynamic limit [15], the variational bound of Guerra [14], and ultimately the matching formula proved by Talagrand [28]. Panchenko’s extension treats general mixed p -spin models, including odd interactions [20]. These results identify the leading order of the mean free energy. They leave the smaller, sample-dependent fluctuations studied here.

The one-sixth fluctuation prediction has a separate physics history. Kondor’s replica expansion [17] and the finite-size prediction of Crisanti, Paladin, Sommers, and Vulpiani [12] preceded the low-temperature large-deviation calculations of Parisi and Rizzo [22, 23]. The predicted $n^{-5/6}$ scale for the physical free-energy density becomes $n^{1/6}$ for $\log Z_n$ at fixed β . To pass from rare free energies to the width of typical fluctuations, Parisi and Rizzo assume that the large-deviation tail matches smoothly onto the central part of the distribution [23, Section VI]. They distinguish the two orders of the replica-number and system-size limits. This matching argument is heuristic: it does not itself provide a variance asymptotic or a limiting law.

Rigorous fluctuation results depend strongly on the regime. Aizenman, Lebowitz, and Ruelle proved order-one Gaussian fluctuations at high temperature [1]. Chatterjee obtained superconcentration and its relation to disorder chaos [7], as well as a constant-scale nonconcentration bound [8]. Chen and Lam’s critical and near-critical bounds concern temperatures approaching the transition [10]. For each fixed $\beta > 1$, Aronow and Lopatto prove

$$c_\beta n^{4/15} \leq \text{Var } F_n(\beta) \leq C_\beta n^{7/15}$$

in [3, Theorem 1.4], and quantitative tail estimates in a separate deviation window in [3, Theorem 1.5]. These bounds allow, but do not establish, variance of order $n^{1/3}$. Recent critical-window limits [11, Theorem 2.1] concern $\beta_n \rightarrow 1$, while the all-temperature results of Chen, Dey, and Panchenko require a nonzero field [9]. Neither hypothesis covers the fixed zero-field model here.

Our companion *The low-temperature Sherrington–Kirkpatrick fluctuation scale* [19], abbreviated *PE*, supplies both its finite-tree comparison toolkit and the fixed-temperature exponent

$$s_n(\beta) = n^{1/6+o(1)}. \quad (1.2)$$

The exponent alone does not identify a limiting distribution or prove convergence of the variance divided by $n^{1/3}$. The additional work here is a quantitative comparison of different sizes, with a summable error. Appendix A lists the precise PE inputs and their hypotheses.

1.2 The limiting law

Theorem 1.1 (Limiting law). *For every fixed real $\beta > 1$, there is a nondegenerate Borel probability measure ν_β such that*

$$X_n(\beta) \implies \nu_\beta \quad \text{as } n \rightarrow \infty \text{ through all integers.}$$

The measure ν_β has mean zero, variance one, and a finite exponential moment in a neighborhood of the origin. It is the unique probability measure with the moments specified in Proposition 1.3 below.

The centering and variance are the exact finite-size quantities. The formula below specifies the moments of ν_β ; no identification with a named distribution is made. The quantitative statement below proves that the variance has a positive limiting prefactor and provides the rate needed for the moment characterization.

For the quantitative result, we use a Gaussian completion convenient for interpolation. Let ζ_n be a centered Gaussian of variance $\beta^2/2$, independent of the couplings, and define

$$F_n^c = F_n(\beta) - n \log 2 + \zeta_n, \quad Y_n = n^{-1/6}(F_n^c - \mathbb{E}F_n^c). \quad (1.3)$$

Different copies below include independent couplings and independent completion variables. The subtraction of $n \log 2$ changes the counting spin prior into the uniform probability prior. The completion changes the covariance into the homogeneous quadratic-overlap covariance used later.

Theorem 1.2 (Moment convergence at the fluctuation scale). *For each fixed $\beta > 1$, there are real numbers $a_k = a_k(\beta)$, $k \geq 1$, with the following property. For every integer $k \geq 1$, there are constants $c_k > 0$ and $C_k < \infty$ such that*

$$|\mathbb{E}Y_n^k - a_k| \leq C_k n^{-c_k}, \quad n \geq 2. \quad (1.4)$$

There are also constants $c > 0$ and $C < \infty$, depending on β , for which

$$\sup_{n \geq 2} \mathbb{E} e^{c|Y_n|} \leq C. \quad (1.5)$$

The limits satisfy $a_1 = 0$ and $0 < a_2 < \infty$.

Theorem 1.1 follows from Theorem 1.2 by moment determinacy and removal of the completion. Indeed,

$$n^{-1/3} \text{Var } F_n(\beta) = \text{Var } Y_n - \frac{\beta^2}{2n^{1/3}} \longrightarrow a_2 > 0,$$

and the centered completion vanishes in L^2 after multiplication by $n^{-1/6}$. The exact finite-size standardization is then justified by Slutsky's theorem. We give the complete deduction, including nondegeneracy, in Section 4. For the physical total free energy $-\beta^{-1}F_n$, the variance constant is a_2/β^2 and its standardized limit is the reflection of ν_β . For its density $-(\beta n)^{-1}F_n$, the variance is asymptotic to $(a_2/\beta^2)n^{-5/3}$.

1.3 A finite-system characterization of the law

The power rate in Theorem 1.2 allows each limiting moment to be recovered from a convergent series of finite-size averages. The following random-size experiment expresses this characterization using independent copies of the finite SK model.

Proposition 1.3 (Moment specification). *Fix $\beta > 1$ and let $C = \sum_{l=2}^{\infty} l^{-2}$. For an integer $k \geq 1$, draw an integer $J \geq 2$ with*

$$\mathbb{P}(J = l) = \frac{l^{-2}}{C}.$$

Conditional on J , draw $k + 1$ independent copies $F_J^{c,0}, \dots, F_J^{c,k}$ of the completed free energy in (1.3). Set

$$t = \log J, \quad P_k = \prod_{i=1}^k J^{-1/6} (F_J^{c,0} - F_J^{c,i}).$$

Conditional averaging over the independent comparison copies gives $\mathbb{E}[P_k \mid J = l] = \mathbb{E}Y_l^k$. The constants in Theorem 1.2 satisfy

$$a_k = C \sum_{h=0}^{\infty} \mathbb{E} \left[P_k \left(\frac{t^h}{h!} - \mathbf{1}_{\{h \geq 1\}} \frac{t^{h-1}}{(h-1)!} \right) \right], \quad (1.6)$$

where the second term is absent for $h = 0$. Each expectation is taken before the series is summed. The series of these scalar expectations is absolutely convergent. We have $a_2 > 0$, and the law in Theorem 1.1 is uniquely determined by

$$\int z^k \nu_\beta(dz) = \frac{a_k}{a_2^{k/2}}, \quad k \geq 1.$$

Section 4 proves the formula and its absolute convergence from the moment rate in Theorem 1.2.

This characterization involves only a fixed prior on finite system sizes and finite Gaussian experiments. It does not require the choice of a subsequence or knowledge of an asymptotic variance. The order in (1.6) is essential: it is not permissible to interchange the unaveraged pointwise series with expectation. Section 4 also realizes each a_k as the expectation of a single integrable random variable obtained from an almost surely finite experiment.

1.4 The size comparison and its main ingredients

The new estimate compares centered log Laplace transforms at different system sizes with a summable error. For a large dyadic size N , put $x = N^{-1/6}$. The comparison gives, for every integer $N' \in [N, 2N]$,

$$\left| \log \mathbb{E} e^{qx^\varepsilon Y_{N'}} - \log \mathbb{E} e^{qx^\varepsilon Y_N} \right| \leq C_q x^{\varepsilon+c'},$$

where $c' > 0$, $q > 0$ is fixed, and $\varepsilon > 0$ is sufficiently small and fixed. The saving c' is independent of sufficiently small ε . Estimates for positive Laplace tests, together with preliminary exponential-tail bounds, recover comparisons of each fixed moment. Their errors can then be summed over dyadic intervals. This is the reduction proved in Section 4.

To obtain the transform estimate, we interpolate through hierarchical Gaussian systems. At each level, a normalized transition kernel specifies the law of the next Gaussian increments; paths can share an initial history and then branch independently. Overlaps are normalized scalar products of the terminal spin configurations. The hierarchy is indexed by an averaging parameter, called mass; its matrix and field clocks record the cumulative variances along this index. A one-site scalar model built from the Parisi minimizer supplies reference overlaps along these clocks.

The scalar input has its own development. Auffinger and Chen proved structural properties of Parisi measures and uniqueness of the Parisi minimizer [4, 5]; Jagannath and Tobasco gave variational support conditions [16]. Zhou established interval support near the critical temperature [29]. We use Lopatto's full-support result for every fixed $\beta > 1$ [18, Theorem 1.1 and Lemma 2.2], in the form developed in PE. The positive initial density needed here is proved in that companion from the scalar identities and full support.

The direct antecedent for our pinned-field and optimized-path comparisons is Aronow and Lopatto [3]. Hierarchical positive laws, interpolation, and cavity calculations also have an established lineage through Ruelle's cascade construction [26] and the cavity variational principle of Aizenman, Sims, and Starr [2]. Here we work with finite normalized transition kernels and prove the quantitative larger-tree and higher-moment estimates required by the size comparison.

These comparisons use normalized probability laws. An extra future branch can be integrated out because its conditional law has mass one; conditioning on an observation from that branch can instead change the law retained in the calculation. PE's appendix "Companion interfaces" proves the polynomial-observation bound and permits an auxiliary common temperature to vary with size; we recall these arguments when verifying their use here. The changed-constraint estimates require the additional local arguments developed in this article. Appendix A lists the imported estimates and the hypotheses retained in their use.

Along a hierarchy we retain the initial disorder as a random input and average over the later Gaussian increments. Ordinary averaging of the resulting continuation gives one pressure; exponential averaging with a small positive parameter s gives the other. Their difference, multiplied by s times the number of sites, is the centered log Laplace transform. We vary the covariance clocks and take finite differences in the number of spins. Proposition 4.1 bounds their combined effect with a power saving beyond the per-site scale x^5 . With $s = qx^{1+\varepsilon}$ and $N = x^{-6}$, multiplication by Ns converts this into the saving in the displayed transform estimate.

The route from N to N' first changes the covariance data at fixed size, then varies the size, and finally reverses the initial changes. Writing $n = Ne^\theta$ and $\ell = e^{\theta/6}$ keeps the endpoint normalization equal to $n^{-1/6}$. Figure 1 shows these three stages.

The comparison rests on four estimates.

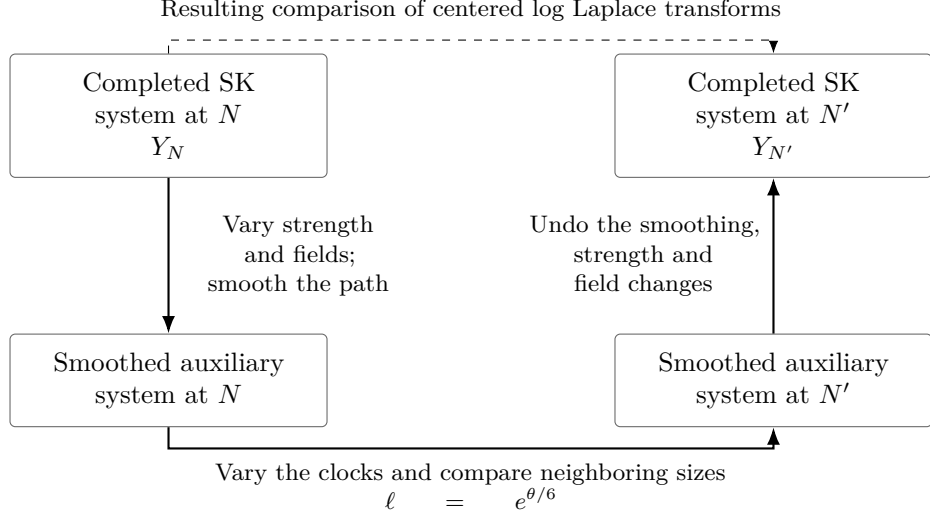


Figure 1: The route from N to an arbitrary $N' \in [N, 2N]$. The displayed endpoints are the centered, rescaled bare systems. The proof bounds the changes along the route and the errors from the small regularizations at its ends. One parameter vector is selected for each finite route and each Laplace test.

1. We solve the linear equations for covariances of overlap errors. The inverse bound remains uniform when the hierarchy is refined or its masses become small. Summation by parts cancels the small-mass denominators. Contributions from branching at an already specified vertex are retained in the bound.
2. We vary the matrix and field covariances to turn pressure comparisons into bounds on mean squared overlap errors. The constrained minimization requires estimates in both the mass coordinate and the quantile's value. A derivative of the minimum controls every minimizer, so the argument does not need to select a minimizer measurably.
3. We obtain higher moments for the actual comparison paths by projecting spin observations onto their revealed Gaussian histories and integrating squared martingale increments. Before integrating a centered quantity over a moving branch point, we show that the paths retained in the calculation have a law independent of that point.
4. We sum the signed pressure coefficients before taking absolute values. Summing the scalar contributions above a fixed cut, combining the quadratic correction, and rescaling size cancel the leading terms of degrees two, three, and four. The remaining increments are summable.

1.5 Consequences

Theorem 1.1 and the variance asymptotic above instantiate the ferromagnetic comparison of Dey and Kang [13, Proposition 1.5]. The comparison does not establish the zero-field law, but it gives the following consequence once that law and its positive variance prefactor are available.

Corollary 1.4 (SK–Curie–Weiss fluctuations below unit coupling). *Fix $\beta > 1$ and $0 \leq \gamma < 1$. Using the same independent standard Gaussian couplings as in (1.1), define the exponent*

and log partition function

$$\mathcal{H}_n^{\text{CW}}(\sigma; \beta, \gamma) = \frac{\beta}{\sqrt{n}} \sum_{1 \leq i < j \leq n} g_{ij} \sigma_i \sigma_j + \frac{\gamma}{2n} \left(\sum_{i=1}^n \sigma_i \right)^2,$$

$$F_n^{\text{CW}}(\beta, \gamma) = \log \sum_{\sigma \in \{-1, 1\}^n} e^{\mathcal{H}_n^{\text{CW}}(\sigma; \beta, \gamma)}.$$

Here γ is the ferromagnetic coupling already in the exponent. As $n \rightarrow \infty$ through all integers,

$$\text{Var } F_n^{\text{CW}}(\beta, \gamma) \sim a_2(\beta) n^{1/3},$$

$$\frac{F_n^{\text{CW}}(\beta, \gamma) - \mathbb{E} F_n^{\text{CW}}(\beta, \gamma)}{\sqrt{\text{Var } F_n^{\text{CW}}(\beta, \gamma)}} \Rightarrow \nu_\beta.$$

Here $a_2(\beta) > 0$ is the constant a_2 above at the same β . The centering and standard deviation are the perturbed model's own exact finite-size quantities.

The comparison controls the added free energy in L^2 uniformly in n . Since the unperturbed standard deviation grows like $n^{1/6}$, this error disappears after either model's exact standardization. The full argument is in Section 11.

This consequence is confined to the displayed zero-field ordered Gaussian model at fixed $\beta > 1$ and $0 \leq \gamma < 1$; it does not cover $\gamma = 1$ or a coupling varying with n .

For the original zero-field model, the variance asymptotic also gives the following consequence of Chatterjee's general Gaussian and replacement inequalities.

Corollary 1.5 (Upper disorder-chaos scales in zero-field SK). *Fix $\beta > 1$ and $n \geq 2$. Put $E_n = \{(i, j) : 1 \leq i < j \leq n\}$ and $M = |E_n| = n(n-1)/2$, and let g' be an independent copy of $g = (g_{ij})_{(i,j) \in E_n}$ in (1.1). Write $\langle \cdot \rangle_{g,h}$ for expectation under the product of that model's Gibbs laws with disorders g, h , and $R = n^{-1} \sum_i \sigma_i \tau_i$ for the overlap. Thus the replicas are independent conditional on their disorders; $\mathbb{E}$ below also averages the disorder and any replacement set.*

For $t > 0$, set $g^t = e^{-t}g + \sqrt{1 - e^{-2t}}g'$. For an integer $0 \leq k \leq M$, let A be independent of g, g' and uniform over the k -element subsets of E_n . Set $g_{ij}^A = g'_{ij}$ on A and $g_{ij}^A = g_{ij}$ otherwise. Then

$$\mathbb{E} \langle R^2 \rangle_{g, g^t} \leq \frac{1}{n} + \frac{2 \text{Var } F_n(\beta)}{\beta^2 n (1 - e^{-t})}, \quad (1.7)$$

$$\mathbb{E} \langle R^2 \rangle_{g, g^A} \leq \frac{1}{n} + \frac{2(M+1) \text{Var } F_n(\beta)}{\beta^2 n (k+1)} + C_\beta n^{-1/2}, \quad (1.8)$$

for a finite constant C_β depending only on β . As $n \rightarrow \infty$, for $t_n \geq 0$ and integers $0 \leq k_n \leq M$, these bounds imply

$$t_n n^{2/3} \rightarrow \infty \implies \mathbb{E} \langle R^2 \rangle_{g, g^{t_n}} \rightarrow 0,$$

$$(k_n/M) n^{2/3} \rightarrow \infty \implies \mathbb{E} \langle R^2 \rangle_{g, g^{A_n}} \rightarrow 0,$$

where A_n has size k_n as above.

Chatterjee's variance identity and replacement inequality bound the mean squared overlap by the free-energy variance. Substitution of the $n^{1/3}$ variance asymptotic gives the displayed sufficient scales; Section 11 checks both normalizations.

These are sufficient annealed perturbation scales, not a sharp transition or a quenched statement. No overlap conclusion is asserted for the SK–Curie–Weiss model in Corollary 1.4.

1.6 Organization and order of limits

Sections 2–3 develop the finite-tree calculus and admissible comparison paths. The pair inverse in Section 5 first supplies the global trial accuracy of Section 6; that accuracy and the local auxiliary comparisons give the genuine-path budgets of Section 7. The analytic inputs for the modified optimizers are proved within that section before their use. Section 8 derives genuine-path statistics from the budgets and uses the pair inverse again in its high-mass covariance equation. Scalar compression in Section 9 and the common parameter order in Section 10 complete Proposition 4.1. The conditional deduction in Section 4 then gives full-sequence moment convergence, positive limiting variance, moment determinacy, the finite-system law specification, and exact finite-size standardization. Section 11 contains the proofs of the two corollaries stated above.

All conclusions are for a fixed $\beta > 1$. Moment orders, expansion orders, and propagation depths are finite and fixed before $N \rightarrow \infty$. Their values may depend on the desired estimate. The exponent used for the small Laplace test is chosen only after the power saving in the increment estimate is established.

2 Hierarchical Gaussian calculus

The size comparison uses two kinds of calculation: expectations under hierarchical probability laws, and signed sums of such expectations produced by covariance differentiation. We define these separately. The resulting finite-order cavity and size expansions express pressure changes in terms of overlap errors relative to a one-site scalar model.

2.1 Normalization and positive tree laws

We refer to [19] as PE; references explicitly marked PE concern that paper. Fix $T = \beta^2 > 1$. As in (1.3), we add an independent spin-constant Gaussian of variance $T/2$ and use the uniform probability prior on spins. The resulting matrix covariance is $mTQ^2/2$, where Q is the overlap defined below. The added Gaussian has bounded moments, and replacing the counting prior by the probability prior shifts only the mean.

We work in dyads $N \leq m \lesssim 2N$ (with small enlargements), m the integer number of sites, and put $x = N^{-1/6}$. All exponents below, including any order of a Taylor expansion, are fixed as $N \rightarrow \infty$; constants can depend on these choices and on T . We use “with subpower loss” or $\lesssim_\circ$ to allow $C_\epsilon x^{-\epsilon}$ for *every* sufficiently small fixed $\epsilon > 0$. “Power small” means $O(x^c)$ for some fixed $c > 0$; for assertions in every fixed L^p a power saving not indexed by p means the exponent is independent of the fixed finite moment order (constant and threshold need not be). Power slack choices in estimates with an indicated hierarchy of exponents are made before using subpower losses.

Estimates in this dyadic notation apply with $N/2 \leq m \leq 3N$; changing comparison scales by bounded factors from this convention is immaterial. Several pieces have local variables (including error majorants and truncation orders) redefined in each calculation. In particular the path martingale X_v below is unrelated to the notation for the standardized free energy, and in Section 7 the letter X without a path subscript is also used for a scale ratio.

Fix a terminal mass $a \in [1, 4]$. The coordinate $v \in (0, a)$ is the replica mass, and the deterministic nonnegative nondecreasing functions b, h are the matrix and spin-field clocks. Jumps are allowed. Their cumulative Gaussian covariance at mass v is

$$m(b(v)Q^2/2 + h(v)Q), \quad Q = m^{-1} \sum_{i \leq m} \sigma_i \sigma'_i.$$

For step clocks, construct independent Gaussian increments with these covariance increases, including any initial increment at mass zero. If H is their sum, the terminal continuation is $a^{-1} \log \int e^{aH}$, with the integral taken against the uniform probability prior. Compute earlier continuations backwards: for an increment at mass $v > 0$, set

$$F_{\text{before}} = v^{-1} \log \mathbf{E} \exp(v F_{\text{after}}),$$

where this expectation averages only that increment. At mass zero use ordinary expectation. Divide the resulting deterministic value by m and subtract $a(b(a-)/4 + h(a-)/2)$; this defines the self-corrected pressure $f_m(b, h)$. Its interval length a will be implicit.

The same continuations define a forward probability law. Reveal each Gaussian increment with its original Gaussian density multiplied by $\exp(v(F_{\text{after}} - F_{\text{before}}))$, and then draw the terminal spin from the Gibbs law of exponent a . The backward recursion makes each transition integrate to one. On a fixed finite rooted tree, branches share the increments before their fork and use conditionally independent forward transitions afterwards. We call this the *positive tree law* and write $\nu[P]$ for its expectation of P , including both fields and terminal spins. The tree topology and clocks are fixed in this expectation. Bounded monotone clocks are obtained by the fixed-site step limits from PE stated below.

A specified observation may lie on either side of a jump or at an intermediate clock value. At a simultaneous jump we use the joint filling

$$(b_t, h_t) = (b_-, h_-) + t(\Delta b, \Delta h), \quad 0 \leq t \leq 1,$$

with mass held fixed, and preserve this filling in step approximations. Endpoint tree laws and mass-integrated identities do not depend on the order of revelation within a same-mass jump. The separate matrix and field contributions to quadratic variation, and observations at intermediate cuts, do use the prescribed filling.

For a single path let $y = \sigma/\sqrt{m}$,

$$\begin{aligned} X_v &= \mathbf{E}_v y, & C_v &= \mathbf{E}_v (y - X_v)(y - X_v)^\top, \\ S_v &= \mathbf{E} |X_v|^2, & B_v &= \mathbf{E} \|\mathbf{E}_v y y^\top\|_{\text{HS}}^2, & D_v^2 &= B_v - S_v^2. \end{aligned}$$

Here $\mathbf{E}_v$ is conditional expectation given the fields revealed through the specified cut, and $\mathbf{E}$ averages the full single-path positive law. Thus X_v, C_v are functions of the random prefix, whereas S_v, B_v, D_v are deterministic statistics of that law. Write H_v for the Hessian of the continuation with respect to a source coupled to y in the downstream Hamiltonian, with the revealed prefix held fixed. Put $L_v = y \cdot X_v - |X_v|^2/2$ and $W_v = (y - X_v)^\top H_v (y - X_v)$. These last two observables also retain the terminal spin.

2.2 Pinned moments and projection

The next estimates distinguish a variation of the deterministic clocks from movement along one fixed revealed path. In the pressure formula, $db(v), dh(v)$ are clock variations in an external parameter. In the susceptibility inequality, dS_v and dh are instead Stieltjes increments along the fixed path, including its prescribed filling at a jump. The source Hessian H_v always keeps the revealed prefix fixed.

Lemma 2.1 (Pressure, source, and single-path estimates). *For the positive laws just defined, the following identities and inequalities hold at every fixed size. The moment bounds in (A1)–(A2) hold for every fixed real $2 \leq p < \infty$, with deterministic $k \in L^2(0, a)$ and zero*

deterministic external field.

$$\begin{aligned}
df_m &= -\frac{1}{4} \int B_v db(v) dv - \frac{1}{2} \int S_v dh(v) dv, \\
vC_v &\preceq H_v \preceq aC_v, \quad dS_v \geq m \mathbf{E} \operatorname{tr} H_v^2 dh, \quad \mathbf{E} W_v = \mathbf{E} \operatorname{tr} C_v H_v, \\
\left\| \int k_v (L_v - \mathbf{E} L_v) dv \right\|_p^2 &\leq c_p \left\| \int k_v^2 W_v dv \right\|_{p/2}.
\end{aligned} \tag{A1}$$

The last line is a moment estimate under the full single-path law, with deterministic weights k_v and zero deterministic external field. More generally, for polynomials P_v of fixed degree with bounded deterministic coefficients,

$$\left\| \int k_v (P_v(L_v) - \mathbf{E} P_v(L_v)) dv \right\|_p^2 \leq c_p \left\| \int k_v^2 P_v'(L_v)^2 W_v dv \right\|_{p/2}. \tag{A2}$$

For the projection bound only, assume that the total clocks are at most a fixed power of $\log N$, and that the positive mass and field length are bounded below by inverse powers of N . If a branch has traversed a pure or mixed clock interval with field length $\Delta h > 0$ after separating from the data determining an anchor vector V , and its mass is at least $v > 0$ throughout that interval, then for every fixed even $p \geq 2$,

$$\|V \cdot (y - X_{\text{end}})\|_p \lesssim_p \frac{\log^2 N}{v\sqrt{m\Delta h}} \|V\|_p \tag{A3}$$

where the vector norm inside L^p is Euclidean. The branch must retain its normalized future kernel conditional on the prefix and outside data; in particular, the bound applies to conditionally independent resamplings.

Proof. Equations (A1) and (A3) are the pressure, source, pinned-moment, and projection estimates of [19, Lemmas 2.1–2.3 and 3.5, equations (1) and (13)]. To match the terminal-mass-one normalization there, put

$$u = v/a, \quad \tilde{b}(u) = a^2 b(au), \quad \tilde{h}(u) = a^2 h(au), \quad \tilde{F} = aF.$$

The conditional spin means and covariances are unchanged, while $H_{au} = a\tilde{H}_u$. The projection denominator is unchanged as well, since $(v/a)\sqrt{ma^2\Delta h} = v\sqrt{m\Delta h}$. Appendix A records the pressure and moment conversions. The fixed-size path-limit result is PE, Proposition 2.14.

For (A2), we include the polynomial argument of PE, Appendix, paragraph “Source and pressure calculus; the scope of pinned moments.” Pin the terminal spin and write the Gaussian increments in terms of standard Gaussian drivers, as in the proof of PE, Lemma 2.3. Let $\mathbf{A}_v$ be the Hessian of the continuation in these drivers and $\mathbf{B}_v$ its mixed Hessian with the normalized spin source. After the transition densities telescope, the negative logarithm of the pinned driver density is the Gaussian quadratic potential minus aH_σ , which is linear, plus $\int_0^a F_v dv$, where F_v is the continuation at the cut v . Consequently

$$\begin{pmatrix} \mathbf{A}_v & \mathbf{B}_v \\ \mathbf{B}_v^\top & H_v \end{pmatrix} \succeq 0, \quad \mathbf{Q} = I + \int_0^a \mathbf{A}_v dv,$$

where $\mathbf{Q}$ is the Hessian of the negative logarithm of the pinned driver density. The driver gradient of L_v is $g_v = \mathbf{B}_v(y - X_v)$. Thus, for every driver direction z , $|z \cdot g_v|^2 \leq (z^\top \mathbf{A}_v z) W_v$. For the polynomial observable set

$$G_P = \int_0^a k_v P_v'(L_v) g_v dv, \quad R_P = \int_0^a k_v^2 P_v'(L_v)^2 W_v dv.$$

Cauchy–Schwarz in v gives $|z \cdot G_P|^2 \leq (z^\top Q z) R_P$, hence $G_P^\top Q^{-1} G_P \leq R_P$. The inverse-Hessian Brascamp–Lieb inequality [6, Theorem 4.1] therefore gives (A2) for $p = 2$. Applying the same variance inequality to the $p/2$ -th absolute power of the centered integral, as in PE, Lemma 2.3, gives the assertion for every fixed $p > 2$.

The zero-field spin gauge transformations act transitively on terminal spins and preserve both L_v and W_v . Because the coefficients of P_v and the weights k_v are deterministic, they also preserve the polynomial integral and R_P . Their conditional laws are consequently independent of the pinned spin, including the mean used for centering. Averaging over that spin yields the full single-path estimate. Bounded-weight approximation and then the fixed- m path limit give the stated range of clocks and weights.

Clocks are bounded for each N , possibly by a power of $\log N$; (A1) is clock independent. For (A3), we apply PE, equation (13), only in this bounded or polylogarithmic clock regime; together with the polynomial mass and field-length bounds, this gives the polynomial coefficient-ratio bounds needed in its moment iteration, as detailed in Appendix A. Our estimates can always be passed from affine finite-step interpolations, or from inserted fixed cuts, by the fixed- m limit assertion, keeping chosen revelation sides and the joint filling at observed intermediate jump cuts, and using the corresponding integrated inequalities. In particular no discretization *rate* in m is used.

The pinned inequalities in this lemma concern the full single-path law. When a tree contains additional branches, we first marginalize the branches not used by the test. Random factors retained on other branches are subsequently handled by unconditional Hölder inequalities; they are not made into random weights in (A2). $\square$

2.3 Signed allocation of replica labels

Covariance derivatives introduce additional replica labels. For a simple example, start with one observed leaf above a cut v , and divide its edge at a level $u \in (v, a)$. An extra label may coincide with that leaf, with coefficient a , or split from it along either edge interval, with coefficient minus the interval length. On a constant summand the total is $a - (a - u) - (u - v) = v$. In a larger tree, attachment to a fork also contributes an atom. We sum all these placements by the signed rule of PE, Section 2, Proposition 2.5, equation (2), and Lemmas 2.6, 2.8, and 2.9. A new label coinciding with an observed leaf has coefficient a ; a new split on an occupied edge has coefficient $-dv$; attachment at an existing mass- v fork with d children has coefficient $-(d - 1)v$. These coefficients multiply expectations under the positive laws just defined. They are not probabilities.

The row-sum rule sums one free label whose summand is constant over all its attachments in a block above a cut v . The result is that constant times v , as in the example. A row whose summand depends on its attachment cannot be collapsed this way. Empty levels can be inserted continuously without changing the calculation. Different infinitesimal fork variables specify different tree vertices even when the covariance clocks have the same value at those vertices.

A sum *per root mass*, denoted by $(\cdot)_\#$, begins without specified old labels. Its first label is an ordinary observation with weight 1, and all subsequent labels use the signed rule. This is the limit of normalized summation as the incoming mass tends to zero. Sums without $\#$ extend whatever old labels are already present. Products of symbolic slot polynomials concatenate all their labels; a coefficient involving only some labels is evaluated on the restriction of the final topology to those labels. The labeled product rules in PE, Proposition 2.5 make this convention associative, commutative, and independent of allocation order. In a per-root product we concatenate first and apply the root normalization only once.

Constant diagonal covariance changes contribute only to the self correction. Thus pressure insertions after that correction use ordered off-diagonal pairs and a factor $1/2$ per insertion. The first insertion gives (A1); for subsequent insertions the constant diagonal terms cancel by the row-sum rule in PE, equation (2). The same rule applies to any affine positive semidefinite interpolation between finite feature models with constant diagonal and fixed a , after subtracting $a/2$ times the terminal diagonal. For total pressures, the inserted covariance change is the total covariance change, rather than its per-site value. Deterministic additive terms common to the two ensembles cancel when we take their pressure difference.

Enlarging a positive-law tree preserves its original marginals. For any fixed number of inserted labels, the total variation of the signed allocation coefficients is uniformly bounded. Taking absolute values in placement order leaves finitely many integrals over new fork masses, existing-fork atoms with bounded multiplicities times their masses, and terminal coincidences. We may therefore reorder the exact signed sum with its topology restrictions first, and then estimate with this absolute allocation measure. This order of operations is essential when a later argument uses cancellation.

2.4 Scalar references and Laplace tests

The signed calculus will compare the matrix model with independent one-site scalar systems. For the reference model take one spin and only a deterministic spin clock $K(v)$. Its continuation can be parametrized by cumulative variance t instead of mass. The mass already reached by time t is $\alpha(t) = |\{v \in (0, a) : K(v) \leq t\}|$. Write $V(t, z)$ for the scalar continuation at field z , and Z_t for its field under the forward positive law. They satisfy

$$V_t = -\frac{1}{2}(V_{zz} + \alpha V_z^2), \quad dZ_t = \alpha V_z(t, Z_t)dt + dB_t, \quad V_{\text{term}} = a^{-1} \log \cosh(az).$$

Set $\Gamma(t) = \mathbf{E}V_z(t, Z_t)^2$, with zero starting field unless otherwise stated. A terminal mass- a extension in scalar time changes earlier continuation values only by a spatial constant. In particular it leaves their derivatives in z unchanged. On a fixed tree let τ_i be the scalar spin at leaf i , and let v_{ij} be the mass where distinct leaves i, j separate. Their positive-law moment is $\nu_K[\tau_i \tau_j] = \Gamma(K(v_{ij}))$. Common covariance insertions and source derivatives of these moments again use the tree calculus. In particular any fixed spatial derivative of positive order of V is bounded uniformly (even with long but finite clocks). Below put

$$q(v) = \Gamma(K(v)), \quad h_*(v) = K(v) - b(v)q(v), \quad d(v) = h(v) - h_*(v).$$

The physical paths used below have $b(v) = b(0+)$ and $h(v) = 0$ for $v < x$. For $0 < s < x$, move the initial matrix revelation from mass zero to mass s . This gives the tilted pressure $f_m^s = f_m(b \mathbf{1}_{v \geq s}, h)$; the untilted pressure is $f_m^0 = f_m(b, h)$. Put $\psi_m(s) = f_m^s - f_m^0$. The factor $\mathbf{1}_{v_{ij} \geq s}$ records whether two tilted branches share the matrix revelation. Accordingly define $Q_{ij}^* = Q_{ij}$ in the untilted ensemble and $Q_{ij}^* = \mathbf{1}_{v_{ij} \geq s} Q_{ij}$ in the tilted ensemble. The mass- s power mean then gives

$$\mathcal{K}_m(s) = s m \psi_m(s) = \log \mathbf{E} \exp\{s(G_m - \mathbf{E}G_m)\}, \quad (\text{A4})$$

where G_m is the random total continuation just after the initial matrix revelation in the untilted path. Each connected subtree with internal splits at $v \geq s$ in a tilted positive law has its untilted marginal reweighted by $\exp(sG_m)/\mathbf{E} \exp(sG_m)$. Distinct such components have independent roots. Products using active branches from more than one component can therefore be bounded by their active-subtree marginal norms without interpreting the forest as sharing an untilted root. Set

$$\Delta = Q^* - q, \quad J = h + bQ^* - K = b\Delta + d$$

on off-diagonal pairs. Scalar paths used as references are *unshifted* in both ensembles. Tests involving matrix-model overlaps only through Q^* are invariant under simultaneous sign flip of all spins on a component (with its high ordinary fields negated); consequently at every pair ij , including below s in the tilted case, $\nu[PQ_{ij}] = \nu[PQ_{ij}^*]$ for these spin tests.

2.5 Finite row and size expansions

We now encode fixed finite Taylor expansions in the row mismatch J . A *jet* means that the series is retained only through a specified finite degree; no convergence of an infinite series is being asserted. In all inserted pairs, the two ends must be different leaves, although labels from different pairs may coincide. Define the symbolic series

$$\Psi_K(J) = \left\langle \exp\left\{\frac{1}{2} \sum J_{cd}\tau_c\tau_d\right\} \right\rangle_K, \quad N_{ij,K}(J) = \left\langle \tau_i\tau_j \exp\left\{\frac{1}{2} \sum J_{cd}\tau_c\tau_d\right\} \right\rangle_K. \quad (\text{A5})$$

To read (A5), first expand the exponential to the required finite degree and allocate its pair labels. For each final topology, the brackets are the positive scalar moment of the labels in that coefficient. Multiply coefficients on their respective restricted topologies before carrying out the signed allocation sum. Thus $\log \Psi$ has the ordinary joint scalar cumulants at each fixed topology as its coefficients. Products, differentiation, and inversion of the series with constant term 1 are all finite-degree polynomial operations under this convention.

For example, in the product of two scalar coefficients $\nu_K[\tau_i\tau_j]\nu_K[\tau_c\tau_d]$, allocate all four labels on one final topology, then evaluate the first moment on its ij restriction and the second on its cd restriction. Their product is not the four-spin moment, even when some labels coincide. The clock derivative and the coefficient linear in the number of added sites will be expressed through the same pressure symbol, namely

$$\mathcal{I} = \log \Psi_K(J) - \frac{1}{4} \sum b_{ij}(Q_{ij}^*)^2. \quad (\text{A6})$$

Independent additive deterministic per-root terms in this symbol will always cancel.

The following Taylor rules require a monotone reference K and bounded $h - K$ and b . A remainder with k insertions is a finite sum over boundedly many allocated labels, possibly integrated over interpolation parameters. Its integrand is a bounded factor times k specified overlap insertions on off-diagonal pairs. We estimate such terms using the absolute allocation measure and a positive probability law.

The intermediate law may still contain one interpolated scalar row. For a nonnegative spin test of the other rows, PE, Lemma 2.9 compares this law to the actual law up to a constant: its Grönwall bound uses the bounded total covariance change of that row. Deleted sites are omitted from an overlap without changing its denominator m . Restoring those sites changes a bounded Lipschitz polynomial of fixed degree by $O(m^{-1})$. Any deterministic topology coefficient or weight multiplying the test also multiplies this error bound. Throughout this interpolation the spin test is fixed, so it contributes no additional derivative.

Lemma 2.2 (Finite cavity quotient). *For overlap-polynomial P invariant as after (A4), on any fixed finite-label tree including ij , one has through order $k - 1$*

$$\nu[PQ_{ij}^*] = \nu[P(N_{ij,K}/\Psi_K)_{<k}] + \text{Rem}_k + O(m^{-1}), \quad (\text{A7})$$

where Rem_k has at least k insertions J , times P .

Proof. By exchangeability, the observed overlap may be represented using one chosen site. Delete that site's contribution from the other tested overlaps, retaining the denominator m ,

and interpolate its Gaussian kernels to those of an independent one-site scalar model with clock K . All other covariance kernels remain fixed. Both endpoints are positive semidefinite, as is their affine interpolation. In the expansion back from the independent endpoint, the covariance insertion is $J_{cd}\tau_c\tau_d$, where J uses the remaining-row overlap and the mask of the actual matrix clock. The omitted terms are $O(m^{-1})$ and constant diagonals. The row covariance change is bounded, so Taylor's integral remainder has the form described above.

At the independent endpoint, averaging the removed spin gives Ψ_K when it is unobserved, and $N_{ij,K}$ when its product on leaves i, j is observed. Expanding the former against the truncated quotient $N_{ij,K}/\Psi_K$ cancels its factor Ψ_K to the retained order. Allocation order symmetry identifies the terms on each final topology and gives (A7). Products here multiply separately averaged scalar coefficients; they do not couple their scalar spins. In particular the linear cavity kernel on a pair cd is

$$\mathbf{M}_{ij,cd} = \frac{1}{2}(\nu_K[\tau_i\tau_j\tau_c\tau_d] - \nu_K[\tau_i\tau_j]\nu_K[\tau_c\tau_d]). \quad (\text{A8})$$

This proof also gives the unestimated remainder representation using one interacting scalar row (the removed site) along the affine interpolation, if symmetries must be used before taking absolute values. □

Lemma 2.3 (Change of scalar reference). *Apply (A7) at a monotone reference K' , then Taylor-expand its deterministic scalar coefficients from K to K' . At any consistent total truncation, the result is the quotient with reference K and combined argument $J = (h + bQ^* - K') + (K' - K)$.*

Proof. PE covariance differentiation, equation (2), represents a derivative of a scalar coefficient by a new pair insertion $(K' - K)_{cd}\tau_c\tau_d/2$. Allocation order symmetry permits us to combine these with the insertions already present in the quotient. The apparent extra factors $\Psi_K(K' - K)$ have no effect on an observed topology whose spin test is independent of the new labels: after allocating these labels last, every positive-degree term is a derivative of the unit scalar moment and sums to zero. For a standalone per-root sum the same terms can remain, but they are deterministic. Taylor's integral remainder has bounded scalar coefficients and powers of $K' - K$; it can also retain ordinary polynomial factors of the original argument $h + bQ^* - K'$. □

Lemma 2.4 (Differentiated row identity). *Let a dot vary b, h, K, q at fixed interval length a , and let $\mathcal{D}$ differentiate (A6) including its scalar coefficients, with the formal rule $\mathcal{D}\Delta = -e\Delta$ for any fixed e . Thus $\mathcal{D}Q^* = \dot{q} - e\Delta$; this rule acts on the symbolic polynomial and does not differentiate a random spin sample. Up to deterministic terms common to the two ensembles and Taylor remainders,*

$$\dot{f}_m^s \equiv (\nu[\mathcal{D}\mathcal{I}])_{\#}. \quad (\text{A9})$$

The same formula applies with superscript $s = 0$. The error and its differentiated weights are as described in the proof below.

Proof. Differentiating a scalar coefficient inserts $\dot{K}_{ij}\tau_i\tau_j/2$. This cancels the $-\dot{K}$ part of $\mathcal{D}J$; terms with no previously attached labels are deterministic. Thus the logarithm contributes

$$\frac{1}{2} \sum \frac{N_{ij,K}}{\Psi_K} (\dot{h} + \dot{b}Q^* + b\mathcal{D}Q^*)_{ij}.$$

Apply (A7) to replace the quotient by Q^* , with its stated error. The term $bQ^*\mathcal{D}Q^*/2$ then cancels the corresponding derivative of the quadratic term in (A6). In particular the choice

of e disappears. The terms left are $\frac{1}{2} \sum \dot{h} Q^* + \frac{1}{4} \sum \dot{b} (Q^*)^2$, precisely the off-diagonal pressure insertions in (A1).

For the error, all tests have bounded degree and at most one differentiated weight. If $\log \Psi$ is truncated after degree k , the residuals in the difference of the two pressure derivatives have at least k insertion factors. They also have one weight at a split, possibly on an additional pair, bounded by a constant times

$$1 + |\dot{h}| + |\dot{b}| + |\dot{K}| + |\dot{q}|;$$

the 1 is omitted when $e = 0$. The $O(m^{-1})$ errors carry the same weights. This follows from the product rule, since differentiating one insertion can remove that factor, and from (A7). If a second reference K' is used, expand the quotient in this gradient first, then recombine by Lemma 2.3. This does not require differentiating K' .

There is one truncation convention to check. Differentiating an insertion lowers its insertion degree by one, whereas differentiating a scalar coefficient does not. Through degree k in the old J factors, the coefficient part of $\mathcal{D}\Psi$ therefore inserts $\frac{1}{2} \sum (N_{ij,K})_{\leq k} \dot{K}_{ij}$, formally including its degree-zero term. In a product with old labels independent of the new placements, that term sums to zero when the new labels are allocated last: it differentiates a unit scalar observation. Standalone per root, its discrepancy is common and deterministic. Thus coefficient differentiation of the logarithm uses the reciprocal jet through degree k , while differentiation of an insertion uses the quotient through degree $k - 1$. All these cancellations take place in the unrestricted signed sums, before imposing split-scale restrictions for estimates.

□

Lemma 2.5 (Finite increments in the number of sites). *At fixed clocks and s , assume that bq^2 is nondecreasing. For every fixed positive integer j , the change from $m\psi_m$ to $(m+j)\psi_{m+j}$, expanded through total order k , is a polynomial in j of degree at most k with zero constant term. Its linear coefficient is the difference of the per-root expectations of (A6), with $\log \Psi$ truncated after degree k . The remainder has the total-pressure bounds given below.*

Proof. For size changes at fixed clocks and s , compare $(m+j)\psi_{m+j}$ to $m\psi_m$, with any fixed positive integer j . Interpolate the enlarged kernel from that of m old spins plus j independent scalar rows at K , to its actual value plus independent, spin-constant cumulative covariance $jbq^2/2$ (assume bq^2 nondecreasing). This addition gives the same pressure shift for both tests. The off-diagonal insertion before the factor $1/2$ is

$$J \sum_{h'=1}^j \tau^{h'} \tau^{h'} - \frac{j}{2} b(Q^{*2} - q^2) + O_j(m^{-1}).$$

Indeed the normalization of the matrix feature changes from m to $m+j$ while field terms are additive. This difference is bounded. Taylor through order k in the total pressure now gives a polynomial in j of degree $\leq k$, constant term zero; its coefficient linear in j is the compared per-root expectation of (A6), $\log \Psi$ truncated after degree k . For the linear coefficient, group the labeled insertions by their assigned scalar row. The scalar rows are independent on each fixed topology, so their series is Ψ_K^j . If C denotes the matrix counterterm $\frac{1}{4} \sum b(Q^{*2} - q^2)$, the combined series is $\Psi_K^j e^{-jC}$. Its coefficient of j is

$$[j](\Psi_K^j e^{-jC}) = \log \Psi_K - C.$$

The deterministic q^2 term cancels between the two tests, leaving (A6). This identity is taken at the same finite total Taylor order as the covariance expansion. Remainders after taking

centered differences can be bounded by $O(m^{-1})$ and products of $k + 1$ factors $|\Delta| + |J|$ at the insertions, under the actual m -spin laws by the same Grönwall comparison. No $(m + j)$ factor is to be put in front of this remainder. $\square$

These rules are identities and bounds for fixed finite Taylor orders, not infinite replica expansions or analytic continuations in a replica count. Differentiation along absolutely continuous paths in (A9) uses pointwise (or integrated) chain rules, with deterministic dotted clocks absolutely integrable including their bounded-density fork averages.

The clock derivative and the coefficient linear in the number of added sites are thus expressed by the same finite pressure symbol (A6). This shared expression permits the size comparison to combine their signed coefficients before estimating the remainders.

3 Scalar reference paths and admissible perturbations

We construct the deterministic clocks used in the pressure comparison. Their first requirement is that the reference field $h_* = K - bq$ be nonnegative and increasing. Since $q(v) = \Gamma(K(v))$, its increment along scalar time $t = K(v)$ is

$$dh_* = (1 - b\Gamma'(t)) dt - q db.$$

The construction first creates a positive margin in $1 - b\Gamma'$, then chooses the matrix variation and the additional field terms within that margin.

We use direct paths while varying the perturbation strength, and smoothed paths while varying the size coordinate. The smoothing keeps the terminal mass fixed and permits size differentiation; a final comparison shows that its pressure cost is smaller than x^5 . Put $\ell = \exp(\theta/6)$, $0 \leq \theta \leq \log 2$. Direct paths have terminal mass $a = \ell$, and smoothed paths have terminal mass $a = 4$. The parameter ℓ is fixed except during a size change. In normalized scalar coordinates, time is $\xi = \ell^2 t$ and mass is v/ℓ . Fix

$$\omega = x^{5+1/4}, \quad M = \log^3(1/x), \quad x^{12} \leq \lambda \leq x^{p_*}, \quad U = \min(1, x\lambda^{-s_*/2}), \quad \lambda_1 = \lambda^{1+\eta}. \quad (\text{P1})$$

The final strength means $\lambda = x^{p_*}$. Choose $s_* < 1/2$ sufficiently close to $1/2$, then $\eta > 0$ sufficiently small also relative to $1/2 - s_*$, and $p_* > 0$ sufficiently small. Later small exponents may be decreased subject to their stated order. Logarithms and constants are absorbed with powers much smaller than p_* times the available gaps between exponents of λ .

3.1 The fixed-temperature scalar reference

For a one-site clock K of terminal mass 1, write $f_{\text{spin}}(K) = f_1(0, K)$ for its self-corrected pressure. Let $z : (0, 1) \rightarrow [0, 1]$ be the deterministic nondecreasing quantile minimizing the fixed-temperature scalar objective

$$f_{\text{spin}}(Tz) + \frac{T}{4} \int_0^1 z(w)^2 dw.$$

The corresponding variance-time quantile is $Tz(w)$. Its distribution function is $A_0(\xi) = |\{w \in (0, 1) : Tz(w) \leq \xi\}|$, and Γ_0 is the scalar response of this clock as defined in Section 2. These are the fixed base objects; the reference clocks below will perturb them.

Lemma 3.1 (Regular scalar profile and a strict top plateau). *At fixed $T > 1$, the scalar minimizing distribution has support $[0, \xi_*]$, satisfies $\Gamma_0(\xi) = \xi/T$ there, and has a positive initial density. Its active-side distribution function has bounded one-sided derivatives of every fixed order. Moreover $p_{\text{top}} = A_0(\xi_* -)$ belongs to $(0, 1)$.*

Proof. We use PE, Lemma 5.8, whose full-support input is [18, Theorem 1.1 and Lemma 2.2]. Its measure has support $[0, \xi_*]$, $\xi_* > 0$; on this support $\Gamma_0(\xi) = \xi/T$. Its hypotheses are precisely fixed $T > 1$, zero field, and the Ising two-spin scalar objective above, with terminal mass 1. The lemma gives $A_0(\xi) \sim \kappa_T \xi/T$ at 0, $\kappa_T > 0$, and continuity of the quantile on the open mass interval. The measure therefore has support down to 0 with no interior gap or atom at 0.

At the minimizing z , vary z to $z + tP_0(z)$ with smooth P_0 supported in $(0, 1)$ (both signs admissible for small t). By scalar (A1) the derivative of the objective is $T \int (z - \Gamma_0(Tz))P_0(z)/2 = 0$, giving self-consistency by continuity throughout the support. On $[0, \xi_*]$ viewed from the active side, A_0 is smooth with bounded one-sided derivatives of all orders, $A'_0 \geq c > 0$ near zero. Indeed $\Gamma''_0 = 0$ in the interior gives $A_0 = \mathbf{E}(V_{zzz})^2/(2\mathbf{E}(V_{zz})^3)$ almost everywhere by PE, Section 5, Lemma 5.3 (in variance time).

The numerator and denominator are continuous moment functions. The positive-Hessian argument of the same lemma, restricted to a bounded field interval of uniformly positive probability, bounds the denominator away from zero throughout the compact active interval. Their ratio therefore gives a continuous representative of A_0 , extending Lipschitzly to both endpoints. Differentiation with the scalar PDE and Itô generator improves this representative one derivative at a time; the positive denominator allows the iteration to every fixed order. The positive initial slope is the initial-density conclusion of PE.

In fact $p_{\text{top}} = A_0(\xi_*-) \in (0, 1)$. Here is a useful argument. Extend by a mass-one annealed clock, and let ρ denote the law of Z_{ξ_*} . The finite measure $w(dz) = \rho(dz)/\cosh z$ is even and log-concave. To see the sign of the potential, take a step approximation ending at ξ_* and telescope its transition weights. After division by $\cosh z$, the joint density of the Gaussian increments is their Gaussian density times the exponential of negative convex continuation potentials. Each coefficient is a nonnegative mass increment; the unused terminal mass similarly multiplies $-\log \cosh z$. Marginalizing the increments at fixed terminal field preserves log-concavity [24]. Fixed-clock approximation, as in PE, Proposition 2.14, passes it to w . The limiting field law has a density because its diffusion has bounded drift over positive time. At subsequent time $\xi_* + d$, the derivative $\mathbf{E}\chi^2$, $\chi = V_{zz}$, is

$$e^{-d/2} \int \text{sech}^3 z (w * G_d)(dz) \leq e^{-d/2} \Gamma'_0(\xi_*),$$

with G_d the centered Gaussian kernel. The inequality follows by even unimodality (translate overlaps of upper level intervals). At $d = 0$ the two sides agree, so the inequality makes the right second derivative strictly negative. The left second derivative is zero by $\Gamma_0(\xi) = \xi/T$. The spatial moments in $\Gamma'' = \mathbf{E}(V_{zzz}^2 - 2A_0\chi^3)$ agree across the endpoint; only the mass changes. Consequently

$$\Gamma''_0(\xi_*) - \Gamma''_0(\xi_*-) = -2(1 - p_{\text{top}})\mathbf{E}\chi^3 < 0,$$

which gives $p_{\text{top}} < 1$. Its positivity follows from the positive initial density and monotonicity of A_0 . Thus the base quantile is constant on a nonempty terminal mass interval. $\square$

3.2 Creating a positive scalar gap

The minimizing scalar reference is marginal: $\Gamma'_0 = 1/T$ on its active support, so in these normalized coordinates the choice $b = T$ leaves no margin for $-q db$. We perturb its density near zero to create a gap of order λt^2 . The matrix construction will add a second margin of order λU^2 .

Choose $z_0 > 0$ small fixed with $A'_0 \geq c > 0$ up to $2z_0$, and let $P(\xi)$ be smooth nonnegative, bounded by ξ , equal to ξ on $[0, z_0/2]$ and zero after z_0 . Adding λP gives the distribution $A_\lambda = A_0 + \lambda P$, which is admissible for sufficiently small λ . Its direct physical clock is $K(v) = t(v) = \ell^{-2} A_\lambda^{-1}(v/\ell)$; it is constant on the top plateau $v/\ell \in [p_{\text{top}}, 1]$.

For the smoothed clock, fix the final strength. Spread the jump $1 - p_{\text{top}}$ linearly over $[\xi_*, \xi_* + \omega]$, add $\omega\xi$ to this mass profile for $0 \leq \xi < M$, and at M jump to total normalized mass $4/\ell$. Denote the resulting profile by $\hat{A}$. Below M it is independent of ℓ , has no support gaps, and satisfies $\omega \leq \hat{A}' \leq C/\omega$ almost everywhere. The physical smoothed clock is $K(v) = \ell^{-2} \hat{A}^{-1}(v/\ell)$. It may have a terminal interval on which its normalized quantile equals M . The positive density below that interval permits the size differentiations used later.

For either reference, changing from normalized to physical coordinates gives

$$V(t, z) = \ell^{-1} \hat{V}(\xi, \ell z), \quad \Gamma(t) = \hat{\Gamma}(\xi),$$

where the normalized mass profile was just given, with terminal mass $j = 1$ or $j = 4/\ell$. Define $b_o = 1/V_{zz}(0, 0)^2$. Since the starting field is zero, $\Gamma'(0) = V_{zz}(0, 0)^2$, so this choice sets the initial gap $1 - b_o \Gamma'(0)$ to zero. The lift below makes that gap positive at every positive time.

Lemma 3.2 (Quantitative scalar lift). *Uniformly (extending by a terminal annealed clock),*

$$\begin{aligned} b_o &\asymp 1, & 1 - b_o \Gamma'(t) &\geq c\lambda \min(t^2, 1) & (t \geq 0), \\ 1 - b_o \Gamma'(t) &\leq C\lambda t^2 & & & (0 \leq \xi \leq \xi_*; \text{ active side}), \\ t(v) &\asymp v, & q(v) &\asymp v & \text{near } 0, \quad q(v) \asymp v \text{ throughout.} \end{aligned} \tag{P2}$$

Proof.

The direct lift. Work in normalized coordinates. Up to ξ_* , positive-order derivatives of continuations change by $O(\rho_1)$, $\rho_1 = \lambda \int P$. Indeed the PDE for their undifferentiated difference has linear drift with bounded smooth spatial coefficients on a fixed time interval and source bounded with derivatives by $C\lambda P$; differentiate its diffusion representation (bounded drift derivatives). Coupling Z 's by common Brownian noise gives L^k change $O_k(\rho_1 \sqrt{\xi})$ for $\xi \leq z_0$, using oddness of the gradients, then $O_k(\rho_1 \sqrt{z_0})$ for $\xi \in [z_0, \xi_*]$; continuation derivatives are unchanged on the latter interval. All constants here are uniform as z_0 is decreased. Thus on $[0, z_0]$,

$$\hat{\Gamma}''(\xi) = -2\lambda P(\xi) \mathbf{E} \hat{\chi}^3 + O(\xi \rho_1)$$

by oddness of third derivatives; $\mathbf{E} \hat{\chi}^3$ stays strictly bounded below. Near zero, $P(\xi) = \xi$, so integration gives a drop of order $\lambda \xi^2$ from $\hat{\Gamma}'(0)$. By time z_0 the accumulated drop is of order ρ_1 , while the integrated error is $O(\rho_1 z_0^2)$. On $[z_0, \xi_*]$ the continuation itself is unchanged, and $\hat{\Gamma}' = 1/T + O(\rho_1 \sqrt{z_0})$. Choosing z_0 small first makes this error smaller than the accumulated drop. This proves both the near-zero order and persistence of the gap. Beyond ξ_* use the annealed inequality just proved (valid also for A_λ). This argument differentiated once in $\log \lambda$ gives $O(\lambda \xi^2)$ for the derivative at fixed small ξ of the normalized gap (the linearized PDE and diffusion estimates are identical). Also $\ell^2 b_o = T + O(\lambda)$.

Smoothing. On any fixed compact time interval, continuing the direct PDE annealed to M with terminal mass j instead of 1 changes values, modulo spatial constants, by $O(e^{-cM})$. Indeed at a fixed time above the support the expectation of $\cosh^{1/j}(j(z+B))$, over the remaining variance, lies between the sum of the two individual exponential contributions

and the same sum times $1 - O(e^{-cM})$ (with the common factor $2^{-1/j}$), by subadditivity and restriction to disjoint half-lines for B . Earlier nonlinear heat transitions preserve uniform additive comparison. Changing the normalized mass to the smoothed one now costs $O(\omega M^2)$ in values, by PDE comparison and bounded gradients. Spatial derivatives on the compact interval thus change by $O_\epsilon(\omega^{1-\epsilon})$, any fixed finite orders, via higher-order difference quotients and uniform spatial derivative bounds; diffusions couple there with the same power accuracy. Consequently the error in the active $\hat{\Gamma}''$ bound above is $o(\lambda)\xi$ even at small ξ (use oddness, the added mass $\omega\xi$, and Brownian coupling at small time with factor $\sqrt{\xi}$).

Beyond the support compare $\hat{\Gamma}'$ to the direct annealed case on a sufficiently long fixed interval. Thereafter use $\hat{\chi}$ between $c(1 - \hat{V}_z^2)$ and $C(1 - \hat{V}_z^2)$, by the scalar case of (A1) since all remaining masses are bounded and bounded below. The martingale diffusion coefficient of $\hat{V}_z$ is $\hat{\chi}$; Itô's formula for $\sqrt{1 - \hat{V}_z^2}$ gives exponential decay of its expectation (localize inside $(-1, 1)$ and use nonnegativity), so $\mathbf{E}\hat{\chi}^2$ eventually stays small. This proves (P2) also with smoothing.

More explicitly, for the scalar conditional mean m_t and $f(m) = \sqrt{1 - m^2}$, localization gives

$$\frac{d}{dt}\mathbf{E}f(m_t) = -\frac{1}{2}\mathbf{E}\frac{\chi_t^2}{(1 - m_t^2)^{3/2}} \leq -c\mathbf{E}f(m_t)$$

once the remaining masses are bounded below. The initial value is at most one, so this bound is uniform in the starting field. It also applies conditionally at the beginning of the long interval and supplies the exponential terminal-datum bound used next. $\square$

The lift provides the gap needed for admissibility. To differentiate pressures later, we also need regularity in the path parameters. There are two different requirements: near zero, derivatives must retain the quadratic vanishing of the scalar gap; over all masses, their absolute integrals must remain controlled despite the small smoothing density ω .

3.3 Regularity in strength and size

Lemma 3.3 (Parameter derivatives of the scalar references). *Let the direct and smoothed scalar references have the parameters in (P1), with $1 \leq \ell \leq 2^{1/6}$; the smoothed reference is at final strength. The following bounds hold uniformly for sufficiently large N . Every derivative order and compact time interval is fixed before $N \rightarrow \infty$.*

For the direct reference, the normalized inverse near zero has uniformly bounded one-sided derivatives of every fixed order. Their $\log \lambda$ derivatives are $O(\lambda)$, and

$$\partial_{\log \lambda} q(v) = O(\lambda \min(v, 1)).$$

At fixed sufficiently small normalized time ξ , the $\log \lambda$ derivative of $1 - \hat{\Gamma}'(\xi)/\hat{\Gamma}'(0)$ is $O(\lambda\xi^2)$.

For the smoothed reference, write $\hat{V}_j$ and $\hat{\Gamma}_j$ to display the terminal mass $j = 4/\ell$. On each fixed compact scalar-time interval $[0, R]$, and for every fixed spatial order $r \geq 1$, there are constants $C_r, c_r > 0$ such that

$$\sup_{0 \leq \xi \leq R, z \in \mathbb{R}} \left| \partial_j \partial_z^r \hat{V}_j(\xi, z) \right| \leq C_r e^{-c_r M}.$$

For a fixed sufficiently small $\xi_0 > 0$, this implies the more precise cancellation

$$\left| \partial_j (\hat{\Gamma}'_j(\xi) - \hat{\Gamma}'_j(0)) \right| \leq C e^{-cM} \xi^2, \quad 0 \leq \xi \leq \xi_0. \quad (\text{P2a})$$

For a fixed sufficiently small mass $v_0 > 0$, the smoothed reference also satisfies, for each fixed $r \geq 0$,

$$\sup_{0 \leq v \leq v_0} (|\partial_v^r t| + |\partial_v^r q| + |\partial_\theta \partial_v^r t| + |\partial_\theta \partial_v^r q|) \leq C_r,$$

with one-sided derivatives at zero. The physical initial spatial derivatives obey the rescaling bound, for each fixed $r \geq 1$,

$$\left| \partial_\theta \left(\ell^{1-r} \partial_z^r V(0, 0) \right) \right| \leq C_r e^{-c_r M}.$$

Over the full mass interval $(0, 4)$, the smoothed clocks t, q are absolutely continuous in θ , with

$$\int_0^4 (|\partial_\theta t(v)| + |\partial_\theta q(v)|) dv \leq CM^C. \quad (\text{P2b})$$

The constants in this integrated bound contain no inverse power of ω .

Proof. For the direct reference, $A'_0(0) > 0$ gives the inverse bounds. The perturbation of its inverse is confined to the initial interval, and differentiating its inverse relation gives the stated $O(\lambda)$ bounds. The linearized PDE and coupled diffusion in the proof of Lemma 3.2 give the bound on $\partial_{\log \lambda} q$. That proof also differentiates the scalar gap only after subtracting its value at zero, which preserves the factor ξ^2 .

For size variations, the normalized smoothed profile below M is independent of ℓ ; only its terminal datum changes through $j = 4/\ell$. The diffusion representation of $\partial_j \widehat{V}_j$ has terminal datum $\partial_j [j^{-1} \log \cosh(jz)]$. It differs from the constant $\log 2/j^2$ by at most $C\sqrt{1 - \tanh^2(jz)}$. The conditional martingale estimate in the preceding proof therefore makes its expectation exponentially small, uniformly in the starting field on a fixed compact time interval. Higher spatial derivatives are uniformly bounded by the step power rules with one bounded smooth terminal insertion and its bounded derivatives; the remaining slots are bounded spin derivatives controlled by PE, Corollary 2.10. If the uniform value error is ε , global finite-difference interpolation between orders zero and any fixed $k > r$ bounds the derivative of order r by $C_{r,k} \varepsilon^{1-r/k}$. This proves the asserted exponential bound uniformly over all starting fields. The calculation differentiates only the terminal datum while the mass profile below M stays fixed.

To obtain (P2a), differentiate the scalar second-derivative identity at fixed normalized time. The spatial derivatives just bounded are exponentially small, $\widehat{V}_z$ and $\widehat{V}_{zzz}$ are odd, and the normalized mass is $O(\xi)$ near zero. Differentiating the Brownian coupling gives a field change $O_p(e^{-cM} \sqrt{\xi})$. Hence $\partial_j \widehat{\Gamma}_j''(\xi) = O(e^{-cM} \xi)$, and integration from zero proves (P2a). Repeated PDE and generator differentiation bounds the response time derivatives near zero, where the normalized mass profile is smooth with bounded derivatives. Applying the same calculation to the equations linearized in j bounds the mixed derivatives $\partial_j \partial_\xi \widehat{\Gamma}_j$ there. Since $A'_0(0) > 0$, composition with the normalized inverse and the exact rescalings $\xi = \ell^2 t$, $\widehat{A}(\xi) = v/\ell$ gives the stated one-sided mass derivatives and their size derivatives. Finally, $\ell^{1-r} \partial_z^r V(0, 0) = \partial_z^r \widehat{V}_j(0, 0)$, and $\partial_\theta j = -j/6$. Thus treating the normalized initial spatial derivatives as constants during a size differentiation incurs only exponentially small errors.

It remains to control these derivatives after integration over all masses. Off the terminal plateau, the inverse relation is $\widehat{A}(\xi) = v/\ell$, with $\xi = \ell^2 t(v)$. Since $\partial_\theta \ell = \ell/6$, differentiation at fixed v gives

$$\partial_\theta t(v) = -\frac{t(v)}{3} - \frac{v}{6\ell^3 \widehat{A}'(\ell^2 t(v))}.$$

The second term can be large pointwise, but under $dv = \ell \widehat{A}'(\xi) d\xi$ its inverse density cancels. Both resulting integrals are $O(M)$. On the terminal plateau, $t = M/\ell^2$ and $\partial_\theta t = -t/3$. Its

moving boundary creates no jump term because the clock is continuous there. In particular $\int |\partial_\theta t(v)| dv \lesssim M^C$, with no inverse power of ω .

For $q(v) = \widehat{\Gamma}_j(\xi)$, there is also dependence on the terminal mass $j = 4/\ell$. We bound it at fixed normalized time ξ by representing $\widehat{\Gamma}_j(\xi)$ as the expectation of the product of two terminal spins whose branches separate at that time. This test is bounded by 1 and does not depend on j . In a step approximation, differentiate the logarithm of its positive tree density, telescoped as in PE, Proposition 2.14. Below M all normalized masses are fixed. The derivative $\partial_j[j^{-1} \log \cosh(jz)]$ is uniformly bounded, and the diffusion representation gives the same bound for interior continuation derivatives. Their coefficients have total variation bounded by the number of observed leaves times the terminal mass, independently of the number of steps. The remaining terminal terms are bounded by $C(1 + |Z_M^1| + |Z_M^2|)$. Hence the logarithmic density derivative has L^1 norm at most $C(1 + M)$, because bounded scalar drift gives $\mathbf{E}|Z_M^i| \lesssim 1 + M$. It follows that

$$|\partial_j \widehat{\Gamma}_j(\xi)| \leq C(1 + M), \quad 0 \leq \xi \leq M,$$

uniformly in the step resolution and in ω . The resulting uniform Lipschitz bound passes to continuous clocks by PE, Proposition 2.14; it supplies absolute continuity and the same derivative bound almost everywhere without requiring convergence of derivatives.

Off the plateau, the chain rule now reads

$$\partial_\theta q(v) = -\frac{v \widehat{\Gamma}'_j(\xi)}{6\ell \widehat{A}'(\xi)} - \frac{j}{6} \partial_j \widehat{\Gamma}_j(\xi).$$

After integration in v , the first term again loses its inverse density. Its integral is bounded by $C \int_0^M \widehat{\Gamma}'_j(\xi) d\xi \leq C$, since $\widehat{\Gamma}_j$ is increasing and lies in $[0, 1]$. The second term has integral $O(1 + M)$. On the plateau only this second term remains. Thus q , like t , has an absolutely integrable size derivative of norm $O(M^C)$. □

3.4 Constructing admissible matrix and field clocks

We now use the scalar gap to allow a varying matrix clock. The positive term from the lift pays for its increase, while a negative constant offset supplies a further margin at the smallest masses. We then add field increments, checking separately the single ramp that can decrease.

Choose a small fixed mass $v_b > 0$, such that inverse densities are regular with bounded derivatives and bounded above and below up to $8v_b$. Take a smooth nondecreasing $G(v)$ equal to zero on $[0, x]$, v^2 on $[8x, v_b]$ and constant positive above $2v_b$, $0 \leq G' \leq Cv$. Use dyadic masses $u = 2^{-h}$ in $[16x, v_b/16]$, a fixed smooth nonnegative ϕ supported on $[1/2, 4]$, positive on $[1, 2]$, and parameters $\gamma = (\gamma_0, \gamma_u)$. These cutoffs are common across path segments on the dyad and independent of ℓ . Take

$$b(v) = b_o + c_b \lambda \left(G(v) - U^2 - c_1 \gamma_0 (G(v) + U^2) - c_1 \sum_u \gamma_u u^2 \phi(v/u) \right), \quad (\text{P3})$$

with sufficiently small fixed $c_1, c_b > 0$. The parameters γ may range in $[-4, 4]$. They are held fixed when taking a positive-law expectation; later we average their coordinates independently over $[1/3, 2/3]$. By bounded overlap, $0 \leq b'(v) \leq Cc_b \lambda v$, $b \asymp 1$, $c_g = -\partial_{\gamma_0} b \asymp \lambda(v^2 + U^2)$ throughout this bounded mass interval. Also $h_* = K - bq$ is nonnegative nondecreasing, with

$$dh_* \geq c\lambda(v^2 + U^2) dt, \quad h_*(v) \lesssim \lambda(v^3 + vU^2) \quad \text{for small } v. \quad (\text{P4})$$

Indeed, along the inverse clock, including constant pieces,

$$dh_* = (1 - b\Gamma'(t)) dt - q db.$$

The lift (P2) supplies the λv^2 gap near zero, and the negative U^2 offset in (P3) supplies λU^2 . The term $q db$ occurs only at small active mass, where $q \asymp v$ and the inverse density is regular; choosing c_b small absorbs it in the positive gap. Outside that interval $db = 0$. At large t , Γ' is small and the gap is bounded below by a constant. This proves the first bound in (P4); integrating its small-time upper counterpart from zero gives the second. We also have for every $z \geq 0$ (with either weight $w(y) = 1$ or $1 - y$)

$$\int_0^1 w(y) [1 - b(v)\Gamma'(t(v) + yb(v)(z - q(v)))] dy \geq c\lambda(v^2 + U^2). \quad (\text{P5})$$

Indeed the first half of the secant lies at time at least $t(v)/2$ since $h_* \geq 0$, hence the base lift there dominates $c\lambda v^2$. The subtracted U^2 offset in (P3) provides the U^2 term unless Γ' itself is small; everywhere the loss relative to these bounds from (P3) is at most $Cc_b\lambda v^2$.

Set $h(v) = 0$ for $v < x$; above that cut use

$$h = h_* + H_{\text{lo}} + \chi_{\text{lin}} H_{\text{lin}} + \chi_e H_e, \quad (\text{P6})$$

$$H_{\text{lo}} = \lambda_1 U^3 \log \frac{\min(v, \ell)}{x}, \quad H_{\text{lin}} = x^3 (\min(v, \ell) - 4v_b)_+.$$

Here H_{lo} supplies additional field increase above the lower cut. The multipliers $\chi_{\text{lin}}, \chi_e$ lie in $[0, 1]$. The linear term is introduced only at final strength and remains during the size comparison. The term H_e is used only on direct paths; it supplies field increase along their top plateau and is removed before switching to smoothed paths. To define it, we join an active-side ramp to a profile on that plateau. Take a sufficiently small fixed $U_0 > 0$, $\tilde{U} = \min(U_0, U)$, a smooth taper $\Upsilon(U)$ equal to 1 on $[0, U_0/2]$, zero on $[U_0, 1]$, with values in $[0, 1]$. Define $E_b > 0$ by

$$p_{\text{top}} - A_0(\xi_* - E_b) = \tilde{U}^4 / E_b^2 = d_-, \quad d_+ = \tilde{U}^4.$$

It exists uniquely near zero by strict increase, with $|dE_b/d\log \lambda| \lesssim E_b$. On the plateau $w = v/\ell \in [p_{\text{top}}, 1]$ set

$$H_e = \lambda_1 \Upsilon(U) \tilde{U}^2 \left((1 - w + d_+)^{-1/2} - (w - p_{\text{top}} + d_-)^{-1/2} \right). \quad (\text{P7})$$

On the active part ramp linearly in $\xi = \ell^2 t(v)$ from zero at $\xi_* - E_b$ to its value at the plateau; below the ramp set $H_e = 0$. Thus on each dyadic strip of distance $\asymp d'$ from an edge of the plateau up to its midpoint (using also one capped strip of width $\asymp d_{\pm}$ adjacent to each edge), the clock increase per unit mass due to H_e , for small U , is comparable to $\lambda_1 U^2 / (d')^{3/2}$. On the active-side ramp its derivative in t may be negative, but its magnitude is $O(\lambda_1)$. This ramp lies at a fixed positive mass, where (P4) gives an increase at least $c\lambda dt$. Since $\lambda_1/\lambda = \lambda^\eta$, the negative part is absorbed for large N . Thus the total field clock remains nonnegative and nondecreasing even where the added defect is negative; all paths are admissible. Both H_e and its $\log \lambda$ derivative are bounded, on the ramp by $C\lambda_1 E_b$, on the plateau by $C\lambda_1 \tilde{U}^2 / \sqrt{d'}$, vanishing away from $U \leq U_0$. In differentiating, ℓ is then fixed and the inverse reference near the ramp does not vary.

3.5 Defect bounds, endpoint tests, and parameter monotonicity

In particular the field defect $d = h - h_*$ satisfies

$$\int d^2/c_g dv \lesssim_{\circ} \lambda^{1+2\eta} U^5 + \chi_e^2 \lambda^{1+2\eta} U^4 + \chi_{\text{lin}}^2 x^6/\lambda. \quad (\text{P8})$$

To verify (P8), separate the part below the cut from the three added field terms. Below x , $d = -h_*$, and (P4) gives

$$\int_0^x \frac{d^2}{c_g} dv \lesssim \lambda U^2 x^3 \lesssim \lambda^{1+2\eta} U^5.$$

The last inequality uses $x/U \lesssim \lambda^{1/12}$ and sufficiently small η . For H_{lo} , integration of $(v^2 + U^2)^{-1}$ contributes $O(U^{-1})$; after squaring its amplitude, the bound is $\lambda^{1+2\eta} U^5$ with logarithmic loss. On the active ramp for H_e , the squared amplitude is at most $C\lambda_1^2 E_b^2$, its mass length is ℓd_- , and $c_g \asymp \lambda$. The identity $E_b^2 d_- = \tilde{U}^4$ therefore gives $O(\lambda^{1+2\eta} U^4)$. On the plateau, the squares of the two reciprocal-square-root terms integrate with only logarithmic loss and give the same bound. Finally, H_{lin} is supported at mass bounded away from zero, where $c_g \gtrsim \lambda$, so its contribution is $O(x^6/\lambda)$. The inequality for the square of a sum combines these contributions with the displayed multipliers.

Strength derivatives. During a direct strength sweep, ℓ, γ are fixed, $(\chi_{\text{lin}}, \chi_e) = (0, 1)$, and a dot means $\partial_{\log \lambda}$. The differentiated defects obey the same localized bounds. On the mask they are $O(\lambda(v^3 + vU^2))$: differentiate the scalar gap only after subtracting its value at zero. Above the mask, differentiating (P6)–(P7) gives the claimed ramp and plateau bounds. In this regime all dotted clocks, $\dot{q}$, and $\dot{b}$ are bounded by $O_{\circ}(\lambda)$.

Size derivatives. On the smoothed size segment, the strength is final, $(\chi_{\text{lin}}, \chi_e) = (1, 0)$, and a dot means ∂_{θ} , with γ fixed. Exact scalar rescaling and Lemma 3.3 give the same small-mass defect bounds for d and $\dot{d}$. In particular, (P2a) retains the factor ξ^2 after differentiating the scalar gap relative to its time-zero value. Above small mass, $d, \dot{d} = O_{\circ}(x^3)$. The other dotted clocks are absolutely integrable with $O_{\circ}(1)$ norms by the global parameter bounds proved above.

Direct comparison endpoints. Rescale a direct path to $[0, 1]$ by $w = v/\ell$. The PE comparison hypotheses at every lower scale $w \gtrsim x$, with $T_m = \ell^2 b_{\circ} \rightarrow T$, take the form

$$\ell^2 b(\ell w) = T_m + o(w^2), \quad \ell^2 h(\ell w) = o(w^3), \quad w \gtrsim x, \quad (\text{P9})$$

because $\lambda U^j \leq x^j \lambda^{1-j s_*/2}$ for $j = 2, 3$. The gaps $1 - j s_*/2 > 0$ make these errors vanish relative to the required powers at the lower comparison scales.

In varying γ it is useful to subtract $\int b q^2/4$ from f_m^0 . Its increasing derivative along any matrix decrease $c = -\partial_{\gamma} b \geq 0$ accompanied by the path prescription satisfies

$$\partial_{\gamma}(f_m^0 - \int b q^2/4) \geq \frac{1}{4} \int c(D_v^2 + (S_v - q)^2) dv, \quad (\text{P10})$$

To check the sign, note that q is fixed under the γ variation. Above the mask the prescribed field satisfies $\partial_{\gamma} h = cq$, so (A1) gives the integrand

$$\frac{c}{4}(B_v - 2qS_v + q^2) = \frac{c}{4}(D_v^2 + (S_v - q)^2).$$

On the mask, $h = 0$ remains fixed and the actual integrand is $c(B_v + q^2)/4$. It exceeds the displayed square by $cqS_v/2 \geq 0$, proving (P10).

3.6 Interchanging the direct and smoothed paths

Proposition 3.4 (Comparison of the two regularizations). *At final strength $\lambda = x^{p*}$, with $(\chi_{\text{lin}}, \chi_e) = (1, 0)$, the direct and smoothed systems satisfy (P11), uniformly with the same parameters γ, ℓ . Their scalar trial gaps*

$$f_{\text{spin}}(K) + \frac{1}{4} \int b^{-1}(K - h)^2 dv - f_m^0,$$

where f_{spin} is the self-corrected one-site pressure at the corresponding terminal mass, also differ by $O(x^{5+c})$. The same fixed $c > 0$ works for these three comparisons, independently of the site count.

Proof. Fix $\lambda = x^{p*}$, $\chi_{\text{lin}} = 1$, and $\chi_e = 0$, and keep γ, ℓ the same in both systems. First extend the direct system to length 4 by a single field jump at mass ℓ . For $v > \ell$, put $K = K_{\text{post}} = M/\ell^2$, $q = 1$, $b = b_{\text{post}} = b(\ell-)$, $h = h_{\text{post}} = K_{\text{post}} - b_{\text{post}} + H_{\text{lo}}(\ell) + H_{\text{lin}}(\ell)$. On this extension $q = 1$ is a deterministic comparison function, rather than a claim about the finite-time scalar response. We first compare these four extended functions to their smoothed counterparts. For fixed $0 < \epsilon < 1/21$, put

$$E_\epsilon = \omega M^2 + \omega^{1-\epsilon} + e^{-c'M}.$$

Then

$$\begin{aligned} & \|K_{\text{hard}} - K_{\text{smoothed}}\|_1 + \|q_{\text{hard}} - q_{\text{smoothed}}\|_1 \\ & + \|b_{\text{hard}} - b_{\text{smoothed}}\|_1 + \|h_{\text{hard}} - h_{\text{smoothed}}\|_1 \lesssim E_\epsilon. \end{aligned}$$

Here and below the norms integrate over $v \in (0, 4)$. For the inverse clocks, the area between monotone quantiles equals the area between their mass profiles; the spread jump and density floor contribute $O(\omega M^2)$. Below mass ℓ , both scalar times remain in a fixed compact interval, so the earlier $O_\epsilon(\omega^{1-\epsilon})$ moment and initial-derivative comparisons control the response at a common scalar time and b_o . The remaining change in q , caused by evaluating the response at different scalar times, is bounded by $\sup |\Gamma'| |K_{\text{hard}} - K_{\text{smoothed}}|$. Its integral is therefore controlled by the inverse-clock estimate.

Above ℓ , the portion before the smoothed terminal plateau has mass length $O(\omega M)$. Outside that strip, the martingale bound gives $q = 1 - O(e^{-c'M})$. Equations (P3) and (P6) then give the asserted L^1 estimates for b, h as well. Since $\omega = x^{5+1/4}$, the choice $\epsilon < 1/21$ gives $(5 + 1/4)(1 - \epsilon) > 5$. With $M = \log^3(1/x)$, this proves $E_\epsilon = O(x^{5+c})$ for a fixed $c > 0$.

We next compute the pressure of this extension, denoted by f_{hard} . Condition on arbitrary earlier Hamiltonian values $(H_\sigma)_{\sigma \in \{-1, 1\}^m}$, and set $D = h_{\text{post}} - h(\ell-)$, $p = \ell/4$. If z has independent standard Gaussian coordinates and Φ is the standard Gaussian distribution function, then

$$\begin{aligned} & \Phi(\ell\sqrt{D})^m e^{m\ell^2 D/2} \sum_{\sigma} e^{\ell H_\sigma} \\ & \leq \mathbf{E}_z \left(\sum_{\sigma} e^{4H_\sigma + 4\sqrt{D} z \cdot \sigma} \right)^p \leq e^{m\ell^2 D/2} \sum_{\sigma} e^{\ell H_\sigma}. \end{aligned}$$

The upper bound is subadditivity for $0 < p \leq 1$. For the lower bound, retain the term indexed by σ on the orthant $\sigma_i z_i \geq 0$ for every i . These orthants are disjoint, and exponential tilting of each Gaussian coordinate gives the factor $\Phi(\ell\sqrt{D})$. Thus the bound is uniform in the conditioned Hamiltonian. After logarithms and division by $m\ell$, the error lies between $\ell^{-1} \log \Phi(\ell\sqrt{D})$ and zero. Here $D = M/\ell^2 - O(1)$, so the error is $O(e^{-c'M})$, independently of m .

Restoring the normalized spin prior multiplies the quantity inside this expectation by 2^{-mp} , whereas the direct partition uses 2^{-m} . Their contribution to the per-site difference is $\log 2(1/\ell - 1/4)$. The Gaussian expectation contributes $\ell D/2$. Combining this with the direct and length-four self corrections, and propagating the resulting uniform additive comparison through the earlier continuations, gives

$$f_{\text{hard}} = f_{\text{direct}} + (\ell - 4)(b_{\text{post}}/4 + h_{\text{post}}/2) + \log 2(1/\ell - 1/4) + O(e^{-c'M}).$$

The calculation also holds for the tilted initial revelation, so its explicit constant cancels in ψ . For the corrected untilted pressure, subtract the added tail $(4 - \ell)b_{\text{post}}/4$ of $\int bq^2/4$. The non-logarithmic correction becomes

$$\frac{\ell - 4}{2}(b_{\text{post}} + h_{\text{post}}) = \frac{\ell - 4}{2}(K_{\text{post}} + H_{\text{lo}}(\ell) + H_{\text{lin}}(\ell)),$$

which is independent of γ , as is the $\log 2$ term. Every earlier power mean preserves a uniform additive comparison, including the tilted initial continuation. Finally, (A1) bounds a pressure change by the L^1 change of its clocks. Applying it to the hard and smoothed systems yields, uniformly,

$$\begin{aligned} \psi_{\text{smoothed}} &= \psi_{\text{direct}} + O(x^{5+c}), \\ (f_m^0 - \int bq^2/4)_{\text{smoothed}} &= (f_m^0 - \int bq^2/4)_{\text{direct}} + \text{const}_{\gamma\text{-indep.}} + O(x^{5+c}). \end{aligned} \quad (\text{P11})$$

For the scalar trial gap we also need stability of its quadratic integral. Although the smoothed K, h can be of order M , their difference is uniformly bounded: above the mask, $K - h = bq - H_{\text{lo}} - H_{\text{lin}}$, while below it $K - h = K = O(x)$. The hard extension has the same bound. Since $b \asymp 1$, the function $(K - h)^2/b$ is uniformly Lipschitz on these paths. Thus its integral changes by $O(E_\epsilon)$ under smoothing, just as the scalar and matrix pressures do.

It remains to compute the hard-extension change. The scalar trial gap $f_{\text{spin}}(K) + \int b^{-1}(K - h)^2/4 - f_m^0$, where f_{spin} is the self-corrected one-site pressure, changes by

$$(4 - \ell)b_{\text{post}}((K_{\text{post}} - h_{\text{post}})/b_{\text{post}} - 1)^2/4 + O(x^{5+c}) = O(x^{5+c})$$

by the same hard-jump computation applied to the scalar pressure. Indeed $K_{\text{post}} - h_{\text{post}} = b_{\text{post}} - H_{\text{lo}}(\ell) - H_{\text{lin}}(\ell)$, so the displayed term is

$$\frac{4 - \ell}{4b_{\text{post}}}(H_{\text{lo}}(\ell) + H_{\text{lin}}(\ell))^2 = O(x^6).$$

At final strength,

$$H_{\text{lo}}(\ell) = O\left(x^3 \lambda^{1+\eta-3s_*/2} \log(1/x)\right) = o(x^3),$$

and $H_{\text{lin}}(\ell) = O(x^3)$. Decreasing the fixed saving c if necessary incorporates this term in $O(x^{5+c})$. The parameters, in particular ℓ , have been kept identical throughout. $\square$

4 Reduction to pressure increments

The comparison estimates and finite pressure expansions will yield one increment bound. We state it first and prove here that it implies both moment convergence and the characterization in Proposition 1.3. The proof of the increment bound is completed in Sections 9 and 10.

Take the coordinates of γ independently and uniformly in $[1/3, 2/3]$, as in (P3), and let $\ell \in [1, 2^{1/6}]$. A direct strength sweep varies $\log \lambda$ in (P1), initially with $(\chi_{\text{lin}}, \chi_e) = (0, 1)$. At final strength, two linear multiplier sweeps pass successively to $(1, 1)$ and $(1, 0)$. A dot denotes differentiation in the parameter of the current segment, always at fixed integer site count m . The size sweep uses the smoothed path at final strength with multipliers $(1, 0)$, and its dot is ∂_θ . For a fixed order r , let $c_0, \dots, c_r$ be the interpolation-differentiation stencil characterized by

$$\sum_{j=0}^r c_j P(j) = P'(0) \quad \text{for every polynomial } P \text{ of degree at most } r.$$

Proposition 4.1 (Stopped pressure increments). *There is a choice of the exponents in (P1) and a constant $c > 0$ with the following property. For every sufficiently small fixed $\varepsilon > 0$, every fixed $q > 0$, and $s = qx^{1+\varepsilon}$, the following bounds hold for every sufficiently large fixed stencil order:*

1. *On each direct segment,*

$$\int \mathbf{E}_\gamma \left[\mathbf{1}_{\{\mathcal{K}_m(s) \leq 3x^\varepsilon\}} |\dot{\psi}_m(s)| \right] |d\vartheta| \lesssim x^{5+c},$$

uniformly for $N/2 \leq m \leq 3N$, where ϑ is that segment's parameter.

2. *On the size segment the corresponding bound holds for*

$$\left| \dot{\psi}_m(s) + \sum_{j=0}^r c_j (m+j) \psi_{m+j}(s) \right|. \quad (\text{I})$$

It also holds when m is a measurable function of θ with values in this integer range, or when the observables are averaged with nonnegative probability weights on these integers depending measurably on θ .

All integrations use absolute path-parameter length. The saving c can be chosen independently of sufficiently small ε . Constants may depend on the fixed test coefficient q , stencil order, and the fixed temperature.

4.1 Exponential augmentation and the stopped change of measure

To transfer moment estimates to the tilted law, we need fixed L^p bounds for its density, which involve Laplace transforms at larger fixed multiples of s . Adding independent exponential noise produces a log-concave law whose centered positive Laplace transform controls its variance and both tails.

At fixed site count, path point, and parameters γ , the transform $\mathcal{K}_m(s)$ is deterministic because its defining disorder expectation has already been taken. Thus the stop $\mathcal{K}_m(s) \leq 3x^\varepsilon$ restricts path data. All density norms below use the untilted initial Gaussian law.

Lemma 4.2 (Augmentation along the physical paths). *Let G_m be the continuation in (A4), and let $\mathcal{E}_{x/2}$ be an independent exponential random variable of rate $x/2$. Along every physical path used above, $G_m + \mathcal{E}_{x/2}$ has a log-concave density. On $\mathcal{K}_m(s) \leq 3x^\varepsilon$, the Radon–Nikodym density*

$$R_m = \frac{e^{sG_m}}{\mathbf{E} e^{sG_m}}$$

has bounded L^p norm for every fixed finite p , uniformly in the path and the allowed site counts for all sufficiently large N .

Proof. First take a finite hierarchy and include all future Gaussian drivers under the normalized positive law. Write the positive revelation masses as $v_1 < \dots < v_L \leq a$, and put $F_0 = G_m$. Let F_i be the continuation after the first i positive-mass revelations, as a function of that Gaussian prefix. In particular F_L is the terminal log partition function divided by a . The transition at v_i contributes $\exp\{v_i(F_i - F_{i-1})\}$, and the terminal Gibbs weight of a fixed spin σ contributes $\exp\{aH_\sigma - aF_L\}$. Multiplying these factors gives the Gaussian driver density times

$$\exp\left\{aH_\sigma - v_1G_m - \sum_{i=1}^{L-1}(v_{i+1} - v_i)F_i - (a - v_L)F_L\right\}.$$

Here H_σ is linear in the drivers and every continuation is convex in the corresponding Gaussian prefix, hence also in the full driver vector when extended independently of its later coordinates. This is the telescoping calculation in the proof of the single-spin pinned moment estimate in [19, Lemma 2.3]. All physical clocks are constant between their initial revelation and mass x . Thus $v_1 \geq x$, or, if empty levels have been inserted, the identical initial-continuation terms combine to a coefficient at least x . Rescaling mass length does not change this coefficient: the product of a mass and its continuation is $(v/a)(aF) = vF$. In particular the smoothed length-four path has the same lower bound.

Before the pin, the future normalized transition kernels and terminal Gibbs probabilities integrate to one. The initial driver therefore has its original Gaussian law. The spin gauge transformations preserve G_m and act transitively on terminal spins, so the law of G_m is also unchanged by the pin. This uses the full joint positive law; no inequality under an arbitrary conditioning is required.

For $z = G_m + \mathcal{E}_{x/2}$, the independent exponential contributes $\exp\{(x/2)(G_m - z)\}$ on the epigraph $G_m \leq z$. At least $x/2$ of the negative convex-continuation coefficient remains. The epigraph is convex, and the joint density in the drivers and z is log-concave. Marginalization therefore gives a log-concave density in z , by the log-concave marginal theorem [24]. Fixed-size path approximation passes the claim to the paths in use: the laws converge weakly, their log-concavity inequalities pass on intervals, and the exponential convolution gives a density and zero mass to interval endpoints.

Let

$$Z = G_m + \mathcal{E}_{x/2} - \mathbf{E}(G_m + \mathcal{E}_{x/2}), \quad d_G^2 = \mathbf{E}Z^2.$$

The one-dimensional log-concave estimate of [19, Lemma 7.7] gives two-sided exponential tails on scale d_G , and constants $c_1 > 0$ such that $\mathbf{P}(Z \geq c_1 d_G) \geq c_1$. On the stopped set,

$$\log \mathbf{E}e^{sZ} = \mathcal{K}_m(s) - \log(1 - 2s/x) - 2s/x \leq 3x^\varepsilon + O(x^{2\varepsilon}).$$

The function $e^t - 1 - t$ is nonnegative, and on the displayed right-hand event it is at least $(sc_1 d_G)^2/2$. Since $\mathbf{E}Z = 0$, it follows that

$$c_1(sc_1 d_G)^2/2 \leq \mathbf{E}(e^{sZ} - 1 - sZ) = e^{\log \mathbf{E}e^{sZ}} - 1 \lesssim x^\varepsilon.$$

Consequently $sd_G \lesssim x^{\varepsilon/2}$. The two-sided exponential tails now bound $\mathbf{E}e^{psZ}$ for every fixed p . Conditional Jensen over the independent centered exponential transfers this bound to $G_m - \mathbf{E}G_m$. Since $\mathcal{K}_m(s) \geq 0$, this proves the asserted bound for R_m . $\square$

The active-component law in (A4) may therefore be estimated by Hölder from its untilted marginal, with an arbitrarily small power loss when the required moment orders are fixed sufficiently large. For a product involving several active components, apply these marginal bounds and Hölder to the finite collection of density factors. Their roots remain independent; no common untilted root is introduced. Outside the stopped set no derivative estimate of this accuracy will be used.

4.2 Mollification in the number of sites

The size path changes both the clocks and the number of sites. At fixed integer m , differentiation of $\mathcal{K}_m(s) = sm\psi_m(s)$ accounts only for the clock change. We average over nearby integers to obtain a differentiable function of a real size center. Differentiating the averaging weights then produces the finite-difference term in (I); a neighboring-site bound will let us remove the averaging at the endpoints.

Let ρ be a fixed nonnegative smooth compactly supported probability density, and set

$$D = x^{-1/2}, \quad n = Ne^\theta, \quad w_m(n) = D^{-1}\rho((m-n)/D).$$

The real center n is frozen on direct segments. Smooth sum–integral comparison gives, for every fixed k ,

$$\sum_m w_m(n) = 1 + O_k(D^{-k}).$$

For example, repeated integration by parts in the Fourier series of the periodization proves the estimate uniformly in n .

Lemma 4.3 (Stencil and neighboring-site errors). *For any sequence Z_m , supported or restricted to the relevant neighborhood of the center,*

$$\frac{d}{dn} \sum_m w_m(n) Z_m = \sum_m w_m(n) \sum_{j=0}^r c_j Z_{m+j} + O_r \left(D^{-r-1} \sup_m |Z_m| \right).$$

At the same clock data, uniformly along the paths under consideration,

$$|\mathcal{K}_{m+1}(s) - \mathcal{K}_m(s)| \leq Cs(1+M). \quad (4.1)$$

Proof. Move the stencil from Z to the weights. The coefficient of Z_m becomes $\sum_j c_j w_{m-j}$. Taylor expansion of the shifted weights, together with the defining polynomial identities of the stencil, leaves the derivative in n . Its remainder has size $O_r(D^{-r-2})$ for each weight and is summed over $O(D)$ sites.

For (4.1), append a zero-input independent spin and interpolate to the completed $(m+1)$ -site model with the adjusted matrix normalization. The change in total cumulative covariance is $O(1+M)$, and the affine interpolation is positive semidefinite. Apply pressure differentiation to both tests in (A4). The spin prior is a probability measure, so the unused spin contributes no additional constant. This gives the displayed bound. $\square$

Proposition 4.4 (Dyadic transform comparison). *Assume Proposition 4.1. There is $c' > 0$, independent of sufficiently small fixed ε , such that for every fixed positive test coefficient q , every integer $N' \in [N, 2N]$, and $s = qx^{1+\varepsilon}$,*

$$\left| \log \mathbb{E} e^{(s/x)Y_{N'}} - \log \mathbb{E} e^{(s/x)Y_N} \right| \lesssim x^{\varepsilon+c'}, \quad Y_m = m^{-1/6}(F_m^c - \mathbb{E}F_m^c).$$

The constants are uniform in the integer N' .

Proof. Use a direct sweep at center N , the switch (P11), the size sweep to $\theta = \log(N'/N)$, the reverse switch, and the reverse direct sweep at center N' . At both ends the regularization strength is x^{12} ; the clocks differ from the bare data $(T/\ell^2, 0)$ by $O_o(x^{12})$. The self corrections cancel in the centered transform.

Apply Fubini to the finite sum of the nonnegative stopped derivative integrals, with the site weights above. Their total expectation in γ is $O(x^{5+c})$, so one choice of γ has this bound up to a fixed constant. The weights can be normalized by their sum, with an error

smaller than any fixed power by increasing the sum–integral order. The weighted form of (I) expressly permits their dependence on θ . The choice is made for this one dyad comparison and one test s ; different tests may use different paths because the bare endpoint laws do not depend on γ .

Write

$$\bar{\mathcal{K}}(\theta) = \sum_m w_m(n) \mathcal{K}_m(s).$$

By (4.1), its active site values have oscillation $O(sD(1+M)) = x^{\varepsilon+1/2-o(1)}$. Since $\mathcal{K}_m \geq 0$ and $\sum_m w_m = 1 + O_k(D^{-k})$, whenever $\bar{\mathcal{K}} \leq x^\varepsilon$, all contributing site counts, and the boundedly many neighboring stencil counts, lie in their individual stops $\mathcal{K}_m \leq 3x^\varepsilon$.

On a size segment use $Z_m = m\psi_m(s)$ in Lemma 4.3. The derivative is

$$\begin{aligned} \frac{d}{d\theta} \bar{\mathcal{K}} &= sn \sum_m w_m \left(\dot{\psi}_m + \sum_{j=0}^r c_j(m+j) \psi_{m+j} \right) \\ &\quad + s \sum_m w_m(m-n) \dot{\psi}_m + O_o(sN^2 D^{-r-1}). \end{aligned}$$

Indeed $Z_m = O_o(N)$, by the pressure bounds in (A1). The dotted-clock L^1 bounds and (A1) make the middle term $O_o(sD)$, after integration as well. The two auxiliary errors have sizes

$$sD = x^{\varepsilon+1/2}, \quad sN^2 D^{-r-1} = x^{\varepsilon-11+(r+1)/2}$$

up to subpower losses. Choose the fixed stencil order large enough to leave a positive saving. Multiplying the increment bounds by $sn = O(x^{-5+\varepsilon})$, and including the switches (P11), gives total drift $O(x^{\varepsilon+c'})$. The bare endpoint approximation costs $O_o(sNx^{12}) = O_o(x^{7+\varepsilon})$. The savings can be decreased independently of ε .

At the bare initial endpoint the fixed-temperature exponent theorem of [19, Theorem 1.1], together with the augmented log-concave estimate, gives $\mathcal{K}_m(s) = o(x^\varepsilon)$ uniformly in the active window. Indeed the variance scale is $x^{-2-o(1)}$ before multiplication by x , and the small-positive-parameter bound gives $O(x^{2\varepsilon-o(1)})$. Continuity on each segment, and the small switch errors, exclude a first violation of $\bar{\mathcal{K}} \leq x^\varepsilon$. Only stopped derivative estimates have entered this argument.

Remove the endpoint weights using (4.1); the cost is again $O(sD(1+M))$. At the final bare endpoint the terminal mass is $\ell = (N'/N)^{1/6}$, while the covariance is T/ℓ^2 . The terminal continuation is therefore distributed as F_N^c/ℓ , with the same physical inverse temperature. Since $x/\ell = N'^{-1/6}$, its x -scaled centered law is exactly $Y_{N'}$. The initial law is Y_N . This proves the claim, including its uniformity throughout the dyad. $\square$

4.3 Moments, positive variance, and full-sequence convergence

Proposition 4.5 (Consequences of the increment bound). *Assume Proposition 4.1. For every fixed integer $k \geq 1$, the moments $\mathbb{E}Y_n^k$ converge at a positive power rate. Their limiting second moment is finite and strictly positive, and the centered variables Y_n have uniformly bounded exponential moments on some constant neighborhood of the origin. They converge weakly to a moment-determinate law. Removing the completion and standardizing by the exact finite-size variance gives the convergence asserted in Theorem 1.1.*

Proof. Fix k . The dyadic comparison gives values of the log Laplace transform at positive arguments. We recover its derivatives at zero by polynomial interpolation, choosing ε after k so that the interpolation loss is smaller than the comparison saving.

At bare endpoints, the exponent theorem, the augmented tail bound, and conditional Jensen give a scale $L_n = x^{-o(1)}$, uniform in the dyad, with $\mathbb{E}e^{|Y_n|/L_n} \leq C$. Hence, for a sufficiently small constant $a > 0$, $M_n(z) = \mathbb{E}e^{zY_n}$ satisfies $|M_n(z) - 1| \leq 1/2$ on $|z| \leq r_n^{\text{an}} := a/L_n$, where its logarithm is analytic and bounded by an absolute constant. Cauchy's estimate on this disk gives

$$\left| \log M_n(z) - \sum_{j=0}^k \frac{(\log M_n)^{(j)}(0)}{j!} z^j \right| \leq C_k (L_n |z|)^{k+1}, \quad |z| \leq r_n^{\text{an}}/2,$$

which is $x^{(k+1)\varepsilon - o(1)}$ at each fixed multiple of x^ε and uses only the preliminary subpower scale.

Interpolate at the $k+1$ nonnegative nodes $0, x^\varepsilon, \dots, kx^\varepsilon$. Proposition 4.4 bounds each nonzero-node difference; at zero the difference is zero. Division by the interpolation scale loses at most $x^{-k\varepsilon}$. Thus every cumulant through order k has dyadic difference bounded by

$$O(x^{c' - (k-1)\varepsilon} + x^{\varepsilon - o(1)}).$$

Take ε sufficiently small after fixing k , for example with $k\varepsilon < c'/2$. Both terms save a fixed positive power. Summing a geometric series over adjacent dyads proves convergence at a power rate on powers of two; the comparison with every integer in the next dyad proves the same conclusion for all integers. Moments are polynomials in finitely many cumulants, so they also have power rates. Enlarge the constants over the finitely many remaining sizes to obtain the bounds for every $n \geq 2$.

Let $v_n = \text{Var}(Y_n)$, and let its finite limit be v . If $v = 0$, its power rate would give $v_n = O(n^{-a})$ for some $a > 0$. The lower bound in the PE exponent theorem gives $v_n \geq n^{-2\eta}$ eventually for every fixed $\eta > 0$, after a harmless adjustment of the exponent notation. Choosing $2\eta < a$ is a contradiction. Hence $v > 0$.

The bounded variances improve the tail scale to a uniform constant. To see the normalization explicitly, put $r_n = n^{-1/6}$ at a bare system of size n , and let E_n be independent exponential noise of rate $r_n/2$. The bare version of the pinning calculation above gives a log-concave density for $F_n^c + E_n$. At a bare path the negative continuation coefficient in that calculation is the terminal mass one, which exceeds $r_n/2$. Thus

$$\tilde{Y}_n = Y_n + r_n(E_n - \mathbb{E}E_n) \quad \text{has variance} \quad v_n + 4.$$

The log-concave tail estimate applies with a common constant scale. Conditional Jensen, using the convex function $z \mapsto e^{t|z|}$, removes the centered independent noise. Consequently, for some $t_0 > 0$,

$$\sup_{n \geq 2} \mathbb{E}e^{t_0|Y_n|} < \infty.$$

Finitely many small sizes can be included by decreasing t_0 . These bounds give tightness and uniform integrability of every fixed power. They also imply moment determinacy, for example through the resulting factorial moment bounds. Every weak subsequential limit therefore has the same moments and is the same probability law. Thus weak convergence holds along the full sequence.

The probability spin prior changes the original free energy only by the deterministic $n \log 2$. The completion is an independent centered Gaussian of variance $T/2$, whose scaled L^2 norm is $O(n^{-1/6})$ and whose scaled variance is $T/(2n^{1/3})$. It follows that the uncompleted scaled variance has the same positive limit v . Removing that Gaussian and dividing by the exact finite-size standard deviation is now justified by Slutsky's theorem. $\square$

4.4 Identification by finite random-size experiments

Proposition 4.6 (Validity of the law specification). *Under Proposition 4.1, the constants in Proposition 1.3 are well defined by an absolutely convergent series after the indicated expectations. They are precisely the limiting centered moments of Y_n , with $a_2 > 0$. The standardized moments $a_k/a_2^{k/2}$ determine one Borel probability measure uniquely. Each a_k is also the expectation of a single integrable random variable constructed from fixed priors and finitely many Gaussian systems almost surely.*

Proof. Fix k , let $m_k(l) = \mathbb{E}Y_l^k$, and denote its power-rate limit by m_k . Recall $C = \sum_{l \geq 2} l^{-2}$, $\mathbb{P}(J = l) = l^{-2}/C$, and

$$P_k = \prod_{i=1}^k J^{-1/6} (F^{c,0} - F^{c,i}), \quad t = \log J,$$

where the completed systems are conditionally independent given J . Conditional on $J = l$ and copy zero, the other k copies average the product to the k -th centered power of copy zero. Thus

$$\mathbb{E}[P_k \mid J = l] = m_k(l).$$

The uniform moment bounds proved above also control every fixed absolute moment of P_k conditional on $J = l$, uniformly in l .

For $\operatorname{Re} z > 0$, put

$$B_k(z) = Cz \mathbb{E}[P_k e^{(1-z)t}] = z \sum_{l \geq 2} m_k(l) l^{-1-z}.$$

If $|m_k(l) - m_k| \leq C_k l^{-a}$ for some $a > 0$, the difference series is holomorphic on $\operatorname{Re} z > -a$, locally absolutely convergent there. The constant part also extends across zero. Indeed, for $\operatorname{Re} z > 0$, the difference between the sum and its integral is

$$\sum_{l \geq 2} \left(l^{-1-z} - \int_l^{l+1} r^{-1-z} dr \right) - \int_1^2 r^{-1-z} dr.$$

On a compact subset of $\operatorname{Re} z > -1$, each summand is bounded by a constant times $l^{-2-\operatorname{Re} z}$. The discrepancy therefore converges locally uniformly and is holomorphic there. Since $z \int_1^\infty r^{-1-z} dr = 1$, multiplication by z removes the integral's pole and gives value one at zero. Combining this with the difference series, B_k extends to $\operatorname{Re} z > -c_k$ for some $c_k > 0$, with $B_k(0) = m_k$.

Its Taylor series about $z = 1$ has radius strictly greater than one. To evaluate it at zero write $z = 1 - w$. Near $w = 0$, differentiation under the expectation is legitimate: for every fixed h and $0 < \delta < 1$, the conditional absolute-moment bound on P_k and $\sum_{l \geq 2} l^{-2+\delta} (\log l)^h < \infty$ give a dominating integrable function on $|w| \leq \delta$. The coefficient of w^h in $(1 - w)e^{wt}$ is

$$\frac{t^h}{h!} - \mathbf{1}_{\{h \geq 1\}} \frac{t^{h-1}}{(h-1)!},$$

with the second term omitted at $h = 0$. Hence

$$B_k(0) = C \sum_{h=0}^{\infty} \mathbb{E} \left[P_k \left(\frac{t^h}{h!} - \mathbf{1}_{\{h \geq 1\}} \frac{t^{h-1}}{(h-1)!} \right) \right] = a_k,$$

and the scalar series is absolutely convergent. The expectation is taken before the infinite sum. Indeed, at each fixed J the bracket series equals $e^t - e^t = 0$; averaging that pointwise

sum would therefore fail to recover the positive second moment. The limiting law from Proposition 4.5 proves existence of the measure with these moments, its positive variance proves $a_2 > 0$, and its exponential moment bound proves uniqueness.

The final construction uses randomized single-term unbiased estimation with replication conditional on the sampled level. For this method, see [25, Section 2, equation (10), and Section 4]. Here the levels index Taylor coefficients; the integrability and almost-sure finiteness proved below do not assert finite expected work or import the SDE complexity guarantees of that reference.

For the final assertion, draw an independent integer $H \geq 0$ with $\mathbb{P}(H = h) = 2^{-h-1}$. Conditional on $H = h$, repeat the whole random-size experiment independently 256^h times, and let $\bar{A}_{k,h}$ be the sample average of

$$A_{k,h} = CP_k \left(\frac{t^h}{h!} - \mathbf{1}_{\{h \geq 1\}} \frac{t^{h-1}}{(h-1)!} \right).$$

Return $W_k = 2^{H+1} \bar{A}_{k,H}$. The estimate $t^h/h! \leq 4^h e^{t/4}$, the prior proportional to l^{-2} , and the uniform conditional moments of P_k give $\|A_{k,h}\|_2 \leq C_k 4^h$, including the shifted term. Independence of the repetitions yields

$$\mathbb{E} \left| \bar{A}_{k,h} - \mathbb{E} A_{k,h} \right| \leq \frac{C_k 4^h}{\sqrt{256^h}} = C_k 4^{-h}.$$

The means $\mathbb{E} A_{k,h}$ are absolutely summable by the Taylor argument. Consequently

$$\mathbb{E} |W_k| = \sum_{h \geq 0} \mathbb{E} |\bar{A}_{k,h}| < \infty, \quad \mathbb{E} W_k = \sum_{h \geq 0} \mathbb{E} A_{k,h} = a_k.$$

All priors are fixed and every experiment uses finitely many finite-size Gaussian systems almost surely. Neither a limiting random input nor a choice of subsequence enters the specification. $\square$

5 Inverting the scalar pair kernel

The cavity equations couple errors carried by different pairs of leaves. To solve them, we need an inverse whose bound survives refinement of the hierarchy and the small masses used in the comparison. The proposition below provides that bound for arbitrary fixed old trees; its proof keeps the signed cancellations before taking absolute values.

A pair function assigns a deterministic number to each pair and each allowed extension of the fixed old-label tree. In the applications, that number is a positive-law expectation of an overlap test. The operator sums below vary the extended topology using signed allocation coefficients; the scalar expectations in its kernel use the positive law on each such topology.

For a finite resolution with terminal mass one, choose mass cuts $0 < p_0 < \dots < p_{R+1} = 1$. A *sheet* r consists of pairs in the same block at cut p_r and different child blocks at cut p_{r+1} ; write K_r for their common scalar time. The resolution records these relations to all fixed old labels, so a pair function need not be constant on a sheet. The ordinary absolute attachment rule means the signed rule of Section 2 with each coefficient replaced by its absolute value. In particular, it retains both the density for a new fork and the atom at an existing fork, as well as terminal coincidences.

Proposition 5.1 (Pair inverse with absolute attachment domination). *The kernel (A8) is considered on extensions of any fixed finite list of old labels. A pair function u_{ab} may*

depend on the topology with those labels but ignores any unused extra labels (is pulled back by restriction). Indices in this section used as pair arguments are distinct leaves. Normalize terminal mass to 1; other bounded terminal masses reduce to this by mass rescaling, scaling cumulative covariances by the square of the mass. Take a monotone step scalar path K , nonnegative bounded-variation step multiplier b , and retain pair levels in a terminal interval of the hierarchy (the whole interval is allowed). Suppose at these levels

$$1 - b_r \mathbf{E} \chi_r^2 \geq e > 0, \quad \chi_r = V_{zz}(K_r, Z_{K_r}).$$

Bounds called polynomial in this section are in $1 + K_{\text{top}} + \|b\|_\infty + \text{var } b + e^{-1}$, independent of the number of steps. On the retained levels

$$u_{ab} \mapsto u_{ab} - \sum_{cd \text{ retained}} \mathbf{M}_{ab,cd} b_{cd} u_{cd} \quad (\text{H1})$$

has an inverse on each finite step resolution. Each output is an integral of inputs over allocations of up to two more labels (some arguments can also be reused), with weights dominated by a polynomial times the ordinary absolute unrestricted attachment rules to the old tree including ab . Restrictions and bounded weights can depend on the topology. This formulation in particular controls new split densities, allowing coincidences and ties to existing forks. It applies with ab using old labels too. We only need symmetric arrays (on antisymmetric ones the subtracted sum is zero).

The proof separates the scalar calculation from the construction of the inverse. We first resolve pair arrays into two sectors and state the scalar response identities they require. With those identities in hand, we construct the inverse and prove its attachment bound. The detailed scalar verification follows afterward.

5.1 Pair resolutions and scalar response

Initially keep the incoming root power $p_0 > 0$ and the clocks strictly ordered. We will remove both restrictions after obtaining bounds independent of the resolution.

The scalar time profile is $\alpha = p_i$ on (K_{i-1}, K_i) , where $K_{-1} = 0$. Add a mass-one terminal interval ending at K_{R+1} ; it leaves all earlier field derivatives unchanged. For a function of a leaf, P_k replaces that leaf by a signed allocation within its p_k -block and divides the result by p_k . The signed row-sum rule therefore gives $P_k 1 = 1$, and $P_{R+1} = I$. Put $E_0 = P_0$ and $E_k = P_k - P_{k-1}$ for $k \geq 1$. These differences separate successive block resolutions. When a pair lies on sheet t , resolve each endpoint only within its own child: the coarsest operator is then $\tilde{E}_{t+1}^t = P_{t+1}$, followed by $\tilde{E}_i^t = E_i$ for $i > t + 1$.

The additive and row-zero sectors For the counting calculation alone, treat the symbols p_i as integer block sizes with sufficiently large integer ratios p_i/p_{i+1} . Thus this temporary integer tree has decreasing block sizes; the increasing real masses are substituted only after the counting identities have been proved. On the integer tree use symmetric off-diagonal pair arrays, with inner product summed over unordered pairs.

Nested block averages on this tree satisfy

$$P_i P_j = P_{\min(i,j)}, \quad E_k E_l = \mathbf{1}_{\{k=l\}} E_k, \quad \sum_{k=0}^{R+1} E_k = I.$$

The coefficient argument below transfers these identities to real masses.

Fix a leaf function ϵ satisfying $E_k \epsilon = \epsilon$. A *sheet-additive* array has value $f_t(\epsilon_a + \epsilon_b)$ when ab lies on sheet t . The coefficients f_t may vary with the sheet. Summing over the second leaf in a row on sheet t gives $d_t^k f_t \epsilon_a$, where

$$d_t^k = p_t - p_{t+1} + p_t \mathbf{1}_{k \leq t} - p_{t+1} \mathbf{1}_{k \leq t+1}, \quad \mu_t^k = -d_t^k.$$

On this integer tree, its squared norm divided by $\sum_a \epsilon_a^2$ is $\sum_t d_t^k f_t^2$. The kernel $\mathbf{M}$ preserves the arrays of this form and acts on the level coefficients f_t by a matrix independent of ϵ . To see this, fix the output pair ab . Tree symmetry makes the sum of a one-leaf input constant on each block away from the paths to a and b . The resulting leaf operators are block projections along those two paths. On the range of E_k , each such projection is either zero or the identity, so the answer is a multiple of $\epsilon_a + \epsilon_b$.

For a general pair array, write $(R_t u)_a = \sum_{b: ab \text{ on sheet } t} u_{ab}$. The additive projection is explicitly

$$(\Pi_{\text{add}} u)_{ab}|_t = \sum_k \frac{(E_k R_t u)_a + (E_k R_t u)_b}{d_t^k}.$$

Subtracting this projection leaves an array whose row sum on each sheet is zero. We call this the *row-zero* part. For a row based at a leaf in a p_t -block, the sum is over all other children of that block together; it need not vanish separately in each child.

Real masses and arbitrary old-label topologies The transfer to real masses concerns coefficients of moment expressions, not Gaussian expectations as functions of an integer replica count. Introduce a formal variable for each pair or four-leaf moment, recording all leaf coincidences and overlap levels. Include products of pair moments as separate expressions. Also leave every input value as a formal variable indexed by its complete genealogy with the old labels. After expanding a block projection or kernel action, the coefficient of each such expression is a rational function of the p_i .

The counts in these coefficients are falling powers: when adding a leaf, one either uses an occupied child or chooses a new child. These are exactly the coefficients in the real-power rule of PE, Section 2. To verify a coefficient identity, clear denominators and regard the branching ratios as independent variables. Integer counting shows that the resulting polynomial vanishes for every sufficiently large integer choice of these variables, also if even ratios are required. Applying the one-variable polynomial identity successively in each variable shows that the polynomial is identically zero. We may therefore substitute the increasing real masses.

In particular, the additive projection is obtained by taking a row sum on sheet t , applying E_k , dividing by d_t^k , and adding the result at the two endpoints. The denominators are nonzero for strictly increasing real masses. Subtracting these projections gives the stated row-zero remainder. Because the input variables retain their entire old-label genealogy, these identities apply to arbitrary pair functions in the proposition, not just sheet-constant functions. The signed real-mass calculation uses these coefficient identities; it makes no assertion about a vector space of negative dimension.

The decomposition reduces inversion to two scalar calculations. On the row-zero sector, the kernel is diagonal in the two child resolutions. On the additive sector, it is a diagonal operator minus a positive Hessian kernel integrated against sheet masses. The next lemma gives both actions and the estimates that will cancel block denominators.

Lemma 5.2 (Scalar response on the two sectors). *Use the finite resolution above, and put $m_t = V_z(K_t, Z_{K_t})$, $\chi_t = V_{zz}(K_t, Z_{K_t})$. Let $\mathbf{E}_t$ condition on the scalar prefix through K_t .*

On the row-zero sector, different sheets do not mix. On sheet r , the multiplier in the child resolutions $\bar{E}_i^r \otimes \bar{E}_l^r$, $i, l \geq r+1$, is

$$g_r^{il} = \mathbf{E}[(\mathbf{E}_r \chi_{i-1})(\mathbf{E}_r \chi_{l-1})].$$

On the additive sector associated with E_k , the action on sheet coefficients is

$$(\mathbf{M}^k f)_r = g_r^k f_r - \sum_t B_{rt}^k \mu_t^k f_t, \quad g_t^k = \mathbf{E}[\chi_t \chi_{\max(k-1, t)}]. \quad (\text{H2})$$

Here $A = K_{k-1}$, with $A = 0$ for $k = 0$, and

$$B_{rt}^k = \mathbf{E}[\text{Hess}_{z, \alpha} V(A, Z_A)[\beta_r, \beta_t]], \quad \beta_t = \begin{cases} (m_t, 0), & t < k, \\ (0, \delta_{K_t}), & t \geq k. \end{cases}$$

The Hessian varies the starting field and the subsequent mass profile. The prefix values m_t , $t < k$, are held fixed during this differentiation, and δ_{K_t} denotes an impulse in the subsequent profile. Expectation is taken after evaluating the Hessian. The matrix B^k is positive semidefinite. Restricting either action to retained levels discards the other input and output sheets.

The kernels B^k, g^k are bounded and Lipschitz in their evaluation times with polynomial constants. Their changes with the resolution index satisfy

$$\|B^{k+1} - B^k\|_\infty + \|g^{k+1} - g^k\|_\infty \lesssim_{\text{pol}} p_k(K_k - K_{k-1}).$$

A successive difference of g_r^{il} in index i is bounded by $C p_i(K_i - K_{i-1})$; a mixed difference is bounded by the product of the corresponding two bounds. Moreover $0 \leq g_r^k, g_r^{il} \leq \mathbf{E} \chi_r^2$. These formulas apply with arbitrary old-label topologies under the coefficient convention just described. For $k = 0$, their Hessian kernel also satisfies

$$|B_{rt}^0| \lesssim C(1 + K_{R+1})^C K_{\min(r, t)}$$

when the earlier evaluation time is small.

We prove the lemma after constructing the inverse. Its role is now explicit: positivity controls the additive resolvent, while its weighted differences cancel the normalizations of the block projections.

5.2 Construction and attachment bound for the inverse

Proof of Proposition 5.1. If there are no retained sheets, the assertion is immediate. Otherwise use the decomposition above. By Lemma 5.2, the gap hypothesis implies

$$\ell_t^k = 1 - b_t g_t^k \geq e, \quad \ell_t^{il} = 1 - b_t g_t^{il} \geq e$$

on each retained sheet. We invert the two sectors, combine their block projections, and only then estimate their attachment coefficients.

The additive resolvent Fix k and retain only the permitted rows and columns. Abbreviate $B = B^k$ and $w = \text{diag}(\mu_t^k b_t / \ell_t^k)$. The diagonal matrix w is nonnegative. Consequently $C = (I + Bw)^{-1}B$ is symmetric and satisfies $0 \preceq C \preceq B$, as is seen by writing it as $B^{1/2}(I + B^{1/2}wB^{1/2})^{-1}B^{1/2}$. The additive-sector inverse is $(\ell^k)^{-1}I + A^k \text{diag}(\mu^k)$, where $A_{rt}^k = -C_{rt}b_t/(\ell_r^k \ell_t^k)$. The bound $C \preceq B$ controls the diagonal entries of C , and positivity gives $|C_{rt}| \leq \sqrt{C_{rr}C_{tt}}$. Thus all entries are polynomially bounded. The identity

$C = B - BwC$, together with $\sum_t w_{tt} = O_{\text{pol}}(1)$, transfers the evaluation-time Lipschitz bounds for B to C and then to A . For A , the relevant distance is $|\Delta K| + |\Delta b|$. Successive resolution indices satisfy the stronger estimate

$$\|A^{k+1} - A^k\|_\infty \lesssim_{\text{pol}} p_k(K_k - K_{k-1} + |b_k - b_{k-1}| + \mathbf{1}_{k=\min(\text{retained})})$$

where $b_{-1} = b_0$. To prove this, note that changing k to $k+1$ changes the sheet measure μ by $p_k(\delta_k - \delta_{k-1})$, with unretained atoms omitted. Use primes for the matrices at $k+1$. Their resolvent identity is

$$C' - C = (I - C'w')(B' - B)(I - wC) - C'(w' - w)C.$$

If both atoms are retained, their contributions to $C'(w' - w)C$ are subtracted before taking absolute values. The evaluation Lipschitz estimates then supply $p_k(K_k - K_{k-1} + |b_k - b_{k-1}|)$. Changes in the kernel and in the diagonal gaps use the preceding bound with the same factor p_k .

At the first retained sheet, the lower atom is absent. Its partner has mass exactly p_k , which accounts for the indicator in the bound. There is only one such boundary term. It still has the factor needed to cancel the block-average denominator.

Absolute attachment domination Fix an output pair ab on a retained sheet r . First apply the diagonal multiplier $1/\ell_r^{il}$ in the two child resolutions. Then add the row correction

$$-\sum_{k,t} A_{rt}^k \{(E_k \text{ row}_t u)_a + (E_k \text{ row}_t u)_b\}, \quad (\text{row}_t u)_c = \sum_{d: cd \text{ on sheet } t} u_{cd}.$$

This is the inverse on both parts of the decomposition. On a row-zero array, the correction vanishes. In each one-leaf summand of an additive array, the other endpoint is constant within its child and therefore has child index $r+1$. The child multiplier on the summand is thus $1/\ell_r^k$. The row sum contributes $d_t^k = -\mu_t^k$, and the minus sign in the correction gives the remaining $A^k \text{diag}(\mu^k)$ part of the inverse.

We now combine the block projections before estimating them. Since $E_0 = P_0$, $E_k = P_k - P_{k-1}$, and $P_{R+1} = I$, the exact one-variable Abel identity is

$$\sum_{k=0}^{R+1} A_{rt}^k E_k = A_{rt}^{R+1} I + \sum_{k=0}^R (A_{rt}^k - A_{rt}^{k+1}) P_k.$$

Thus the only terminal term is an identity operator. In every other term, the coefficient difference has a factor p_k , canceling the denominator of P_k .

Here is the corresponding pointwise comparison with attachment densities. Write $c \sim_k a$ if c belongs to the p_k -block of a , and let $\mathcal{S}(dc)$ denote the ordinary signed attachment of c to the old tree, including ab . By definition,

$$(P_k F)_a = \frac{1}{p_k} \int \mathbf{1}_{\{c \sim_k a\}} F_c \mathcal{S}(dc).$$

A subsequent row sum attaches d to the tree that now includes c ; denote that signed rule by $\mathcal{S}_c(dd)$. Relative to these two successive attachment rules, the nonterminal correction at the endpoint a has coefficient

$$\kappa_a(c, d) = -\sum_{k=0}^R \sum_{t \text{ retained}} \frac{A_{rt}^k - A_{rt}^{k+1}}{p_k} \mathbf{1}_{\{c \sim_k a\}} \mathbf{1}_{\{cd \text{ on sheet } t\}}.$$

For a fixed allocation, the pair cd belongs to at most one sheet. Put $\rho_k = K_k - K_{k-1} + |b_k - b_{k-1}| + \mathbf{1}_{k=\min(\text{retained})}$. The weighted difference estimate therefore gives

$$|\kappa_a(c, d)| \lesssim_{\text{pol}} \sum_{k=0}^R \rho_k \mathbf{1}_{\{c \sim_k a\}} \leq K_R + \text{var } b + 1.$$

This bounds the coefficient at each placement relative to the unrestricted absolute rule $|\mathcal{S}|(dc)|\mathcal{S}_c|(dd)$. The endpoint b has the same bound. The terminal identity term requires just the row attachment of d , with a polynomially bounded coefficient.

For the child multiplier, abbreviate $D_{il} = 1/\ell_r^{il}$, $i, l = r+1, \dots, R+1$. Set $\Delta_i D_{il} = D_{il} - D_{i+1, l}$ and $\Delta_l D_{il} = D_{il} - D_{i, l+1}$. Applying the same Abel identity in both positions gives all endpoint terms explicitly:

$$\begin{aligned} \sum_{i, l=r+1}^{R+1} D_{il} \tilde{E}_i^r \otimes \tilde{E}_l^r &= D_{R+1, R+1} I \otimes I \\ &+ \sum_{i=r+1}^R \Delta_i D_{i, R+1} P_i \otimes I + \sum_{l=r+1}^R \Delta_l D_{R+1, l} I \otimes P_l \\ &+ \sum_{i, l=r+1}^R \Delta_i \Delta_l D_{il} P_i \otimes P_l. \end{aligned}$$

There is no separate lowest-child average: it is included in the difference sums. With $\Delta K_i = K_i - K_{i-1}$, the estimates for g_r^{il} and the gap $\ell_r^{il} \geq e$ give

$$\begin{aligned} |\Delta_i D_{i, R+1}| &\lesssim_{\text{pol}} p_i \Delta K_i, & |\Delta_l D_{R+1, l}| &\lesssim_{\text{pol}} p_l \Delta K_l, \\ |\Delta_i \Delta_l D_{il}| &\lesssim_{\text{pol}} p_i p_l \Delta K_i \Delta K_l. \end{aligned}$$

The single differences cancel one block denominator. The coefficient of the two-attachment term is bounded pointwise by

$$\sum_{i, l=r+1}^R \frac{|\Delta_i \Delta_l D_{il}|}{p_i p_l} \mathbf{1}_{\{c \sim_i a\}} \mathbf{1}_{\{d \sim_l b\}} \lesssim_{\text{pol}} \left(\sum_{i=r+1}^R \Delta K_i \right)^2.$$

The identity term uses no new label; the two edge sums use one; and the last sum uses at most two. All are thus bounded by the claimed unrestricted absolute attachment rules.

The allocation order identities in Section 2 allow the block and old-topology restrictions to be retained throughout these equalities. We discard them only in the pointwise inequalities above. The summable coefficients depend on total clock length, total variation of b , and the single retained boundary, not on the number of steps. Existing-fork atoms and coincidences keep their ordinary attachment weights. This also explains why the argument controls new split densities, rather than only the total mass of the inverse.

Limits of finite resolutions At a fixed finite resolution, let strictly separated clocks approach flat clocks and then let $p_0 \downarrow 0$. The combined coefficients above stay bounded after their block denominators have been canceled. Taking subsequences of these finitely many coefficients preserves the inverse identity and its absolute bounds. A strict gap persists after an arbitrarily small weakening.

The same resolution operators can be applied to continuum placements: retain the indicators of membership in the blocks at each resolution cut, while keeping the exact old-label positions. To check this, insert empty steps between the cuts. Integer counting

of additional slots depends only on their coarse genealogy with the old labels, so refining a group of children does not change that count. The real-mass coefficient identity then gives the same conclusion. Consequently the pointwise comparison with the unrestricted attachment measure remains valid for continuum placements, including atoms at old forks.

For a nonstep path, apply the inverse at finite resolutions and fixed system size. PE, Proposition 2.14 gives convergence of positive expectations at each fixed placement, preserving prescribed cuts and excluding only a null set of new placements. The inverse kernels need not themselves converge: their common absolute majorant bounds the error made by replacing a bounded step-path observable with its limit, and dominated convergence makes that error tend to zero. This is the form used in the applications.

Thresholds, prescribed levels, and specified sides of jumps can be included in every approximation. Sample simultaneous path values and, if needed, weaken the strict gap slightly; at fixed bounded clocks, the scalar PDE derivatives and moments are uniformly continuous under L^1 approximation of the mass profile in common time. Finally, for sheet-only inputs the $k = 0$ formula (H2) passes to the limit with $\mu^0 = 2 dv$. $\square$

5.3 Verification of the scalar response formulas

Proof of Lemma 5.2. We first calculate the row-zero action by conditional independence. The additive action then comes from a colored covariance variation. Finally, explicit scalar Hessian kernels give positivity and the weighted estimates used in the inverse.

The row-zero action Fix an input p_t -block on the integer tree and condition on its shared scalar prefix. Within a child define $C_{ac} = \mathbf{E}_t[\tau_a \tau_c] - m_t^2$, and set $C_{ac} = 0$ when the two leaves are in different children. If the output leaves also lie in different children of the input block, conditional independence gives the following four-spin moment for an input pair cd on sheet t :

$$m_t^4 + m_t^2(C_{ac} + C_{ad} + C_{bc} + C_{bd}) + C_{ac}C_{bd} + C_{ad}C_{bc}.$$

The constant and one-end terms vanish when summed against a row-zero input. The disconnected subtraction in (A8) vanishes for the same reason: the input pair moment is constant on this block and sheet. The two remaining products give $C \otimes C$; the two orderings of the input pair cancel the factor $1/2$ in (A8). If the output does not occupy two different children of this same block, each kernel term instead depends on at most one input end and its sum is zero. This proves that different sheets do not mix.

The conditional matrix C is diagonal in the child resolutions. Its values at successive overlap levels are $L_j = \mathbf{E}_t m_j^2 - m_t^2$, with $L_t = 0$ and $L_{R+1} = 1 - m_t^2$. The eigenvalue at index i is $\sum_{j=i}^{R+1} p_j(L_j - L_{j-1}) = \mathbf{E}_t \chi_{i-1}$, by the conditional source-Hessian formula. Multiplying the two conditional eigenvalues and averaging the prefix gives g_t^{il} . Finally, C has the same constant row sum in each child. Its tensor action and the child projections therefore preserve the row-zero condition used above.

This calculation transfers to real masses by the coefficient rule above: the entries of C are differences of pair moments, and products under a shared prefix are four-leaf moments with independent descendant branches. Keeping the complete old-label genealogy in the input variables preserves the formula for arbitrary pair functions.

A colored covariance calculation To compute the matrix in (H2), it suffices to compute its quadratic form on an E_k sector. For $k \geq 1$, color half of the children at depth k in each parent $+$ and half $-$, and give every descendant the sign ϵ of its child. This leaf function

lies in the range of E_k . For real powers the corresponding expression has two child integrals, each raised to $p_{k-1}/(2p_k)$. For $k = 0$, use the constant $+$ function.

Multiply each Gaussian increment on a leaf's path by $1 + \theta \epsilon h_j$ at depth j , including increments inherited from above the colored fork. Choose $h_j(K_j - K_{j-1}) = f_j - f_{j-1}$, where $f_{-1} = 0$, and choose any terminal value f_{R+1} . Strictly separated clocks make this choice possible. Write $G_j = \sum_{s \leq j} h_s^2(K_s - K_{s-1})$, $G_{-1} = 0$, and $s_t = p_{t+1} - p_t$.

Differentiate the log root partition divided by p_0 twice at zero. Covariance acceleration contributes

$$G_{R+1} - \sum_{t \geq k} s_t G_t \mathbf{E} m_t^2 - p_k G_{k-1} \mathbf{E} m_{k-1}^2$$

with the last term omitted for $k = 0$. The remaining curvature is $\sum_t d_t^k f_t(\mathbf{M}^k f)_t$. Indeed, on sheet t the first covariance variation is $(\epsilon_a + \epsilon_b)f_t$, and the coefficient of θ^2 is $\epsilon_a \epsilon_b G_t$. The spin-square diagonal is constant and contributes no covariance Hessian. On an integer complete tree, ordinary differentiation with $\sum_a \epsilon_a^2 = p_0$ gives the claimed contraction, pattern by pattern in the moments.

For real powers, carry the inherited $+$ and $-$ inputs as a two-component Gaussian field until the color separation, where the two child powers are multiplied. Gaussian covariance differentiation and the labeled product rule assign two hits to the same or different children with the same falling-power coefficients as in the integer calculation. Only shared increments enter a covariance. At $\theta = 0$, every normalized transition has the unperturbed scalar law, so the moment factors are the ones required above.

The disconnected product comes from differentiating the root logarithm. Its two pair sums are independent symbolic sums; on a common extended topology, each moment is evaluated on its own restriction. The allocation identity from Section 2 preserves their product. All coefficients therefore agree by the real-mass argument, with singular Gaussian covariance matrices handled by continuity. This identifies the derivative with the contraction before we evaluate it.

Evaluation of the colored Hessian We now use the increasing real masses and evaluate the derivative under the unperturbed positive scalar law. Continue to write $A = K_{k-1}$, and temporarily put $u = m_{k-1}$, $F = f_{k-1}$, $p = p_k$, and $\chi = \chi_{k-1}$. When $k = 0$, take $A = 0$, $F = 0$, and evaluate u, χ at the initial state. Let $\mathcal{Z}$ be the sum of the Gaussian inputs before color separation, each weighted by h_j , $j < k$.

For a positive-color child, let V_θ be the continuation whose times, including its starting and ending times, have moved to $K_t + 2\theta f_t + \theta^2 G_t$. Dots below mean derivatives at $\theta = 0$. For $k > 0$, the first derivatives from the two opposite colors cancel pointwise before the fork. The second derivative of the normalized root value is consequently

$$\mathbf{E}[\ddot{V} + 2\mathcal{Z}\dot{V}_z + \mathcal{Z}^2\chi].$$

For $k = 0$ it is the root continuation derivative with $\mathcal{Z} = 0$.

Set $\eta = 2\sum_{t \geq k} s_t f_t \delta_{K_t}$, $J = D_\alpha V_A[\eta] = \sum_{t \geq k} s_t f_t \mathbf{E}_A m_t^2$. Differentiating the moved times gives (at fixed starting field)

$$\begin{aligned} \dot{V} &= f_{R+1} - F(\chi + pu^2) - J, \\ \ddot{V} &= G_{R+1} - G_{k-1}(\chi + pu^2) - \sum_{t \geq k} s_t G_t \mathbf{E}_A m_t^2 - 2 \sum_{t \geq k} s_t f_t^2 \mathbf{E}_A \chi_t^2 \\ &\quad + 4F^2 \partial_{AA} V - 4F \partial_A J + \text{Hess}_{\alpha\alpha} V_A[\eta, \eta]. \end{aligned}$$

To account for the terms, moving a step location changes the profile by minus its jump times the displacement. Its first response is $\mathbf{E}_A m_t^2/2$, whose time derivative is $\mathbf{E}_A \chi_t^2/2$. These

give $-J$ in the first line and the two sums over $t \geq k$ in the second. The remaining terms differentiate the starting time, the annealed terminal constant, and the profile twice.

The impulses can also be obtained by integrating over the short intervals swept out by the moving steps. The Hessian kernels below are continuous when the impulse times meet. At fixed bounded clocks, the continuation and its needed derivatives are continuous under L^1 profile perturbations, by their bounded-coefficient diffusion equations. This justifies the impulse limit. Extend the initial constant-mass interval slightly when differentiating the starting time; strict clock separation keeps all other steps outside that interval.

The prefix is sampled under its normalized positive law, rather than under the original Gaussian measure. Multiplying its successive density factors gives a telescoping exponential. Gaussian integration by parts therefore differentiates both the endpoint test and this density. For Y depending on the endpoint field, the first two identities are

$$\begin{aligned}\mathbf{E}[\mathcal{Z}Y] &= \mathbf{E}(FY_z + HY), \\ \mathbf{E}[(\mathcal{Z}^2 - G_{k-1})Y] &= \mathbf{E}[F^2Y_{zz} + 2FHY_z + (H^2 + N')Y], \\ H &= pFu - \sum_{t < k} s_t f_t m_t, \quad N' = pF^2\chi - \sum_{t < k} s_t f_t^2 \chi_t.\end{aligned}$$

Here H and N' are the first and second contributions from differentiating the tilted density. The starting-time PDE gives $4V_{AA} = \chi_{zz} + 2p\chi^2 + 4pu\chi_z + 4p^2u^2\chi$ and $-4J_A = 2J_{zz} + 4puJ_z$. Insert these and the two integration-by-parts identities into the second derivative. After subtracting the covariance acceleration displayed above, the contraction becomes

$$-pF^2\mathbf{E}\chi^2 - \sum_{t < k} s_t f_t^2 \mathbf{E}(\chi\chi_t) - 2 \sum_{t \geq k} s_t f_t^2 \mathbf{E}\chi_t^2 + \mathbf{E}[\text{Hess}_{z,\alpha} V_A[\xi, \xi]],$$

where $\xi = (pFu + \sum_{t < k} s_t f_t m_t, \eta)$, and the Hessian is evaluated as a bilinear form. For $t < k$, $\mu_t^k = s_t + p_k \mathbf{1}_{t=k-1}$; for $t \geq k$, $\mu_t^k = 2s_t$. Thus $\xi = \sum_t \mu_t^k f_t \beta_t$, and the three negative sums are $-\sum_t \mu_t^k g_t^k f_t^2$. Using $d_t^k = -\mu_t^k$, this is the quadratic form of (H2). Polarization recovers the matrix action, which is symmetric with respect to the sheet inner product.

Positivity and weighted kernel estimates The estimates must retain the mass factors that will cancel the denominators in P_k . First, B^k is positive semidefinite. The scalar continuation is jointly convex in its starting field and its subsequent mass profile. Indeed, for a fixed bounded adapted control, the controlled terminal field is affine in these variables, the terminal value is convex, and the control cost is linear in the profile. Taking the supremum preserves convexity; completing the square verifies this control representation (PE, Section 5). The same calculation defines the scalar PDE and control formula for every bounded nonnegative profile, without imposing monotonicity. A strictly positive step profile therefore admits both signs of every bounded profile direction, so its full Hessian is nonnegative. Approximate impulses by such directions, and only then average the quadratic form over the prefix. The coefficients m_t , $t < k$, in β_t are held fixed during this differentiation.

Here are explicit kernels for the impulse limit and its estimates. Put $u(y) = V_z(y, Z_y)$, $L^y(s) = \mathbf{E}_s[u(y)^2]/2$ for $s \leq y$, and let $L_z^y(s)$ denote its derivative with respect to the field at time s , evaluated at Z_s . The subscript z is a field derivative; the separate symbol z' below is a time. Order the two evaluation times as $y \leq z'$, and keep the starting time $A = K_{k-1}$ visible. If both times precede A , the kernel is $\mathbf{E}[u(y)u(z')\chi(A)]$. If $y \leq A \leq z'$, it is $\mathbf{E}[u(y)L_z^{z'}(A)]$. If both times follow A , it is

$$\mathbf{E}[u(y)L_z^{z'}(y)] + \int_A^y \alpha(s) \mathbf{E}[L_z^y(s)L_z^{z'}(s)] ds.$$

These formulas come from the second derivative of the diffusion PDE. For bounded directions η, η' , its mixed source is $\eta u D_z D_\alpha V[\eta'] + \eta' u D_z D_\alpha V[\eta] + \alpha D_z D_\alpha V[\eta] D_z D_\alpha V[\eta']$. Letting the directions concentrate at the two indicated times gives the kernels above. When the two times coincide, the first term has the common limit $\mathbf{E}[u(y)^2 \chi(y)]$. The formulas also agree at $y = A$ and $z' = A$, since $L_z^A(A) = u(A)\chi(A)$. Thus no additional boundary term appears when an evaluation time crosses A . For $k = 0$, the initial field is $Z_0 = 0$. Field parity, bounded derivatives, and the Brownian moment bound at the earlier time give $|B_{rt}^0| \lesssim C(1 + K_{R+1})^C K_{\min(r,t)}$ when that time is small.

The kernels B^k, g^k are bounded and Lipschitz in the evaluation times, with polynomial constants. The spatial derivatives needed for this claim are uniformly bounded: the source and attachment rules express u, χ , their products, and the marked conditional expectations as finite signed spin moments, even after any fixed number of common-field derivatives.

For the kernel with both times after A , varying the later time uses $\partial_{z'} L^{z'}(s) = \mathbf{E}_s \chi(z')^2 / 2$. Varying the earlier time also differentiates $u(y)$ and $L_z^{z'}(y)$; the latter has drift $-\alpha \chi L_z^{z'}$. For an evaluation time before A , condition the later factors on that time and apply Itô's product rule with $u(y)$. The bounded derivatives just described control all terms. The same argument applies to g^k , using the drift $-\alpha \chi^2$ of χ .

Moving the starting time from K_{k-1} to K_k gives the stronger estimate

$$\|B^{k+1} - B^k\|_\infty + \|g^{k+1} - g^k\|_\infty \lesssim_{\text{pol}} p_k(K_k - K_{k-1}).$$

On this interval $\alpha = p_k$. In the three cases above, the derivative with respect to A is respectively $-\alpha \mathbf{E}[u(y)u(z')\chi(A)^2]$, $-\alpha \mathbf{E}[u(y)\chi(A)L_z^{z'}(A)]$, and $-\alpha \mathbf{E}[L_z^y(A)L_z^{z'}(A)]$. They remain bounded when a marker crosses A , where the formulas match. Integration over the interval supplies the factor $p_k(K_k - K_{k-1})$.

For the child kernels, set $a_i = \mathbf{E}_t \chi_{i-1}$, and write $\chi(s) = V_{zz}(s, Z_s)$ for the continuous scalar-time process. Its supermartingale drift gives

$$a_i - a_{i+1} = p_i \int_{K_{i-1}}^{K_i} \mathbf{E}_t \chi(s)^2 ds.$$

Thus a mixed difference of $g_t^{il} = \mathbf{E}[a_i a_l]$ is the expectation of the product of the two differences. This proves the stated single and mixed bounds. The same product bound holds for $(1 - b_t g_t^{il})^{-1}$, as used above: interpolate the two factors a_i, a_l linearly and differentiate $F(z) = (1 - b_t z)^{-1}$ twice. The interpolated kernel is bounded above by $\mathbf{E} \chi_t^2$, so the gap bounds both derivatives of F . This includes $i = l$; flat clock intervals follow by continuity. Positivity and the same supermartingale drift give $0 \leq g_t^k, g_t^{il} \leq \mathbf{E} \chi_t^2$, completing the scalar estimates. $\square$

6 Global scalar-trial accuracy

The local comparison budget needs an initial error smaller than x^4 . We obtain it first for the unmasked field before adding seeds. The pair inverse upgrades a coarse comparison to this strict saving; only convergence of the common temperature is needed at the coarse stage.

Proposition 6.1 (Accuracy before field seeding). *For the direct paths, use the matrix b of (P3) and the unmasked, unseeded field $h_* = t - bq$ in the untilted system. Fix $T > 1$ and the exponent choices of Section 3. For m in the stated dyad, the following bound is uniform over the allowed λ, ℓ, γ . There is a saving $c_G > 0$ independent of the allowed p_* and of s_* sufficiently close to $1/2$, such that for all sufficiently large N ,*

$$0 \leq F_* - f_m^0 \leq x^{4+c_G}, \quad F_* = f_{\text{spin}}(t) + \frac{1}{4} \int b q^2 dv. \quad (\text{G1})$$

Here $f_m^0 = f_m(b, h_*)$, and $f_{\text{spin}}(t)$ is the one-site scalar pressure with the same terminal mass ℓ . The size threshold may depend on the fixed exponent choices.

Proof. Replacing the matrix by the field bq gives a nonnegative square-error interpolation, which proves the first inequality. Throughout the proof normalize ℓ to 1 as in (A1): multiply the pressure by ℓ , the clocks by ℓ^2 , and use mass v/ℓ . The size rescaling is held fixed; below θ denotes a new interpolation parameter. We will remove the matrix in a constrained interpolation. The next two inputs control its endpoints: PE bounds the cost of an initial small removal, while scalar coercivity bounds the endpoint at which the matrix has been removed completely.

The finite-propagation comparison used below We use PE, Proposition 6.6, in its ordinary all-mass case $I = [0, 1]$ (PE, Definition 6.1 and equations (3)–(6)). Its parameters are a small fixed power slack $m_{\text{PE}} > 0$, $\delta_{\text{PE}} = m_{\text{PE}}/10000$, and scales satisfying

$$0 < d = O(1), \quad m^{-2/3+m_{\text{PE}}}d \lesssim E \leq m^{-1/2}d^2, \quad a_{\text{PE}} = \sqrt{m}E/d, \quad v_* = (mE)^{-1}.$$

Here d is PE’s scale parameter, not our field defect. The bare clocks b_0, h_0 must be bounded, nonnegative and nondecreasing, with a fixed positive matrix margin after the comparison; d^2 has a fixed positive gap below T . At open-interval masses $z \gtrsim a_{\text{PE}}$, with fixed comparison constants, the endpoint hypotheses are $b_0(z) = T + o(z^2)$ and $h_0(z) = o(z^3)$.

At strength $0 < \lambda_{\text{PE}} \leq d^2$, this comparison removes $e(r) = \lambda_{\text{PE}}f(\lambda_{\text{PE}}r/d^3)$ from the matrix, adds $re(r)$ to the field, and adds $\int e(r)r^2/4$ to the objective. It minimizes over onto quantiles of $[0, 1]$ whose Lipschitz inverses satisfy

$$k(r) \leq \alpha'(r) \leq M_{\text{PE}}(\alpha(r)), \quad k(r) = \frac{m^{\delta_{\text{PE}}}v_*^2}{(r + m^{2\delta_{\text{PE}}}v_*)^2},$$

$$M_{\text{PE}}(v) = m^{\delta_{\text{PE}}} \left(\frac{v + v_*}{\min(v_*, a_{\text{PE}}^3/d^2)} + \frac{d^3}{E}(v + v_*)^2 \right).$$

The fixed smooth taper f equals 1 near zero and is of order $(1 + \log t)/t$ at infinity. Its logarithmic decay slope $p(t) = -tf'(t)/f(t)$ increases from 0 towards 1, with $1 - p \gtrsim (1 + \log(1 + t))^{-1}$. Provided the changed paths are admissible, the minimum exceeds the bare pressure by at most $m^{20\delta_{\text{PE}}}E$, and each minimizing radius is comparable to mass above $m^{\delta_{\text{PE}}}\sqrt{m}E/\sqrt{\lambda_{\text{PE}}}$. In our use, the matrix margin and taper monotonicities ensure admissibility.

A common varying base temperature The preceding conclusion remains valid when the endpoint hypotheses use a common $T_m \rightarrow T > 1$, with T_m in a fixed compact subset of $(1, \infty)$ and held fixed throughout each parent/child tree at size m . The scale restrictions and propagation slack stay the same. This sequential extension is proved in [19, Appendix: Companion interfaces, paragraph “A common prelimit temperature”]. We recall its argument to specify how it applies here without a rate of convergence for T_m .

First, the constraints and conditional moment estimates in PE, Section 3 use bounded clocks and fixed comparability constants; its endpoint Taylor hypothesis enters only the subsequent profile argument. At the regular calibration cut in PE, Proposition 4.1, use $b_* = T_m + O(W^2)$, $K_* = T_m S_* + O(W^3)$, and set $J_0 = T_m \chi^2 - 1$ in PE (26)–(27). The proof divides by $\chi K_* \asymp W$ and absorbs $\eta_n |J_0| x_{\text{test}}^2 / W^2$. These operations use the actual T_m , so they introduce no $T_m - T$ error; the conclusion PE (22) likewise uses the actual scalar clock.

In the small-scale proof in PE, Section 5 take time K/T_m , subtract K/T_m in the scalar compactness and uniform scalar linearization lemmas, and put $((b_0 - T_m)S + h_0)/x_{\text{test}}^3$ for

the endpoint remainder at test scale x_{test} . All prelimit identities and bounds there are then identical (scalar derivative and strict curvature bounds use only compact temperature and total time bounds); in the rescaled ODE/potential limits T_m converges to T .

In the macroscopic direct result (PE, Section 5, Proposition 5.9) the actual endpoint convergence and limiting variational problem are at T , so its fixed-temperature scalar-minimizer input needs no uniform version. This gives the same positive upper/lower bounds on limiting slopes for the profile contradictions.

Child data, covariance margins (choose with room at the limit), and moving barriers in PE, Section 6 then use PE (4) with that common T_m , and the fixed finite depth subsequence proof applies unchanged. Thus the parameter replacement invokes no theorem uniform up to the critical temperature or change of the fixed propagation slack.

Scalar coercivity We first establish a coercive comparison. If $K' = h_* + bz$ is an admissible scalar path with $0 \leq z \leq C$, then

$$f_{\text{spin}}(K') + \frac{1}{4} \int bz^2 \geq F_* + c \int_0^{Q_0} \left(\int_s^{Q_0} (\alpha_{K'} - \alpha_t) \right)^2 ds, \quad (\text{G2})$$

where Q_0 is a fixed clock bound, after annealed extension of the time distributions. Use the base optimal feedback martingale $u_s = V_z(s, Z_s)$ as control in both scalar problems, as verified in PE, Section 5. This gives the lower first variation $\frac{1}{2} \int \Gamma(s)(\alpha_{K'} - \alpha_t)(s)ds$, together with the terminal log cosh convexity remainder.

Write $P(s) = \int_s^{Q_0} (\alpha_{K'} - \alpha_t)$ and $A = \int_0^{Q_0} P(s)^2 ds$. The terminal displacement is $Y = \int (\alpha_{K'} - \alpha_t)u = \int P(s)\chi_s dB_s$, where $\chi_s = V_{zz}(s, Z_s)$. On bounded clocks, $\mathbf{E}\chi_s^2 \geq c$, while χ_s is bounded, so $\mathbf{E}Y^2 \geq cA$ and $\mathbf{E}Y^4 \leq CA^2$. Also Y is deterministically bounded from its original integral. Choose a fixed compact set for the base terminal argument with complement probability at most $c^2/(4C)$. Cauchy-Schwarz leaves at least $cA/2$ of $\mathbf{E}Y^2$ on this set. The second derivative of log cosh is bounded below along every displaced segment there. Its convexity remainder is therefore at least $c'A$, as required in (G2).

The first variation plus penalty change cancels linearly, leaving $\frac{1}{2} \int dv b(z - q)^2 \int_0^1 (1 - y)[1 - b\Gamma'(t + yb(z - q))]dy \geq 0$ by (P5), without needing monotonicity of z . The lower bound on $\mathbf{E}\chi_s^2$ here follows also by comparison to the base T scalar path on a fixed compact including its annealed extension as in (P2).

A constrained interpolation Choose $e_0 = x^{d_g}$, where the small exponent $d_g > 0$ will be fixed below, and set $d_{\text{PE}} = e_0^{1/4}$. Apply PE with $d = d_{\text{PE}}$, strength e_0 , and error parameter $E = x^{4-d_g/16} d_{\text{PE}} = x^{4+3d_g/16}$. Taking its fixed power slack sufficiently small relative to d_g satisfies PE (3) and makes $m^{20\delta_{\text{PE}}} E \leq x^{4+c}$. The endpoint scale is $a_{\text{PE}} \asymp x^{1-d_g/16} \gtrsim x$. Also $e_0/d_{\text{PE}}^3 \rightarrow 0$, so the taper is identically 1: this comparison removes the constant e_0 and inserts $e_0 r$. Equations (P2)–(P4) and (P9) verify the endpoint and admissibility hypotheses with the common base $T_m \rightarrow T$ just discussed. The optimized value, including $\int e_0 r^2/4$, is consequently within x^{4+c} of the bare pressure.

We can pass to onto inverse paths $\alpha : [0, 2] \rightarrow [0, 1]$ with constant bounds $\kappa = x^{4+1/4} \leq \alpha' \leq M_c = x^{-10}$, at upper-cost loss $O(\kappa + 1/M_c)$. Indeed any radius law on $[0, 1]$ can be approximated in first transport moment to that accuracy, by a slight bounded-density spreading within the enlarged interval and mixing a uniform floor. Cost follows by (A1). Keep these latter constraints fixed and minimize

$$\Phi(\theta) = \min_r \{f_m(b - e, h_* + er) + \int er^2/4\}, \quad e = (1 - \theta)e_0 + \theta b, \quad 0 \leq \theta \leq 1.$$

The density box is compact at fixed size, so a minimizer exists. Equation (A1) gives monotonicity and Lipschitz continuity of Φ , as well as its envelope upper derivative at a minimizer. All paths are admissible. At the two endpoints, $\Phi(0) \leq f_m(b, h_*) + x^{4+c}$ by the PE comparison and $\Phi(1) \geq F_*$ by (G2). At every intermediate θ , trying a constrained approximation to q and replacing the remaining matrix by the field $(b-e)q$ gives $\Phi(\theta) \leq F_* + O(\kappa + M_c^{-1})$.

Fix θ and a minimizing r . The objective is an averaged pressure, so this choice, its inverse α , and all intervals chosen from them are deterministic before taking the positive-law expectation. Evaluate S, B, D at this changed path. A subscript in radius s means the cut at mass $\alpha(s)$. Put

$$\tilde{b} = b - e, \quad K^{\text{emp}} = h_* + er + \tilde{b}S.$$

This empirical scalar clock replaces each remaining matrix overlap by its mean S . It is nonnegative and nondecreasing.

It remains to show that Φ increases by at most $O(x^{4+c})$. Its envelope derivative is bounded by the integral of $D^2 + (S-r)^2$. We first use the minimizing density constraints to control $S-r$, then compare K^{emp} to t . Deterministic radius buffers will give high moments of $Q-S$; the pair inverse will improve these to the required integrated bound on D^2 . The final envelope estimate combines these four steps.

First variation gives a potential H with $H'(s) = e(\alpha(s))(s - S_{\alpha(s)})$. Indeed the variation against $\delta\alpha$ is $-\int H'\delta\alpha/2$, by changing variables in (A1). The inverse is bi-Lipschitz at fixed size, and the required path moments are continuous under these perturbations by PE, Section 2. After integration by parts this is a linear functional of the density variation. Exchanging density between two locations, while keeping total mass fixed, gives a constant shift of H for which $\alpha' = \kappa$ on $H > 0$ and $\alpha' = M_c$ on $H < 0$.

A component of $H < 0$ has radius width at most M_c^{-1} . Its endpoint signs and monotonicity put S between its two radius endpoints; at exterior boundaries use $0 \leq S \leq 1$. On $H = 0$, $S = s$ almost everywhere. We call the complement of the positive components *nonfloor*. Thus $|S - s| \leq M_c^{-1}$ there, while the floor has total mass at most 2κ . Integrating H' on a negative component also gives $H \geq -C(e_0 + \theta)M_c^{-2}$.

We will use the following local consequence. Within distance $O(h)$ of a nonfloor point s_0 , where $M_c^{-1} \ll h$, $S = s_0 + O(h)$, provided one further interval of length h fits on each required side. For if S exceeded this range, monotonicity would keep $s - S$ of one sign and of size $\gtrsim h$ on such an interval. Since $e \asymp e_0 + \theta$, integrating H' to or from s_0 , where $H(s_0) \leq 0$, would force a negative value of size $\gtrsim (e_0 + \theta)h^2$. This contradicts the lower bound for H .

A quantitative coarse clock comparison The empirical scalar clock is a full mass path. Its time distribution will be denoted by $\alpha_{K^{\text{emp}}}$, with field time as its argument. We first need the coarse estimate

$$\|\alpha_{K^{\text{emp}}} - \alpha_t\|_1 \leq x^{c_1}, \quad (\text{G3})$$

where $c_1 > 0$ is independent of sufficiently small d_g . Here and below these are scalar time profiles, not optimizer inverse paths. There is a monotone $0 \leq z \leq 1$ with

$$f_{\text{spin}}(h_* + er + \tilde{b}z) + \int \tilde{b}z^2/4 \leq f_m(\tilde{b}, h_* + er) + x^{c_0}$$

for some absolute $c_0 > 0$. If $\sup \tilde{b} \lesssim x^{1/10}$, take $z = 0$ and use the pressure Lipschitz bound in (A1). Otherwise $\tilde{b} = (1 - \theta)(b - e_0)$ has $b_* := \inf \tilde{b} \gtrsim x^{1/10}$, since $b - e_0 \asymp 1$. We use the quantitative bounds in the proof of PE, Section 5, Lemma 5.7, as follows.

In that scalar-trial interpolation, denote its interpolation time by τ , its inverse floor and cap by $\eta_m = m^{-1/8}$ and $Q_m = m^{1/8}$, and its optimizer by z_τ . In this paragraph the observables refer to this inner optimizer, with clocks $((1-\tau)\tilde{b}, h_* + er + \tau\tilde{b}z_\tau)$. The mismatch satisfies $\int (S - z_\tau)^2 dv \leq \eta_m + Q_m^{-2}$. For $\tau \geq \varepsilon$, after also discarding masses below ε , the covariance trace satisfies

$$\int_{\varepsilon}^1 \mathbf{E} \operatorname{tr} C_v^2 dv \leq R := \frac{Q_m}{mb_*\varepsilon^3},$$

and the variance of an unnormalized L -interval integral is at most $CQ_m/(mb_*\varepsilon^2)$. Partition the mass interval where S varies by at most Δ , and discard end strips of width ρ . There are at most $C(1 + \Delta^{-1})$ strips. The omitted masses and times therefore cost $O(\varepsilon + \rho(1 + \Delta^{-1}))$. On every retained mass, the earlier and later L -averages give

$$\| |X_v|^2 - S_v \|_2 \leq C\Delta + \frac{C}{\rho} \left(\frac{Q_m}{mb_*\varepsilon^2} \right)^{1/2} =: A.$$

Finally use $D_v^2 = \operatorname{Var}(|X_v|^2) + \mathbf{E} \operatorname{tr} C_v^2 + 2\mathbf{E} X_v^\top C_v X_v$, with the last term bounded in integrated mass by $2R^{1/2}$. The integrated envelope derivative is consequently at most

$$C\{\eta_m + Q_m^{-2} + \varepsilon + \rho(1 + \Delta^{-1}) + A^2 + R + R^{1/2}\}.$$

With $\varepsilon = \Delta = x^{1/100}$ and $\rho = x^{2/100}$, each term saves a fixed power of x : for example, $R \lesssim x^{128/25}$ and $A \lesssim x^{1/100} + x^{509/200}$. These estimates use only bounded monotone actual clocks and the lower bound on b_* , so they are uniform over our minimizing r . The scalar endpoint of this interpolation supplies the asserted trial. We now return to the observables of the outer minimizing r .

Put $K' = h_* + er + \tilde{b}z$ and $\bar{z} = (er + \tilde{b}z)/b$. The exact penalty decomposition is

$$er^2 + \tilde{b}z^2 = b\bar{z}^2 + \frac{e\tilde{b}}{b}(r - z)^2.$$

Apply (G2) to $K' = h_* + b\bar{z}$. The trial inequality and $\Phi(\theta) \leq F_* + O(\kappa + M_c^{-1})$ then bound both the squared primitive in (G2) and $\int e\tilde{b}(r - z)^2 dv$ by $O(x^{c_0})$, after decreasing c_0 .

To pass from the primitive to the profiles, write $g = \alpha_{K'} - \alpha_t$ and $P(s) = \int_s^{Q_0} g$. The total variation of g is bounded. Averaging over intervals of length a gives $\|g\|_1 \leq C(a + a^{-1}\|P\|_1)$; boundary intervals have the same $O(a)$ cost. Since $\|P\|_1 \lesssim x^{c_0/2}$, choosing $a = x^{c_0/4}$ gives a power-small profile error. Moreover the cap/floor mismatch and Cauchy–Schwarz give

$$\int \tilde{b}|S - z| dv \lesssim \kappa + M_c^{-1} + e_0^{-1/2} x^{c_0/2}.$$

For monotone clocks, the L^1 distance between their quantiles equals that between their time distributions. Thus this is also the cost of replacing K' by K^{emp} . Choose d_g sufficiently small relative to c_0 ; a fixed smaller exponent $c_1 > 0$ now proves (G3) independently of d_g . Decrease c_1 once more, if needed, to cover the bounded-clock scalar derivative comparison used at the end of the proof; all remaining exponents are chosen after this choice.

Integrated pinned moments and rough concentration Choose $\delta = x^{a_g}$ with $0 < a_g \ll c_1$, put $h = \delta^2$, and take d_g sufficiently small relative to a_g . Near a nonfloor point $s_0 \geq \delta$, the local potential bound gives $S = s_0 + O(h)$, and hence $K^{\text{emp}} \geq c\delta$. Such a clock cannot occur at mass $\ll \delta$: monotonicity and the regular initial slope of t would give an area of order δ^2 between the two time distributions, contradicting (G3), since $c_1 > 2a_g$. Thus these neighborhoods and all later cuts have mass $\gtrsim \delta$. They have room for the required buffers, since $s_0 \leq 1 + M_c^{-1}$, while the radius interval extends to 2.

For fixed $p \geq 2$, on a radius window I there (within $[0, 5/4]$) we have

$$\left\| \int_I W ds \right\|_{p/2} \lesssim_{p,\circ} x^6 / (e_0 \delta^2). \quad (\text{G4})$$

A reusable residual estimate. We isolate the deterministic blocking calculation also used in Sections 7 and 8. It applies to the full single-path positive law and does not require an optimizing radius.

Lemma 6.2 (Residual blocks and integrated increments). *Fix a positive path law with terminal mass $a \in [1, 4]$, and parametrize a deterministic increasing family of revealed cuts by s . At a cut s , let branches $\mathbf{a}, \mathbf{b}$ be conditionally independent continuations with their normalized positive kernels, given the common prefix. Partition their futures into the same k deterministic blocks, the last ending at the terminal spin. Write Δ_j for the martingale increments, so that $y - X_s = \sum_{j=1}^k \Delta_j$, and let $\mu'_j \leq a$ be the largest mass in block j . Evaluating W_s on branch $\mathbf{a}$, we have*

$$W_s \leq k \sum_{i,j} \mu'_j \mathbf{E}_{\mathbf{b}}[(\Delta_i^{\mathbf{a}} \cdot \Delta_j^{\mathbf{b}})^2 \mid \mathbf{a}]. \quad (6.1)$$

Conditioning on $\mathbf{a}$ retains its whole branch and the common prefix; only continuation $\mathbf{b}$ is averaged. For any deterministic interval J , any increment length $l > 0$, and $q \geq 1$, provided all indicated cuts exist, put $s_- = \inf J$ and $s_+ = \sup J + 2l$. Then

$$\left\| \int_J |X_{s+2l} - X_{s+l}|^2 ds \right\|_q \leq C_q l \left\| |X_{s_+} - X_{s_-}|^2 \right\|_q. \quad (6.2)$$

The same estimate holds with increment $X_{s+l} - X_s$. For any nonnegative process V_s , not necessarily adapted,

$$\left\| \int \mathbf{E}_s V_s ds \right\|_q \leq C_q \left\| \int V_s ds \right\|_q. \quad (6.3)$$

Proof of the lemma. The step Hessian rule writes H_s as the conditional sum of future increment covariances weighted by their masses; the last term is the terminal Gibbs covariance multiplied by a . Conditional orthogonality gives

$$H_s \preceq \sum_j \mu'_j \mathbf{E}_s \Delta_j \Delta_j^{\mathbf{T}}.$$

Expanding only branch $\mathbf{a}$'s residual and applying Cauchy–Schwarz gives (6.1). In particular $H_s \preceq a C_s$. The fixed-size approximation in (A1) passes this statement to filled clocks.

For the integrated estimate, the shifted-grid identity is

$$\int_J |X_{s+2l} - X_{s+l}|^2 ds = \int_0^l \sum_{j: u+jl \in J} |X_{u+(j+2)l} - X_{u+(j+1)l}|^2 du.$$

At each fixed u the increments are disjoint. Apply the martingale square-function and Doob inequalities to $X - X_{s_-}$, and integrate in u , to obtain (6.2); for $q = 1$ martingale orthogonality suffices. The earlier increment has the same proof.

For (6.3) with $q > 1$, pair with a nonnegative Z of unit $L^{q'}$ norm. Move the conditional expectation onto Z , bound $\mathbf{E}_s Z$ by its supremum, and apply Doob and Hölder. For $q = 1$, the two expectations are equal. $\square$

Application to the integrated pinned moment. We return to the normalized terminal mass $a = 1$ and prove (G4). Here H_s , the source Hessian, is distinct from the optimizer potential $H(s)$. The lemma also applies after an increment on branch $\mathbf{b}$ is averaged back to its fork, by (6.3). We use $\mu'_j \leq 1$, or directly $H_s \preceq C_s$, giving

$$W_s \leq \mathbf{E}_{\mathbf{b}}[(y^{\mathbf{a}} - X_s) \cdot (y^{\mathbf{b}} - X_s))^2 \mid \mathbf{a}].$$

For the residual decomposition in (G4), fix a small constant $L_* > 0$ with $s + L_* < 2$ for $s \in I$. Choose $l_0 = 2^{-J_*} L_*$ of arbitrarily small inverse-polynomial size. Use the initial block $X_{s+l_0} - X_s$, the blocks $X_{s+2l} - X_{s+l}$ for $l = l_0, 2l_0, \dots, L_*/2$, and the terminal residual $y - X_{s+L_*}$. There are $k = O(\log(1/x))$ blocks and their sum is exactly $y - X_s$.

For any pair other than two initial blocks, project the block with the later starting distance, denoted $l_{\max}$. The anchor A is the other branch's increment. Write $A = |A|\hat{A}$, with $\hat{A} = 0$ when $A = 0$, and apply (A3) to $\hat{A}$. Conditional on the common prefix, the whole other branch is exterior data, so fixing this anchor leaves the projected branch's normalized future kernel unchanged. Equal starting distances may be resolved in either order. The gap after the fork has field length at least $e_0 l_{\max}$ and mass at least $c\delta$. For a finite block, conditional Jensen converts the residual estimate at its start into the same estimate for its increment.

Fix a small $\epsilon > 0$. At a sufficiently high fixed even moment, Markov's inequality gives

$$\mathbf{P} \left\{ |\hat{A} \cdot \Delta_{\text{later}}|^2 > \frac{x^{6-\epsilon}}{e_0 \delta^2 l_{\max}} \right\} = O(x^K)$$

for any prescribed fixed K . Indeed (A3) gives the same square scale with $x^6 \log^4 N$; taking the moment order large relative to K/ϵ supplies this tail bound. The factor $|A|^2$ remains in the subsequent integral. On the exceptional event the unnormalized squared product is bounded, since $|y|, |X| \leq 1$. Jensen before and after conditional averaging shows that its integral has $L^{p/2}$ norm $O(x^{2K/p})$. Choose K large enough to make this negligible.

By (6.2)–(6.3), the earlier squared anchor integrated over I has $L^{p/2}$ norm at most $C_p l_{\max}$. If both blocks are terminal, boundedness gives the same assertion since $l_{\max} = L_*$ is fixed. Thus each projected pair contributes at most $C_p x^{6-\epsilon}/(e_0 \delta^2)$. For two initial blocks, bound one by a constant and integrate the other's square; the cost is $O_p(l_0)$, negligible by the choice of l_0 . Summing the logarithmically many pairs in (6.1) proves (G4).

Earlier and later buffers. We next turn the integrated pinned estimate into control at individual cuts. The following bracket argument also records what is needed when the later applications use different radius windows.

Lemma 6.3 (Buffer brackets and a common past). *Fix a full single-path positive law with terminal mass $a \in [1, 4]$ and deterministic revealed cuts. Let w be one such cut, and let ρ_- , ρ_+ be deterministic probability measures supported on cuts before and after w , respectively. Write*

$$A_- = \int L_v \rho_-(dv), \quad A_+ = \int L_v \rho_+(dv).$$

Suppose that, for a deterministic number $\bar{S}$,

$$|S_v - \bar{S}| \leq \eta \quad \text{on the two buffers}, \quad \|A_{\pm} - \mathbf{E}A_{\pm}\|_p \leq \varepsilon_p, \quad 2 \leq p < \infty.$$

The bound on S_v is required only almost everywhere for the two buffer measures. Then

$$\| |X_w|^2 - \bar{S} \|_p + \left\| \int |X_w - X_v|^2 \rho_-(dv) \right\|_p \leq C(\eta + \varepsilon_p).$$

If two cuts w, w' use the same earlier measure ρ_- , supported before both, and each has a later buffer satisfying the same hypotheses with the same $\bar{S}, \eta, \varepsilon_p$, then

$$\| |X_w - X_{w'}|^2 \|_p \leq C(\eta + \varepsilon_p).$$

Proof of the lemma. Since $2\mathbf{E}L_v = S_v$, the buffer assumptions give $|2\mathbf{E}A_{\pm} - \bar{S}| \leq \eta$. The martingale identities are

$$\begin{aligned} 2\mathbf{E}_w A_- &= |X_w|^2 - \int |X_w - X_v|^2 \rho_-(dv), \\ 2\mathbf{E}_w A_+ &= \int \mathbf{E}_w |X_v|^2 \rho_+(dv) \geq |X_w|^2. \end{aligned}$$

Thus $|X_w|^2$ lies between $2\mathbf{E}_w A_-$ and $2\mathbf{E}_w A_+$, each of which is within $\eta + 2\varepsilon_p$ of $\bar{S}$ in L^p , by conditional Jensen. The first identity then also bounds its nonnegative past squared-increment integral. For two cuts, integrate

$$|X_w - X_{w'}|^2 \leq 2|X_w - X_v|^2 + 2|X_{w'} - X_v|^2$$

against their common earlier measure and use the two preceding bounds. $\square$

Apply the lemma with normalized radius averages over buffers of length comparable to h , before and after cuts near s_0 . Their measures are deterministic because the optimizer was fixed before sampling. For an interval of radius length h , the mass weight in (A1) is $k(v) = 1/[h\alpha'(r(v))]$. Consequently (G4) gives

$$\|A_{\pm} - \mathbf{E}A_{\pm}\|_p \lesssim_{p,\circ} \frac{x^3}{\sqrt{e_0\delta h\sqrt{\kappa}}} = x^{7/8-d_g/2-3a_g} O_{p,\circ}(1) = O_p(h).$$

The last estimate holds by taking the fixed exponents small before choosing the subpower loss. The local potential bound gives $S_v = s_0 + O(h)$ throughout these buffers. Lemma 6.3 therefore applies with $\bar{S} = s_0$, $\eta = O(h)$, and $\varepsilon_p = O_p(h)$. It bounds both $\| |X_w|^2 - s_0 \|_p$ and the averaged past squared increments by $O_p(h)$. Using the same earlier buffer for two nearby cuts gives $\| |X_w - X_{w'}|^2 \|_p = O_p(h)$.

Now take two branches splitting at s_0 , and project their terminal spins to cuts a distance comparable to h later. Their squared norms are $s_0 + O_{L^p}(h)$, and their increments from the common ancestor have squared norms $O_{L^p}(h)$. Polarization therefore puts their projected inner product within $O_{L^p}(h)$ of s_0 . The two projection errors in (A3) are at most $O_{p,\circ}(x^3/(\delta\sqrt{e_0h}))$, so in particular $\|Q - S\|_p = O_p(\delta)$ at this nonfloor split.

For splits below radius δ , choose instead a nonfloor cut at radius comparable to a sufficiently large fixed multiple of δ . To find it, take a mass band of that scale. Equation (G3) and monotonicity put K^{emp} between comparable multiples of δ on an interior band; the discarded floor has mass at most $2\kappa \ll \delta$. At a nonfloor point of this band, $h_* = o(\delta)$ and $r = S + O(M_c^{-1})$, so $K^{\text{emp}} = br + o(\delta)$ and $r \asymp \delta$, as needed. The same buffers give $\| |X|^2 \|_p = O_p(\delta)$ there. Project both branches to this cut, using its last comparable subinterval for (A3); Cauchy–Schwarz bounds their projected product by $O_p(\delta)$. Since $S \lesssim \delta$ at the original cut, we obtain, after weakening the exponent, $\|Q - S\|_p \lesssim_p x^{a_g/2}$ at almost every nonfloor split, for every fixed moment order.

Centered inversion and absorption At the sampled clock K^{emp} , the matrix multiplier $\tilde{b}$ has the gap required in (H1), of size at least ce_0 . To check this, (G3) and the bounded-clock PDE estimates compare scalar derivatives at a common time with error $O(x^{c_1})$. For $v \geq \delta$,

monotonicity and the same area argument give $K^{\text{emp}}(v) \gtrsim v$. The positive part of $b(v) - b_0$ is at most $C_{c_b} \lambda v^2$, which is absorbed by (P2) when c_b is small. Below δ this positive part is $o(e_0)$. Finally, replacing b by $b - e$ gains $e\Gamma'$: wherever this is not comparable to e_0 , Γ' is small and the original gap is bounded below by a constant. The common-time error is $o(e_0)$, so the gap persists in both cases.

Fix the old pair ij and use (A7) to first order about K^{emp} , with test $T_{ij} = Q_{ij} - S_{ij}$. The row mismatch is exactly

$$J = (h_* + er) + \tilde{b}Q - K^{\text{emp}} = \tilde{b}(Q - S).$$

The test and the deterministic centers S remain fixed throughout this cavity interpolation; the actual row and the independent scalar row at K^{emp} are its positive-semidefinite endpoints. The unknown pair function is the deterministic, topology-indexed expectation $u_{ab} = \nu[T_{ij}(Q_{ab} - S_{ab})]$. It is unchanged on adding unused labels, by positive-law marginalization. The scalar degree-zero term, including any deterministic offset on a given input topology, vanishes because $\nu[T_{ij}] = 0$. The linear terms give (H1) with multiplier $\tilde{b}$. Its errors are $O(x^6)$ and products of T_{ij} with at least two $Q - S$ insertions.

Apply the inverse, evaluate at $ab = ij$, and then integrate the old fork mass. Absolute attachment domination bounds new fork densities by a polynomial in e_0^{-1} ; a fork reused at ij is now integrated as well. Truncate one error insertion at $x^{a_g/4}$. Its higher fixed moments, just proved off the floor, make the exceptional probability $O(x^6)$ by Markov's inequality, and all factors are bounded there. To charge a floor placement, follow its fork to the first time that vertex was introduced. It is either the original ij fork, whose mass is now integrated, or a fresh fork with bounded density. Later reuses retain the ordinary bounded atom weights and introduce no new fixed mass. The floor mass is $O(\kappa)$, so the finite union of these placements also costs $O(\kappa)$ under the dominated allocation measure. On the retained event this leaves $x^{a_g/4}|T_{ij}(Q_{ab} - S_{ab})|$. Cauchy-Schwarz bounds its expectation by $x^{a_g/4}D_{v_{ij}}D_{v_{ab}}$; the bounded fork measures and $2D_v D_w \leq D_v^2 + D_w^2$ give the integrated estimate

$$\int D_v^2 dv \leq O(e_0^{-C})(\kappa + x^6 + x^{a_g/4} \int D_v^2 dv).$$

For nonstep paths, perform this inversion first at fixed m on step approximations, using each approximation's own S and K^{emp} in the cavity formula. Their moments converge at almost every placement and in mass L^1 . The common-time scalar comparison and monotone lower clock bound persist, so the sheet gap remains positive when b, e are sampled from their paths. Pass to the limit inside the dominated integrated error bounds, as in the [fixed-resolution limit argument](#) of Proposition 5.1.

Now choose d_g so small that $Cd_g < a_g/4$ and $Cd_g < 1/4$. The first inequality absorbs the variance term on the right; the second makes $e_0^{-C}\kappa$ smaller than x^4 by a fixed power. At a minimizing r , (A1) gives the envelope upper derivative

$$\Phi'(\theta) \leq \frac{1}{4} \int (b - e_0) \{D_v^2 + (S_v - r(v))^2\} dv = O(x^{4+c}).$$

Here the mismatch integral is $O(\kappa + M_c^{-2})$, by the floor mass and the nonfloor cap bound. Integrate over $0 \leq \theta \leq 1$, and use $F_* \leq \Phi(1)$ and $\Phi(0) \leq f_m^0 + x^{4+c}$. Decreasing the fixed saving exponent if necessary proves (G1). □

7 Weighted comparison budgets

The purpose of this section is to turn optimized pressure comparisons into uniform second-moment budgets for the genuine paths. The global comparison gives both parts of (C2)

directly. To prove the remaining low-mass part of (C1), we first obtain coarse comparability for auxiliary minimizers and then strengthen it to the accuracy required by a local pressure comparison. The local auxiliary optimizers are used only to prove those budgets. Their density constraints will change during the argument, so their regularity and concentration estimates are established anew before any one-site closure is applied. All constants and all finite iteration counts are fixed before the size limit. Parameter choices are recorded where they first become necessary; the final order is collected in Section 10.

7.1 Averaging targets and the global comparison

Unlettered equation numbers in references to source estimates are those of [19]. Its numbered statements use the source numbering. The estimates of this section carry labels (C1)–(C20). Whenever the density constraints differ from those in [19], we state the replacement analytic inputs and repeat the relevant proof argument below.

Write $A_v = B_v - 2q(v)S_v + q(v)^2$ for the actual untilted second moment about the reference. Fix $k_c = 0.03$. In what follows take $s_* = 1/2 - \tau$, $s_0 = s_* - \tau_0$, with

$$0 < \tau_0 \ll \eta \ll \tau \ll c_G, \quad \tau \ll 10^{-4}$$

($1 - 2s_0$ can in particular be made arbitrarily small relative to $c_G/12$). The final-strength exponent p_* can be taken sufficiently small after these choices.

Proposition 7.1 (Uniform weighted comparison budgets). *The averaging bounds needed are, on all segments of Proposition 4.1, uniformly,*

$$\mathbf{E}_\gamma \left[\max_{U \lesssim u \lesssim 1} u \int_{v \approx u} A_v dv \right] \lesssim_\circ U^4, \quad (\text{C1})$$

where one can use ordinary dyads with fixed comparable enlargements, and on size paths in addition

$$\int (v^2 + U^2) \mathbf{E}_\gamma A_v dv \leq x^{4+c}, \quad f_{\text{spin}}(K) + \int (K - h)^2 / (4b) - f_m^0 \leq x^{4+c}. \quad (\text{C2})$$

The second assertion of (C2) is pointwise, with some $c > 0$ independent of sufficiently small p_* .

Reduction to localized comparisons. We first obtain the size-path bounds from the global comparison. Fix a direct endpoint and an unseeded, unmasked path whose global parameter is smaller by a fixed $\delta > 0$. To reach the endpoint, remove δc_g from the matrix clock and add $\delta c_g q + d$ to the field clock. Both paths are admissible. Completing the square along their affine interpolation and using (G1) bounds

$$R = f_m^0 - \int b q^2 / 4 + \int_{v \geq x} dq / 2$$

below by $f_{\text{spin}}(K) - O_\delta(\mathcal{B})$; the corresponding upper bound follows by the trial replacing the remaining matrix to reach K . Here, with subpower losses harmless,

$$\mathcal{B} \lesssim_\circ x^{4+c_G} + \lambda^{1+2\eta} U^5 + \chi_e^2 \lambda^{1+2\eta} U^4 + \chi_{\text{lin}}^2 x^6 / \lambda.$$

The lower comparison costs $\int d^2 / (\delta c_g)$, and the upper comparison costs $\int d^2 / b$. Omitting the mask costs $O(\lambda U^2 x^3)$. The linear restoration is independent of γ , so (P10) can be integrated between the global parameter endpoints. For some fixed $b_s > 0$, this gives

$$\int (v^2 + U^2) \mathbf{E}_{\gamma_0} A_v dv \lesssim x^4 \lambda^{-2s_0 + b_s}$$

also when the global interval of averaging is any fixed interior interval leaving crossing room. We can choose b_s, τ_0 sufficiently small relative to η , with further room to absorb errors $x^4 u \lambda^{k_c/3-1-o(1)}$ on the right. At final strength with edge off this argument and (P11) give both parts of (C2), with savings independent of further decreases in p_* : even the differentiated estimate uses only $O_\circ(x^{4+c_G}/\lambda + \lambda^{2\eta} U^5 + x^6/\lambda^2)$ plus the switch cost from (P11) divided by λ .

For (C1), the remaining task is a direct-path dyad u of (P3) with $U \lesssim u \leq \lambda^{b_\ell}$, where $b_\ell > 0$ is sufficiently small relative to b_s . Bounded overlaps already suffice when U is large. Fix all parameters other than γ_0, γ_u , and start at $\gamma_u = 0$. At this endpoint we minimize over an auxiliary quantile r : the matrix removal is $C(r) > 0$, the field addition is $C(r)r$, and the penalty is $\int C(r)r^2/4$. We will specify its density constraints below. Writing $X = x/u$, it is enough to prove the following two assertions for the final constraints:

- (i) averaged excess of this optimized value over f_m^0 at most $O(x^4 u \lambda^{1-2s_0})$;
- (ii) for the used optimizers, on $\text{supp}(c_u)$, $c_u = -\partial_{\gamma_u} b$, one has $C(r) \gg c_u$ and

$$\mathbf{E}_{\gamma_0} \left[\sup |r - q|^2 \right] \lesssim (u X^2 \lambda^{-s_0})^2. \quad (\text{C3})$$

To see this, compare a minimizing auxiliary path to the actual endpoint $\gamma_u = 1$. Starting at that actual path, the square interpolation removes $C(r) - c_u$ and adds $C(r)r - c_u q$. The increase of the restored pressure R between the two endpoints is therefore at most the optimizer excess plus $\int C(r)c_u(r - q)^2/[4(C(r) - c_u)]$. Averaging this bound controls the nonnegative derivative in (P10); summing over dyads costs at most a logarithm. Before differentiating on a smoothed size path, transfer the same pressure difference by (P11). Its cost is $O(x^{5+c})$, since only the restoration $-\int b q^2/4$ affects the difference. Fixed rescalings of mass change these bounds only by bounded factors.

The second requirement is an estimate for every optimizer outside one common null set of the global parameter. It is enough to construct a measurable scalar majorant for all these optimizers; no choice of an optimizer needs to be integrated. The remaining subsections establish these two local assertions and hence complete (C1). Both parts of (C2), with a saving independent of the local strengthening depth, have already been proved above. $\square$

7.2 Local auxiliary comparisons and coarse profiles

For the local argument normalize each direct path to length 1, reusing notation for its matrix and field coefficients multiplied by ℓ^2 , the scaled scalar clock and moments, and writing v for normalized mass. Thus $q(v)$ here is the old $q(\ell v)$, and q and q^{-1} have bounded derivatives and positive slopes on fixed initial intervals. All pressure bounds (with their corrections) change by bounded normalization factors. The scale u of (C3) need not be rescaled: its relevant mass range is still contained in a fixed compact positive band of scale u . Fix such a dyad, with $\gamma_u = 0$. In particular

$$x \lambda^{-s_*/2} \lesssim u \leq \lambda^{b_\ell}, \quad X = x/u, \quad \Lambda = \lambda^{k_c}, \quad A = X^2 \Lambda^{-1/3}, \quad a_{\text{goal}} = X^2 \lambda^{-s_0}. \quad (\text{C4})$$

Thus $U = x \lambda^{-s_*/2}$ for large sizes here. Use an arbitrarily small fixed $\zeta > 0$ (depending on previously required power margins including p_*, b_ℓ , decreased as needed). Apply [19, Proposition 6.6] with

$$d_{\text{PE}} = c_d u \Lambda^{1/3}, \quad E = x^{4-\zeta} d_{\text{PE}}, \quad C(r) = C_0 f(C_0 r / d_{\text{PE}}^3), \quad C_0 = u^2 \Lambda,$$

i.e. at strength C_0 ; f here is the PE taper of (6), constant 1 near 0. Choose the fixed c_d sufficiently large to have this constant profile on $[0, c_I u]$ for an arbitrarily large fixed c_I , in particular well beyond the compact range of scale u where (C3) will be tested.

PE power slack (its small parameter in (3), and δ) is chosen sufficiently small relative to ζ . Write $k(r)$ for the PE floor (5), $v_* = (mE)^{-1} \asymp x^\zeta uA$. These parameters satisfy (3) since $X \lesssim \lambda^{s_*/2}$, and $C_0 \leq d_{\text{PE}}^2 = o(1)$. Conditions (4)–(6) with base $T_m \rightarrow T > 1$ hold by (P3)–(P9); all masses are changed, with no radius wall, and a fixed positive covariance margin throughout. Use the common-base version explained at (G1). The optimized excess over the actual untilted pressure is thus $\lesssim x^{4-2\zeta} u\Lambda^{1/3}$.

The modified optimizations below, at this same strength and with the same objective $f_m(b - C(r), h + C(r)r) + \int C(r)r^2/4$, use inverse constraints on $[0, 1]$

$$\bar{k}(r) \leq \alpha'(r) \leq M_j, \quad \bar{k}(r) = k(r) + \kappa \mathbf{1}_{r \in I}, \quad 0 \leq \kappa \lesssim \lambda^{2s_0} \Lambda^{-2/3}. \quad (\text{C5})$$

Here I/u is a fixed closed subinterval of $(0, c_I)$, and $M_j = 2^{j+1}x^{-100}$, with only finitely many fixed stages $j \geq -1$. The inverse distribution α maps onto $[0, 1]$; its inverse $r = \alpha^{-1}$ is the forward quantile. The initial choice $\kappa = 0$, $j = -1$ contains the original PE class. All these constraints are deterministic and independent of γ_0 . Compactness and the segment continuity in (A1) give a minimizer. In the local argument, the symbols S, B, D, X_v, W_v refer to the changed path of any such minimizer, unless stated otherwise.

Lemma 7.2 (Modified optimizer geometry). *For every minimizer in (C5), the first-variation potential can be normalized so that its derivative is*

$$\mathcal{H}' = C(r)[(2-p)r - 2(1-p)S - pB/r], \quad p = -rC'/C. \quad (\text{C6})$$

The minimizing density equals the cap M_j where $\mathcal{H} < 0$ and the floor $\bar{k}$ where $\mathcal{H} > 0$; the remaining points are contacts. Each cap has radius width at most M_j^{-1} . Outside the floor, apart from a null set of contact exceptions, $|S - r| \lesssim_\circ D + O(x^{90})$. At contacts where C is constant, $S = r$; on every cap contained in that constant region, $|S - r| \leq M_j^{-1}$. These conclusions are the parts of the geometry of [19, Lemma 3.3, equations (9)–(10)] that will be used below.

Proof. The first variation minimizes a linear functional of α' , subject to its prescribed lower and upper bounds and total mass one. The multiplier for total mass adds a constant to the primitive $\mathcal{H}$. With this choice, the minimizing density equals M_j where $\mathcal{H} < 0$, and $\bar{k}$ where $\mathcal{H} > 0$. These are the cap and floor regions, respectively; the remaining points are contacts. Differentiation of inverse monotone maps is justified as at (G1); the radii at which a jump is crossed form a null set.

The monotonicities of S, B, p , together with the zero initial wall, give the endpoint signs, almost-everywhere contact relations, and the cap bounds in PE (9)–(10). Each cap has radius width at most M_j^{-1} . We call cap and contact points *nonfloor points*, discarding the null set of contact exceptions. There $|S - r| \lesssim_\circ D + O(x^{90})$; on a component in the constant part of C , the stronger bound is $|S - r| \leq M_j^{-1}$. Also $S \leq r$ at almost every contact. On a negative cap, S cannot exceed its right endpoint radius: otherwise $B \geq S^2$ and monotonicity make $\mathcal{H}' < 0$ on a terminal portion of the cap, contradicting its endpoint sign. At the outer endpoint the same argument uses $S \leq 1$.

□

Put

$$y_c = x^{-10\zeta}(X\Lambda^{-1/6} + X^{3/4}\lambda^{0.12}), \quad w_c = y_c u.$$

Here and in this local argument δ on a PE power denotes δ_{PE} as recalled at (G1). In particular $a_{\text{PE}} \asymp x^{1-\zeta}$; $w_c/u \ll \sqrt{\Lambda}$, x/w_c is power small, and $w_c \geq m^\delta \sqrt{m}E/\sqrt{C_0}$. Indeed $X \lesssim \lambda^{s_*/2}$, $\lambda \leq x^{p^*}$ allow the prefactor $x^{-10\zeta}$ in the first assertion if ζ is small enough. Also $C(w_c) = C_0$, $C_0 w_c^2 \gtrsim x^{-18\zeta} mE^2$.

Proposition 7.3 (Comparability and the calibrated initial slope). *For every optimizer in (C5), uniformly,*

$$r(v) \asymp v \quad (w_c \leq v \leq 1), \quad \sup_{cu \leq v \leq Cu} |r(v) - q(v)| = o(u) \quad (\text{C7})$$

for each fixed $0 < c < C$ (fixed numeric multiples, not the removal profile).

The proof has two uses for concentration estimates. During the comparability contradiction they are available only outside a possible irregular region; after comparability is proved they apply with the threshold w_c . We first give this conditional argument, without assuming the conclusion of (C7).

7.2.1 Analytic estimates conditional on outer regularity

Fix an optimizer of (C5) and a scale $w \geq w_c$ with $w = o(u)$. Suppose that, for fixed positive constants, its quantile is comparable to mass above $C_{\text{reg}}w$. Thus $r(v) \asymp v$ there; by monotonicity $r(v) = O(v + w)$ everywhere. We call bands lying beyond a sufficiently large fixed multiple of this threshold regular. The scale bounds used in the argument are

$$w \geq m^\delta \sqrt{mE}/\sqrt{C_0}, \quad C(w)w^2 \gtrsim m^{1+\delta}E^2, \quad m^{2\delta}v_* \ll w.$$

In both applications below these follow from $w \geq w_c$. The assumption $w = o(u)$ places the enhanced-floor patch I in the regular region. All assertions in the next lemma are conditional on this fixed optimizer and these outer bounds. No minimizer is selected as the parameters vary.

Lemma 7.4 (Moments under outer regularity). *For every fixed $2 \leq p < \infty$, at every specified revelation cut and fork,*

$$\|X_v\|_p \lesssim_p \sqrt{v + w}, \quad \|Q_v\|_p \lesssim_p v + w \quad (\text{C8})$$

where Q_v is the overlap of a pair splitting at mass v . On a regular small radius band $J \asymp v$,

$$\left\| \int_J W_{\alpha(r)} dr \right\|_{p/2} \lesssim_{p,\circ} x^6 / C(v).$$

When a preceding regular projection interval of comparable scale fits, the synchronized two-branch residual has norm $O_{p,\circ}(x^3/(v\sqrt{C(v)}))$. The constants are uniform over optimizers with the stated outer bounds.

Proof.

Raw moments under outer regularity. At regular dyads of scale v (mass and radius comparable beyond a sufficiently large fixed multiple of w) one has field slope at least $C(v)/O(\log(1/x))$ in radius, and

$$mC(v)k(v)v^4 \gtrsim m^\delta C(v)v^2/(mE^2)$$

diverges polynomially. We have $S = O(v + w)$ by looking ahead to a nonfloor point (a band of mass comparable to regular v has floor mass $o(v)$) or by monotonicity below regular scales. We will prove the raw bounds stated above by an absorption argument, following PE (12).

Fix the optimizer, all path parameters, and $l \geq 1$. Define

$$Q_{\text{raw}} = \max \left\{ 1, \sup_t \frac{\|X_t\|_{4l}}{\sqrt{v(t) + w}} \right\},$$

where t ranges over deterministic revelation cuts, including the specified intermediate cuts, and $v(t)$ is their mass. At fixed size this is a finite deterministic number: $w > 0$ and $|X_t| \leq 1$. We will bound it uniformly, without assuming an upper bound on the mass of a cap. On a regular small radius band $J \asymp v$, we claim

$$\left\| \int_J W_{\alpha(r)} dr \right\|_l \lesssim_{l,\circ} Q_{\text{raw}}^2 x^6 / C(v). \quad (\text{C9})$$

Fork an independent continuation at each radius r . Decompose both residuals into future martingale increments, using the weighted Hessian representation in Lemma 6.2. Retaining its mass weight is necessary here. First use dyadic distances from a sufficiently small inverse power of size up to a fixed fraction of v . Then use doubling scales up to a fixed positive radius and the terminal layer. For a pair of layers with largest scale $z \gtrsim v$, the Hessian weight is $O(z)$; for two short layers it is $O(v)$.

For short layers, let h' be the starting distance of the later one. Unless both are the first increment, (A3) bounds the squared product by the earlier squared anchor times $x^{6-\epsilon}/(v^2 C(v) h')$, apart from a negligible tail. Here one first uses arbitrarily high fixed moments for the normalized anchor, then truncates; boundedness of the actual increments controls the discarded tail. The shifted-grid and dual conditional-expectation estimates used for (G4) bound the integral of the earlier squared increment by $O_l(h' Q_{\text{raw}}^2 v)$ in L^l . Thus the three factors in a short-layer contribution are

$$\underbrace{v}_{\text{Hessian weight}} \underbrace{\frac{x^{6-\epsilon}}{v^2 C(v) h'}}_{\text{projection}} \underbrace{h' Q_{\text{raw}}^2 v}_{\text{integrated anchor}} = \frac{x^{6-\epsilon} Q_{\text{raw}}^2}{C(v)}.$$

Two first increments are negligible by the square-sum estimate and the choice of their inverse-power length.

For a larger layer of scale z , there is a preceding regular interval of length and mass comparable to z . Projection costs $x^{6-\epsilon}/(z^3 C(z))$, and the squared opposite anchor has L^l norm $O_l(Q_{\text{raw}}^2 z)$. Its integrated contribution is therefore at most

$$z \frac{x^{6-\epsilon}}{z^3 C(z)} Q_{\text{raw}}^2 z |J| = \frac{x^{6-\epsilon} Q_{\text{raw}}^2 |J|}{z C(z)} \lesssim \frac{x^{6-\epsilon} Q_{\text{raw}}^2}{C(v)}.$$

The last step uses $|J| = O(v)$ and $z C(z) \gtrsim v C(v)$, also at the fixed outer cutoff. Summing the logarithmically many layer pairs proves (C9).

By (A1) and the floor lower bound, the centered average of L over a future radius window of length comparable to v has L^{2l} norm at most $\varepsilon_m v Q_{\text{raw}}$, where $\varepsilon_m \rightarrow 0$ with a power saving. At scale v , its bound is $O_\circ([m C(v) k(v) v^4]^{-1/2})$; the polynomial divergence proved above absorbs the subpower loss. Conditional expectation back to an earlier cut bounds $|X|^2/2$ by this average. Its deterministic mean is $O(v)$. For a cut inside the nonregular region, choose the window at a fixed larger regular scale comparable to w ; for a regular small cut, choose it at a fixed multiple of that cut's scale. Macroscopic cuts use $|X| \leq 1$. Taking the supremum over these deterministic cuts gives

$$Q_{\text{raw}}^2 \leq C_l + \varepsilon_m Q_{\text{raw}}.$$

Finiteness of Q_{raw} now allows absorption and proves its uniform bound. For a small fork, project both branches into the same type of later window and telescope the remaining synchronized layers, as in PE (16). If v is the window scale, the residual is $O_{p,\circ}(x^3/(v\sqrt{C(v)})) \ll v$. This proves (C8), and the PE (16) postprojection bounds whenever the required preceding regular intervals fit. $\square$

Lemma 7.5 (Second moments with an enhanced floor). *Under the same outer-regularity hypotheses, on regular mass dyads, with $B_f = X^{3/2} \lambda^{s_0/2} \Lambda^{-5/12}$,*

$$(\text{avg } D^2)^{1/2} \lesssim_\circ (x^6/(C(v)v))^{1/3} + x^3/(v\sqrt{C(v)}) + m^{\delta/2} v_* + \mathbf{1}_{v \gtrsim u} u B_f. \quad (\text{C10})$$

The average is normalized by the length of the mass dyad. The indicator denotes a fixed band covering the enhanced floor, when present. The constants are uniform over optimizers with the stated outer bounds.

Proof.

Second moments and the enhanced floor. For the nonfloor contribution to (C10), use the whole-cap trace and bracket estimates PE (20)–(21), with cap width $\leq M_j^{-1}$, (C6) giving the same ΔS and contact hypotheses, and the admissible changed clocks giving (A1). Enlarge bands to include complete caps with comparable masses by regularity and the small radius width. This gives the first two terms. Floor mass from k uses simply (C8).

For a nonzero extra floor, use sliding radius windows on I , where $\alpha' \geq \kappa$ and $S \lesssim u$. Equation (A1), integrated in radius, gives

$$\int \mathbb{E} W \, dr \lesssim x^6/C_0, \quad \int \mathbb{E} \text{tr } C_v^2 \, dr \lesssim x^6/(C_0 u).$$

Here C_v denotes the spin covariance and the masses are comparable to u . Let $\bar{L}_{r,h}^+$ and $\bar{L}_{r,h}^-$ be normalized averages on the following and preceding radius windows of length $h = t'u$. Applied in mass coordinates, their deterministic weight is $1/(h\alpha')$. Let I_h omit radius strips of width h at the ends of I , so both windows remain where $\alpha' \geq \kappa$. Equation (A1) and Fubini give

$$\int_{I_h} \text{Var}(\bar{L}_{r,h}^\pm) \, dr \lesssim \frac{1}{\kappa h} \int_I \mathbb{E} W \, dr.$$

The single inverse power of h comes from integrating the moving window, whose multiplicity is at most h .

For two ordered cuts $i < j$, use the identity

$$L_j - L_i = (y - X_j) \cdot (X_j - X_i) + \frac{1}{2} |X_j - X_i|^2.$$

Its noise term has second moment at most

$$(\mathbb{E} \text{tr } C_j^2)^{1/2} (S_j - S_i + C'(D_i + D_j)).$$

Average this identity to the right and left of the tested cut. The radius-window kernels

$$K_h^\pm(i, j) = h^{-1} \mathbf{1}_{\{0 < \pm(j-i) < h\}} \mathbf{1}_{\{i \in I_h\}}$$

are supported in $I \times I$, and each marginal has density at most one. Monotonicity and $\text{osc}_I S = O(u)$ give $\iint K_h^\pm(S_j - S_i)^2 \, di \, dj = O(hu^2) = O(t'u^3)$. Also $\iint K_h^\pm(D_i + D_j)^2 \leq 4 \int_I D^2$. Define

$$\mathcal{D} = u^{-3} \int_I D^2 \, dr, \quad \mathcal{T} = u^{-3} \int_I \mathbb{E} \text{tr } C_{\alpha(r)}^2 \, dr \lesssim A^3.$$

Cauchy–Schwarz therefore bounds the integrated noise, divided by u^3 , by $C\sqrt{\mathcal{T}(t' + \mathcal{D})}$. Combining the side-average variance bound, the raw endpoint strips, and $D^2 \leq 4 \text{Var } L + \mathbb{E} \text{tr } C_v^2$, we obtain

$$\mathcal{D} \leq C \left(t' + \frac{A^3}{\kappa t'} + \mathcal{T} + \sqrt{\mathcal{T}(t' + \mathcal{D})} \right).$$

Young's inequality absorbs the part involving $\mathcal{D}$: $C\sqrt{\mathcal{T}(t' + \mathcal{D})} \leq (\mathcal{D} + t')/2 + C'\mathcal{T}$. Thus the two-sided sliding estimate is

$$\frac{1}{u^3} \int_I D^2 dr \lesssim t' + A^3/(\kappa t') + A^3, \quad 0 < t' \leq c.$$

Boundary strips of width h use the raw moments. Choose t' to balance its first two terms, or use the raw bound if that choice exceeds the allowed interval. After multiplying by the extra density κ , its contribution is $O(A^{3/2}\sqrt{\kappa}) \leq O(B_f^2)$. This supplies the final term of (C10). $\square$

7.2.2 Calibration and closure under strong removal

In addition to the outer-regularity hypotheses, suppose now that $C(w) \gtrsim w^2$, and set

$$W = \max(w, \sqrt{C(w)}) \asymp u\sqrt{\Lambda} = o(u).$$

The endpoint estimates (P9) are assumed on the regular outer region. This stronger setting is used only in the strong-removal branch of the profile contradiction.

Lemma 7.6 (A common calibration cut). *On compact target bands with $cw \leq v \leq Cw$ and $r = O(w)$,*

$$\int D^2 dv = o(w^3), \quad \int D^2 dr = o(w^3), \quad \int_0^{C'W} D_v dv = o(w^2)$$

for every fixed C' . There is a deterministic nonfloor cut $$ in a regular band of scale W , above any prescribed fixed multiple of w , for which $S_* \asymp W$ and*

$$D_* \leq x^{c_*} w^2 / W, \quad \|Q_* - S_*\|_p \leq C_p x^{c_*} W$$

for every fixed p and some $c_ > 0$ independent of p . The cut is chosen from deterministic second moments and geometry before spin sampling; it is the same cut for all these moment estimates.*

Proof.

Inner mass and radius estimates. For $v \gtrsim W$ the right side of (C10) is polynomially smaller than w^2/W ($vC(v) \gtrsim W^3$); for the extra floor this uses

$$uB_f W / w_c^2 \lesssim \lambda^{s_0/2 + k_c/12 - 0.24}.$$

On regular bands between a large fixed multiple of w and scale W , (C10) saves a power relative to $w^2/(W^2 v)^{1/3}$. The extra floor is absent there. We also need

$$\int D^2 = o(w^3)$$

on bands with mass between fixed positive multiples of w and radius $O(w)$, separately in mass and radius. In radius, apply the sliding estimate with $S = O(w)$, field slope C_0 , and floor lower bound $k(O(w))$. Its errors vanish because $mC_0 k(O(w))w^4 \gg 1$; narrow domains and boundary strips use (C8).

In mass, the floor contributes $o(w^3)$, since its mass is $o(w)$ and the raw scale is $O(w)$. On caps and contact, use PE (20)–(21) with that same raw scale. If a cap crosses the lower

cutoff cw from below $cw/2$, use its portion above $cw/2$. This portion is a fixed fraction of its total mass, because outer comparability puts its upper endpoint at $O(w)$. The whole-cap bound on ΔS and the constant cap density give the trace estimate on that portion. Thus the trace bound is always used with a positive lower mass cutoff. The condition $mC_0w^4 \gg 1$ finishes the mass estimate. Summing these bands and the regular intermediate estimates, and using (C8) below them, gives $\int_0^{C'W} D dv = o(w^2)$ for every fixed C' .

A good calibration cut. Choose a regular band of scale W , above any required fixed multiple of w . The constant removal, (C6), and (C10) give a nonfloor cut $*$ there with $S_* \asymp W$ and D_* polynomially smaller than w^2/W . This selection uses deterministic moments of the fixed optimizer and parameters. It is made before sampling spins and independently of a moment order.

At this same cut, $\|Q_* - S_*\|_p \ll W$ for every fixed p , with a saving exponent independent of p . To prove this, take earlier and later radius buffers of length $h' = x^{c_\theta}W$, with fixed $c_\theta > 0$ sufficiently small. Integrating (C6) to or from $*$ shows that S varies by $O(h')$ on fixed enlargements: use $\mathcal{H}(*) \leq 0$, the local lower bound $\mathcal{H} \geq -O(C_0/M_j^2)$, and monotonicity of S . Equations (C9) and (A1) make the centered averages on these buffers polynomially smaller than W . The past/future brackets and common-past comparison in Lemma 6.3 then give the same improvement for $\|X\|^2 - S_*\|_p$ and for squared path increments in L^p . Finally project both leaves to a central cut a fixed fraction of h' beyond their fork. Equations (A3) and (C8), with synchronized residual layers, bound its residual by $O_{p,\circ}(x^3 \sqrt{W/h'} / (W \sqrt{C_0}))$. The two normalized errors just used are bounded, up to subpower losses, by

$$[mC_0k(W)W^4]^{-1/2}(W/h'), \quad [mC_0W^4]^{-1/2}(W/h')^{1/2},$$

respectively. Both save a power when c_θ is small enough, independently of the fixed moment order. This radius-window argument uses the floor lower bound $mC(v)k(v)v^4 \gg 1$; it does not require the upper-cap bound used in PE's original mass-window proof. $\square$

Set

$$\widehat{K} = h + C(r)r + (b - C(r))S,$$

and let $\Gamma_{\widehat{K}}$ be the scalar moment curve for this entire clock path.

Proposition 7.7 (Closure for the modified constraints). *Along any fixed sequence of optimizers with the strong-removal hypotheses above and fixed outer-comparability constants, the following estimate holds for targets i with $cw \leq v_i \leq Cw$ and $r_i \leq Cw$,*

$$\frac{|S_i - \Gamma_{\widehat{K}}(\widehat{K}_i)|}{w^3} \leq C'\{\varepsilon_m + (D_i/w)^\vartheta\}, \quad \varepsilon_m \longrightarrow 0, \quad \vartheta > 0. \quad (7.1)$$

The constants and modulus are common to the fixed target band. Thus this difference is $O(w^3)$ throughout the band, and is $o(w^3)$ along any moving target with $D_i = o(w)$. Since the target-band integrals of D^2 are $o(w^3)$, the same modulus gives, separately in the two coordinates,

$$\frac{1}{w} \int \frac{|S_i - \Gamma_{\widehat{K}}(\widehat{K}_i)|}{w^3} dv_i = o(1), \quad \frac{1}{w} \int \frac{|S_{\alpha(r)} - \Gamma_{\widehat{K}}(\widehat{K}_{\alpha(r)})|}{w^3} dr = o(1). \quad (7.2)$$

Each integral is restricted to the target band just described.

Proof. The constraints (C5) differ from those in the statement of PE’s closure proposition. We therefore use its proof, and verify the replacement inputs below. A *block* here consists of leaves sharing through a fixed continuity cut of mass comparable to W . A boundary edge joins two such blocks. The tests used in that proof are the fixed finite products of centered overlap factors and projected factors in [19, Definition 4.2]; their natural raw size is denoted by p_0 . The constants may depend on that fixed degree. All centers are the actual moments S , held fixed during removal of one site.

- (i) *Projection estimates and row changes.* Use the selected star $*$ and a regular continuity cut at a large fixed multiple of W to form the blocks. For each projection at scale v , outer comparability supplies a preceding radius interval of length comparable to v , after the low fork, with masses comparable to v . Its field length is at least $C(v)v/O(\log m)$. Above a sufficiently large fixed multiple of W , $vC(v) \gtrsim W^3$. Define $z_v = w^3/(W^{3/2}\sqrt{v})$. The postprojection bound and the star-anchor version of (A3) then read

$$\frac{x^3}{v\sqrt{C(v)}} \lesssim (x/w)^3 z_v, \quad \frac{x^3\sqrt{W}}{\sqrt{v^3 C(v)}} \lesssim (x/w)^3 \frac{w^3}{Wv}.$$

The factor $(x/w)^3$ is power small. These are the two boundary estimates in PE (13), (16), and its Section 4, “Analytic input and finite-degree tests”. They also hold for conditional replacements. The raw bounds (C8), the good-star bound, and the second-moment estimates above supply its remaining inputs (12), (17)–(19), with m, w, W in place of n, x, W .

The clock $\widehat{K}$ is monotone, bounded, and $O(v + w)$. Above the regularity threshold this follows from outer comparability, (C8), and (P9); below it use monotonicity. Consequently the scalar row sum χ in that proof is bounded above and below by positive constants, and its two-spin coefficient at distinct high mass v is $O(v)$. The bounded-row comparison of [19, Lemma 4.3] applies: all clocks remain bounded and genuine, and the ordinary field increments on the remaining sites do not change. Their raw projection norms also follow by conditional Jensen from positive overlap tests.

- (ii) *The finite-degree expansion and calibration.* The row interpolation replaces the cavity-spin kernel by

$$h + C(r)r + (b - C(r))Q^-,$$

where Q^- is the overlap with the deleted site omitted but with the original normalization. Its difference from $\widehat{K}$ is $(b - C(r))(Q^- - S)$, up to the $O(m^{-1})$ common-noise convention. Every row interpolation preserves positive covariance. Thus the row expansion in [19, Lemmas 4.5, 4.7 and 4.8], (23)–(25), has the same algebra and the same centers. Here is how its errors are bounded after substitution.

A high error integrated with bounded split density, including repeated vertices in one dyad, costs a power-small multiple of w^2/W . The error at $*$ has the same bound. Ordinary low factors cost $O_p(W)$, or $O_p(w)$ if their split is at the target scale. A varying low error costs $\int_0^{C'W} D = o(w^2)$; an error at the old target costs $D_i + O(m^{-1})$, with positive powers of D_i/w after moment interpolation. Deleting the row costs $(mw)^{-1} = o(w^3)$. The observed-block truncation step uses variations power-smaller than Wz_v and p_0z_v/w , which were supplied in (i). These estimates justify the truncation before any signed sums are bounded absolutely.

For completeness, suppress the common raw test factor p_0 . In the terms with three low insertions, an integrated error is $o(w^2)$ and its other low factors cost $O(W^2)$ or $O(W^3)$. If the error is at the old target and $w \ll W$, one additional factor has scale w . When necessary, this factor is obtained by turning off revelation below a smaller block

cut of scale w , then restoring it once; the zero-field sign symmetry cancels the term before restoration. High remainders contain either one high error and two low factors, or two high errors and one low factor. Their bounds are power-small multiples of Ww^2 and w^4/W , respectively. Moments slightly above two suffice for the integrated errors; their interpolation loss is paid by the strict power saving.

If a third block is already observed, first project the fresh endpoint in the other target block to the star and collapse its scalar row to χ . Next replace the factor at the second leaf of the third block by its value at the first leaf. The postprojection estimate in (i) gives a power-saved $w^3/(Wv)$ for this replacement. Its scalar coefficient is $O(v)$, and the other low factor costs $O(W)$, so the total is $o(w^3) \leq o(Ww^2)$. After this replacement that leaf's scalar row also collapses to χ . These are the two row reductions in [19, Lemma 4.7, observed-third-block case].

Let $v_{\text{blk}} \asymp W$ denote the continuity cut defining the blocks. The tests have no overlap factor internal to an old block; its old internal relationships occur only in fixed scalar coefficients and the prescribed topology. Only the unweighted attachment of the first leaf in the third block remains. At its fresh-at-cut value the summand is independent of the deeper attachment, so the signed row-sum rule, including all old-fork atoms and coincidences, gives the factor $v_{\text{blk}} = O(W)$. At the star the baseline product difference is $O(WD_*)$; its contribution is therefore

$$O(v_{\text{blk}}WD_*) = O(W^2D_*) = o(Ww^2).$$

Restoring the deeper attachment uses the boundary comparison of [19, Lemma 4.5]; its high-layer bound is

$$\varepsilon_m v(Wz_v)(z_v/w) = \varepsilon_m w^5/W^2 \leq \varepsilon_m Ww^2, \quad \varepsilon_m \rightarrow 0,$$

where ε_m has the fixed power saving supplied by (i). If the third block is unobserved, its varying low error costs $W \int_0^{O(W)} D = o(Ww^2)$; a star attachment has the same entry factor $O(W)$. At the old target the two low factors have scale w , with the positive-power modulus in D_i/w described above. Thus both observed and unobserved third blocks obey the required remainder bound.

For the leading scalar-row replacement, we spell out the test difference in [19, Lemma 4.8, proof of the first replacement]. By linearity take an admissible monomial

$$P_a = HY_a^k E_{oa}, \quad P_c = HY_c^k E_{oc}, \quad Y_d = X_* \cdot y_d - S_*, \quad E_{od} = Q_{od} - S_j,$$

where H uses only exterior leaves, the incoming edge has split $j \in \{i, *\}$, and the edge factor is omitted if absent. The leaf c is a fresh relative of a in its block. Let $\mathcal{F}$ be the complete target-trunk prefix through $*$, excluding exterior continuations and subsequent target innovations. Define ΔP_{ac} by averaging $P_a - P_c$ over the entire exterior joint continuation, conditional on $\mathcal{F}$ and the joint future of a, c . The signed exchange identity is

$$\sum_c \gamma_{ac} \nu[(Y_c - Y_a)P_a] = \frac{1}{2} \sum_c \gamma_{ac} \nu[(Y_c - Y_a)\Delta P_{ac}],$$

where γ_{ac} is the scalar two-spin coefficient with revelation below the block cut moved to that cut, and the sum uses the signed attachment rule within the block. This is the source's first symmetrization, followed by conditional expectation.

Put $\mathfrak{s}_i = w$, $\mathfrak{s}_* = W$, and let p_H be the raw scale of H . Then $p_0 = p_H W^k \mathfrak{s}_j$ with the incoming edge and $p_0 = p_H W^k$ without it. The exterior anchor $V = \mathbb{E}[Hy_o \mid \mathcal{F}]$ obeys

$$\|V\|_p \leq \|H\|_{2p} \|Q_j\|_p^{1/2} \lesssim_p p_H \sqrt{\mathfrak{s}_j}.$$

This is the conditional Cauchy–Schwarz estimate in that same source proof. It uses the ordinary split- j marginal; no exterior terminal spin is conditioned upon. The target projection powers stay outside this averaging and outside the anchor. For a regular innovation interval before a high split of scale v , write the unit-anchor bound as

$$\pi_v = O_{p,\circ} \left(\frac{x^3}{\sqrt{v^3 C(v)}} \right) \leq \varepsilon_m \frac{w^3}{W^{3/2}v},$$

with ε_m power small. Then $\|Y_c - Y_a\|_p \lesssim \sqrt{W}\pi_v$. Telescoping $P_a - P_c$, a changed projection power costs $p_0\pi_v/\sqrt{W}$; the changed incoming edge, after the exterior average, costs $p_0\pi_v/\sqrt{5_j}$. Apply projection before Hölder with the remaining powers, as in PE’s test-difference estimate. Thus

$$\|\Delta P_{ac}\|_p \lesssim \frac{\varepsilon_m p_0}{v} \left(\frac{w^3}{W^2} + \frac{w^{5/2}}{W^{3/2}} \right).$$

The norm is of the exterior-averaged difference, not of the original incoming-overlap difference. At a distinct split, the attachment variation per dyad and γ_{ac} each contribute $O(v)$. Consequently the symmetrized error per dyad is at most

$$Cv^2 \left(\frac{\varepsilon_m w^3}{Wv} \right) \frac{\varepsilon_m p_0}{v} \left(\frac{w^3}{W^2} + \frac{w^{5/2}}{W^{3/2}} \right) \lesssim \varepsilon_m^2 p_0 W w^2,$$

since $w \leq W$. The saving pays the dyadic sum. For a connected high insertion, one $O(v)$ placement factor suffices, because its integrated high error supplies the additional saved w^2/W ; this remains true when the vertices share a dyad.

This verifies the remainder estimates in PE (24)–(25). No variation of the optimizer occurs in this finite-tree argument. We spell out how the finite recursion preserves their uniform target modulus. For a block $\mathcal{A}$ at the chosen continuity cut v_{blk} , let ν^{sc} be the scalar law with revelation below v_{blk} moved to that cut, and put $\gamma_{ac} = \nu^{\text{sc}}[\tau_a \tau_c]$. The connected insertion of PE (23), paired with a test P , is

$$\mathcal{L}(P, a) = \frac{1}{2} \sum_{\substack{c,e,f \in \mathcal{A} \\ e \neq f}} (\nu^{\text{sc}}[\tau_a \tau_c \tau_e \tau_f] - \gamma_{ac} \gamma_{ef})(b - C)_{ef} \nu[P(Q_{ef} - S_{ef})].$$

Here S_{ef} is the actual moment at the split of e, f , and all three added labels use the signed within-block attachment rule. The coefficients and expectations are evaluated on the same final topology. This formula defines the insertion through its pairing; no positive probability interpretation of the signed sum is used.

At the star, $\widehat{K}_* = T_m S_* + O(W^3)$ and the changed matrix is $T_m + O(W^2)$. Put $J_0 = T_m \chi^2 - 1$. The recursion of [19, Lemma 4.9] divides by $\chi \widehat{K}_* \asymp W$. The remaining pairing of the same degree at the partner root is removed by applying the row identity to two fresh interchangeable roots absent from P . Their equal leading pairings have total coefficient $2\chi \widehat{K}_*$; every remaining term adds a star factor at cost $O(W^{-1})$. At depth k , accumulated coefficients are $O_k(W^{-k})$ and every unresolved term has k new centered star factors. Lemma 7.6 therefore bounds its terminal contribution by $O_k(p_0 x^{kc_*})$. Choose k fixed so that $kc_* > 2$, using $w \geq w_c \gg x$; then this is $o(w^2 p_0)$. Only after choosing k do we fix the finite degrees and Hölder orders. The intermediate errors give

$$|\mathcal{L}(P, a)| \leq Cw^2 p_0 \left\{ \varepsilon_m + (D_i/w)^\vartheta + \varepsilon_m |J_0|/W^2 \right\}, \quad \varepsilon_m \rightarrow 0,$$

with the target term absent for star-only tests. There are only finitely many target exponents, so their positive minimum gives one ϑ for the whole fixed target band.

Return to the finite-degree row identity with $P = 1$, as in [19, Lemma 4.10]. Its surviving high corrections are pairings against one star factor, of raw size $p_0 = W$. Apply the preceding bound to those pairings and use $\Gamma_{\widehat{K}}(\widehat{K}_*) = \chi^2 \widehat{K}_* + O(W^3)$. Indeed, the second scalar coefficient is a two-edge path through an unobserved third block: its $O(W)$ entry factor multiplies two low factors of size $O(W)$, and bounded finite-label variation controls the third-order remainder. This gives

$$S_* = \chi^2 T_m S_* + O(W^3) + O(\varepsilon_m W w^2 (1 + |J_0|/W^2)).$$

Dividing by $S_* \asymp W$ gives

$$|J_0| \leq C W^2 + \varepsilon_m w^2 + \varepsilon_m |J_0| w^2 / W^2.$$

Since $w \leq W$, the last term is absorbed. Hence $J_0 = O(W^2)$, and the calibrated bound is

$$|\mathcal{L}(P, a)| \leq C w^2 p_0 \{\varepsilon_m + (D_i/w)^\vartheta\}.$$

The common base T_m has remained fixed throughout the finite tree; no rate for $T_m - T$ enters these estimates.

- (iii) *Return from the calibration scale to the target.* Repeat the mean expansion with the block cut at a large fixed multiple of w . The two regular-dyad bounds make high errors power-smaller than w , and all low factors now have raw scale w . An old-target error contributes a positive power of D_i/w , while the remaining errors tend to zero uniformly over the target band. The surviving mean equation at this smaller block cut is

$$S_i - \Gamma_{\widehat{K}}(\widehat{K}_i) = 2(b_i - C(r_i))\chi_w \mathcal{L}_w(Y_a^{(i)}, a) + \text{Rem}_i, \quad Y_a^{(i)} = X_i \cdot y_a - S_i.$$

Here $\mathcal{L}_w$ is the same connected pairing with block cut of scale w , and χ_w its scalar row sum. The remainder is bounded by $C w^3 \{\varepsilon_m + (D_i/w)^\vartheta\}$, uniformly in the fixed target band. The changed matrix coefficient is $b_i - C(r_i)$. To use the calibrated insertion, first remove insertion edges between this cut and the calibration block cut v_{blk} . Their cost is

$$O(w) \int_0^{C'W} D = o(w^3).$$

Next change the high scalar kernels; above a sufficiently large fixed $c'W$, the cost is

$$O(wW) \int_{c'W}^1 D = o(w^3).$$

These are precisely the two cutoff changes in PE's "Return to the target cut". Allocate the error pair first relative to the sole old root a in its block, and then the summed leaf c ; this gives the bounded split density used in these integrals without a high-fork atom. After the scalar-kernel change, independent block flips force c into a 's block, while the error endpoints share a block; if that block is different, the connected scalar covariance vanishes. The remaining pairing is therefore $\mathcal{L}(Y_a^{(i)}, a)$ at the calibration-scale block cut v_{blk} . The test $Y_a^{(i)}$ has raw scale $p_0 = w$ and is admissible by its fresh-branch representation. Its calibrated bound is therefore $C w^3 \{\varepsilon_m + (D_i/w)^\vartheta\}$; $b - C$ and χ_w are bounded. This proves (7.1). The already proved target-band D^2 bounds then give both integrals in (7.2).

For example, when $0 < \vartheta \leq 2$, Hölder bounds its discrepancy term by a constant times $(w^{-3} \int D^2)^{\vartheta/2}$; a larger exponent may first be decreased using the raw bound $D = O(w)$. $\square$

Proof of Proposition 7.3. We first exclude a failure of comparability, using a PE child for weak removal and Proposition 7.7 for strong removal. The final step identifies the initial slope by a macroscopic transfer.

Profile arguments allow subselection. If loose fixed comparability fails, choose an offending mass $w \geq w_c$ comparable to the supremum of offending masses. This supplies the outer regularity above a fixed multiple of w . Choose the loose constants strictly outside the bounds of the PE direct profile limits. The scale estimates displayed before Lemma 7.4 follow from $w \geq w_c$.

Step 1: weak removal and feasible barriers. First suppose $C(w) = o(w^2)$. Apply PE finite propagation to a child acting from the changed path of this optimizer, with the data d', E' of [19, Lemma 6.2] at w , and ordinary PE constraints. The data verification in that lemma uses (3) and the parent strength, cutoff and outer bound (not the parent's optimization constraints); hence (3)–(4) hold in the child. It has nonnegative monotone bare paths with bounded clocks and a fixed positive all-mass matrix margin, choosing its fixed d'/w small. Its final error is at most $m^{20\delta} E'$; write z for a final minimizing quantile and $\mathcal{D}(z) = d'^2 f(z/d')$. The movement inequality of [19, Lemma 6.3] applies with parent-removal C to any trial satisfying (C5): its proof only uses feasibility, optimality, the slope bound on C and the square interpolation. We explain the corresponding barriers to handle the modified floor. Here $d' = c_0 w$, $E' = \min(m^{-1/2} d'^2, m^{-100\delta} C(w) w^3)$, $c_0 > 0$ sufficiently small fixed. In particular the parent changed endpoint still tests against the common T_m by the estimates of [19, Lemma 6.2], since $r = O(y)$ on comparable mass bands of scale $y \gtrsim w$, $r = O(w)$ below that.

Suppose that r and z are separated by fixed thresholds whose gap is comparable to y , on a mass interval of length comparable to y , where $y \gtrsim w$. Macro intervals lie strictly inside $(0, 1)$. Failure of agreement in normalized measure produces such an interval by monotonicity, and outer regularity gives $r \leq C'y$ there. Here agreement means that, on each fixed comparable mass band, the mass of $\{|r - z| > \epsilon y\}$, divided by y , tends to zero for every fixed $\epsilon > 0$.

The forward constraints are $l \leq r' \leq 1/\bar{k}(r)$, with $l = M_j^{-1}$. We have $\sup \bar{k} = o(1)$ and $\int_0^1 \bar{k} = o(w)$, the latter using $w \gtrsim u\sqrt{\Lambda}$. Reserve fixed short portions at the two ends of the separation interval for the following barriers. If z is above r , choose an intermediate threshold H and an early subinterval $[a, b]$, with $b - a \asymp y$, and set

$$b_{\uparrow}(v) = lv + \begin{cases} 0, & v \leq a, \\ H(v - a)/(b - a), & a < v < b, \\ H, & v \geq b. \end{cases}$$

Choose $H \asymp y$ in the threshold gap, leaving fixed distance from its ends. The rise has fixed slope $s = l + H/(b - a)$, and eventually $s \sup \bar{k} \leq 1$. Also $l + H < 1$, so $\max(r, b_{\uparrow})$ preserves both endpoints and both slope bounds.

If z is below r , use $b_{\downarrow}(v) = H + lv$ until a late point a . Thereafter solve

$$\int_{H+la}^{b_{\downarrow}(v)} \bar{k}(t) dt = v - a$$

until this curve meets $1 - (1 - v)l$, and follow that line after the meeting. The ramp has derivative $1/\bar{k}$. Since $l \sup \bar{k} < 1/2$, its mass length is at most $2 \int_0^1 \bar{k} = o(w)$, so it fits in the reserved part of the interval. The trial $\min(r, b_\downarrow)$ again preserves both endpoints and both bounds. In both constructions choose the threshold and reserved portions as in [19, Lemma 6.4], so every moved point stays distance $\gtrsim y$ from z . Upward motion has radii $O(y)$, with $r \asymp y$ if $y/w \rightarrow \infty$; downward motion has both parent radii comparable to y .

If $w \geq \lambda^{k_c/4}u$, full clipping already satisfies the strict movement fraction (displaced distance less than $|z - r|\mathcal{D}(z)/(\mathcal{D}(z) + C(r))$). For $y/w \rightarrow \infty$, $y\mathcal{D}(y) \gtrsim w^3 \gg yC(y)$ since $yC(y) \lesssim_o u^3\Lambda$; at $y \asymp w$, $C(0) \ll w^2$. Thus for $z \leq O(y)$ the ratio of strengths is sufficient with our placement margin, and for $z/y \rightarrow \infty$ use monotonicity of $z\mathcal{D}(z)$ to permit any fixed $O(y)$ displacement.

In the remaining case $w < \lambda^{k_c/4}u$ we need separation tests only up to $y \leq m^{2\delta}w \ll u$, since the child is already comparable above $m^\delta d'$ by finite propagation. All active radii are now below the enhanced patch. Use the partial interpolation of [19, Lemma 6.4] with fraction a small constant times $\min(1, \mathcal{D}(y)/C(y))$, forward for upwards, affine in $\int_0^r k$ for downwards. On the active portions $\bar{k} = k$; feasibility and the strict fraction proof in that lemma use just l as lower slope, convexity of the PE k in its integrated coordinate, $C(r) \asymp C(y)$ for $r \asymp y$, $C(0) = o(w^2)$ at $y \asymp w$, and monotonicity of $z\mathcal{D}(z)$. Hence they are unchanged.

For either trial write $\hat{r} = r + \omega(z - r)$. The movement inequality of [19, Lemma 6.3] states

$$\int C(r)\omega(z - r)^2 dv \leq 4m^{20\delta}E', \quad 0 \leq \omega < \frac{\mathcal{D}(z)}{C(r) + \mathcal{D}(z)}.$$

The constructions just verified make its left side at least $cy^3 \min(C(y), \mathcal{D}(y)) \gg m^{20\delta}E'$. Thus no fixed threshold separation can occur. The resulting agreement in normalized measure gives the child outer regularity above w when $w = o(1)$, and agreement on compact positive bands of scale w . Its direct-test hypotheses hold there with $\mathcal{D}(w) \asymp w^2$ and $mE'^2 \lesssim w^4$. Apply [19, Theorem 5.1 and Proposition 5.9] to the child. Its continuous limiting profile transfers to the parent pointwise: bracket any tested mass between two nearby continuity masses, use agreement in measure to select comparison points on either side, and then use monotonicity of r, z . Shrinking the bracket gives the child's loose bounds also at the offending point. Near mass one use $r \leq 1$ as well. This excludes a weak-removal violation.

Step 2: strong removal and the profile contradiction. Suppose $C(w) \gtrsim w^2$. Then $w \leq W = \max(w, \sqrt{C(w)}) \asymp u\sqrt{\Lambda} = o(u)$, so the conditional analytic hypotheses hold. The bare endpoints satisfy (P9) on the regular exterior. Lemma 7.6 and Proposition 7.7 therefore provide the common calibration cut and the closure in both coordinates required next.

We use this closure to identify the scaled optimizer profile, following PE (28)–(29). On each compact band in the scaled radius, $p = 0$ and $C = C_0$. Cap widths remain negligible after multiplication by C_0/w^3 , and the floor mass on each such band is $o(w)$. Equation (C6) has no additional state-dependent factor there.

Subselect the distribution of r/w under mass divided by w . It converges locally to a Radon measure on $[0, \infty)$, whose cumulative function we denote by U_1 . Outer regularity gives $U_1(t) \asymp t$ for large t . The scalar clock $\widehat{K}/(T_m w)$ has the same limiting distribution. Indeed on compact positive bands of scaled mass, $r, S = O(w)$, $|S - r| \leq M_j^{-1}$ almost everywhere off the negligible floor, and $((b - T_m)S + h)/w^3 = o(1)$. Very small scaled masses contribute at most their length. Monotonicity and the comparison on large fixed bands exclude far scaled masses from any fixed clock band.

The profile contradiction. Set

$$g_m(t) = \frac{\Gamma_{\widehat{K}}(T_m w t) - w t}{w^3},$$

extending the scalar clock if necessary. Let a_2, a_4 be subsequential limits of its second and fourth spatial continuation derivatives at zero time and field. The scalar derivative formula and [19, Lemma 5.3], applied to the bounded normalized clocks, give

$$g_m(0) = 0, \quad g_m'' \longrightarrow At - BU_1(t) \quad \text{in } L^1_{\text{loc}}, \quad A = T^3 a_4^2, \quad B = 2T^2 a_2^3.$$

Both constants have fixed positive lower and upper bounds, and the second derivatives are locally bounded. To see the limit, oddness and bounded time and space derivatives give $V_{zzz}(T_m w t, Z) = a_{4,m} Z + O_p(w^{3/2})$, while $\mathbb{E} Z^2/w \rightarrow Tt$. The rescaled clock mass converges at continuity points of U_1 .

Choose a nonfloor point of regular scale comparable to w . Such points exist because the local floor mass is small. There $S \asymp w$ and $\widehat{K} - T_m S = o(w^3)$, since $C_0|S - r|/w^3 = o(1)$. Equation (7.1) bounds g_m at an argument bounded above and below by positive constants. Together with $g_m(0) = 0$ and the curvature bound, this bounds its initial slope. Pass to a C^1_{loc} limit g . The mass estimate in (7.2) allows nonfloor approximants to every support point, giving $g = 0$ on the support; at zero use $g(0) = 0$.

Above the support infimum, subselect further so that $S_{\alpha(wt)}/w \rightarrow s(t)$ almost everywhere. On compact intervals there, $\alpha(wt) \asymp w$. The closure identity is

$$\frac{C_0(S_{\alpha(wt)} - wt)}{T_m w^3} = g_m(\widehat{K}(\alpha(wt))/(T_m w)) + \text{err}.$$

The error is bounded and tends to zero in local $L^1(dt)$, by the radius estimate in (7.2) and the bare-clock bound. If $C_0/w^2 \rightarrow h_1 \in (0, \infty)$, it gives

$$g(s(t)) = h_1(s(t) - t)/T, \quad \mathcal{H}'(wt)/w^3 \longrightarrow 2h_1(t - s(t)).$$

If $C_0/w^2 \rightarrow \infty$, it instead gives $S_{\alpha(wt)} = wt + o(w)$ and $\widehat{K}/(T_m w) = t + o(1)$; the derivative limit is then $-2Tg(t)$. These derivative sequences are locally bounded and converge in local L^1 .

On an interior support gap, the potential $\mathcal{H}(wt)/w^4$ therefore has a nonnegative limiting primitive that vanishes at both ends. Nonfloor approximants pin the endpoint values to zero; for a negative cap, use its nearest right zero within M_j^{-1} . The same cap-width bound excludes a negative excursion inside the gap. If the left endpoint is a first atom, choose approximants after a fixed positive amount of scaled mass has accumulated. The derivative bound then holds between them and the gap, even when the atom is at zero. All these estimates are used only where scaled radii are bounded and scaled masses are bounded above and below.

Monotonicity gives $s(t)$ in the closed gap interval almost everywhere. A positive derivative of g at the left endpoint would force the limiting primitive to decrease immediately to its right. For finite h_1 , the identity $g(s(t)) = h_1(s(t) - t)/T$ forces $s(t) > t$, whether $s(t)$ is near that endpoint or farther into the gap. For infinite strength, the conclusion follows from the derivative $-2Tg(t)$. Both contradict nonnegativity and the zero endpoint value. Reflection excludes a positive derivative at the right endpoint as well. But on a gap of length L , the curvature is $g''(t) = At - \text{const}$ and the endpoint values are zero, so the sum of endpoint derivatives is $AL^2/6 > 0$. No gap is possible.

Above the support infimum l , we thus have $g = 0$ and $U_1(t) = At/B$. If $l > 0$, then $g'' = At$ below l and $g(l) = g'(l) = 0$, contradicting $g(0) = 0$. Consequently $U_1(t) = At/B$

throughout. This is the linear quantile limit PE (28), with the same fixed slope bounds. It contradicts our choice of loose comparability constants and proves the first part of (C7).

The first part of (C7) has now been proved for every optimizer. In particular Lemmas 7.4 and 7.5 can henceforth be used with $w = w_c$ and fixed uniform comparability constants. This is the version of (C8)–(C10) used in the strengthening argument below; it does not invoke the strong-removal closure again.

Step 3: identification of the initial slope. Repeat the weak-removal transfer at $w = u$, now without assuming a violation. We identify the child's linear slope by its macroscopic clock limit. First, $S_{\text{child}} - z \rightarrow 0$ in mass, by PE (18), geometry, and regularity on each fixed interior band. The required error estimates are

$$m\mathcal{D}(v) \gtrsim mu^3, \quad E'/d'^3 \lesssim m^{-1/2}/u, \quad m^{\delta/2}/(mE') \rightarrow 0.$$

Compare z to a further ordinary PE minimizer at final strength, with fixed small removal parameter $d'' > 0$ and $E'' = m^{-1/2}d''^2$. Its macroscopic endpoint hypothesis PE (4) follows from (P9) and the vanishing preceding removals; all-mass admissibility and a positive covariance margin remain. Finite propagation gives error $m^{2\delta}E'' \ll u^3$, and [19, Proposition 5.9] gives the further minimizer's limiting quantile q_T .

The ordinary PE barriers transfer this limit to z . At a fixed interior mass scale, full clipping is feasible with the child's minimum slope $1/M(v)$; both forced lengths are $o(1)$. The movement fraction holds because $\sup \mathcal{D} = o(1)$, whereas the newest removal is bounded below. A fixed threshold separation would cost $\gtrsim \inf \mathcal{D} \gtrsim u^3$, contradicting the propagation error. Thus $h_{\text{child}} + b_{\text{child}}S_{\text{child}} \rightarrow Tq_T$ in macroscopic L^1 .

Bounded-clock PDE stability, as at (G3), now gives convergence of the scalar spatial derivatives at zero. Equivalently one can use drift comparison after L^1 convergence of the inverse-clock distributions. The slope formula A/B in PE (29), together with [19, Lemma 5.8], therefore identifies the child's inverse-profile slope as κ_T . Agreement transfers its linear quantile limit to r ; meanwhile $q(v)/v \rightarrow 1/\kappa_T$ by (P2) and the normalized inverse construction. This proves the second part of (C7). On compact positive radius bands of scale u inside the constant-removal region, it also gives $S - r = o(u)$ uniformly, by (C6), monotonicity, and the local $o(u)$ floor mass. $\square$

7.3 Finite strengthening and common parameter envelopes

Use the notation

$$K_* = h_* + bq, \quad d = h - h_*, \quad K_s = h + C(r)r + (b - C(r))S, \quad \Gamma_* = \Gamma_{K_*}, \quad \Gamma_s = \Gamma_{K_s}$$

in the normalized local argument (s on K_s, Γ_s here denotes the sampled path, not a tilt). Let $\beta_0(r) = q^{-1}(r)$ locally near the patch; it and q have slopes between fixed positive constants there. Choose finitely many nested compact intervals scaled by u , strictly interior to the constant profile and at positive radii of order u , each containing with room the required image of the mass band by (C7). Their fixed spacings can be as small as needed for the number of stages.

Take

$$a_{-1} = 1, \quad a_j = \max(a_{\text{goal}}, \lambda^{-0.12}A^{1/3}a_{j-1}^{2/3}), \quad \kappa_j = (A/a_{j-1})^2 \quad (j \geq 0). \quad (\text{C11})$$

These scales are nonincreasing, $a_0 = o(1)$, and reach a_{goal} in a bounded fixed number of steps: the iterated term has fixed point $A\lambda^{-0.36} \ll a_{\text{goal}}$, while $A \gtrsim \lambda^{2/p*}$. Use (C5) with the indicated floor on the next nested patch I_j , dropping the previous extra floor.

Lemma 7.8 (Finite strengthening). *On a slightly smaller interval interior to I_j , at each stage,*

$$|\alpha - \beta_0| \lesssim ua_j Z \quad (\text{C12})$$

for almost every global parameter, simultaneously for all its optimizers, where $Z \geq 1$ has bounded averaged square in that parameter. For each fixed choice of the other path data, the exceptional set is independent of the optimizer and may be taken common to the finitely many stages. At successive stages that parameter may be restricted to slightly smaller fixed interior intervals, always leaving room beyond $[1/3, 2/3]$; parameters not explicitly varied need no restriction beyond those of (C3). At stage -1 the analogous assertion holds by (C7) with $Z = 1$.

We prove the lemma by induction. First we show that the preceding-stage bound pays the new optimization and supplies a common global norm bound. We then rule out a violation at the new stage by a buffered contradiction. The comparison cost therefore depends only on the already proved accuracy.

Lemma 7.9 (Stage cost and an envelope common to all minimizers). *Suppose the preceding stage satisfies (C12). At the new stage the average optimized excess is at most $O(x^{4-2\zeta}u\Lambda^{1/3})$. For almost every global parameter, every minimizer simultaneously satisfies (C13) below, with a common measurable majorant Z whose averaged square is bounded.*

Proof. Let the old optimizer have radius law $kdr + \nu$. On a fixed enlargement of I_j , convolve the fraction $c'\kappa_j$ of ν with the symmetric uniform kernel of radius $\sigma = \min(c_{\text{small}}u, C_{\text{big}}ua_{j-1}Z)$. Choose the fixed constants, depending on the stage, so that all motion stays inside the preceding controlled band and the constant part of C . Every radius interval of length 2σ used here carries mass $\gtrsim \sigma$. This follows from (C12) and the positive lower slope of β_0 ; if the fixed cutoff determines σ , use (C7) instead. Subtracting the density k does not affect this lower bound. Taking c' sufficiently large therefore supplies density at least κ_j throughout I_j . The convolved fraction is $o(1)$, so the new upper density bound is at most twice the old one.

Write $r_{\text{old}}, r_{\text{new}}$ for the two quantiles. Conditional Jensen shows that the new radius law dominates the old in convex order and has the same mean. For every $a' \in [0, 1]$, the upper-tail quantile formula

$$\int_{a'}^1 r(v) dv = \inf_{s'} \{(1 - a')s' + \mathbb{E}(r - s')_+\}$$

therefore gives a nonnegative upper-tail integral of $r_{\text{new}} - r_{\text{old}}$. Integration by parts against the nondecreasing moment S_t at each forward interpolation time yields

$$\int_0^1 S_t(v)(r_{\text{new}}(v) - r_{\text{old}}(v)) dv \geq 0.$$

Since all motion occurs where $C = C_0$, (A1) makes the pressure change nonpositive. Only the penalty must be paid; it is $O(C_0\kappa_j u\sigma^2) \lesssim x^4 u\Lambda^{1/3} Z^2$. Averaging and starting from the bound before (C5) gives the asserted cost at every stage.

We next obtain a norm bound common to every new minimizer. Write $g = \gamma_0$, let $F_j(g, \alpha)$ be the objective with the inverse profile fixed, and put $V_j(g) = \min_{\alpha} F_j(g, \alpha)$. The admissible density box is independent of g . Equation (A1), followed by its fixed-size path limit, makes each frozen objective differentiable and bounds its derivative uniformly. The minimum V_j is therefore Lipschitz.

At a differentiability point of V_j , fix any minimizing α . The function $g' \mapsto F_j(g', \alpha) - V_j(g')$ is nonnegative and vanishes at g . Its two one-sided derivatives give

$$\partial_g F_j(g, \alpha) = V_j'(g).$$

The exceptional set is the nondifferentiability set of the single function V_j ; it is independent of the minimizer.

Restore the scalar terms by setting

$$\mathcal{R}_j(g) = V_j(g) - \frac{1}{4} \int b q^2 dv + \frac{1}{2} \int_{v \geq x} d(v) q(v) dv.$$

For each minimizer define

$$\mathcal{N}^2 = \int (v^2 + U^2) [D_v^2 + (S_v - q)^2] dv.$$

The aligned matrix and field variation in (P10) gives $\mathcal{R}'_j(g) \geq c\lambda\mathcal{N}^2$. Above the mask this is the completed square $c_q[D^2 + (S - q)^2]/4$; below it the field stays fixed and the extra term is nonnegative. Thus the same derivative bounds $\mathcal{N}$ for every minimizer at g .

Choose an upper parameter endpoint outside the common null set in a fixed positive-length margin beyond the next averaging interval, where the optimized excess is at most a fixed multiple of its average on that margin. At the lower endpoint use the nonnegativity of the excess. Integrating $\mathcal{R}'_j$ and using the direct pressure comparison gives

$$c\lambda \mathbf{E}_{\gamma_0} \mathcal{N}^2 \leq \mathbf{E}_{\gamma_0} \mathcal{R}'_j \lesssim \mathcal{B} + x^{4-2\zeta} u \Lambda^{1/3}.$$

This notation denotes a bound by the common measurable derivative, not integration of a chosen optimizer. The endpoint depends only on the stage and the fixed path data.

Choose $0 < h_e \ll \min(1, c_G)$, with $p_* \ll h_e$. The bound on $\mathcal{B}$, with ζ sufficiently small, now permits the following explicit common majorant. Off the common null set take

$$Z(g)^2 = 1 + \frac{\mathcal{R}'_j(g)}{c\lambda x^4 \lambda^{-2s_0}},$$

using the lower-bound constant c above, and set $Z = 1$ on that null set. If $\lambda = x^{p_*}$, $u \leq x^{h_e}$, and $\chi_e \leq x^{h_e}$, use the smaller denominator $c\lambda x^{4+h_e/2}$ instead. The integrated derivative bound gives $\mathbf{E} Z^2 \lesssim 1$ in both cases. Thus $Z \geq 1$, and simultaneously for every minimizer outside that set,

$$\mathcal{N} \leq x^2 \lambda^{-s_0} Z, \quad \mathcal{N} \leq x^{2+h_e/4} Z \quad \text{if } \lambda = x^{p_*}, \quad u \leq x^{h_e}, \quad \chi_e \leq x^{h_e}. \quad (\text{C13})$$

Only direct paths occur here. The majorant may change at the next stage. Restricting to slightly smaller deterministic parameter intervals leaves room beyond $[1/3, 2/3]$, since there are only finitely many stages. The only use of the preceding-stage accuracy was to pay for the new floor; no estimate at the new stage was assumed. $\square$

7.3.1 Buffered geometry for a possible violation

Fix the new stage and a sequence of nonexceptional global parameter values contradicting (C12). Choose a radius r_0 and discrepancy $d_0 = uL = |\alpha(r_0) - \beta_0(r_0)|$, with $L \geq a_j Z$ and $L \rightarrow 0$ by (C7). We may arrange

$$|\alpha - \beta_0| = O(d_0) \quad \text{within radius } b_{\text{buf}} = u(\sqrt{L} + \lambda^\rho) \text{ on either side of } r_0.$$

Here $\rho > 0$ is fixed and sufficiently small relative to the power gaps below. To obtain this buffer, whenever its discrepancy bound fails, move to a point where the discrepancy has at least doubled and repeat. There are only logarithmically many moves. The sum of the $u\sqrt{L}$ travel bounds is dominated by the last one, which is $o(u)$ by (C7); the sum of the $u\lambda^\rho$ bounds is also $o(u)$. Thus all selected points stay in the interior of the patch.

Put

$$h_1 = u \max(L, \sqrt{\lambda}), \quad G_1 = \min(\max(L^2, \lambda), \Lambda), \quad P_1 = LG_1.$$

The buffer contains unboundedly many lengths h_1 . On each smaller fixed fraction, every cut has nonfloor points within $O(d_0)$ on both sides: otherwise the small floor density contradicts the positive lower slope of β_0 and the discrepancy bound. Equation (C6) and monotonicity then give $|S - r| = O(d_0)$, while the oscillations of S and α on a radius interval of length l are $O(d_0 + l)$. At nonfloor points, $|S - r| \leq M_j^{-1}$. Every later buffer restriction is by a fixed proportion; the finite number needed in the moment iteration is specified there.

To compare scalar curvatures, define in this buffer

$$k_*(r) = \Gamma_*^{-1}(r), \quad \tilde{r}(v) = \Gamma_*(K_s(v)), \quad \tilde{\alpha}(r) = |\{v : \tilde{r}(v) \leq r\}|.$$

Near the tested band the defect d consists only of the logarithmic seed, so

$$|d| \lesssim_\circ u^3 X^3 \lambda^{1+\eta-3s_*/2} = o(u^3 P_1).$$

The comparison uses $G_1 \geq \sqrt{\lambda}L$ and $L \geq X^2 \lambda^{-s_0}$. Split the latter lower bound at $\sqrt{\lambda}$ and $\sqrt{\Lambda}$; in each range the ratio has a power saving. At the $\sqrt{\lambda}$ breakpoint the saving is $1/4 + O(\tau + \eta)$ in λ . Equation (P2) and monotone bracketing now give $\tilde{r}(\alpha(r)) = r + O(d_0)$ and $\tilde{\alpha} - \beta_0 = O(d_0)$.

Extend the scalar clocks annealed to a common bounded time interval, and write their distribution difference in mass length as $\delta\alpha_t = |\{K_s \leq t\}| - |\{K_* \leq t\}|$. We claim

$$\int \min(u, t) |\delta\alpha_t| dt = o(d_0).$$

The one-dimensional inverse-transport bound, together with (C7), (C8), and (P4)–(P7), bounds the left side by $\lesssim \int \min(u, v + w_c) |K_s - K_*| dv$. Here the low clock is controlled because $\lambda_1 U^3 \log(1/x) \lesssim w_c$. Masses $v \leq O(w_c)$ contribute $O(w_c^3) \ll d_0$. Above them, the terms $b(S - q)$ and d contribute $O(\sqrt{u} x^2 \lambda^{-s_0} Z)$, by (C13) and (P8).

It remains to bound $C(r)(r - S)$. Off the floor, use (C6) and the same norm. The original floor contributes $O(uC_0 \int k)$ by boundedness. On the extra floor, at $r \asymp u$, the $S - q$ part is again controlled by the norm. Fubini and (C7) bound the remaining $|r - q|$ integral by

$$O(\kappa_j) \int_{I_j} |\alpha(r) - \beta_0(r)| dr \lesssim \kappa_j \int_{v \asymp u} |r(v) - q(v)| dv,$$

on a comparable enlargement. Its extra-floor portion can be absorbed, since $\kappa_j = o(1)$; its other portions have just been bounded. Multiplication by uC_0 proves the claimed transport estimate, with a power margin for subpower losses.

Lemma 7.10 (Scalar curvature in a buffer). *By the scalar second derivative identity we now have uniformly a.e. in the inner buffer*

$$\frac{d^2}{dr^2} [\Gamma_s(k_*(r)) - r] = -A_*(\tilde{\alpha}(r) - \beta_0(r)) + o(d_0), \quad (\text{C14})$$

where A_* converges to a positive constant $(2(V_{*,zz}(0,0))^3 k'_*(0)^2)$.

Proof. Let Q_0 be a fixed bound for the extended clocks. At time $t \asymp u$, PDE drift comparison bounds the difference of continuation derivatives in sup norm by $O(\int_t^{Q_0} |\delta\alpha_v| dv)$; odd derivatives gain an additional factor $|z|$. For a bounded smooth even observable, changing the forward diffusion costs at most

$$C \int_0^t v \left(|\delta\alpha_v| + \int_v^{Q_0} |\delta\alpha_{v'}| dv' \right) dv.$$

Indeed the drift difference is odd and bounded by the indicated coefficient times $|Z_v|$. The propagated spatial gradient is also odd with bounded derivative, and $\mathbb{E}Z_v^2 = O(v)$. These comparisons require only bounded spatial drift derivatives on the bounded time interval.

Now use $\Gamma'' = \mathbb{E}V_{zzz}^2 - 2\alpha_t \mathbb{E}V_{zz}^3$. The two masses are $O(u)$, and their difference is $O(d_0)$. Oddness in the first term and the preceding transport estimates leave, to order d_0 , only the mass-distribution difference in the second term. Multiplying by $k_*'^2$ gives the displayed coefficient A_* . The chain-rule term involving $k_*'' = O(\lambda u)$ uses the same comparison for $\Gamma'_s - \Gamma'_*$ and is $o(d_0)$. This proves (C14). $\square$

7.4 Buffered moments, free forks, and power margins

Write $\delta_i = Q_i - S_i$ at a fork. For a radius interval $J = [a, b]$ in the inner buffer with $|J| \asymp d_0$, define the two box integrals by

$$\text{avg}_{J,r} F = \frac{1}{d_0} \int_a^b F_{\alpha(r)} dr, \quad \text{avg}_{J,v} F = \frac{1}{d_0} \int_{\alpha(a)}^{\alpha(b)} F_v dv.$$

Endpoints have zero measure in these integrals. Both use the scale d_0 as denominator. In particular the second is not normalized by the mass of the interval: that mass is $O(d_0)$, but can be arbitrarily small. In box assertions, avg denotes either operation, or their sum, uniformly over moving intervals of this kind. An explicitly subscripted dyadic average such as $\text{avg}_{v \sim w}$ instead divides the mass integral by the length of that dyadic band.

Lemma 7.11 (Moments on moving boxes). *Uniformly inside the buffer and for these averages, for every fixed moment order, we have*

$$\|\delta_i\|_p \lesssim_p d_0, \quad \text{avg} \|\delta_i\|_2^2 \lesssim d_0^2 \lambda^{0.14}, \quad \text{avg} \|Q_i - |X_i|^2\|_2^2 \lesssim_{\circ} \lambda^{-\rho} u^2 A^3. \quad (\text{C15})$$

Proof. Fix the optimizer and the moment order p . Choose a finite sequence of nested buffers $B_0 \supset B_1 \supset \dots \supset B_{n_{\text{buf}}}$, with successive boundaries separated by fixed fractions of b_{buf} . Their number will be fixed below, before taking the size limit. Define

$$\mathcal{Q}_k = \max \left\{ 1, \sup_{t, t' \text{ in } B_k} \frac{\|X_t - X_{t'}\|_p^2}{d_0 + |r(t) - r(t')|} \right\}.$$

The cuts in this supremum are deterministic and lie on one path. Boundedness gives finiteness at fixed size, and (C8) gives $\mathcal{Q}_0 = O_p(L^{-1})$. To estimate cuts in B_{k+1} , use only windows lying in B_k . If such a window J has radius length $\Delta \gtrsim d_0$, the weighted-layer calculation for (C9) gives

$$\left\| \int_J W dr \right\|_{p/2} \lesssim_{p,\circ} u^3 A^3 \Delta (\mathcal{Q}_k + \lambda^{-\rho}).$$

For layers ending within radius distance $O(u)$ and still at comparable scales, projection after distance t' costs $O_{\circ}(x^6/(C_0 u t'))$ times the earlier squared increment, including the

Hessian weight. The normalized-anchor truncation in Paragraph 6 retains that increment; it requires no higher moment of $\mathcal{Q}_k$. When $t' \lesssim \Delta$, (6.2) bounds its integral by $O_p(t' \mathcal{Q}_k \Delta)$, using terminal increments on an interval of length $O(\Delta)$ inside B_k . When $t' \gtrsim \Delta$, its pointwise norm is $O_p(t' \mathcal{Q}_k)$, giving the same integrated bound. Layers leaving B_k start at distance $t' \gtrsim u \lambda^\rho$ and use (C8); this accounts for $\lambda^{-\rho}$. The first inverse-power increments are negligible by the square-sum estimate. Outer layers at scale $v \gtrsim u$ cost $O_{p,\circ}(x^6/(vC(v)))$ per unit radius length, which also fits the displayed bound.

Using $\alpha' \geq \kappa_j$, normalized L averages on such intervals have centered norm relative to Δ at most $O_{p,\circ}((A/L)^{3/2} \kappa_j^{-1/2})(\mathcal{Q}_k + \lambda^{-\rho})^{1/2}$. The prefactor, before subpower losses, is at most $\lambda^{0.18}$ by (C11). For two cuts in B_{k+1} , take earlier and later windows of length comparable to $\Delta = \max(d_0, \text{their radius separation})$. The brackets in Lemma 6.3 then control both centered square norms and their squared increment. The deterministic oscillation of S costs $O(\Delta)$. After absorbing subpower losses and the $\lambda^{-\rho}$ term, this is the recurrence

$$\mathcal{Q}_{k+1} \leq C_p(1 + \lambda^{0.1} \sqrt{\mathcal{Q}_k}).$$

There is a fixed K with $\mathcal{Q}_0 \lesssim_p \lambda^{-K}$. After n_{buf} iterations its potentially growing term is bounded by a constant times $\lambda^{0.2(1-2^{-n_{\text{buf}}})-K} 2^{-n_{\text{buf}}}$. Choose the fixed n_{buf} so that this exponent is positive. This proves bounded ratios on $B_{n_{\text{buf}}}$, and the same brackets give $\| |X_i|^2 - S_i \|_p \lesssim_p d_0$.

Project both leaves to radius distance comparable to d_0 after their fork. The synchronized later layers cost $\lesssim_{p,\circ} u A^{3/2}/\sqrt{L} \ll d_0$. Polarization controls the projected inner product, since the squared distance of the two branches through their common prefix is $O_p(d_0)$. This proves the first assertion of (C15).

Second moments in radius and in mass. For the second bound, over each box we have

$$\int \mathbf{E} W \, dr \lesssim u^3 A^3 d_0, \quad \int \mathbf{E} \text{tr} C_v^2 \, dr \lesssim u^2 A^3 d_0.$$

These estimates follow from (A1), the S oscillation bound, and constant removal strength. Use earlier and later windows of length $h = t' d_0$. The sliding variance estimate proved at (C10) is now

$$\int_{J_h} \text{Var}(\bar{L}_{r,h}^\pm) \, dr \lesssim \frac{1}{\kappa_j h} \int_{J^+} \mathbf{E} W \, dr.$$

Here J_h omits endpoint strips of width h , so the windows stay inside J ; J^+ merely provides the available integral upper bound. Discard those strips with the pointwise moment bound. Apply the absorption calculation at (C10) on height scale d_0 , now with trace term $\mathcal{T} \lesssim A^3/L^2$ and variance term $(A/L)^3/(\kappa_j t')$. It gives

$$\int D^2 dr / d_0^3 \lesssim t' + (A/L)^3/(\kappa_j t') + A^3/L^2.$$

Take $t' = (A/L)^{3/2}/\sqrt{\kappa_j} \leq \lambda^{0.18}$. This proves the radius estimate and also the mass estimate on the floor, whose density is bounded above. To treat the remaining mass, enlarge endpoint caps to whole caps. Total mass and S oscillation remain $O(d_0)$. Equation (C6) and PE (20), with $p = 0$, give the same integrated W and trace bounds on caps and contact.

Order this measure by the deterministic coordinate $s(v) = \int^v \mathbf{1}_{\text{nonfloor}}(t) \, dt$, and take windows of s -length $h = t' d_0$. The optimizer and its floor set were fixed before sampling; hence the window indicators are deterministic weights in (A1). Fubini now bounds the integrated centered variance by $h^{-1} \int \mathbf{E} W \, dv$, with no floor-density denominator. The ordered identity and monotonicity control the noise and mean changes as in radius. If the

whole nonfloor measure has length less than two windows, the endpoint-strip bound alone suffices. This proves the second assertion of (C15) in both measures.

The alias error. For the final bound of (C15), it remains by (A1)'s covariance convention to bound $\mathbf{E}(X_i \cdot (y - X_i))^2$. Telescope X_i into past layers ending at dyadic distances from the current radius, down to x^{40} . For an increment ending at distance t' after starting at distance $\asymp t'$, with $x^{40} \lesssim t' \ll u$, (A3) costs for its squared product with $y - X_i$ at most $O_\circ(x^6/(u^2 C_0 t'))$ times the layer's squared norm in expectation. That norm has box average $O(t' \lambda^{-\rho})$. In radius this follows for $t' \leq d_0$ by sliding the S variation, for $d_0 \leq t' \ll b_{\text{buf}}$ pointwise, and otherwise by (C8).

In mass, the floor uses the radius estimate, and around a nonfloor evaluation the S oscillation at distances $O(t')$ within the buffered patch is $O(t')$, by (C6) and the lower potential bound $-O(C_0/M_j^2)$, as at the good cut. The unprojected last past increment thus has negligible squared norm on average by the same facts, and the initial anchor, projected from fixed comparable distance within constant strength, costs $O_\circ(u^2 A^3)$. Summing proves the last assertion. $\square$

Here are needed integrated estimates on free forks. Choose a sufficiently large fixed c_H so $v \geq c_H u$ in mass lies beyond the extra floor, and away from the observed buffer with room for prior comparable projection intervals.

Lemma 7.12 (Integrated errors at free forks). *Uniformly,*

$$\begin{aligned} \int_0^{c_H u} D_v dv &\lesssim_\circ u^2 T_1, & T_1 &= A + B_f + y_c^2, \\ \int_0^{c_H u} D_v^2 dv &\lesssim_\circ u^3 T_2, & T_2 &= A^2 + B_f^2 + A^{3/2} y_c + A y_c^{5/3}, \\ (\text{avg}_{v \sim w} D_v^2)^{1/2} &\lesssim_\circ u A & (w &\geq c_H u). \end{aligned} \tag{C16}$$

Proof. Regular bands use (C10), with $y_c \gtrsim \sqrt{A}$; the inner region $v \lesssim w_c$ uses (C8) for the first line, also on floor (of mass $O(uA)$) for the second.

For contact and comparable-endpoint caps in inner dyads $v/u \asymp t' \leq O(y_c)$, (20)–(21) with constant field slope lower bound C_0 and raw scale $u y_c$ give for the dyadic mass average of D^2/u^2 at most

$$O_\circ((A^3 y_c/t'^2)^{2/3} + A^3/t'^2)$$

apart from negligible cap-width and strip errors. Clip also by $O(y_c^2)$; summation weighted by t' costs $O_\circ(A^{3/2} y_c)$.

For noncomparable-endpoint caps meeting the region the upper masses are $O(u y_c)$ by (C7) and the width bound. Slice each such cap geometrically into units at masses comparable to their lengths, of scale $u t'$. On one unit the whole-cap $\Delta S \leq M_j^{-1}$ trace proof and constant cap density cost an additional factor $O(y_c/t')$ in both trace bounds of (20). Use one height bin there: in (21)'s proof the squared mean-oscillation term costs negligible since S varies by at most M_j^{-1} , including for reverse ordering (its fourth moment estimate uses (C8)). Thus the same average is bounded by $O_\circ(A^3 y_c/t'^3)$, again clipped by $O(y_c^2)$. There are boundedly many such units per mass dyad by disjointness; summation gives $O_\circ(A y_c^{5/3})$. All band calculations can stop at $v \sim x^{10}$ with negligible cost. This proves (C16).

For completeness, the additional broad-cap cost can be seen directly. Writing $y = y_c$, its normalized mass sum is bounded by

$$\sum_{t' \text{ dyadic}} t' \min\{y^2, A^3 y/(t')^3\}.$$

The two terms meet at $t'_0 = Ay^{-1/3}$. Below this scale the geometric sum is at most $Cy^2t'_0$; above it the sum is at most $CA^3y/(t'_0)^2$. Both equal $O(Ay^{5/3})$. If the crossover is outside the finite summation range, the corresponding one-sided geometric bound is only smaller. The original-floor raw contribution is at most $O(Ay^2)$, absorbed in this same term since $y = o(1)$. The analogous crossover for comparable caps is $t'_0 = A^{3/2}/y$, giving $O(A^{3/2}y)$.

The trace conversion on a sliced cap uses the lower mass bound of that slice. The whole-cap numerator is divided by its smaller slice mass, which is precisely the factor y/t' already paid above. No trace estimate is applied below its lower mass cutoff. These are mass estimates; the radius estimates used in Lemma 7.11 and in the coarse profile proof were proved separately by radius slides. $\square$

Lemma 7.13 (High projection and centered primitives). *Work in the buffered violation constructed above, with $d_0 = uL$, $L \geq a_{\text{goal}}Z$, $h_1 = u \max(L, \sqrt{\lambda})$, and $G_1 = \min(\max(L^2, \lambda), \Lambda)$. Here $Z \geq 1$ is the deterministic majorant in the global parameter from Lemma 7.9; it is not a random variable under the positive spin law. For a high dyad $v \sim w \geq c_H u$, replace an overlap at the observed buffer fork by the inner product of the two branches projected to a fixed cut of order w , after their separation and strictly before the dyad. In each fixed moment the cost is at most*

$$O_{p,o}(z_w), \quad z_w = x^3 \min\{(w\sqrt{C(w)})^{-1}, \lambda^{-1/2}w^{-2}\}. \quad (\text{C17})$$

The integral of $L_v - \mathbb{E}L_v$ over a high dyad, against any deterministic weight bounded by one, has L^2 norm $o(u^2G_1)$, with a fixed power saving. This includes the plateau for every edge multiplier used on the comparison path.

Proof. The first option in (C17) is the postprojection bound obtained after (C9). The actual field gives the second option: $dh \gtrsim \lambda w^2 dv$ on comparable active intervals by (P4) and bounded active density; the edge ramp loss is absorbed there. Even at the plateau or a macroscopic band, a preceding active interval is available at a comparable scale. Telescope both branches in regular layers to obtain the stated minimum.

We next estimate the centered primitive. On the active part, (A1) gives

$$\left\| \int k_v(L_v - \mathbb{E}L_v) dv \right\|_2 \lesssim_o x^3 \min\{C(w)^{-1/2}, \lambda^{-1/2}/w\} \lesssim_o u^2 X^3 \Lambda^{-1/3} \lambda^{-1/6}.$$

For the first bound integrate $\mathbb{E}W$ using PE (20) on comparable caps and contact, and in radius on the floor. Complete a partial cap before estimating it. This option also holds on the plateau. The second option uses the actual-field slope just given; balancing the two in w/u gives the last expression. Its ratio to u^2G_1 is power small by (C18) below.

For the plateau, first suppose $\chi_e > 0$. From (C13), $\mathcal{N} \leq x^2 \lambda^{-s_0} Z \leq U^2 Z$. On a plateau strip of width d' , Cauchy–Schwarz and monotonicity compare S to the constant reference q on neighboring intervals of length comparable to d' . The denominator is $\sqrt{d'}$, so the resulting oscillation bound is

$$\text{osc } S \lesssim \mathcal{N}/\sqrt{d'} \lesssim ZU^2/\sqrt{d'}.$$

On the capped left strip, use also the preceding active interval of mass comparable to d_- . Its reference varies by at most $O(E_b) = O(U^2/\sqrt{d_-})$, which fits the same bound. On the capped right strip, boundedness suffices, because its width is $d_+ = U^4$ and $U^2/\sqrt{d_+} = 1$.

The edge slope on this strip is $dh/dv \gtrsim \lambda_1 \chi_e U^2/(d')^{3/2}$. Thus (A1) bounds the squared primitive norm there by

$$O_o \left(\frac{x^6 (d')^{3/2} ZU^2}{\lambda_1 \chi_e U^2 \sqrt{d'}} \right) = O_o \left(\frac{x^6 Z d'}{\lambda_1 \chi_e} \right).$$

Summing the dyadic strips gives the edge bound $O_{\circ}(x^3 \lambda_1^{-1/2} \chi_e^{-1/2} \sqrt{Z})$. When $\chi_e = 1$, the auxiliary-removal bound at $w \asymp 1$ is $O_{\circ}(x^3 u^{-3/2} \Lambda^{-1/2})$. Taking the smaller bound, and using $\min(a, b) \leq a^{2/3} b^{1/3}$, gives

$$O_{\circ}(u^2 X^3 \Lambda^{-1/3} \lambda_1^{-1/6} Z^{1/6}).$$

Equation (C18) again makes its ratio to $u^2 G_1$ power small, including the dependence on Z .

Other values of χ_e occur only at $\lambda = x^{p*}$, with $\chi_{\text{lin}} = 1$. The violation bound supplies the useful lower bound

$$u^2 G_1 \geq u^2 \sqrt{\lambda} L \geq x^2 \lambda^{1/2-s_0} Z.$$

There are three cases.

- (a) If $\chi_e \geq x^{h_e}$, the edge bound divided by this lower bound is at most

$$O_{\circ}\left(x^{1-h_e/2} \lambda_1^{-1/2} \lambda^{s_0-1/2} / \sqrt{Z}\right) = o(1).$$

- (b) If $\chi_e < x^{h_e}$ but $u \geq x^{h_e}$, the linear seed gives primitive norm $O(x^{3/2})$ by (A1) and boundedness of S . Since $G_1 \geq \lambda$, its ratio to the target is at most $O(x^{3/2-2h_e} \lambda^{-1}) = o(1)$.
- (c) If both $\chi_e < x^{h_e}$ and $u < x^{h_e}$, remove strips of width $\delta = x^{2+h_e/8}$ at the two plateau ends. Their primitive norm is $O(\delta)$ by boundedness, and their ratio to the target is

$$O\left(x^{h_e/8} \lambda^{s_0-1/2} / Z\right) = o(1).$$

On the remaining plateau, the stronger part of (C13) and the same neighboring-interval argument give

$$\text{osc } S \lesssim x^{2+h_e/4} Z / \sqrt{\delta} = x^{1+3h_e/16} Z.$$

The linear slope is $dh/dv \gtrsim x^3$, so (A1) gives primitive norm at most $O(x^{2+3h_e/32} \sqrt{Z})$. Its ratio to the target is $O(x^{3h_e/32} \lambda^{s_0-1/2} / \sqrt{Z}) = o(1)$.

All displayed ratios have fixed positive power margins because $Z \geq 1$, h_e is small and fixed, and $p_* \ll h_e$. This proves the plateau assertion in every case. $\square$

Lemma 7.14 (Strict power margins). *For clarity, here is some power accounting used here and below (write $\ll$ in such estimates for polynomial gains). Uniformly under (C4), (C11), $a_{\text{goal}} Z \leq L \leq 1$, we have*

$$T_2, AT_1, X^3 A \Lambda^{-1/3} \lambda^{-1/6}, \lambda^{-2\rho} A^3 \ll P_1, \quad \lambda^{0.06} T_1, LT_1, X^3 \Lambda^{-1/3} \lambda_1^{-1/6} Z^{1/6} \ll G_1. \quad (\text{C18})$$

Proof. Write $X = \lambda^{\xi}$, with $\xi \geq s_*/2 - o(1)$. For terms whose numerator does not depend on L , monotonicity of G_1 and $P_1 = LG_1$ reduces the estimate to $L = a_{\text{goal}}$. For LT_1/G_1 , also check $L = \sqrt{\lambda}$ and $L = 1$; these are the remaining possible extrema of the piecewise ratio, and the former requires $\xi \geq s_0/2 + 1/4$. The Z -dependent term is treated separately below. Finally $AT_1 \lesssim T_2$; inserting the powers of A shows that the $X^3 A$ and $\lambda^{-2\rho} A^3$ ratios have larger margins than the principal ratios computed next, for sufficiently small ρ .

The tables below give the three principal ratio calculations. At $s_* = s_0 = 1/2$, $\eta = 0$, $k_c = 0.03$, and with the arbitrarily small ζ -loss suppressed, write $e(F)$ for the power of λ in

F . At $L = a_{\text{goal}}$, $e(P_1) = 2\xi - 1/2 + e(G_1)$, and the required piecewise functions are

F	range of ξ	$e(F)$
G_1	$[1/4, 103/400]$	$3/100$
	$[103/400, 1/2]$	$4\xi - 1$
	$[1/2, \infty)$	1
T_1	$[1/4, 99/200]$	$2\xi - 1/100$
	$[99/200, \infty)$	$3\xi/2 + 19/80$
T_2	$[1/4, 1/2]$	$11\xi/3 - 11/600$
	$[1/2, 57/50]$	$13\xi/4 + 19/100$
	$[57/50, \infty)$	$3\xi + 19/40$

The first two pieces for T_2 come from $Ay_c^{5/3}$, and the last from B_f^2 . Subtracting the denominator exponents shows that all three minima occur at $\xi = 1/2$:

ratio	term attaining the minimum	minimum exponent
T_2/P_1	$Ay_c^{5/3}/P_1$	$63/200 = 0.315$
$\lambda^{0.06}T_1/G_1$	$\lambda^{0.06}B_f/G_1$	$19/400 = 0.0475$
$X^3\Lambda^{-1/3}\lambda^{-1/6}/G_1$	$X^3\Lambda^{-1/3}\lambda^{-1/6}/G_1$	$97/300$

On the final intervals the relative exponents have positive slopes. Thus strictly positive margins persist for the prescribed small τ, η, τ_0 , and for sufficiently small $\rho, \zeta/p_*$, and fixed-order moment interpolation losses, chosen in the stated parameter order. The number of strengthening stages is already fixed at this point. None of these choices requires decreasing p_* again.

The dependence on an unbounded Z in the primitive estimate can also be checked directly. Put $Y = X\sqrt{Z} = \lambda^\xi$ and $L_0 = Y^2\lambda^{-s_0} \leq L \leq 1$, so $\xi \geq s_0/2$. Since G_1 is nondecreasing in L , the same piecewise calculation gives

$$\frac{X^3\Lambda^{-1/3}\lambda_1^{-1/6}Z^{1/6}}{G_1(L)} \leq Z^{-4/3}\lambda^{d_*}, \quad d_* = \frac{97}{300} - \frac{3}{2}(\tau + \tau_0) - \frac{\eta}{6} > \frac{1}{4}.$$

The minimum occurs at $\xi = s_0/2 + 1/4$. Thus a large majorant only improves this ratio; no pointwise upper bound on Z was used. □

7.5 Quantitative one-row closure

Proposition 7.15 (Buffered one-row closure). *In any of the boxes of (C15) we claim*

$$\text{avg } |S_i - \Gamma_s(K_s(i))| = o(u^3 P_1). \quad (\text{C19})$$

Proof. Apply (A7) through degree two at K_s , with the untilted law, the two observed labels a, b , and test 1. Each mismatch insertion is $(b - C)\delta$. Linear terms vanish termwise after averaging spins under the actual law. For the remainders, expand the cavity-spin test to order two, then each quotient term of degree $j \leq 2$ to order $2 - j$, from the independent-row endpoint to the actual row. Taylor's integral remainder has three mismatch factors on a fixed finite extension of the tree. It retains $\tau_a\tau_b$, either in the interpolated test or in one deterministic scalar coefficient; every additional cavity spin occurs in an inserted pair.

After deleting the row, nonnegative spin-only bounds transfer by the bounded-covariance Grönwall rule, also during the bounded positive semidefinite row changes below. Deletion itself costs $O(m^{-1}) \ll u^3 P_1$, since $P_1 \geq \sqrt{\lambda} a_{\text{goal}}^2$. Keep deleted overlaps during sign

arguments, so the other mismatch factors use only the remaining rows. Changing this convention costs another $O(m^{-1})$. Moving a revelation cutoff means an affine positive semidefinite interpolation of the row before applying PE (2). It does not change the conditioning in a term already being estimated.

The allocation rule after (A1) supplies all fork measures. Distinct free vertices away from the observed fork have bounded joint densities in their mass coordinates, by sequential attachment. Repeated attachments at one vertex use its single variable. Ordering restrictions are deterministic and may be retained or dropped when taking bounds. An inserted pair is *pinned* if its two leaves split at the observed fork. These give the only split atoms, since diagonal pairs are omitted. Every other pair is called low or high according as its split mass is below or above $c_H u$.

- (i) Terms with no pinned pair: two low errors cost $O_o(\int_0^{c_H u} D^2)$ by marginalization and Cauchy–Schwarz, whether at the same vertex or not. One low and one high cost $O_o(u^3 T_1 A)$ using distinct-vertex densities. Any extras here can be dropped.

If all errors are high one gains an extra $O_p(u)$ low covariance. Indeed turn off the relevant cavity-row revelation below $c_H u$, retaining its later cumulative covariance by a jump there. This includes the scalar row law containing $\tau_a \tau_b$ if that factor occurs in a deterministic coefficient: restoring that law costs $O(u)$ by PE (2) since $K_s = O(u)$ there. Products can be treated factor by factor. With low revelation off, the term containing those two spins vanishes by sign flips of a whole branch component (negating the attached row Gaussians if interacting); they are in distinct components and all other pairs are internal.

In an interpolated interacting row this is an admissible PSD interpolation (move the cumulative row-specific field and rest-row couplings, or their affine mixture with the scalar row, to the jump). Spin-only tests on remaining rows are invariant. The new pair for restoring this row costs $O_p(u)$ at each low split by (C8) and bounded row transfer, since its coefficient, apart from $O(1/m)$, is bounded by $O(|Q_v^{\text{deleted}}| + u)$. Take large fixed moments here and slight higher-than-second moments on the two high errors, costing arbitrarily small power losses by boundedness and (C16). Old topology and weights remain fixed during restoration. Thus all-high terms cost $O_o(u^3 A^2)$.

- (ii) At degree at least three with pinned errors: for three pinned factors use one (C15) averaged RMS saving and pointwise higher orders on two, paying $O(\lambda^{0.07} d_0^3)$ (or with arbitrarily small extra losses); $\lambda^{0.07} L^2 \ll G_1$.

For two pinned and one free factor use the pointwise costs d_0^2 times $O_o(uA + u^2 T_1)$. With one pinned and two free, use a high fixed moment for the pinned error and second moments with slight power loss for the others. The latter two integrated cost at most $O_o(u^2 A^2 + u^3 T_2 + u^3 A T_1)$. These bounds and the preceding ones are sufficient by (C18).

- (iii) At degree two with one pinned error, a low free error costs $O_o(d_0 \lambda^{0.07} u^2 T_1)$, using second moments and the average at the observed fork. For a high free error, fix its dyad $v \asymp w$ and project the pinned overlap as in (C17). The error, including the free-vertex mass length, is

$$O_o(w z_w u A), \quad w z_w \lesssim u^2 X^3 \Lambda^{-1/3} \lambda^{-1/6}.$$

It is $o(u^3 P_1)$ by (C18).

For the projected term we must keep its joint law fixed as v varies. Fix the placement topology and all other free vertex coordinates. Use a projection cut fixed throughout the dyad, prune the pinned branches at that cut, and retain one path through the high vertex to its terminal spin. The sharing of these retained data is independent of v : any

sharing through the high fork already extends beyond the projection cut. All discarded branches have normalized continuation kernels. Let F be the projected centered pinned factor; it is measurable with respect to the retained earlier data. Conditional on those data and the high prefix, the high pair has independent continuations with conditional mean X_v . Hence

$$\mathbb{E}[F(Q_v - S_v)] = \mathbb{E}[F(|X_v|^2 - S_v)] = 2\mathbb{E}[F(L_v - \mathbb{E}L_v)].$$

The second equality conditions the retained terminal spin on its prefix. The scalar coefficients and allocation restrictions are bounded deterministic weights k_v , and $\|F\|_2 = O(d_0)$. Lemma 7.13 therefore gives

$$\left| \mathbb{E} \left[F \int k_v 2(L_v - \mathbb{E}L_v) dv \right] \right| \leq 2\|F\|_2 \left\| \int k_v (L_v - \mathbb{E}L_v) dv \right\|_2 = o(d_0 u^2 G_1) = o(u^3 P_1).$$

This uses unconditional Cauchy–Schwarz on the fixed marginal law. If other slots attach at the same high vertex, there is still only one integration in its mass. Fubini and the power margins permit summation over topologies and dyads.

Two insertions pinned at the observed fork. It remains to treat both degree-two pairs pinned. Write each centered overlap as $m_i + \eta_{\text{pair}}$ with $m_i = |X_i|^2 - S_i$ on the common prefix and $\eta_{\text{pair}} = Q_i - |X_i|^2$. Single η terms vanish by conditioning.

For two η 's the coefficients individually gain $O(u)$: either a third child has entered at the fork (coefficient $O(u)$), or both pairs run across the two occupied children. In the latter case the total number of scalar spins, including old spins, in each child is odd; at least one scalar moment factor has a sign cancellation on turning off scalar revelation through this cut (keep later cumulative clock via a jump after the observation, or take its immediate right limit). Indeed the two children then have independent sign symmetry. If using cuts immediately to the right for a fixed placement with the same children, they separate from all further distinct fork vertices eventually; the restoring bound by PE (2) is uniform $O(u)$. Thus (C15)'s final bound and (C18) suffice.

The factor $\mathbb{E}m_i^2$ is common to all pinned topologies, so extract it before summing coefficients. We claim that the remaining signed scalar coefficient, before the common $(b - C)^2$, is $\Gamma_s''(K_s(i))/2$ at almost every buffered cut.

To identify it, compress a small neighborhood of the observed mass into a clock jump at that mass, retaining the observation at its original intermediate clock value. At a continuity cut there is room to move this value in both directions: K_s is strictly increasing locally because its field contains $C_0 r$. On an already extended old tree, increase the common subincrement at the distinguished vertex by variance ϵ , and decrease the following subincrement in every occupied child by the same amount. The covariance direction is then one for pairs entering different children of that vertex and zero for same-child pairs, diagonals, and pairs outside its subtree. Total leaf variances and all other split covariances stay fixed. Thus this is exactly the pinned direction, even when earlier insertions have occupied more than two children.

In the quotient expansion of (A5), the degree-two scalar terms are

$$N_2 - N_1 \Psi_1 + N_0(\Psi_1^2 - \Psi_2).$$

In every term with positive denominator degree, allocate one whole such Ψ moment last. Previously placed factors pull back from the induced old genealogy, and the extracted prefix expectation is independent of placement. The complete signed sum of that denominator moment is therefore a covariance derivative of the constant-one observation on this genealogy,

hence zero. This cancellation requires the full signed sum; it is not a row-sum identity for a nonconstant restricted test. Coincidences within the permitted children remain part of that sum.

Only N_2 remains. Its coefficient includes the Taylor factor $1/2!$. The two subincrements at the same positive mass compose into the original scalar transition, so moving their intervening split changes only the evaluation time of Γ_s . The coefficient is consequently $\Gamma_s''/2$. This calculation introduces no additional split atom: a new attachment at a same-mass subdivision without old divergence has zero coefficient, because that edge has zero length and one continuing child. A different vertex with the same numerical mass likewise creates no atom at the observed fork.

Finally shrink the compressed neighborhood. Bounded-clock path limits with the cut retained pass the aggregated coefficients to the original path. Other freely varying fork masses in the shrinking interval have vanishing absolute weight. The scalar identity $\Gamma_s'' = \mathbb{E}V_{zzz}^2 - 2\alpha_t \mathbb{E}V_{zz}^3$ uses exactly the fork mass at the intermediate observation. At the limit this remains its mass, since K_s grows strictly locally and the spatial derivatives and path moments converge. Null contact and jump exceptions do not affect either box integral. This proves the claimed coefficient identity.

By the drift estimates at (C14), $\Gamma_s''(K_s(i)) = O(d_0 + \lambda u)$. The m_i^2 average by (C15) is at most $O(d_0^2 \lambda^{0.14})$; their product suffices since $\lambda^{0.14} L(L + \lambda) \ll G_1$. This completes (C19).

In particular the error budget remains relative to the removal strength:

$$\frac{u^3 P_1}{d_0 h_1^2} = \frac{G_1}{\max(L^2, \lambda)} \asymp \min(1, R_n)$$

with the notation used next. This observation will be needed when R_n tends to zero; an absolute $o(1)$ error would not suffice. $\square$

7.6 Buffered rigidity and completion of the budgets

Proof of Lemma 7.8. The ratio R_n defined below compares the removal strength with the curvature scale of the rescaled box. We first treat $R_n \rightarrow \infty$, with a second rescaling if the box shrinks, and then bounded R_n , including $R_n \rightarrow 0$. The closure error is $o(\min(1, R_n))$; its relative form is essential when calibration at support points requires division by R_n .

Use $r = r_0 + h_1 z$, $\theta = d_0/h_1$ (local notation, unrelated here to size-sweep time), and indices n along the growing systems. Put

$$\begin{aligned} m_n(z) &= (\alpha - \beta_0)(r)/d_0, & \tilde{m}_n(z) &= (\tilde{\alpha} - \beta_0)(r)/d_0, \\ Y_n(z) &= (S - r)/d_0, & \varphi_n(z) &= (\tilde{r}(\alpha(r)) - r_0)/h_1, \\ g_n(z) &= [\Gamma_s(k_*(r)) - r]/(d_0 h_1^2). \end{aligned}$$

The first three functions are bounded on compact sets with a common bound, and we can take $\beta'_0 \rightarrow c_\beta > 0$, $c_q = 1/c_\beta$ in these local ranges. We have on compacts

$$g_n'' = -A_* \tilde{m}_n + o(1), \quad \varphi_n = z + \theta(Y_n + o(1)),$$

the curvature statement a.e. (also distributionally, since the scalar first derivatives of Γ are absolutely continuous) with A_* now its positive limiting value. Here φ_n is a nondecreasing map. Indeed Taylor expand $\tilde{r}(v)$ at $K_*(v)$, $v = \alpha(r)$; the argument increment is $b(S - q) - C_0(S - r) + d$ and $\Gamma_*'' = O(\lambda u)$, so the quadratic error is $O(\lambda u d_0^2) = o(u^3 P_1)$. Also $q(v) - r = d_0(c_q m_n + o(1))$. Hence (C19), with $d = o(u^3 P_1)$, gives

$$\begin{aligned} g_n(\varphi_n) &= R_n Y_n + D_n(Y_n - c_q m_n) + e_n, \\ R_n &= \Gamma'_*(K_*(v))C_0/h_1^2, & D_n &= [1 - b(v)\Gamma'_*(K_*(v))]/h_1^2. \end{aligned} \tag{C20}$$

where $e_n = o(\min(1, R_n))$ in box-averaged absolute value (both measures as before) on all moving boxes of length comparable to θ in compact z -ranges. By (P2)–(P5), $D_n \asymp \lambda u^2/h_1^2 \lesssim 1$, $D_n = o(R_n)$; the pointwise lift pays the small v^2 offset in (P3), and $U \lesssim u$. On compacts D_n varies by $o(1)$, R_n by $o(R_n)$, by the active derivative bounds ($b' = O(\lambda u)$, $\Gamma_*'' = O(\lambda u)$). We subselect repeatedly.

We have uniform limiting bounds on g_n, g_n' independent of compact interval. Indeed intervals of length a sufficiently large fixed multiple of θ contain nonfloor physical mass $\gtrsim d_0$ by the buffer bounds on m_n ; there $R_n|Y_n| \leq O(R_n/(d_0 M_j)) = o(1)$. Thus (C20) calibrates bounded values at bounded spacings, sufficient by the curvature bound. Take a C_{loc}^1 limit g . Recall $\mathcal{H}' = 2C_0(r - S)$ in radius and

$$\mathcal{H}(\alpha' - \beta'_0) \leq 0, \quad \mathcal{H} \geq -O(C_0/M_j^2)$$

locally since β'_0 lies strictly between floor and cap. On nonfloor $\mathcal{H} \leq 0$.

Unbounded rescaled removal. Suppose first $R_n \rightarrow \infty$. Equation (C20) gives $Y_n = o(1)$ in box-averaged radius, hence uniformly on compacts (a violation persists over a fraction of a box on one side, by monotonicity of S). Then $\tilde{m}_n - m_n \rightarrow 0$ in local L^1 : $\tilde{r}(\alpha(r)) - r = o(d_0)$, and by Fubini the area between distributions on each such interval of length $O(h_1)$ costs $o(d_0)$ times $O(h_1)$ mass (crossings are local by monotonicity). Write

$$J_n(z) = \Gamma'_*(K_*(\alpha(r_0))) \mathcal{H}(r)/(2d_0 h_1^3), \quad \ell_1 = \lim c_q D_n.$$

The variational sign gives $J_n m_n' \leq 0$. Equation (C20) gives $J_n' = -R_n(0)Y_n = -g_n - \ell_1 m_n + o(1)$, where the error tends to zero in the moving radius-box integrals. Nearby nonfloor calibrations give $J_n = O(\theta)$, with a common local bound; the normalized M_j^{-2} potential error is $o(\theta)$. Take a locally uniform limit J and a weak-star limit $m_n \rightarrow m_\infty$. Then $J' = -g - \ell_1 m_\infty$ and $g'' = -A_* m_\infty$.

For clarity, the energy estimate follows by two integrations by parts. On a fixed interval, integrate $J_n m_n' \leq 0$, substitute the equation for J_n' , and use $m_n = -g_n''/A_* + o(1)$. This gives

$$0 \geq [J_n m_n] + \int g_n m_n + \ell_1 \int m_n^2 + o(1), \quad \int g_n m_n = -[g_n g_n'/A_*] + \int g_n'^2/A_* + o(1).$$

Consequently

$$\int (g_n'^2/A_* + \ell_1 m_n^2) \leq [g_n g_n'/A_* - J_n m_n]_{\text{left}}^{\text{right}} + o(1).$$

All boundary factors are uniformly bounded, independently of the fixed interval. Passing to the limit and then enlarging the interval first shows that $g' \in L^2(\mathbb{R})$. Bounded curvature makes g' uniformly continuous, hence it tends to zero at both ends. If $\theta \rightarrow 0$, then $J = 0$, so both boundary terms vanish as the endpoints tend to infinity. If θ has positive limit, the measure $d(c_\beta z + (\lim \theta)m_\infty)$ has support unbounded in both directions. Nonfloor approximants give $J = 0$ on that support, because the floor density vanishes. Send the endpoints to infinity through its support; again both boundary terms vanish. The energy estimate now gives $g' = 0$, and $g'' = -A_* m_\infty$ gives $m_\infty = 0$. Finally $J' = -g$ and boundedness of J give $g = 0$. For positive limiting θ , monotonicity of α then gives the pointwise contradiction $|m_n(0)| = 1$.

Unbounded removal: the shrinking-box subcase. If $\theta \rightarrow 0$, we have $\ell_1 > 0$ since $h_1^2 = \lambda u^2$. Put $K_n(t) = J_n(\theta t)/\theta$, $M_n(t) = m_n(\theta t)$. They are bounded on compacts with uniform limiting bounds, $K_n' = -\ell_1 M_n + o_{L_{\text{loc}}^1}(1)$ by the moving box estimate and

$g = 0$, $K_n M'_n \leq 0$. M_n converge on subselection in local L^1 and at continuity points by monotonicity with the deterministic trend added. Now $\ell_1 \int M_n^2 \leq -[K_n M_n] + o(1)$ on fixed intervals. The limit thus has finite global squared norm; sending endpoints to infinities through continuity points where it tends to zero makes that norm zero. Again monotonicity gives the pointwise contradiction.

Bounded rescaled removal, including a vanishing limit. Suppose now that R_n is bounded. Then $\theta = 1$. After adding their deterministic trends, m_n, Y_n are monotone; subselection therefore gives almost-everywhere and local L^1 limits. The transport estimate also gives $\tilde{m}_n - m_n \rightarrow 0$ in local L^1 . Indeed on nonfloor points the displacement $\tilde{r}(\alpha(r)) - r$ is $o(d_0)$; floor crossings travel only $O(d_0)$ and have density $o(1)$. Fubini bounds the area between the inverse distributions on a compact interval by $o(d_0 h_1)$. It follows that $g'' = -A_* m_\infty$. The normalized potential $\mathcal{H}(r)/(2C_0 d_0 h_1)$ has derivative $-Y_n$. Its lower bound and nearby nonfloor references give a bounded nonnegative limiting function $\bar{J}$.

We next calibrate at the support of $\mu = d(c_\beta z + m_\infty)$. The measures $\mu_n = d\alpha/h_1$ in this coordinate converge vaguely to μ . Every fixed neighborhood of a support point has positive limiting mass. Its floor mass is $o(1)$, while the mass integral of $|e_n|/R_n$ tends to zero. Hence it contains a nonfloor point with $e_n/R_n \rightarrow 0$. At this point $Y_n = o(1)$ and $D_n/R_n \rightarrow 0$, so (C20) gives $g_n(\varphi_n) = o(R_n)$. Shrink the fixed neighborhoods and choose a diagonal subsequence. This produces approximants to each support endpoint used below. At a contact the potential is zero; in a cap its nearest zero is within M_j^{-1} , whose normalized error vanishes. We conclude that $\bar{J} = g = 0$ on $\text{supp } \mu$, even if $R_n \rightarrow 0$.

A floor interval can still have positive rescaled radius length; only its mass has vanished. We therefore retain the separate radius estimate for e_n/R_n on any such support gap. After a further subsequence, for almost every radius coordinate z ,

$$s(z) = \lim \varphi_n(z) = z + Y(z), \quad g(s(z)) = (\lim R_n)Y(z), \quad \bar{J}' = -Y.$$

The function s is nondecreasing. On a support gap (a', b') , monotonicity and the endpoint approximants give $s \in [a', b']$. The gap is bounded, and g'' is affine there with slope $A_* c_\beta > 0$.

Suppose $g'(a') > 0$. If $\lim R_n > 0$, then $g(s) = (\lim R_n)(s - z)$ forces $s(z) > z$ almost everywhere just to the right of a' : this holds if s is near a' , where g is positive to its right, and also if s is farther inside the gap. Thus $\bar{J}' = -Y < 0$, contradicting $\bar{J}(a') = 0$ and $\bar{J} \geq 0$.

If $R_n \rightarrow 0$, then $g(s) = 0$, and the sign requires the relative error. Fix an almost-everywhere point $z > a'$ where the radius error divided by R_n tends to zero. If $s(z) = a'$, take nonfloor approximants $z_n^a \rightarrow a'$ as above. Eventually $z_n^a < z$, so $\varphi_n(z_n^a) \leq \varphi_n(z)$. Both images lie near a' , where $g'_n > 0$. Since the value at the approximant is $o(R_n)$ and R_n is asymptotically constant on compact intervals,

$$\liminf_n \frac{g_n(\varphi_n(z))}{R_n(z)} \geq 0.$$

Divide (C20) by R_n . It gives $Y(z) \geq 0$, contrary to $Y(z) = a' - z < 0$. Every other zero of g is bounded away from a' in this neighborhood, and hence gives $s(z) > z$. Again $Y > 0$ almost everywhere just to the right, contradicting the potential sign. Reflection excludes $g'(b') > 0$ in both cases.

But $g(a') = g(b') = 0$ and the affine curvature imply

$$g'(a') + g'(b') = A_* c_\beta (b' - a')^2 / 6 > 0.$$

At least one endpoint derivative must be positive. Thus no support gap exists.

Finally, the boundedness of m_∞ makes the support of $d(c_\beta z + m_\infty)$ unbounded in both directions. With no gaps it is all of $\mathbb{R}$, so $g = 0$ and $m_\infty = 0$. In each regime where the normalized discrepancies converge locally in L^1 to zero, the contradiction at the originally selected point is pointwise: monotonicity of the inverse distribution, after adding back its deterministic trend, gives for fixed continuity points $\pm h$

$$-Ch + o(1) \leq m_n(0) \leq Ch + o(1).$$

Here one may take $C = c_\beta/\theta_\infty$ when $\theta \rightarrow \theta_\infty > 0$, and $C = c_\beta$ in the bounded removal case $\theta = 1$. Letting $h \downarrow 0$ contradicts $|m_n(0)| = 1$. For the second-rescaling sequence the identical argument is applied to M_n , whose deterministic trend has slope tending to c_β . No pointwise relative error at the selected point is used.

This proves (C12), completing the finite induction. $\square$

Completion of Proposition 7.1. At its last stage the radius assertion converts by (C7) to (C3); the averaged excess $O(x^{4-2\zeta}u\Lambda^{1/3})$ suffices there. This completes the local comparison required for (C1).

Here are the final exponent checks. The local pressure derivative has scale λu when it is used to bound $u \int_{v \asymp u} A_v dv$. The optimized excess therefore contributes

$$x^{4-2\zeta} \lambda^{k_c/3-1} = x^4 \lambda^{-2s_0} (x^{-2\zeta} \lambda^{k_c/3-1+2s_0}).$$

The parenthesized factor is power small because $k_c/3 > 1 - 2s_0$ and ζ is chosen after p_* . The discrepancy term in (C3) contributes $u^4 a_{\text{goal}}^2 = x^4 \lambda^{-2s_0}$. Since $U^4 = x^4 \lambda^{-2s_*}$, the remaining factor $\lambda^{2\tau_0}$ pays dyadic logarithms and the permitted small losses. At the smallest dyad $u \asymp U$, the switch error from (P11), divided by λu and then by U^4 , is at most

$$x^{c'} \lambda^{5s_*/2-1},$$

which has a strict saving. Larger dyads only improve this estimate. The global budget handles the remaining dyads $u \geq \lambda^{b_\ell}$, as in the initial reduction. Both parts of (C2) were already obtained from the global comparison, with a final-strength power independent of subsequent sufficiently small choices of p_* . Thus the local induction does not enter the choice of the later size Taylor order. $\square$

8 Statistics for the genuine paths

The cavity equations require moments of the random overlap Q and of its centered version $Q - S_v$. We derive them from two outputs of the comparison argument. The mass-averaged budget (C1) first gives deterministic averaging windows for the single-path observable L_v . A useful window must contain enough field increase for projection while keeping the variation of S_v small. Balancing these requirements gives the pointwise and integrated low-mass estimates below, including the direct-path estimates needed during the strength comparison.

On size paths, the pointwise scalar-trial excess in (C2) has a separate role: it gives uniform coarse accuracy in every fixed moment. That argument uses the single-path inequalities and the scalar gap, and does not use the refined low-mass estimates. Its power saving therefore fixes the Taylor order before the final choice of p_* . We then combine the refined low-mass estimates with the pair inverse to close the high-mass covariance equation. Throughout, S_v is the untilted deterministic mean, including when a tilted law is tested.

8.1 Scales and the pointwise estimates

Fix the path parameters and the site count m . Unless stated otherwise, expectations and L^p norms in this section are then under the genuine untilted positive law. Strength variations use the direct path with multipliers $(0, 1)$. The comparison budget (C1) allows us to choose a deterministic number $1 \leq Z \leq x^{-C}$, depending on these fixed data, such that $\mathbf{E}_\gamma Z^4 \lesssim_\circ 1$ and, with $Y = UZ$,

$$\int_{v \gtrsim r} (S_v - q(v))^2 dv/r \leq CY^4/r^2 \quad (r \gtrsim U). \quad (\text{D1})$$

For example, let B be the maximum dyadic budget appearing in (C1) and take $Z = 1 + C(B/U^4)^{1/4}$. This explains both the fourth parameter moment and the pointwise bound (D1). The same estimate holds on fixed comparable enlargements of the dyads. Since $(S_v - q(v))^2 \leq A_v$, the budget indeed controls the mean error. The number Z is random only when the parameters γ are averaged; it is fixed under every positive-law expectation below.

We first assume $Y = o(1)$, with $U = x\lambda^{-s_*/2}$. Large Y and multiplier changes are treated separately. We will also use a **low-region variant**: the path is a size path, $Y = UZ = o(1)$ satisfies (D1) at small masses, and refined estimates are requested only below a sufficiently small fixed mass. Put

$$z_r^0 = \frac{x^3}{\sqrt{\lambda_1} r^3}, \quad g_0 = \lambda^{(3s_*-1-\eta)/2}, \quad g_B = g_0/Z^{3/2}.$$

Thus $z_r^0 = g_0(U/r)^3$. These are dimensionless projection scales, not scalar fields.

In envelope estimates with arbitrarily small power losses, the envelope itself may be chosen for the desired loss: more precisely we use a geometric slack $b_\epsilon = x^\epsilon$ with $\epsilon > 0$ arbitrarily small, and lose fixed powers of b_ϵ , as well as subpower factors. Power losses in the resulting applications, involving any fixed moment/expansion orders, can thus all be taken as small as desired. We write $\lesssim_\circ$ with this convention. Exponents in such estimates below do not cost an extra power of Z unless explicitly shown.

Proposition 8.1 (Pointwise moments). *Assume the direct-path hypotheses above and $Y = o(1)$, or their low-region size-path variant. For every fixed $1 \leq p < \infty$ the following estimates hold:*

$$\begin{aligned} \| |X_v|^2 \|_p + \| L_v \|_p + \| Q_{\text{split at } v} \|_p &\lesssim_p v + Y; \\ \| Q_{\text{split at } v - q(v)} \|_p + \| 2L_v - q(v) \|_p + \| |X_v|^2 - q(v) \|_p &\lesssim_{p,\diamond} e(v) \quad (v \gtrsim Y). \end{aligned} \quad (\text{D2})$$

In the low-region variant the second line is only low. e is deterministic conditional on the parameters, with, on each such dyad $v \asymp r$,

$$x^C \leq e(v) \lesssim r, \quad \int e(v)^2 dv/r \lesssim_\circ m_r^2, \quad m_r = Y^2/r.$$

On those regions $|d(v)| + |\dot{d}(v)| \lesssim_\circ \lambda r^2 e(v)$, by the corresponding path estimates. At $v \lesssim Y$ use $r = e = Y$ in this last assertion and (D2).

The envelope e depends on the deterministic path data and on the comparison majorant Z , not on a sample from the positive path law. We first construct it, including the windows needed in the proof.

8.2 Deterministic widths and averaging windows

All window lengths in this construction are measured in z ; mass integrals are converted by $dv = G_z dz$. On the active interval of a direct path, set $z = \xi \in [0, z_*]$, where $z_* = \xi_*$. Extend this coordinate increasingly over the plateau by

$$dz = U^2[(1 - w + d_+)^{-3/2} + (w - p_{\text{top}} + d_-)^{-3/2}] dw, \quad w = v/\ell,$$

and write $G_z = dv/dz$. The function $q(v(z))$ is Lipschitz. On the active interval G has bounded derivatives of every fixed order and is bounded above and below near zero, although it may vanish elsewhere. On the plateau put

$$\mathfrak{h}(z) = \max \left(\frac{U^2}{\sqrt{w - p_{\text{top}} + d_-}}, \frac{U^2}{\sqrt{1 - w + d_+}} \right).$$

The explicit coordinate derivative makes this width Lipschitz and gives $G_z \mathfrak{h}^3 \asymp U^4$. At the two ends of the plateau its width is comparable respectively to E_b and to one.

The extended z interval has bounded length. Partition it into dyadic bands $z \asymp r$ near zero and finitely many macroscopic intervals above, placing a boundary at the active endpoint. On each band its length is comparable to r , and $r \asymp \min(z, 1) \asymp v$. Fixed comparable enlargements may be used for buffers. We work on bands $r \gtrsim Y$. On their active parts (P4) gives a lower field slope, with the negative ramp slope absorbed as in (P7). On their plateau parts the formulas for dz/dw and H_e give $dh/dz \gtrsim \lambda_1$; these masses are macroscopic. Thus throughout the construction,

$$dh \geq c\lambda_1 r^2 dz. \tag{D3}$$

Write $f(v) = S_v - q(v)$ here; this f is not a pressure. Monotonicity, (D1), and $q(v) \lesssim v$ imply $S_v \lesssim v + Y$.

The active width has three requirements: enough mass for projection, enough room for the mean error, and compatibility with the endpoint ramp. The next four functions impose these requirements. Put $r(z) = \min(z, 1)$. Reflect the active G_z evenly at the endpoints for defining integrals outside, and add an artificial constant x^K , K arbitrarily large fixed. Use

$$H_0(z) = \min(b_\epsilon r(z), l_0), \quad l_0^2 \int_{z-l_0}^{z+l_0} (G_{s,\text{refl}} + x^K) ds = U^6 / r(z)^3.$$

Add the cone envelope

$$H_f(z) = \sup_{s \text{ active}} [\min(b_\epsilon r(s), |f(v(s))|/Z^2) - A|s - z|]_+$$

with a sufficiently large constant A . Near the upper active endpoint, writing $d_* = z_* - z$, $E = E_b$, add also the two functions

$$(E/2 - d_*)_+, \quad [2 \min(E, b_\epsilon) - (d_* - E)_+]_+.$$

Take the maximum of the four functions as $\mathfrak{h}(z)$. We only need this width at $z \gtrsim Y$. Finally set $e(v(z)) = \min(r(z), Z^2 \mathfrak{h}(z))$.

We will repeatedly compare the mass of neighboring intervals in the z coordinate. On the active side, if J and J' have comparable length $l \leq Cb_\epsilon$, lie within $O(l)$ of one another, and stay inside a fixed comparable band, then

$$\sup_J G \leq C \left(\frac{1}{|J'|} \int_{J'} G dz + x^K \right).$$

Here and below K may be increased to make additive errors negligible. To prove this, expand the smooth nonnegative function G to a fixed order at the scale l . Choose that order so that the Taylor remainder is at most x^K . On the rescaled bounded intervals, polynomial norm equivalence bounds the supremum by the average on any comparable subinterval; nonnegativity permits replacing the polynomial's absolute average by that of G , with the same remainder. The constants depend on the fixed Taylor order, not on the interval. For a reflected interval meeting an endpoint, restrict to its part on the original side, whose length is comparable. This gives the same comparison.

Lemma 8.2 (Width construction). *The preceding construction gives a deterministic envelope with the size, integral, and defect bounds stated in Proposition 8.1. The random moment bounds (D2) will be proved in the next subsection. The widths and windows have the following additional properties.*

- (i) *At comparable points, $\mathfrak{h}(z') \lesssim \mathfrak{h}(z) + |z' - z|$, with comparability of widths when $|z' - z|$ is a sufficiently small fraction of $\mathfrak{h}(z)$. Also $x^C \leq \mathfrak{h} \lesssim r$, $e \gtrsim b_\epsilon Y^2/r$, $|f| \lesssim b_\epsilon^{-1}e$, and $\int_{\text{band}} G_z(\mathfrak{h}/r)^2 dz/r \lesssim_\diamond (U/r)^4$.*
- (ii) *Unless $Z^2 \mathfrak{h} \gtrsim b_\epsilon r$, there are past and future windows of z -length comparable to $\mathfrak{h}$, adjacent or at that distance, each with average G_* there satisfying*

$$\sup G_z \lesssim G_*, \quad G_*(\mathfrak{h}/r)^3 \gtrsim (U/r)^6. \quad (\text{D4})$$

They can be taken small fixed fractions of the indicated scales. From such a nontrivial point there is also remaining forward z -length at least a small fixed macroscopic constant (for taking longer layers up to a macroscopic mass), with (D3) holding using each successive scale.

- (iii) *Partition any full band as above into consecutive tiles of lengths comparable to $\mathfrak{h}$, with this width and e comparable over each tile. Writing $\mu_I = \int_I G_z dz/r$,*

$$\sup_I G_z \lesssim b_\epsilon^{-C}(r\mu_I/|I| + x^K), \quad \Delta_I S \lesssim b_\epsilon^{-1} \min(r, Z^2|I|), \quad \sum_I \sup_I (e/r)^2 \lesssim Z^2. \quad (\text{D5})$$

The variation assertion holds also on comparable-scale enlargements of these tiles within the path.

Proof.

The intrinsic width. The map $l \mapsto l^2 \int_{z-l}^{z+l} (G_{s,\text{refl}} + x^K) ds$ is strictly increasing. Ball inclusion therefore compares its solution at neighboring centers. Together with the cap $b_\epsilon r$, this proves the stated quasi-Lipschitz bound for H_0 and comparability within a small fraction of H_0 . Boundedness of G gives

$$H_0 \gtrsim \min(b_\epsilon r, U^2/r).$$

The adjacent-interval comparison at this radius also gives $G_z H_0^3 \lesssim U^6/r^3$, up to an arbitrarily small additive error. Since G is bounded, the interpolation

$$G_z H_0^2 = (G_z H_0^3)^{2/3} G_z^{1/3} \lesssim U^4/r^2$$

proves the squared width budget after integration over a band of z -length $O(r)$ and division by r^3 .

The deviation cone. For each source point s , the function $[\min(b_\epsilon r(s), |f(v(s))|/Z^2) - A|s - z|]_+$ is A -Lipschitz. Their supremum is therefore Lipschitz too. At its source, a height of order l implies $|f| \gtrsim Z^2 l$. If f is positive, monotonicity of S and Lipschitz continuity of $q(v(z))$ preserve half this deviation on a forward interval of length cl ; for a negative deviation use the backward interval. Thus each source of height l supplies a one-sided interval J_s with

$$Z^4 l^2 \int_{J_s} G_z dz \lesssim \int_{J_s} f(v(z))^2 G_z dz.$$

Its cone is supported within $O(l)$ of s . The adjacent-interval comparison bounds the mass of that support by the mass of J_s , with negligible additive error. At each dyadic height $l \geq x^K$, choose separated source points to cover the cone superlevel set. The corresponding intervals have bounded overlap at that height. Summing the preceding inequality, and then the logarithmically many heights, gives

$$\int_{\text{band}} G_z H_f(z)^2 dz \lesssim_\diamond Z^{-4} \int_{\text{enlarged band}} f(v)^2 dv \lesssim_\diamond U^4/r.$$

Sources of height below x^K contribute negligibly. The lower boundary causes no difficulty because $l \leq b_\epsilon r$. If a positive source's forward interval reaches the upper active endpoint, its deviation persists through the plateau. The macroscopic version of (D1) then gives $Z^4 l^2 \lesssim Y^4$, or $l \lesssim U^2$; these sources already obey the same macroscopic squared width budget.

The plateau and its active edge. The reference q is constant on the plateau. A positive deviation of $S - q$ persists towards its upper end; a negative deviation persists towards its lower end. Integrate (D1) over the corresponding endpoint strip, whose mass is comparable to the distance to that endpoint. Away from the capped strips, the result is $|f| \lesssim Z^2 \mathfrak{h}$. Inside the upper d_+ strip, $\mathfrak{h} \asymp 1$ and the bound follows from boundedness of S, q . For a negative deviation inside the lower d_- strip, the backward interval must instead use the active side. The last active z -interval of length $E = E_b$ has mass comparable to $d_- = U^4/E^2$. Either $|f| = O(Z^2 E)$ already, or the Lipschitz variation of q over that interval is too small to remove half the negative deviation. In the latter case (D1) gives

$$|f|^2 \frac{U^4}{E^2} \lesssim Y^4, \quad \text{hence} \quad |f| \lesssim Z^2 E,$$

which is the required lower-edge bound.

The explicit plateau formula also gives $\int_{\text{plateau}} \mathfrak{h}^2 dv \lesssim_\diamond U^4$ by integrating $U^4/(d + d_\pm)$ from each endpoint. The first extra active function has height $O(E)$ on a region of mass $O(U^4/E^2)$, so its squared cost is $O(U^4)$. The second has height at most $2 \min(E, b_\epsilon)$; its part inside the last E costs the same. For its additional layer of length $2 \min(E, b_\epsilon)$, compare adjacent active intervals at that length. Their mass is controlled by the preceding last- E region, giving again $O_\diamond(U^4)$. These two functions also majorize the active ramp defect and its parameter derivative divided by λ , with at most a b_ϵ^{-1} loss. The bounds in (P6)–(P8) then give the claimed defect estimates. Below Y , those estimates use only the low and mask bounds from (P6), (P8).

Matching the two coordinates at the join. Suppose first that $E \lesssim b_\epsilon$, and let $d_* = z_* - z$ on the active side. We claim, for $d_* \leq C b_\epsilon$, that

$$|f(v(z))|/Z^2 \lesssim E + d_*.$$

For $d_* \lesssim E$, use the last active interval of length E or the adjacent plateau, in the direction in which the deviation persists. The preceding mass calculation proves the claim. For $d_* \gtrsim E$, an interval of length comparable to d_* fits on the active side in either direction. Unless the claimed bound already holds, the deviation persists on that interval. Adjacent interval comparison bounds below its mass by that of the last active length E , up to negligible error. Applying (D1) as above would then give $|f| \lesssim Z^2 E$, proving the claim in this case too.

With A sufficiently large, sources of the cone obeying this linear bound imply $H_f(z) \lesssim E + d_*$. The same bound holds for H_0 : a ball of radius $C(E + d_*)$ contains the last active length E , and its squared radius times that mass is at least a constant times U^4 , which exceeds the intrinsic target U^6/r^3 on this macroscopic band. When $E > b_\epsilon$, the corresponding upper bounds follow from the caps in H_0, H_f . The explicit endpoint functions provide the lower bound near the join. Thus at distances much smaller than E , both the active and plateau widths are comparable to E . Their separate quasi-Lipschitz bounds consequently remain valid across the join.

Windows and tiles. Consider first an active point with $d_* \geq E$ for which $Z^2 \mathfrak{h} < b_\epsilon r$ up to the fixed threshold in (ii). The intrinsic cap is then inactive. Its defining equation and $\mathfrak{h} \geq H_0$ give

$$\mathfrak{h}^2 \int_{z-C\mathfrak{h}}^{z+C\mathfrak{h}} G_s ds \gtrsim U^6/r^3,$$

with negligible additive error. Choose a past interval and a future interval, each a small fixed fraction of $\mathfrak{h}$ and within $O(\mathfrak{h})$ of z . Since $\mathfrak{h} \lesssim E + d_* \lesssim d_*$, sufficiently small fixed fractions fit wholly on the active side. Near zero the cap $\mathfrak{h} \lesssim b_\epsilon r$ also leaves room for the past window. Adjacent-interval comparison gives, for each chosen window, $\sup G \lesssim G_*$ and $G_* \mathfrak{h}^3 \gtrsim U^6/r^3$, which is (D4).

For active points with $0 \leq d_* \leq E$, the endpoint functions and the nontrivial accuracy requirement force $E \lesssim b_\epsilon$ and $\mathfrak{h} \asymp E$. The plateau width is likewise comparable to E in a sufficiently small fixed E -neighborhood of the join. The mass of the last active interval of length E is comparable to U^4/E^2 , so comparable active windows there have average density $G_* \asymp U^4/E^3$. The adjoining plateau windows in that small neighborhood have the same bound. Hence $G_* \mathfrak{h}^3 \asymp U^4 \gtrsim U^6$, as required on this macroscopic band. Outside that neighborhood on the plateau, the explicit width formula allows small fixed fractional windows wholly within the plateau and makes G comparable on each of them. The identity $G \mathfrak{h}^3 \asymp U^4$ gives the same conclusion. Finally the terminal d_+ strip has macroscopic z -length and width comparable to one. A point requiring nontrivial accuracy must precede that strip, leaving the fixed forward length needed for longer projection layers.

Tiling can for instance use equal bounded small steps in the integral of $dz/\mathfrak{h}$ on each band (resizing the steps by bounded factors to fit). Width comparability holds by the local property. On an active tile longer than b_ϵ , subdivide into b_ϵ -size pieces for (D5); otherwise use the adjacent-interval bound directly. For a tile I of length comparable to $\mathfrak{h}$, use $S = q + f$, the Lipschitz bound on $q(v(z))$, and $|f| \lesssim b_\epsilon^{-1} e$ on its enlargement. This gives

$$\Delta_I S \lesssim b_\epsilon^{-1} \min(r, Z^2 |I|).$$

Finally, with $a_I = \sup_I e/r$, width comparability and $e = \min(r, Z^2 \mathfrak{h})$ imply

$$a_I^2 \lesssim \frac{Z^2 |I|}{r}, \quad \sum_I a_I^2 \lesssim Z^2,$$

since the disjoint tiles fill an interval of length $O(r)$. The same argument applies to fixed comparable enlargements. Combining the squared budgets for the four defining widths also

gives

$$\int e(v)^2 dv/r \leq Z^4 \int \mathfrak{h}(z)^2 G_z dz/r \lesssim_\diamond Y^4/r^2 = m_r^2.$$

The lower intrinsic width yields $e \gtrsim b_\epsilon Y^2/r$ for $r \gtrsim Y$, including when either cap is active.

The low-region construction. Choose a small fixed working interval and a larger fixed buffer interval on which the size-path inverse is regular. Use $z = v$ there, so $G = 1$ and (D3) holds. Define

$$\mathfrak{h} = \max\{\min(b_\epsilon r, Y^2/r), H_f\}, \quad e = \min(r, \mathfrak{h}),$$

where the cone now uses $|f|$ in place of $|f|/Z^2$. The preceding cone argument gives squared width budget $(Y/r)^4$ on the working interval. The intrinsic lower width is now $\min(b_\epsilon r, Y^2/r)$, which is at least the direct-path lower width because $Y \geq U$. Consequently the window bound (D4) remains valid. The pointwise and variation bounds use $e = \min(r, \mathfrak{h})$, and the tile sum is now $\sum_I \sup_I (e/r)^2 \lesssim 1$; these are precisely the later bounds in which $Z_h = 1$.

If more room is needed for buffers, start a fixed Lipschitz ramp only beyond the working interval, increasing $\mathfrak{h}$ to a macroscopic value before the end of the buffer interval. No integrated width estimate is used on that auxiliary ramp. Longer layers stop after a fixed small macroscopic length within the regular region. The defect bounds are the low size-path bounds of (P8).

Measurability of the construction. All windows can be chosen measurably and deterministically conditional on the path parameters. For example, fix the fractional lengths and casewise windows constructed above, using the prescribed comparable band enlargements for buffers, and tile by equal increments of $\int dz/\mathfrak{h}(z)$. The cone supremum may be formed over a countable dense set together with the specified one-sided values of the monotone function S . Its value is unchanged, since q is continuous and the one-sided limits of S exist. The implicit intrinsic radius is the monotone solution of its displayed integral equation. These constructions are measurable functions of the deterministic path data. In particular, none of the windows is selected after observing a spin sample. $\square$

8.3 Proof of the pointwise moments

Proof of Proposition 8.1. All estimates in this proof use the untilted positive law with fixed path data. Lemma 6.2 supplies the conditional block decomposition (6.1) and its integrated-increment bounds. At each deterministic fork cut, partition the residual on each of two conditionally independent continuations into future martingale increments, including a last residual ending at the terminal Gibbs spin. A block pair contributes its squared inner product, weighted by the largest mass in the Hessian block. We may bound that weight by the largest mass in either block and project the block with the later starting point.

For the opposite increment write $A = |A|\hat{A}$, setting $\hat{A} = 0$ when $A = 0$. Conditional on the common prefix, its whole branch is exterior data for the projected continuation. Applying (A3) to $\hat{A}$ therefore preserves the projected branch's normalized future kernel. The factor $|A|^2$ remains in the subsequent integral. Fix a permitted loss $\varepsilon > 0$. At each fixed fork, declare an exception when the squared projection exceeds $x^{-\varepsilon}$ times its squared scale in (A3). Choose the projection moment sufficiently high after ε and the required negligible exponent K are fixed. Markov's inequality then makes this exception have probability $O(x^K)$. The factor $x^{-\varepsilon}$ is included in the permitted losses below. Bounded spins, integration over the fork coordinate, and conditional Jensen make its contribution negligible in the integral norm. This does not require an event simultaneous over all real forks, or a higher-moment bound

for $|A|$ proportional to the bootstrap constant. These operations can first be performed with inserted deterministic cuts and then passed to the fixed-size path limit.

Raw-moment bootstrap. Fix a large even exponent p and set

$$B_0 = \max \left(1, \sup_v \frac{\| |X_v|^2 \|_p}{v + Y} \right).$$

This number is finite at each fixed size because $|X_v| \leq 1$ and $Y > 0$. At a small scale $r \gtrsim Y$ in the regular inverse region, let J be a fork window of z -length comparable to r . We claim

$$\left\| \int_J W_z dz/r \right\|_{p/2} \lesssim_{p,\circ} (z_r^0)^2 r^3 B_0. \quad (\text{D6})$$

Use an initial block of arbitrarily small inverse-polynomial length, then blocks starting at doubled distances from the fork, and finally a terminal residual. Stop the doubling at a fixed small macroscopic distance, with a full preceding interval still inside the regular region. Consider a block pair other than two initial blocks, and let lr be its later starting distance. If $l \lesssim 1$, the projection gap has mass at least cr and field length at least $c\lambda_1 r^3 l$. Its squared projection coefficient, before the mass weight, is consequently $O_\circ(x^6/(\lambda_1 r^5 l))$.

The earlier squared anchor has the integral bound

$$\left\| \int_J |A_z|^2 dz/r \right\|_{p/2} \leq C_p l B_0 r.$$

Indeed, (6.2) integrates translated grids of disjoint increments, and the raw bootstrap bounds the terminal squared norm on the enlarged window by $C B_0 r$. If that anchor belongs to the independently resampled continuation, apply (6.3) to its nonnegative squared increment to average it back to the fork. Thus the contribution from this block pair is at most

$$\underbrace{Cr}_{\text{mass weight}} \underbrace{C_{p,\circ} \frac{x^6}{\lambda_1 r^5 l}}_{\text{squared projection coefficient}} \underbrace{C_p l B_0 r}_{\text{integrated squared anchor}} \lesssim_{p,\circ} (z_r^0)^2 r^3 B_0.$$

The two initial blocks instead use the sliding estimate without projection. Choose their length so small that the resulting term is negligible after the polynomial prefactors.

For a later starting distance comparable to $r' \gtrsim r$, use a full gap at masses comparable to r' , with field length at least $c\lambda_1 (r')^3$. This includes a macroscopic r' for the terminal block. The mass weight, squared projection coefficient, and raw squared anchor now cost respectively

$$Cr', \quad C_{p,\circ} \frac{x^6}{\lambda_1 (r')^5}, \quad C_p B_0 r'.$$

Their product is $O_{p,\circ}((z_{r'}^0)^2 (r')^3 B_0)$, no larger than the right side of (D6). The normalized fork window has bounded length. Summing the logarithmically many block pairs in (6.1) proves (D6). The sliding and conditional integration bounds are uniform in the inserted grid sizes, so this argument uses no size-uniform path approximation rate and no derivative of S .

Let $L_J = |J|^{-1} \int_J L_u du$ for a future mass window J of width comparable to r . The future bracket identity in Lemma 6.3 gives $|X_v|^2 \leq 2\mathbf{E}_v L_J$, while $\mathbf{E} L_J = O(r)$ by the deterministic mean bound. Equations (A1) and (D6) give

$$\|L_J - \mathbf{E} L_J\|_p^2 \lesssim r^{-1} \left\| \int_J W_u du/r \right\|_{p/2} \lesssim_{p,\circ} (z_r^0)^2 r^2 B_0.$$

Conditional Jensen therefore yields $\| |X_v|^2 \|_p \lesssim_p r(1 + z_r^0 \sqrt{B_0})$. For $v \lesssim Y$ use a future window at scale Y ; at macroscopic masses use $|X_v| \leq 1$. Taking the supremum in the definition of B_0 now gives $B_0 \leq C_p(1 + g_0 \sqrt{B_0})$, including the harmless subpower factors. The fixed power saving in g_0 absorbs these factors, and this quadratic inequality bounds B_0 uniformly.

For L, Q , project endpoints into a future interval at comparable scale $v + Y$ and then telescope both branches in future doubled layers with terminal block, as above. Each tail product uses (A3) on the later layer (with comparable preceding clock fitting strictly after the fork); the opposite anchor has norm $O_p(\sqrt{r'})$ at worst at scale r' , so these tails cost $O_{p,\diamond}(z_{r'}^0 r')$. For L the anchor X_v already precedes the gap. This gives the first line of (D2). It suffices here too to telescope at small scales.

Refined-moment bootstrap. For this argument write $Z_h = Z$ on direct paths and $Z_h = 1$ in the low-region size-path variant. Let $B_1 \geq 1$ be the least constant for which

$$\| |X_{z'} - X_z|^2 \|_p \leq B_1 Z_h^2 (|z - z'| + \mathfrak{h}(z))$$

holds for comparable nearby cuts in the bands $r \gtrsim Y$ and their buffers. The supremum defining B_1 is finite at fixed size because $\mathfrak{h} \geq x^C$; take it simultaneously over the bands to avoid artificial interior boundaries. At $z \asymp Y$ the raw bound already suffices with a b_ϵ^{-1} loss, as it does on the large-width auxiliary ramp. Cuts whose separation is not a small fraction of their scale can likewise use the raw bound.

Let J be one of the windows in (D4) near z_0 , and put $h_0 = \mathfrak{h}(z_0)/r$, so $|J| \asymp r h_0$. We claim

$$\left\| \int_J W_z dz/r \right\|_{p/2} \lesssim_{p,\diamond} (z_r^0)^2 r^3 h_0 B_1 Z_h^2.$$

For a block pair whose later starting distance is lr with $l \ll 1$, the same mass and projection factors as above apply. The integrated earlier anchor satisfies, in both ranges of l ,

$$\left\| \int_J |A_z|^2 dz/r \right\|_{p/2} \leq C_p l B_1 Z_h^2 r h_0.$$

When $l \geq h_0$, the bootstrap and quasi-Lipschitz width bound give $\| |A_z|^2 \|_{p/2} \leq C B_1 Z_h^2 l r$ pointwise; integration contributes $|J|/r \asymp h_0$. When $l < h_0$, apply (6.2) after subtracting the initial martingale value on an enlargement of J . Its terminal squared increment is bounded by $C B_1 Z_h^2 r h_0$, and the sliding estimate contributes l . The conditional integration inequality handles a resampled anchor in exactly the same way as in the raw argument. Hence each such pair costs

$$\underbrace{C r}_{\text{mass weight}} \underbrace{C_{p,\diamond} \frac{x^6}{\lambda_1 r^5 l}}_{\text{squared projection coefficient}} \underbrace{C_p l B_1 Z_h^2 r h_0}_{\text{integrated squared anchor}} \lesssim_{p,\diamond} (z_r^0)^2 r^3 h_0 B_1 Z_h^2.$$

The pair of initial blocks again has negligible cost. Longer layers use the already closed raw bound. Their product at scale $r' \gtrsim r$ is $O_{p,\diamond}((z_{r'}^0)^2 (r')^3)$ pointwise, and the fork window contributes h_0 . Since $B_1 Z_h^2 \geq 1$ and $(z_{r'}^0)^2 (r')^3 \leq (z_r^0)^2 r^3$, these layers obey the same bound. The forward length in Lemma 8.2 supplies their full projection gaps, including the terminal one. Summing the blocks proves the claim.

The mass of this window is comparable to $G_* r h_0$, and $\sup G \lesssim G_*$. Apply (A1) with the constant weight $1/(G_* r h_0)$ on the window. The squared centered norm of its mass average is at most

$$C_{p,\diamond} \frac{G_* r (z_r^0)^2 r^3 h_0 B_1 Z_h^2}{(G_* r h_0)^2} = C_{p,\diamond} \frac{(z_r^0)^2 r^2 B_1 Z_h^2}{G_* h_0} \lesssim_{p,\diamond} r^2 B_1 Z_h^2 h_0^2.$$

In the last step we used $z_r^0 = g_0(U/r)^3$ and $G_* h_0^3 \gtrsim (U/r)^6$ from (D4). We now apply the bracket identities of Lemma 6.3. Earlier and later averages bracket $|X_z|^2$. To see also how they control increments, let $s \leq z \leq z'$ and use

$$2\mathbf{E}_z L_s = |X_z|^2 - |X_z - X_s|^2.$$

Average s over one common past buffer before z . A future buffer after each of z, z' bounds its squared norm from above; the common past identity then bounds the averages of $|X_z - X_s|^2$ and $|X_{z'} - X_s|^2$. For every such s ,

$$|X_{z'} - X_z|^2 \leq 2|X_{z'} - X_s|^2 + 2|X_z - X_s|^2.$$

The means of these buffer differences cost at most $Cb_\epsilon^{-1}Z_h^2(|z' - z| + \mathfrak{h}(z))$ by the deterministic variation bound. Their centered parts have the bound just proved, with conditional Jensen where needed. Consequently

$$\| |X_{z'} - X_z|^2 \|_p \leq C_{p,\diamond}(1 + \sqrt{B_1})Z_h^2(|z' - z| + \mathfrak{h}(z)).$$

If either cut has $Z_h^2 \mathfrak{h} \gtrsim b_\epsilon r$, the raw bound already gives this inequality, using the quasi-Lipschitz width bound to compare the two cuts. The lower band and the auxiliary ramp are handled in the same way. The supremum defining B_1 therefore satisfies $B_1 \leq C_{p,\diamond}(1 + \sqrt{B_1})$, so it is bounded with the permitted loss. Applying the same two-sided brackets at a single cut gives $\| |X_v|^2 - q(v) \|_p \lesssim_{p,\diamond} e(v)$.

Finally project the endpoints of an overlap, or the terminal endpoint in L , to a cut a distance comparable to $\mathfrak{h}$ later. Polarization about the common ancestor expresses the projected inner product through the squared norms and squared differences just bounded. The remaining tails, estimated by (A3) and longer layers, cost at most $O_{p,\diamond}(z_r^0 r / \sqrt{h_0})$. The lower width $h_0 \gtrsim b_\epsilon(U/r)^2$ gives

$$\frac{z_r^0 r}{\sqrt{h_0}} = r h_0 \frac{g_0(U/r)^3}{h_0^{3/2}} \lesssim b_\epsilon^{-3/2} g_0 r h_0 \lesssim_\diamond e(v)$$

at every nontrivial cut. This completes (D2). $\square$

8.4 Integrated variances and centered polynomials

The pointwise envelope locates the regions of small overlap error. For the pressure expansion we also need a gain after integration in mass, including on regions where that envelope is relatively large. We first use the source inequality and the deterministic variation of S on each tile to bound the integral of W . Mass averaging then gives integrated overlap variances, and the polynomial single-path inequality gives centered integral bounds for the later allocation argument.

Continue with (D1)–(D6); band integrals without limits below are over $v \asymp r$, $r \gtrsim Y$. Put $Z_h = Z$ for the direct path, $Z_h = 1$ in the low-region variant. The following estimates concern active overlaps (unmasked).

Lemma 8.3 (Integrated variance estimates). *For every fixed integer $k \geq 0$, the following bounds hold. All quantities on the right are untilted; in particular this convention applies to S, D :*

$$\begin{aligned} \mathbf{E} \int W e(v)^{2k} dv/r &\lesssim_\diamond (z_r^0)^2 Z_h^2 r^3 \begin{cases} 1 & k = 0, \\ m_r^2 r^{2k-2} & k \geq 1, \end{cases} \\ \int \mathbf{E}[X^\top C_v X + \text{tr } C_v^2] dv/r &\lesssim_\diamond (z_r^0)^2 Z_h^2 r^2, \quad \int D_v^2 dv/r \lesssim_\diamond z_r^0 Z_h r^2. \end{aligned} \tag{D7}$$

Proof. On a tile I in a band $v \asymp r$, the inequalities $v \mathbf{E} W \leq \mathbf{E} \operatorname{tr} H^2$ and $dS \geq m \mathbf{E} \operatorname{tr} H^2 dh$ from (A1), together with (D3), imply

$$\int_I \mathbf{E} W_z dz/r \lesssim \frac{x^6}{\lambda_1 r^4} \Delta_I S \lesssim_\diamond (z_r^0)^2 r^3 Z_h^2 \frac{|I|}{r}.$$

Here $Z_h = Z$ for the direct construction and $Z_h = 1$ for the low-region construction, exactly as in their respective variation bounds. Multiply by the supremum of G on I and use (D5) to obtain

$$\int_I \mathbf{E} W_z G_z dz/r \lesssim_\diamond (z_r^0)^2 r^3 Z_h^2 (\mu_I + x^K).$$

There are only polynomially many tiles because their widths have a polynomial lower bound; increasing K absorbs all additive errors. Summing gives the first line of (D7) when $k = 0$. When $k \geq 1$, e is comparable on each tile and $e \lesssim r$, so

$$\sum_I \sup_I e^{2k} \mu_I \lesssim r^{2k-2} \int e^2 dv/r \lesssim_\diamond r^{2k-2} m_r^2.$$

This proves the weighted version. Since $H_v \succeq v C_v$, $\mathbf{E} \operatorname{tr} C_v^2 \leq \mathbf{E} W_v/v$; the same bound also proves the trace part of the second line.

The mixed covariance term. Telescope X_z backwards over dyadic distances lr , down to an arbitrarily small inverse-polynomial length, and retain an oldest anchor a fixed small fraction of r earlier. For each nonterminal increment, the gap after its end allows (A3) against $y - X_z$. Its squared projection coefficient is at most $C_\diamond (z_r^0)^2 r/l$, multiplied by the increase of S over $[z - 2lr, z - lr]$.

We need the integral of that increase against $G_z dz/r$. If $lr \gtrsim \mathfrak{h}(z)$, the deterministic variation estimate bounds it pointwise by $C_\diamond Z_h^2 lr$. If lr is shorter than a tile I , Fubini for the positive measure dS gives

$$\int_I [S(z - lr) - S(z - 2lr)] dz \leq lr \Delta_{I^+} S,$$

where I^+ is a fixed enlargement. Multiply by $\sup_I G/r$, use $\Delta_{I^+} S \lesssim_\diamond Z_h^2 |I|$, and sum the tile mass bounds from (D5). In both cases the weighted integral is $O_\diamond(r Z_h^2 l)$. Multiplication by the projection coefficient therefore costs $O_\diamond((z_r^0)^2 r^2 Z_h^2)$. The last tiny increment is treated without projection by the same sliding estimate, choosing its length to dominate all polynomial prefactors. The oldest anchor uses a comparable field gap from (D3) and its raw moment from (D2). Only logarithmically many layers occur. Summing proves the claimed mixed covariance bound.

Mass averaging. Let $h = r z_r^0 Z_h$ be a width in the mass coordinate. If h is comparable to or larger than r , the raw bound $D_v = O_p(r)$ already proves the desired estimate. Otherwise bracket $|X_v|^2$ by mass averages of L on its two sides. The points within distance $O(h)$ of a band edge can be discarded at cost $O(r^2 h/r)$ using (D2). For the remaining points, monotonicity and $\Delta S \lesssim r$ give the same cost for the integrated squared mean error. More explicitly, a sliding increment of S is at most Cr , and its integral in dv/r is $O(h)$; its squared integral is therefore $O(rh)$. By (A1) and Fubini, the centered averages contribute at most $Ch^{-1} \int \mathbf{E} W dv/r$. Thus

$$\int \operatorname{Var}(|X_v|^2) dv/r \lesssim_\diamond rh + \frac{(z_r^0)^2 Z_h^2 r^3}{h} \lesssim_\diamond r^2 z_r^0 Z_h.$$

The first two parts of the lemma control the other terms in the exact conditional-covariance identity

$$D_v^2 = \text{Var}(|X_v|^2) + 2\mathbf{E}X_v^\top C_v X_v + \mathbf{E} \text{tr} C_v^2.$$

This proves the last assertion of (D7). $\square$

Lemma 8.4 (Large-envelope localization). *There is also a localized bound for the direct path. For any sufficiently small fixed $\delta' > 0$, on all the bands in (D7) together,*

$$\int_{\{e(v) \geq x^{\delta'}\}} D_v^2 dv \lesssim_\diamond x^{-C_0 \delta'} z_1^0 Y^2 Z^2 \quad (\text{D8})$$

with an absolute finite loss exponent C_0 independent of δ' .

Proof. Retain every tile meeting $\{e \geq x^{\delta'}\}$. Such a tile has $r \gtrsim x^{\delta'}$, and width comparability makes e at least a constant times $x^{\delta'}$ throughout it. Write $a_I = \sup_I e/r$ and recall that its mass is $r\mu_I$.

Bracket within a retained tile using mass windows of width h . The pointwise error bound in (D2) gives squared accuracy $O_\diamond(r^2 a_I^2)$ near the tile edges. Since their mass is $O(h)$, this costs $O_\diamond(r a_I^2 h)$ in dv/r . Monotonicity and $\Delta_I S \lesssim_\diamond r a_I$ give that same bound for the interior mean errors. The tile estimate for W in the preceding proof gives centered cost $O_\diamond((z_r^0)^2 r^3 Z^2 \mu_I / h)$. Balance these two terms by taking

$$h = \frac{r z_r^0 Z \sqrt{\mu_I}}{a_I}, \quad r a_I^2 h + \frac{(z_r^0)^2 r^3 Z^2 \mu_I}{h} \lesssim r^2 z_r^0 Z a_I \sqrt{\mu_I}.$$

If h does not fit in the tile, use the raw squared error on the whole tile instead. Its cost $r^2 a_I^2 \mu_I$ is no larger than the displayed bound when $h \gtrsim r \mu_I$. This also handles overlapping buffers. Tiles of negligible mass may first be discarded so the additive x^K errors in the W bound do not enter the denominator.

The trace contribution is bounded by the tile W estimate divided by r . For the mixed term use Cauchy–Schwarz and the raw fourth moment of X on the tile. These two contributions together cost at most $O_\diamond(r^2 z_r^0 Z \mu_I)$. To sum them and the variance bound, the squared envelope budget gives on the retained tiles

$$\sum_I \mu_I \lesssim_\diamond x^{-2\delta'} \frac{Y^4}{r^2} \lesssim_\diamond x^{-C\delta'} Y^4, \quad \sum_I a_I \sqrt{\mu_I} \leq \left(\sum_I a_I^2 \right)^{1/2} \left(\sum_I \mu_I \right)^{1/2} \lesssim_\diamond Z x^{-C\delta'} Y^2.$$

The second inequality uses (D5). Converting dv/r back to dv multiplies a band bound by r , and $r^3 z_r^0 = z_1^0$. Since $Y = o(1)$ and $Z \geq 1$, the sum of the mixed and trace terms is also bounded by $x^{-C\delta'} z_1^0 Y^2 Z^2$. Summing the logarithmically many retained bands gives (D8), with a fixed exponent C_0 independent of δ' . $\square$

Lemma 8.5 (The lower band). *At $v \leq CY$ we will discard $v < x$ or treat it by raw bounds. For dyads of size $w \lesssim Y$ restricted to $v \geq x$, put $\epsilon_w = x^3 \sqrt{Y} / (\sqrt{\lambda_1} \max(w, U)^{3/2})$, $E_0 = g_B Y^2 = x^3 \sqrt{Y} / (\sqrt{\lambda_1} U^{3/2})$. Then*

$$\int_{v \gtrsim w, v \geq x} \mathbf{E} W dv \lesssim \epsilon_w^2, \quad \int_{v \gtrsim w, v \geq x} D_v^2 dv \lesssim_\diamond Y \epsilon_w. \quad (\text{D9})$$

Proof. On these dyads, $v dh/dv \gtrsim \lambda_1 \max(v, U)^3$ and $S \leq CY$. The source inequality in (A1), integrated against the total increase of S , therefore gives

$$\int \mathbf{E} W dv \lesssim \frac{x^6 Y}{\lambda_1 \max(w, U)^3} = \epsilon_w^2.$$

If $w < \epsilon_w/Y$, the raw bound gives directly $\int D_v^2 dv \lesssim Y^2 w \leq Y \epsilon_w$. In the other case use mass slides of width $h = \epsilon_w/Y$. The mean and boundary errors cost $O(Y^2 h)$, while the centered error costs $O(\epsilon_w^2/h)$; both equal $O(Y \epsilon_w)$. Also $\int \mathbf{E} \text{tr} C_v^2 dv \lesssim \epsilon_w^2/w \leq Y \epsilon_w$. Cauchy–Schwarz with the raw bound for $|X|^2$ bounds the mixed term by $CY \epsilon_w$. This proves (D9).

The largest ϵ_w is E_0 , so summing dyads shows that $\int_{x \leq v \lesssim Y} D_v^2 dv/Y \lesssim_\circ E_0$. We shall also use the following first-moment consequence:

$$\begin{aligned} Y \int_{v \asymp w, v \geq x} D_v dv &\lesssim_\circ Y(wY \epsilon_w)^{1/2} \\ &= \frac{x^{3/2} Y^{7/4} w^{1/2}}{\lambda_1^{1/4} \max(w, U)^{3/4}} \lesssim_\circ x^3 \lambda^{-C} Z^{7/4}. \end{aligned}$$

The expression in w is largest at $w \asymp U$; substitution of $Y = UZ$ and $U = x\lambda^{-s_*/2}$ proves the last inequality. $\square$

Aliases and projected endpoints. We next specify how the estimates apply inside a larger finite topology. For a pair split at v , its overlap may be used either at the terminal leaves or projected, i.e. using endpoint posterior means at later cuts where the branches are separate. These estimates also hold in a larger fixed topology. The error replacing any such overlap by $|X_v|^2$, or $|X_v|^2$ by $2L_v$ along a single path through the fork, has squared L^2 norm bounded by $C\mathbf{E}[X^\top C_v X + \text{tr} C_v^2]$, by branch independence and conditional contraction. For example, conditional independence at the split gives

$$\mathbf{E}[(Q - |X_v|^2)^2 \mid \mathcal{F}_v] = 2X_v^\top C_v X_v + \text{tr} C_v^2, \quad \mathbf{E}[(2L_v - |X_v|^2)^2 \mid \mathcal{F}_v] = 4X_v^\top C_v X_v.$$

Projecting either endpoint to a later cut is conditional expectation on an already separate branch, so conditional contraction only reduces these bounds. Thus (D7), (D2) control these *alias* errors, while D_v^2 bounds errors relative to S_v without the replacements. Products assigned to one and the same fork have the **same** ancestor value $|X_v|^2$.

Lemma 8.6 (Centered polynomial integrals). *Fix a dyad $v \asymp r$, $r \gtrsim Y$, and a deterministic polynomial P_v of total degree $j \geq 1$ in the centered inserted value. Assume its remaining deterministic powers are bounded by the corresponding powers of $e(v)$. Then the centered integral using $2L_v$ on one path satisfies*

$$\left\| \frac{1}{r} \int [P_v(2L_v - S_v) - \mathbf{E} P_v(2L_v - S_v)] dv \right\|_2 \lesssim_\circ z_r^0 r Z_h \begin{cases} 1 & j = 1, \\ m_r r^{j-2} & j \geq 2. \end{cases} \quad (\text{D10})$$

Here the coefficient of degree i can be any deterministic function bounded by $Ce(v)^{j-i}$.

Proof. Write $P_v(t) = \sum_{i=0}^j c_i(v) t^i$, with $|c_i(v)| \leq Ce(v)^{j-i}$. On the event $|2L_v - S_v| \leq x^{-\epsilon'} e(v)$, differentiation gives $|P'_v(2L_v - S_v)| \lesssim x^{-C_j \epsilon'} e(v)^{j-1}$. Equations (A2) and (D7), with $k = j - 1$, therefore give

$$\left\| \frac{1}{r} \int (P_v - \mathbf{E} P_v) dv \right\|_2^2 \lesssim_\circ \frac{1}{r} \mathbf{E} \int W e^{2j-2} dv/r \lesssim_\circ (z_r^0)^2 Z_h^2 r^2 \begin{cases} 1, & j = 1, \\ m_r^2 r^{2j-4}, & j \geq 2. \end{cases}$$

The factors of two in the argument of P_v only change constants. Taking square roots proves (D10). For the truncation, first fix the allowed loss $\epsilon' > 0$, then a finite moment order large enough that (D2) makes the tail negligible; W is bounded. Choose the geometric slack in (D2) last, small enough for that moment order. The same truncation of the other factors at a fork, followed by Cauchy–Schwarz in dv/r , bounds its alias errors by the same expression.

At $x \leq v \leq CY$ a centered **linear** integral of $2L_v - S_v$ with bounded coefficients instead costs $O_\circ(E_0)$ before any mass normalization, by (D9) and (A1). When there are no other random factors in the expectation of that linear term, no alias error is required even with an initial density tilt: conditional averaging reduces the overlap to $|X_v|^2$ and then $2L_v$ exactly. $\square$

Stopping, forests, and the order of truncation. For a tilted active path on the stop, (A4) represents its expectation as an untilted expectation multiplied by a density having every fixed moment required here. Hölder therefore transfers the integrated second moments and squared errors in (D7)–(D10). The needed norm is slightly above L^2 . Interpolation between its L^2 bound and its deterministic bound loses an arbitrarily small power, since the target scale is polynomial in x and the path observables are bounded. The same observation applies to the low linear centered integral.

In a finite product, first truncate the other factors using their fixed moments from (D2); their deterministic envelope bounds then cost only the chosen small power loss, and their discarded tails are negligible. These operations concern active subtrees or bounded deterministic values at masked pairs, as specified in (A4). They do not by themselves give a common marginal law when the topology varies. Such a law must first be constructed for the centered integral and its retained tests. For a forest, apply the change of measure separately on each active component. A bounded number of component densities has every required fixed moment on the stop, so Hölder handles their product. Neither this transfer nor (A2) is an assertion about a common untilted root for the whole forest. When other random path factors remain, first realize the centered integral and those factors under one fixed marginal law, apply the unconditional centered estimate, and only then apply Hölder. The existence of that marginal when a fork coordinate varies is proved explicitly in Lemma 8.12.

Corollary 8.7 (Projection to a common low cut). *One more consequence of (A3), (D2) is the following: for $Y \lesssim c'$ up to a sufficiently small fixed scale, a pair split well before c' (say mass $\leq c'/8$) changes in overlap by at most $O_{p,\circ}(z_c^0 c')$ in every fixed moment upon projecting down from later cuts to mass c' .*

Proof. Use future doubling layers up to a small macroscopic mass before the long terminal block, hitting the later layer in each product via (A3) with a full gap after the fork (also for projected rather than terminal ends). This is the two-branch raw argument of (D6). $\square$

8.5 Coarse paths and multiplier changes

Lemma 8.8 (Coarse and switch contributions). *For sufficiently small fixed $\kappa > 0$, the direct-strength contribution from $Y > x^\kappa$ satisfies the direct part of Proposition 4.1. The two final-strength multiplier changes also satisfy its required power-saving bound.*

Proof.

The large- Y event. Choose $\kappa > 0$ small relative to the power gains below. Markov's inequality and $\mathbf{E}_\gamma Z^4 \lesssim_\circ 1$ give

$$\mathbf{P}_\gamma(Y > x^\kappa) \lesssim_\circ U^4 x^{-4\kappa}.$$

We first bound the contribution at each parameter value on this event. Here U retains its cap $\min(1, x\lambda^{-s_*/2})$. Since $v dh/dv \gtrsim \lambda_1 U^3$ for $v \geq x$, the source estimate and $0 \leq S \leq 1$ give

$$\int_{v \geq x} \mathbf{E} W dv \lesssim e_1^2, \quad e_1 = \frac{x^3}{\sqrt{\lambda_1} U^{3/2}}.$$

With unit raw bounds, mass slides of width e_1 then cost $O(e_1)$ in both their mean and centered parts. The trace term on $v \geq \max(e_1, x)$ costs $O(e_1)$ by division by v ; below e_1 use the unit raw bound. Cauchy–Schwarz on dyads handles the mixed covariance term. Hence

$$\int_{v \geq x} D_v^2 dv \lesssim_\circ e_1.$$

The dotted clocks in the strength derivative are $O_\circ(\lambda)$. In the pressure difference from (A1), the portion $v < x$ therefore costs $O_\circ(\lambda x)$. At the other splits expand Q^2 around the common untilted mean S_v . The quadratic centered term uses the preceding D_v^2 estimate. For the linear term, conditional averaging replaces $Q - S_v$ first by $|X_v|^2 - S_v$ and then by $2L_v - S_v$ exactly; there is no alias error in its expectation. The centered integral estimate from (A1) costs $O_\circ(e_1)$. Transfer these estimates to the stopped tilted law by (A4). The stopped derivative is consequently $O_\circ(\lambda(x + e_1))$ at each parameter value.

After parameter averaging, apart from $x^{-4\kappa}$, the two costs are bounded by

$$\lambda x U^4 \leq x^5 \lambda^{1-2s_*}, \quad \lambda e_1 U^4 = x^3 \lambda^{(1-\eta)/2} U^{5/2} \lesssim x^{3+(1-\eta)/s_*}.$$

For the second bound, if $U = 1$ then $\lambda \leq x^{2/s_*}$ and the claim follows directly. If $U < 1$, substitution gives $x^{11/2} \lambda^{(1-\eta)/2-5s_*/4}$. The exponent of λ is negative for s_* sufficiently close to $1/2$; using $\lambda \geq x^{2/s_*}$ gives the same bound. Since $s_* < 1/2$ and η is chosen small, both terms leave a fixed power saving beyond x^5 . Choose κ to preserve it after $x^{-4\kappa}$, subpower losses, and strength integration.

The multiplier changes. In varying χ_{lin} and then χ_e at final strength, U is power small. Use the direct active and plateau coordinates and tiles of (D5) on macroscopic bands, also when $Y > x^\kappa$. The deterministic width properties there only use (D1), S, q bounded, and do not require $Y = o(1)$. The plateau lower slope in (D3) comes from $\chi_e H_e$; for $\chi_e > 0$ its use therefore costs χ_e^{-1} . The active part has at least the same weaker bound. Consequently for $H = H_{\text{lin}}$ or H_e , respectively,

$$\mathbf{E} \int H^2 W dv \lesssim_\circ \chi_e^{-1} x^6 \lambda_1^{-1} Z^2 \begin{cases} x^6, \\ \lambda_1^2 U^4. \end{cases}$$

Indeed both are supported at masses bounded below, the first kernel is uniformly $O(x^3)$, the second bounded by $O_\circ(\lambda_1 \mathfrak{h})$; apply the tile expectation bound of (D7) with that same loss χ_e^{-1} and the integrated squared width estimate. By (A1), (A4) the derivative difference along each multiplier costs the square root of this bound on the stopped set (again exact linear conditional averaging); $\chi_e^{-1/2}$ is integrable. Both $x^6/\sqrt{\lambda_1}$ at final strength and $x^3\sqrt{\lambda_1}U^2$ allow power saving beyond x^5 . This handles the multiplier changes. $\square$

8.6 Uniform coarse accuracy on size paths

The pointwise trial excess in (C2) provides an independent starting estimate on size paths. We turn it into an integrated bound for $|S - q|$ by a convex variation of the mass coordinate, then use the scalar gap to make that bound uniform. The final moment upgrade uses the same block and buffer identities as above with unit raw bounds. Its purpose is to supply one positive exponent for every fixed moment order before choosing a Taylor degree.

Proposition 8.9 (Uniform all-moment accuracy). *Here $\lambda = x^{p_*}$; assume p_* sufficiently small. Uniformly at every parameter value on smoothed size paths in the untilted system there is a fixed $\delta_* > 0$, available independently of further decreases in p_* , such that*

$$\sup_v \|Q - q(v)\|_p \leq C_p x^{\delta_*} \quad (\text{D11})$$

for a pair split at v , for **every** fixed p .

Proof. Use both coordinates $v, t = t(v)$. Up to $t_M = M/\ell^2$, before the terminal constant piece beginning at v_M , both coordinate derivatives are bounded by C/ω ; the two are inverse, with $v_M = \ell + O(\omega M)$. For $v > x$ we have $dh/dt \gtrsim \lambda v^2$ by (P4).

Freezing beyond the wall. At $t \geq t_M/2$ and thereafter $1 - q$ and $1 - S$ are smaller than every power of x . In fact for the scalar spin the conditional mean martingale above a fixed macroscopic time has diffusion coefficient between $c(1 - V_z^2)$ and $C(1 - V_z^2)$, giving exponential decay of Γ' as after (P2). Thus $dh/dt \geq c > 0$ above a sufficiently large fixed time (with b constant there). For each actual spin, the conditional mean $m_i = \sqrt{m}(X_v)_i$ has martingale bracket at least $c(1 - m_i^2)^2 dh$ there, by the i -th diagonal source-Hessian bound. Indeed the field contributes diffusion matrix mH_v to all these coordinates (the source-Hessian and clock identities in [19, Section 2]); matrix innovations are independent. More explicitly, since the masses here are bounded below, $H_{ii} \geq c(1 - m_i^2)/m$. The field contribution to quadratic variation is at least $m^2 H_{ii}^2 dh \geq c(1 - m_i^2)^2 dh$; the independent matrix innovations only increase it. With $dh/dt \geq c$, localized Itô calculus for $F(u) = \sqrt{1 - u^2}$ gives

$$\frac{d}{dt} \mathbf{E} F(m_i(t)) \leq -c \mathbf{E} \frac{(1 - m_i(t)^2)^2}{(1 - m_i(t)^2)^{3/2}} = -c \mathbf{E} F(m_i(t)).$$

Thus its expectation decays exponentially, uniformly in i, m . The calculation works first across filled finite increments at masses bounded below, then by approximation. Averaging proves the assertion; since $Q \leq 1$, $\nu Q = S$, it gives (D11) in this region already.

Integrated fluctuation control. Let $\mathfrak{a} = 1/200$, $V'_0 = x^{\mathfrak{a}}$, with $p_* < \mathfrak{a}/100$. On $[V'_0, v_M]$, (A1) gives

$$\int \mathbf{E} W dt \lesssim x^6 / (\lambda (V'_0)^3).$$

Let $R = x^6 / (\omega \lambda (V'_0)^3)$. The bounds on the two coordinate derivatives give

$$\int \mathbf{E} W dv = \int \mathbf{E} W \frac{dv}{dt} dt \lesssim R, \quad \int \mathbf{E} W \left(\frac{dt}{dv} \right)^2 dv = \int \mathbf{E} W \frac{dt}{dv} dt \lesssim R.$$

The second integral is the one required by (A1) for a t -average, since that average has weight dt/dv when expressed in mass. Mass slides and t -slides can therefore each use width $R^{1/2}$, with unit raw bounds. They give

$$\int_{V'_0}^{v_M} D_v^2 dv + \int_{t(V'_0)}^{t_M} D_{v(t)}^2 dt \leq x^{1/4}. \quad (\text{D12})$$

Use the bracket identities of Lemma 6.3, with raw unit bounds, monotone S , and width $(x^6 / (\omega \lambda (V'_0)^3))^{1/2}$ in each coordinate, using dt/dv in the t -averages in (A1). The $\text{tr } C^2$ terms cost at most $Cx^6 / (\omega \lambda (V'_0)^4)$ and Cauchy suffices for $X^\top C X$ (coordinate lengths at most $O_\circ(1)$). All these bounds have the indicated slack.

A convex mass variation. We use the trial gap in (C2) to control the integral of $|S - q|$ in mass. Let ϕ be a smooth bump of either sign, supported between V'_0 and ℓ , and set

$$\frac{1}{m_\theta(v)} = \frac{1}{v} - \theta \frac{\phi(v)}{v^2}, \quad 0 \leq \theta \leq 1.$$

The bumps used below have small slope and $|\phi|/v \ll 1$, making m_θ an increasing bi-Lipschitz change of mass. Reassign each clock increment formerly at v to $m_\theta(v)$, keeping the increment itself, the terminal length, and the terminal self correction fixed. Let $F(\theta)$ be the resulting pressure.

The exponential variational formula expresses each step power mean as a conditional supremum of expectation minus relative entropy divided by the step mass. After iteration, the complete pressure is a supremum over the choices of transition laws; the entropy coefficients are the reciprocal masses $1/m_\theta(v)$, which are affine in θ . Thus F is convex. The initial mass-zero expectations are unchanged. Passing from step clocks to the fixed-size limit preserves convexity. The segment formula (A1) gives

$$F'(0) = \frac{1}{4} \int (Bb' + 2Sh')\phi \, dv.$$

The clocks are Lipschitz on the bump at fixed size, so this chain rule also follows directly by approximating with continuous observation cuts.

In the new mass coordinate $w = m_\theta(v)$, compare to the fixed scalar trial $K(w)$. Square completion along the affine interpolation to that trial gives

$$F(\theta) \leq f_{\text{spin}}(K) + P(\theta), \quad P(\theta) = \int \frac{(K(w) - h(v))^2}{4b(v)} \, dw, \quad v = m_\theta^{-1}(w).$$

At $\theta = 0$, (C2) bounds this trial excess by x^{4+c} . Convexity therefore yields

$$F'(0) \leq F(1) - F(0) \leq x^{4+c} + P(1) - P(0).$$

Differentiating the penalty and returning to the old coordinate gives

$$P'(\theta) = \frac{1}{4} \int (z_\theta^2 b' + 2z_\theta h') \partial_\theta m_\theta(v) \, dv, \quad z_\theta = \frac{K(m_\theta(v)) - h(v)}{b(v)}.$$

If the oscillation of K on the swept support is at most x^{3a} , then $z_\theta - z_0 = O(x^{3a})$. Also $\partial_\theta m_\theta = \phi(1 - \theta\phi/v)^{-2} = \phi[1 + O(\sup |\phi/v|)]$. These are the two errors in comparing $P'(\theta)$ to $P'(0)$. Integrating that comparison in θ gives

$$\int [(B - z_0^2)b' + 2(S - z_0)h']\phi \leq 4x^{4+c} + C(x^{3a} + \sup |\phi/v|) \int (b' + h')|\phi|. \quad (\text{D13})$$

Here $z_0 = q - d/b$ is bounded and $|d| \lesssim_\circ x^3$ throughout, by (P6).

Partition $[V'_0, \ell]$ into cells of mass width $l = x^{5a}$. Give each cell a fixed enlargement, with uniformly bounded overlap. Discard the boundary cells and every cell whose enlargement has K -oscillation greater than x^{3a} . Since K is nondecreasing and its total increase is $O(M)$,

$$\#\{\text{discarded interior cells}\} x^{3a} \lesssim M, \quad |\text{discarded mass}| \lesssim l + Ml/x^{3a}.$$

On each retained enlargement, Lipschitz continuity of Γ gives q -oscillation $O(x^{3a}) = o(x^a)$. If $|S - q| \geq Cx^a$ somewhere in its cell, monotonicity of S preserves that sign, at half the threshold, on a one-sided interval of length comparable to l in the enlargement: use the forward side for a positive deviation and the backward side for a negative one.

Choose a smooth bump ϕ on this one-sided interval, with that sign, height cl , slopes bounded by a sufficiently small fixed constant, and $\int |\phi| \asymp l^2$. Its swept support remains in the good enlargement. Moreover $|\phi|/v \lesssim l/V'_0 = x^{4a}$; the small slope and this bound make all m_θ increasing and bi-Lipschitz. Since $|d| \lesssim_\circ x^3$, $b \asymp 1$, and $q \gtrsim V'_0$ on this support, $z_0 = q - d/b \geq 0$ and

$$(S - z_0)\phi \gtrsim x^a |\phi|.$$

Put $I_\phi = \int h' |\phi|$. The error on the right of (D13) is at most

$$C(x^{3a} + x^{4a})(1 + x^{-a})I_\phi = O(x^{2a})I_\phi = o(x^a I_\phi),$$

because $b' \lesssim h'/V'_0$.

To estimate the remaining term on its left, use the exact identity

$$B - z_0^2 = (S - z_0)(S + z_0) + D_v^2.$$

The first summand has favorable sign after multiplication by $b'\phi$. The second can have unfavorable sign only for a negative bump. If b' meets the bump, its support lies in the regular low region, where $b' \lesssim \lambda$ and $h' \gtrsim \lambda(V'_0)^2$. By (D12),

$$\int D_v^2 b' |\phi| \lesssim \lambda x^{1/4}, \quad x^a I_\phi \gtrsim \lambda x^{3a} l^2.$$

Their ratio is $O(x^{1/4-8a}) = o(1)$. If the bump lies outside that low region, $b' = 0$ and there is no variance error. Finally $h' \gtrsim x^3$ everywhere on the bump: use the regular low inverse before $4v_b$, and H_{lin} thereafter. Thus

$$x^a I_\phi \gtrsim x^{3+11a} \gg x^{4+c},$$

contradicting (D13). Every retained cell therefore has $|S - q| \lesssim x^a$. Since S, q are bounded, the low interval of length V'_0 and the discarded cells contribute $O(x^a)$ and $O(l + Ml/x^{3a})$, respectively. The strip between ℓ and v_M has mass $O(\omega M)$, so its contribution is $O(\omega M) = o(x^a)$, including its part before the frozen region. The remaining tail is controlled by the freezing estimate for $t \geq t_M/2$. This proves

$$\int |S - q| dv \lesssim x^a. \tag{D14}$$

From integrated means to pointwise means. Set $K_S = h + bS$, nondecreasing, and write Γ_S for its scalar response (time variable), extending both profiles as needed by terminal annealed clocks. Use one common terminal extension for the two monotone clocks. The area between their graphs can be integrated first in mass or first in time, giving

$$\int |\alpha_{K_S}(t) - \alpha_K(t)| dt = \int |K_S(v) - K(v)| dv \lesssim x^a$$

by (D14) and $K_S - K = bf + d$. Thus

$$\sup_t |\Gamma_S(t) - \Gamma(t)| \lesssim (1 + M)^C x^a$$

on the needed times. To see the polynomial constant, interpolate inverse profiles with evaluation time fixed; $\Gamma(y)/2$ is the root impulse derivative in the profile. Its derivative uses the [post/post Hessian kernel](#) in the proof of Lemma 5.2, which is uniformly $O(1 + M)$ for mixtures as well: the spatial derivative $L_z^y(s)$ of the conditional scalar pair moment divided

by two is bounded by the labeled source derivative rule (also directly by telescoping the conditional transition log-density when shifting the field, with bounded gradient of each scalar continuation). Impulse formulas pass from step profiles as there. Extend to one common terminal time $O(M+1)$.

At reference K_S , the returning mismatch is exactly $h + bQ - K_S = b(Q - S)$. Apply (A7) with $P = 1, k = 1$. Its absolute attachment measure has only one prescribed fork: the tested pair's split at v . An insertion tied there costs D_v ; a freely placed insertion has bounded mass density and costs $\int D_u du$. The row-deletion contact contributes $O(m^{-1})$. Consequently

$$|S_v - \Gamma_S(K_S(v))| \lesssim D_v + \int D_u du + O(m^{-1}).$$

All row changes are bounded uniformly. Integrate over $0 < t < t_M$ at $v = v(t)$, using (D12), discarding $v < V'_0$ at cost $O((1+M)V'_0)$ since the low inverse is regular, and using the frozen estimate for post-wall attachments. Combine with the response comparison and $|d| \lesssim_\circ x^3$; we get $\int |S_{v(t)} - \Gamma(t + b(v)(S_v - q(v)))| dt \lesssim (1+M)^C x^a$. To use this residual estimate, write $f = S - q$ and integrate the scalar derivative along the segment from t to $t + bf$:

$$S - \Gamma(t + bf) = f \int_0^1 [1 - b\Gamma'(t + ybf)] dy.$$

The average in brackets is at least $c\lambda v^2$ by (P5). On $v \geq x^{a/10}$, division by this gap therefore gives, allowing the preceding subpower factors,

$$\int_{v(t) \geq x^{a/10}} |f(v(t))| dt \lesssim_\circ x^{4a/5 - p_*} \leq x^{a/2}.$$

Here $p_* < a/100$ leaves room for the weakened exponent. The function $S_{v(t)}$ is nondecreasing, whereas $\Gamma(t)$ is Lipschitz. A deviation of height d therefore persists at height at least $d/2$ on a one-sided t -interval of length comparable to d . The integral bound forces $d^2 \lesssim x^{a/2}$, giving $|f| \lesssim x^{a/4}$ above mass $2x^{a/10}$. Both-sided buffers fit until the frozen region, which was treated separately. Below this cutoff, monotonicity and $q(v) \asymp v$ near zero give $S_v + q(v) \lesssim x^{a/10}$. Consequently $\sup_v |S_v - q(v)| \lesssim x^{a/10}$.

The all-moment upgrade. Fix a finite moment order p . First take a cut of mass $v \geq 4x^{a/10}$ and scalar time $t \leq t_M/2$. Its small t -neighborhoods stay above V'_0 . Apply Lemma 6.2 with unit row bounds. For a later block at t -distance l , the gap has field length at least $c\lambda(V'_0)^2 l$ and mass at least V'_0 . Its squared projection coefficient and the integrated earlier anchor cost are

$$\frac{C_{p,\circ} x^6}{\lambda(V'_0)^4 l} \quad \text{and} \quad C_p l,$$

respectively. Stop the doubling layers at a fixed positive t -distance and then use a terminal block. Summing gives

$$\left\| \int W dt \right\|_{p/2} \lesssim_{p,\circ} \frac{x^6}{\lambda(V'_0)^4}$$

on each bounded t -window under consideration.

Put $\rho = x^{a/4}$. Choose a past buffer and a future buffer of width comparable to ρ on the two sides of a smaller neighborhood of the cut. They remain at mass at least V'_0 and below t_M ; any part past $t_M/2$ is already controlled by freezing. The global mean bound and Lipschitz continuity of Γ give, for two buffer times t_1, t_2 ,

$$|S_{v(t_2)} - S_{v(t_1)}| \lesssim x^{a/10} + |t_2 - t_1| \lesssim x^{a/10},$$

since $|t_2 - t_1| = O(\rho)$ and $\rho = x^{a/4} = o(x^{a/10})$. Applying (A1) to a normalized t -average, with $dt/dv \lesssim 1/\omega$, gives

$$\|L_{\text{buffer}} - \mathbf{E}L_{\text{buffer}}\|_p \lesssim_{p,\circ} \frac{x^3}{\sqrt{\omega\lambda(V'_0)^2\rho}}.$$

This is smaller than $x^{a/10}$ for our parameter choices. The two-sided brackets and the common-past estimate in Lemma 6.3 consequently give

$$\| |X_t|^2 - q(v(t)) \|_p + \| |X_t - X_{t'}|^2 \|_p \lesssim_{p,\circ} x^{a/10}$$

for cuts t, t' in the smaller neighborhood.

For an overlap split there, project both endpoints to a cut a fixed fraction of ρ later. The two projected branches have the preceding norm and increment bounds, so polarization about their common ancestor controls their inner product. Equation (A3), now even with unit anchors, bounds the residual projection errors using that ρ -length gap. For a smaller split, choose instead a central cut at mass a sufficiently large fixed multiple of $x^{a/10}$. Its lower comparable mass interval lies strictly after the fork and supplies a full projection gap; it has the buffers just constructed. Its squared posterior norm has L^p size $O_p(x^{a/10})$. Together with the frozen region, these estimates prove (D11), for example with $\delta_* = a/50$. The stopped transfer (A4) gives the same coarse accuracy for shifted active overlaps. $\square$

Corollary 8.10 (A deterministic cap on the low scale). *For (D1) and the low statistics on size paths we may cap Y at $x^{\delta_*/3}$, decreasing p_* if necessary, and still write $Y = UZ$. Indeed (D11) gives $|S_v - q(v)|^2 \lesssim x^{2\delta_*}$, whereas on every bounded band*

$$x^{2\delta_*} \lesssim x^{4\delta_*/3}/r^2.$$

Thus the capped scale still satisfies (D1). Its new majorant $Z = \min(Z_{\text{old}}, x^{\delta_/3}/U)$ is at least one and no larger than the old majorant, so its fourth parameter moment is preserved.*

8.7 High-mass closure at final strength

The coarse estimate fixes the Taylor order in the cavity equation. The sharper low-mass integral bounds then control its sources below a cutoff, while the pair inverse controls the stochastic linear terms above that cutoff. Keeping one prescribed high pair as a centered test will preserve its second norm through this calculation.

Proposition 8.11 (High-mass means and variances). *Take Z as at the end of (D11); by (C2) take also $Z_g \geq 1$, $\mathbf{E}_\gamma Z_g^2 \lesssim 1$ with $\int v^2 \nu[(Q - q)^2] dv \leq x^{4+c} Z_g^2$ untilted. Choose*

$$V_i = x^{a_i} \quad (0 \leq i \leq 2), \quad p_* \ll a_2 \ll a_1 \ll a_0 \ll \min(c, \delta_*).$$

There are fixed $b_H > 0, C_H < \infty$, with $b_H \gg C_H a_1$, $a_1 \gg C_H a_2$, for which

$$\sup_{v \geq V_1} |f(v)| \leq x^{2+b_H} (Z^2 + Z_g), \quad D_{H,i} := \sup_{v \geq V_i} D_v \leq x^{3-C_H a_i} Z^2 \quad (i = 1, 2). \quad (\text{D15})$$

We first isolate the centered low-source estimate needed to invert the high equations. Its topology assumptions are part of the statement.

Lemma 8.12 (Low sources with a retained high pair). *Consider an allocation with bounded degree and bounded topology weights. The old prescribed topology is either absent or consists of one pair splitting at mass at least V_i ; in particular it contains no prescribed low fork.*

Call designated factors at $v < V_i$ low. Each low factor is either stochastic, of the form $\delta = Q - S_v$, or deterministic, bounded by $O_\diamond(e(v))$, with $e = Y$ for $v \lesssim Y$. There is at least one stochastic low factor and no stochastic factor at a higher mass. Deterministic factors at higher masses may be included in the coefficient. If there is exactly one low factor, assume that its coefficient is $O_\diamond(v)$. Projected pairs whose endpoints are already on separate branches are allowed, with the convention following (D9).

There are two conclusions. In the untilted law, test the product by $P = Q_{ab} - S_{ab}$ for the prescribed high pair. Alternatively, compare its tilted and untilted expectations on the stop, with no P , using $\delta^* = Q^* - S_v$ in the tilted law and the same deterministic coefficients in both laws. After signed integration the respective bounds are

$$O_\diamond(x^3 \lambda^{-C} Z^2) \mathcal{P} + O(x^K), \quad \mathcal{P} = D_{ab} \text{ or respectively } 1. \quad (\text{D16})$$

The negligible error exponent K can be arbitrarily large.

Extra auxiliary slots are permitted. On each fixed tree shape the absolute joint density of freely placed distinct fork vertices is bounded; deterministic weights and restrictions may depend on all their times. An additional multiplier depending only on the high times can be factored out and integrated separately. Ties to a low vertex reuse its freely integrated coordinate, by the signed rule; no initially free low fork requires an atomic time measure. This distinction concerns the underlying allocation measure. The old high fork may carry its prescribed atom. A new low fork is instead a free coordinate with bounded density; later attachments to it reuse that coordinate. Merely having the same mass as a different vertex supplies no additional atom. The argument below retains this distinction before discarding deterministic restrictions in absolute estimates.

Proof.

The region below Y . At $v < x$, a stochastic insertion costs $O_p(Y)$. If it is the only low factor, the required coefficient is $O(v)$, and the integral is $O(x^2 Y)$. A second factor at the same vertex gives $O(xY^2)$ instead. At a distinct low vertex its integrated envelope is at most $O_\diamond(Y^2)$, giving $O_\diamond(xY^3)$. All three are bounded by $O_\diamond(x^3 \lambda^{-C} Z^2)$, since $Y = x \lambda^{-s^*/2} Z \leq 1$. Additional factors are bounded after truncation.

For $x \leq v \lesssim Y$, retain one stochastic factor in second norm. In the sole-factor case its coefficient is at most CY ; in the tied case the other envelope is $O(Y)$; in the distinct-vertex case its integrated envelope is $O_\diamond(Y^2) \leq O_\diamond(Y)$. All three cases are therefore bounded by

$$Y \int_{x \leq v \lesssim Y} D_v dv \lesssim_\diamond x^3 \lambda^{-C} Z^{7/4} \leq x^3 \lambda^{-C} Z^2,$$

using (D9) and summing dyads. Cauchy–Schwarz with P supplies its factor D_{ab} ; the remaining factors are truncated at their deterministic envelopes. Tilted active moments transfer on the stop as already specified.

Alias errors above Y . For stochastic insertions at $v \gtrsim Y$, first replace $Q - S_v$ by $|X_v|^2 - S_v$. On a dyad r , the second line of (D7), with $Z_h = 1$, gives the following integral of the alias's second norm:

$$A_r := \int_{v \gtrsim r} \|Q - |X_v|^2\|_2 dv \lesssim_\diamond z_r^0 r^2 = \frac{x^3}{\sqrt{\lambda_1} r}.$$

The sole-factor coefficient cancels this denominator: $r A_r \lesssim x^3 / \sqrt{\lambda_1}$. If a second factor is tied to the same fork, Cauchy–Schwarz in mass uses its integrated envelope budget and gives

$$\int_{v \gtrsim r} e(v) \|Q - |X_v|^2\|_2 dv \lesssim_\diamond z_r^0 r^2 m_r = \frac{x^3 Y^2}{\sqrt{\lambda_1} r^2} \leq \frac{x^3}{\sqrt{\lambda_1}}.$$

For a second distinct vertex, integrate its envelope separately; the cost is at most $A_r \int_{v < V_i} e(v) dv \lesssim_\diamond x^3 Y^2 / (\sqrt{\lambda_1} r)$, which is sufficient because $r \gtrsim Y$ and $Y \leq 1$. Truncation handles any further factors, and Cauchy–Schwarz with P again retains D_{ab} .

Centered integration in a fixed marginal. Process the remaining stochastic vertices in decreasing time order. At the current vertex, replace every tied overlap by the same ancestor value $|X_v|^2 - S_v$, and then replace its power by the corresponding power of $2L_v - S_v$ on one continuation through that vertex. The alias estimates just proved bound these changes. Next replace this polynomial by its deterministic *untitled* mean. The mean is bounded by the corresponding envelope power.

The centered error in this last replacement is estimated by (D10). Its bound has the same three forms as the alias bounds: a sole linear factor uses the extra coefficient v ; a tied product uses the m_r gain; a factor at a distinct vertex uses the integrated envelope there. Earlier random ancestor factors are truncated at their deterministic envelopes before this estimate. To apply (D10), we must realize the entire moving- v integral and its retained random tests under a single fixed marginal law. The next construction verifies that requirement.

More explicitly, fix the tree shape, its time order, every other distinct fork coordinate, and every projection cut. If an old leaf descends through the moving vertex, retain that leaf as the continuation defining L_v . Both leaves of the old pair then lie in the same child until their prescribed higher split. The retained old pair consequently has its original, fixed split law. If no old leaf descends through the moving vertex, the chosen continuation separates from the retained old tree at a different, already fixed vertex. In both cases normalized continuations of all other children integrate to one. Earlier random ancestor factors are evaluated before the moving coordinate; later stochastic factors have already been replaced by deterministic means. Projected old endpoints are evaluations of the same retained paths at fixed later cuts. Auxiliary labels with no remaining spin observable can be deleted by the same consistency rule.

Thus the entire varying- v integral lives under one marginal law. We do not condition the centered estimate on the retained random test P . Let I denote the centered integral, and truncate the product R of the earlier factors at its deterministic envelope product M_R . Then

$$|\mathbf{E}[PRI]| \leq M_R \|P\|_2 \|I\|_2 = M_R D_{ab} \|I\|_2.$$

Equation (D10) is applied to I under the fixed marginal, before this Cauchy–Schwarz inequality. The discarded tails are negligible by the previously specified moment truncation. In the tilted comparison, transfer and Hölder require a norm only slightly above L^2 , with the already permitted small power loss. A stochastic tie at the moving vertex is one polynomial power of its ancestor value; it does not add an independent coordinate or create a second atom. In the comparison version all moving stochastic times here are active; the same fixed marginal reasoning and (A4) suffice. After successive replacements (split ties as one power), the deterministic product vanishes against centered P , or cancels across ensembles. This proves (D16). $\square$

Remark 8.13 (An earlier pair retained above a moving fork). The same marginal construction will be needed in Section 9 when a retained pair splits at a fixed mass $u < v$, rather than above the moving fork. At most one endpoint of that pair can descend through v . Choose its path for L_v , if such an endpoint exists, and retain the other endpoint. After the alias replacement the other child at v is unneeded and can be pruned. The retained paths still split at u , so their law is independent of v . Even if the earlier pair has already been projected to a fixed cut $c > v$, its projected endpoints are fixed-time observables of these same retained

paths. The centered inequality remains unconditional, followed by Hölder with the earlier random factor. Figure 2 records this configuration.

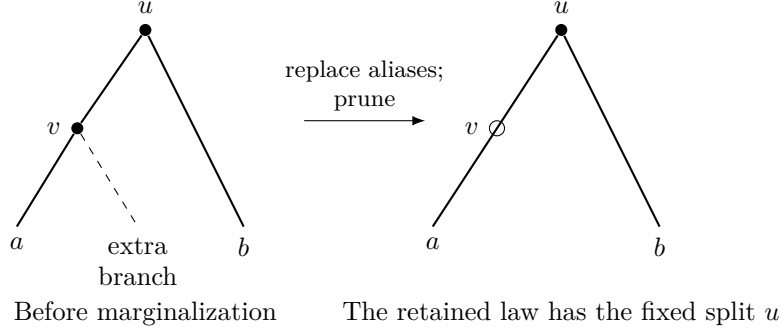


Figure 2: The earlier-pair configuration in the centered integration argument. The pair (a, b) splits at the fixed mass $u < v$. Replacing the aliases and integrating out the extra branch leaves the two paths on the right. Their joint law is independent of v , which now marks an evaluation point rather than a retained fork. Values at fixed projection cuts remain observables of these paths. The single-path estimate is applied under this fixed law before Hölder’s inequality handles the other random factors.

Proof of Proposition 8.11.

The covariance equation. Fix an old pair ab splitting at or above V_i , where $i \in \{1, 2\}$, and put $P = Q_{ab} - S_{ab}$. For every tested pair jk at or above the same threshold, define

$$u_{jk} = \nu[P(Q_{jk} - S_{jk})].$$

These are pair functions on extensions of the old labels ab ; unused labels are ignored by marginalization. Apply (A7) at the reference K , expanding $J = b\delta + bf + d$, where $\delta = Q - S$. Choose a fixed truncation degree large enough that (D11) makes the true Taylor remainder $O(x^K)$ for any needed fixed K . The row-deletion error remains $O(m^{-1}) = O(x^6)$. Every all-deterministic term vanishes against P .

Retain the high stochastic linear term on the left. The equations have the form

$$u_{jk} - \sum_{cd: v_{cd} \geq V_i} \mathbf{M}_{jk,cd} b_{cd} u_{cd} = R_{jk}^{\text{low}} + R_{jk}^{\text{nonlinear}} + O(x^K) + O(x^6).$$

The first source includes the stochastic linear term below V_i ; the second contains at least two insertions. This left side is exactly (H1), because the positive-law expectation of each inserted pair with P depends only on the indicated labels. By (P5), its gap on retained levels is at least $c\lambda V_i^2$. Proposition 5.1 therefore inverts it with cost $O(x^{-Ca_i})$, once $p_* \ll a_i$ is chosen. Evaluate the result at $jk = ab$. The inverse adds only a bounded number of freely attached labels. Their attachment densities and existing-fork weights are bounded after removing the inverse cost; they do not turn a new low fork into a prescribed atom.

As explained in the [fixed-resolution limit argument](#), these operations can first be made on resolutions keeping the tested cut and threshold. At fixed system size gaps persist on fine approximations; inverse domination with block restrictions allows replacing path expectations and coefficients outside that inverse by continuous-path ones with errors tending to zero (empty step intervals can be refined for attachment). Our bounds apply to bounded measurable topology weights from that inverse after its cost factor is removed.

Consider a nonlinear term containing a high stochastic factor δ_{cd} . Keep P and that factor untruncated. There is at least one further small factor: truncate it at $x^{\delta_*/2}$, and bound all other factors uniformly. Equation (D11) controls the discarded tails in arbitrarily high fixed moments. Cauchy–Schwarz on the two untruncated factors gives

$$|\nu[P\delta_{cd} \text{ remaining factors}]| \lesssim x^{\delta_*/2} \|P\|_2 \|\delta_{cd}\|_2 + O(x^K) \leq x^{\delta_*/2} D_{ab} D_{H,i} + O(x^K).$$

This preserves the exact second-moment factors; no higher norm proportional to D_{ab} is assumed. Every remaining source contains only low stochastic factors, so Lemma 8.12 applies, including to the low linear term.

The coefficient of a sole low factor. If just one low insertion occurs at v , its scalar coefficient indeed costs $O(v)$: in every term of the finite quotient it belongs to a spin pair moment (possibly with other pairs), and all other pairs in that moment, including the tested pair if present, are high. Immediately across the low branching its two ends thus give two different groups of odd counts. Conditional on prefixes up to that cut, group expectations on independent continuations are odd smooth bounded functions (for the odd groups) with bounded gradients by the scalar source rules. Their product costs $O(K(v)) = O(v)$ using Brownian moments and bounded drift. More explicitly, the two odd group functions $F_1(z), F_2(z)$ vanish at zero and have uniformly bounded derivatives, so $|F_1(z)F_2(z)| \leq C|z|^2$. The shared scalar prefix field at time $K(v)$ has second moment $O(K(v))$ by its Brownian representation with bounded drift. Hence this scalar coefficient is $O(K(v)) = O(v)$. Additional labels and coincidences within the groups do not change the odd parity or the derivative bound.

Combining these estimates after inversion gives

$$D_{ab}^2 \lesssim_\diamond x^{-Ca_i} (x^{\delta_*/2} D_{ab} D_{H,i} + x^3 \lambda^{-C} Z^2 D_{ab} + x^6).$$

The final x^6 is the row-deletion contact error. Increasing the Taylor degree does not remove it. Take the supremum over ab and write $D = D_{H,i}$. Choose the power losses so that $Cx^{\delta_*/2-Ca_i} < 1/2$. Absorption then leaves

$$D^2 \lesssim_\diamond x^{3-Ca_i} \lambda^{-C} Z^2 D + x^{6-Ca_i}.$$

If $D^2 \leq AD + B$ with $D \geq 0$, then $D \leq A + \sqrt{B}$ up to a fixed constant. Since $Z \geq 1$ and $p_* \ll a_i$, the factors λ^{-C} and the permitted small losses can be included in a fixed $C_H a_i$. This proves $D_{H,i} \leq x^{3-C_H a_i} Z^2$ for $i = 1, 2$.

The longitudinal equation. For $r \geq V_1$, apply (A7) without a random test. The longitudinal identity (H2) separates the linear term into its diagonal and its integral part. The resulting scalar equation is

$$[1 - b(r)\Gamma'(K(r))]f(r) = \Gamma'(K(r))d(r) + R_{\text{int}}(r) + R_{\text{nonlinear}}(r) + O(x^6).$$

The diagonal is in field units; returning from normalized mass changes only bounded constants. Its lower bound is

$$1 - b(r)\Gamma'(K(r)) \geq c\lambda V_1^2.$$

For the integral term, the post kernel bound in Section 5 gains the earlier time $K(u) \lesssim u$ when u is small. It gives

$$|R_{\text{int}}(r)| \lesssim (1 + M)^C \left(\int |d| du + \int_{u < V_0} u |f(u)| du + \int_{u \geq V_0} |f(u)| du \right).$$

The low dyadic envelope budget and the raw bound below Y imply $\int_{u < V_0} u |f(u)| du \lesssim_\diamond V_0 Y^2$. For the last integral use (C2) and the majorant Z_g : Cauchy–Schwarz, followed by a harmless weaker cutoff bound, gives $\int_{u \geq V_0} |f(u)| du \lesssim x^{2+c/2-a_0} Z_g$. Thus the three terms on this right side are

$$O_\diamond(x^3), \quad O_\diamond(x^{2+a_0-p_*s_*} Z^2), \quad O(x^{2+c/2-a_0} Z_g).$$

For a nonlinear allocation with a factor at the prescribed fork r , keep that factor and truncate one of the others at $x^{\delta_*/2}$. Its contribution is

$$O(x^{\delta_*/2})(|f(r)| + x^3 + D_{H,1}).$$

If no factor is pinned at r , integrate one factor in second norm. The low envelope and the high comparison budget give

$$\int \|Q - q(u)\|_2 du + \int |d(u)| du \lesssim_\diamond Y^2 + x^{2+c/2-a_0} Z_g + x^3.$$

A further factor contributes $x^{\delta_*/2}$ after truncation, even when it is tied to the same vertex. The negligible tails use a fixed higher moment from (D11).

Divide by $c\lambda V_1^2$. The coefficient of $|f(r)|$ is at most $Cx^{\delta_*/2-p_*-2a_1}$, up to the fixed polynomial and small power losses, and is absorbed. The remaining leading exponents before division are

$$3, \quad 2 + a_0 - p_*s_*, \quad 2 + c/2 - a_0, \quad 2 + \delta_*/2 - p_*s_*.$$

The already proved variance bound controls the extra $x^{\delta_*/2} D_{H,1}$ term. Choose a_0 small relative to c, δ_* , then choose b_H to be a sufficiently small fixed fraction of a_0 . Next take a_1 sufficiently small, choose $a_2 \ll a_1$, and only then take p_* sufficiently small that the loss from the gap, $(1+M)^C$, and every permitted small power loss is less than the remaining margins above $2 + b_H$. This also ensures $b_H \gg C_H a_1$. We obtain uniformly for $r \geq V_1$

$$|f(r)| \leq x^{2+b_H} (Z^2 + Z_g),$$

as required. This choice of $a_2 \ll a_1$, followed by p_* , preserves both this bound and the covariance absorption. $\square$

Remark 8.14 (The powers of the parameter majorants). The parameter moment used throughout is $\mathbf{E}_\gamma Z^4 \lesssim_\diamond 1$. The direct construction uses $e = \min(r, Z^2 \mathfrak{h})$, whereas its low-region size analogue incorporates Y in the width and uses $e = \min(r, \mathfrak{h})$. This is the reason for $Z_h = Z$ in the direct integrated bounds and $Z_h = 1$ in the low-region bounds: no extra power of Z is implicit in $\lesssim_\diamond$. In particular the right side of (D8) is $x^{-C_0 \delta'} z_1^0 U^2 Z^4$ up to the permitted losses. The lower-band factor $Z^{7/4}$ is bounded by Z^2 , and the low-source and high-variance estimates require only Z^2 . The high-mean estimate additionally uses Z_g with $\mathbf{E}_\gamma Z_g^2 \lesssim 1$. Hence the combinations needed later, Z^4 and $Z_g Z^2$, are integrable by Cauchy–Schwarz. These powers are also valid for path parameters and site counts that vary measurably: the majorants and all windows are chosen pointwise from deterministic path data, before taking a positive-law expectation.

Parameter order. The final-strength saving c from (C2) and the exponent δ_* in (D11) are fixed before the final choice of p_* . Hence the row Taylor degree in the covariance equation is fixed using δ_* , and can make the remainder, apart from m^{-1} contacts, smaller than x^6 by an arbitrary fixed power. The inverse then has a fixed polynomial loss in $(\lambda V_i^2)^{-1}$ and $1 + M$. Choose a_0 relative to c, δ_* , next a_1 relative to a_0 , next a_2 relative to a_1 , and only

then p_* relative to all these gaps. This gives $b_H \gg C_H a_1$ and $a_1 \gg C_H a_2$ without a degree depending on a later choice of p_* . The local strengthening depths and their power slacks may then be fixed as in Section 10. The geometric loss exponent used in the width construction is chosen last, after the finite moment and truncation orders have been specified.

9 Finite pressure expansion and power count

We now estimate the finite pressure polynomial that represents the increment in Proposition 4.1. The moment bounds control its high-degree terms. For degrees two, three, and four, we must first sum the signed scalar coefficients: the required saving appears only after this sum and, on a size path, after applying the size derivative.

All estimates are on the stopped set. On direct strength paths we restrict to $Y \leq x^\kappa$, since Lemma 8.8 treats its complement. On size paths we use the cap of Corollary 8.10. Group masses up to a fixed multiple of Y into one lower band, assigned scale $r = Y$, and use dyadic bands $v \asymp r$ above it. Fixed changes in these endpoints affect only constants. Write

$$\delta = Q^* - S_v, \quad f = S_v - q(v), \quad \Delta = \delta + f.$$

Here S_v is the untilted mean, used as the same deterministic centering in both ensembles. When specified, Q^* is replaced by the masked inner product of posterior means at a common later cut. The allocation estimate below permits these projected factors. On size paths we apply it only to factors in the low region of (D2). The bounds on $Q - q$ and $S - q$ give the same envelope bound for δ by the triangle inequality; on the lower band it is $O_p(Y)$.

9.1 The pressure polynomial and its derivatives

Choose a fixed Taylor degree J_0 , allowing different degrees for size and direct variations. The size degree can be chosen from (D11) before any final decrease of p_* . Take the size stencil order at least this degree. Lemma 2.5 expresses the change from m to $m + j$ as a polynomial in j . The stencil extracts its linear coefficient, which is the per-root expectation of the whole symbol (A6). The path derivative adds ∂_θ of that symbol. Thus the operator for the size increment is $\mathcal{O} = 1 + \partial_\theta$, with the unit acting on every term. For a direct strength increment the operator is instead $\mathcal{O} = \partial_{\log \lambda}$.

These operators act on formal polynomials: on a size path assign $\partial_\theta \Delta = -\Delta/6$, and on a direct path hold Δ fixed. This is the rule in Lemma 2.4, not a derivative of a random overlap or of its law. Scalar coefficients receive their ordinary parameter derivatives. After the exact cancellation of the singleton cumulant with the linear counterterm, (A6) becomes, up to deterministic terms,

$$-\frac{1}{4} \sum b \Delta^2 + \sum_{j=2}^{J_0} \frac{1}{2^j j!} \sum_{\text{pairs}} \kappa_K(\tau_{a_1} \tau_{b_1}, \dots, \tau_{a_j} \tau_{b_j}) \prod_{i=1}^j (b \Delta + d)_{a_i b_i}. \quad (\text{E1})$$

where the pairs are ordered and off diagonal, and κ_K is their joint scalar cumulant at fixed topology. We must bound the difference of per-root positive-law expectations after applying $\mathcal{O}$ to (E1). The degree counts below always refer to the base pair factors in this polynomial. Coefficient differentiation may allocate an additional scalar pair, whose clock weight is treated separately.

For high masses on size paths, (D15) gives the needed gain. For low masses, the remaining task is to obtain a coefficient of size λw^{5-j} for a product of j factors at scales at most w . This is why the next lemma uses that particular power of w . Its extra ratio for two distinct quadratic vertices records the smaller split scale. We will obtain these coefficient bounds by

summing the signed scalar contributions of the polynomial; before doing so, we prove the estimate that converts them into an increment bound.

9.2 A centered allocation estimate

Lemma 9.1 (Centered allocation estimate). *Fix $j \geq 2$ pair factors. Each factor is either δ , possibly projected, or a deterministic function bounded by $O_\diamond(e(v))$; use $e = Y$ on the lower band. Fix the bands, their order, and every projection cut. Each projection cut lies after the split of its pair. Let $s = 1, \dots, l$ index the distinct vertices to which these factors are assigned, with multiplicities k_s and band scales r_s , and put $w = \max_s r_s$. Suppose that the deterministic weights are common to both ensembles and, on each of boundedly many topology shapes, are bounded by*

$$O_\diamond(\lambda w^{5-j}) \prod_{s=1}^l dv_s/w, \quad (\text{E2})$$

times measures of bounded volume in the other integration variables. These are product majorants, so deterministic restrictions may be retained. If $j = l = 2$, assume in addition the factor $O_\diamond(\min_s r_s/w)$. More explicitly, if H is this deterministic weight and T_i^{tilt}, T_i^0 are the pair factors in the two ensembles, we estimate

$$\int H \left\{ \nu^{\text{tilt}} \left[\prod_{i=1}^j T_i^{\text{tilt}} \right] - \nu^0 \left[\prod_{i=1}^j T_i^0 \right] \right\}.$$

The weight, deterministic factors, cuts, and centering S are common; only the positive law and the masked overlap vary. The absolute value of this difference is bounded by

$$O_\diamond(\lambda Y^5(x/Y + g_B)) + O(x^K) \quad (\text{E3})$$

where K can be arbitrarily large. Indeed the two contributions are

$$\lambda Y^5 \frac{x}{Y} = x^5 \lambda^{1-2s_*} Z^4, \quad \lambda Y^5 g_B = x^5 \lambda^{(1-\eta)/2-s_*} Z^{7/2}.$$

Both powers of λ are positive. Since $\mathbf{E}_\gamma Z^4 \lesssim_\circ 1$, the averaged bound is $O(x^{5+c'})$ for some $c' > 0$, after taking the allowed losses smaller than these gains.

Proof. Set $a_s = r_s/Y$, $A = w/Y$, and $m_{r_s} = Y/a_s$. The ratios a_s and A are at least one. At a vertex carrying k factors, (D2) and the integrated envelope bound give

$$P_k = m_{r_s}^{\min(2,k)} r_s^{k-\min(2,k)}$$

after integration in dv_s/r_s . Thus one or two factors use the first or second envelope moment, and each further factor costs r_s . Changing dv_s/w to $(r_s/w)dv_s/r_s$ in (E2) shows that the multiplier relative to λY^5 is

$$F = A^{5-j-l} \prod a_s^{1+k_s-2\min(2,k_s)}$$

with the additional factor $\min_s a_s/A$ when $j = l = 2$. The exponent of an individual a_s is 0 for a single factor, -1 for a double factor, and $k_s - 3$ for $k_s \geq 3$. Consequently

multiplicities	F or an upper bound for F
$l = 1$	A
$(1, 2)$	$1/a_{\text{duplicate}}$
$(1, 1)$	$\min(a_1, a_2)$
all other cases with $l \geq 2$	A^{-1} .

If an assigned vertex lies below x , its integration uses only the fraction x/Y of the lower band. In this case at least one $a_s = 1$, and every entry in the table is at most a constant. Raw bounds in the two ensembles therefore give the first contribution in (E3).

We may now assume that all assigned vertices are at least x . Fully deterministic products cancel, so there is a stochastic factor. Suppose first that all stochastic factors occur at one vertex s . At that vertex, replace the overlaps by the common ancestor norm squared and then by $2L_v$ on a path through the fork. The alias errors are controlled by (D7). Center the resulting polynomial at its untilted expectation and apply (D10), treating the other times and restrictions as deterministic coefficients. After truncating the other factors at their envelopes, the cost is $O_\diamond(z_{r_s}^0 r_s Z)$ times 1 for $k_s = 1$, or times $m_{r_s} r_s^{k_s-2}$ for $k_s \geq 2$. Relative to P_{k_s} , this gives at least the gain g_B/a_s . On the lower band the same gain follows from conditional averaging and the linear bound after (D9) when there is one stochastic power, and from the integrated D_v^2 bound when there are two or more. Multiplying by the appropriate entry for F gives $O_\diamond(g_B)$.

If stochastic factors occur at several vertices, select two of them at scales aY, bY . Their integrated second moments about S , from (D7) and (D9), are bounded by $O_\diamond(g_B Y^2/a)$ and $O_\diamond(g_B Y^2/b)$. First apply unconditional Cauchy–Schwarz to the two selected stochastic factors at fixed masses, and truncate the remaining factors at their deterministic envelopes. At a selected vertex of scale $r = aY$ and multiplicity $k \geq 2$, Cauchy–Schwarz in mass then gives

$$\int e(v)^{k-1} D_v \frac{dv}{r} \lesssim r^{k-2} \left(\int e(v)^2 \frac{dv}{r} \right)^{1/2} \left(\int D_v^2 \frac{dv}{r} \right)^{1/2} \lesssim_\diamond r^{k-2} m_r (g_B Y^2/a)^{1/2}.$$

Dividing by the raw budget $P_k = m_r^2 r^{k-2}$ gives the factor $O_\diamond(\sqrt{g_B a})$; for $k = 1$, division of $(\int D_v^2 dv/r)^{1/2}$ by $P_1 = m_r$ gives the same factor. Thus the two selected vertices improve the raw product by $O_\diamond(g_B \sqrt{ab})$. Since $\sqrt{ab} \leq A$, this proves the claim whenever $F \lesssim A^{-1}$.

There are two exceptional multiplicity patterns. For (1, 2), let a_d be the duplicate scale and a_s the singleton scale. If both factors at the duplicate are stochastic, selecting them uses $\int D_v^2 dv/r$ at that vertex. Relative to its raw budget m_r^2 , this gains $O_\diamond(g_B a_d)$, so $F g_B a_d = g_B$. If only one is stochastic, selecting it together with the singleton gives

$$F g_B \sqrt{a_s a_d} = g_B \sqrt{a_s/a_d},$$

which suffices when $a_s \lesssim a_d$. In the remaining case the later singleton is linear and stochastic. Integrating it as a centered path integral gives g_B/A , so the product with $F = 1/a_d$ is again at most g_B . For (1, 1), both factors must be stochastic. The lower-band case has $a_1 = a_2 = 1$ and follows directly from Cauchy–Schwarz. Otherwise the later linear factor has scale A and gives g_B/A ; multiplying by $F = \min(a_1, a_2)$ gives at most g_B .

For completeness, the last centered integration has a fixed marginal law even though its fork moves. There is only one earlier stochastic pair to retain. At most one of its leaves descends through the later fork, so choose that leaf for the L -path when it is present. After the alias replacement the other child of the moving fork has no observable and integrates to one. The retained pair always splits at its fixed earlier time. A prior projection merely evaluates these same two paths at fixed later cuts. If neither endpoint descends through the moving fork, choose a continuation whose separation from the retained tree is at another fixed vertex. In either case the moving time occurs only in the L -observable and in deterministic kernels. Apply (A2), or its consequence (D10), unconditionally under this marginal and then apply Hölder with the earlier random factor. This is the construction in Remark 8.13; it does not require an inequality conditioned on that factor. Deterministic ties belong to the kernels, and stochastic ties form one polynomial power. The truncation and tilted-law transfers follow the conventions after (D10). This proves (E3). $\square$

9.3 Taylor remainders and separation of the masses

On size paths, (D11) makes each insertion small in every fixed moment. A sufficiently high fixed degree therefore bounds the Taylor remainders using the absolute allocation measure. The differentiated clock weights in (A9) have absolute integrated norm $O_o(1)$. Their placements have bounded densities, including when the added pair attaches to an existing integration variable; there is no prescribed fork in this pressure calculation. Thus they cost only a subpower factor.

The same weighted accounting applies to row contacts and normalization errors. Since $m \asymp N = x^{-6}$, their contribution is bounded by

$$O(m^{-1}) \left(1 + \|\dot{b}\|_{L^1} + \|\dot{h}\|_{L^1} + \|\dot{K}\|_{L^1} + \|\dot{q}\|_{L^1}\right) = O_o(x^6).$$

Here the norms denote the absolute mass integrals with the bounded-density allocation weights from (A9). The stencil has finitely many fixed coefficients. Moreover, Lemma 2.5 gives a total-pressure remainder, so no further factor m multiplies these errors. This proves that (A7)–(A9) and the stencil reduce the size increment to (E1) with the required saving. On direct paths the differentiated weights carry the additional factor $O_o(\lambda)$.

For a direct variation, start with the monotone reference $K_S = h + bS$. Apply (A7) with insertions $b\delta$, expanding the pressure gradient of (A9) through order $J_0 - 1$. By (P8), the test multiplying the cavity moment has degree at most one and weight $O_o(\lambda)$. In the true Taylor remainder distinguish two cases. If at most one insertion has $e(v) \geq x^{\delta'}$, at least $J_0 - 2$ insertions have smaller envelope. Their moment bounds from (D2) supply an arbitrarily large fixed power of x once $\delta' > 0$ is fixed and J_0 is large enough. Choose the truncation losses first so that they do not use up this gain.

If two insertions have $e(v) \geq x^{\delta'}$, apply Cauchy–Schwarz and (D8) to those insertions, and bound the others uniformly. The selected vertices have bounded absolute density even if the insertions are tied. The one-row interpolated laws are comparable to the genuine laws by (A7); the transfer to tilted active components uses (A4) and the stopping conventions after Lemma 8.6. Before averaging over γ , the cost is

$$O_o\left(\lambda x^{-C_0\delta'} z_1^0 Y^2 Z^2\right) = O_o\left(x^{5-C_0\delta'} \lambda^{(1-\eta)/2-s_*} Z^4\right).$$

The fourth moment of Z therefore gives the averaged bound

$$O_o(x^{5-C_0\delta'} \lambda^{(1-\eta)/2-s_*}),$$

which saves a power when $C_0\delta' < p_*((1-\eta)/2 - s_*)$, with room for the remaining losses. We choose δ' this small, also relative to κ , and then choose the direct Taylor degree.

Next expand the deterministic quotient coefficients from K to K_S , keeping the same total truncation degree. These scalar Taylor remainders have bounded common coefficients and at least J_0 pair factors in all. Each factor is either $K_S - K = bf + d$ or $b\delta$. Split the pressure-gradient test into its deterministic and linear stochastic parts as well. We need only the difference between the two ensembles, so Lemma 9.1 applies once its weight condition is checked. Every designated factor has its envelope bound, and the density has prefactor λ . With j factors at l distinct vertices, changing to the normalized measure costs w^l . Since $j + l \geq 5$,

$$\prod_s dv_s = w^l \prod_s dv_s/w \lesssim w^{5-j} \prod_s dv_s/w.$$

Thus the remainder has the weight (E2) and is bounded by (E3).

The retained terms recombine by Lemma 2.3. Differentiating a scalar moment toward K_S inserts $(K_S - K)\tau_a\tau_b/2$. Allocation symmetry permits these insertions within products,

with each scalar moment evaluated on its own subtopology. Insertions into a unit moment sum to zero when attached to the observed topology. Use these identities in the unrestricted allocation sums, before dividing terms into mass bands for estimates. The quotient retained through degree $J_0 - 1$ is therefore the quotient at K with argument $J = (K_S - K) + b\delta = b\Delta + d$.

By (A9), differentiating the logarithm through degree J_0 produces this gradient. Any additional highest-degree coefficient derivatives have J_0 factors and an $O_\diamond(\lambda)$ weight; after splitting the factors into deterministic and stochastic parts, the same calculation puts them in (E2). Pure deterministic terms cancel. This establishes (E1) on the whole remaining direct path, including masses at which a pointwise small-insertion argument was unavailable.

For size changes, first consider terms with a base pair at mass at least V_2 . Call a base factor *high* if its mass is at least V_1 ; recall that $V_1 < V_2$. Count the square term as two base factors on the same pair. Expanding $\Delta = f + \delta$, every high deterministic factor, including d or $\dot{d}$, is bounded by $O_\diamond(x^{2+b_H}(Z^2 + Z_g))$. Stochastic high factors have the variance bounds (D15); all factors also have the all-moment smallness (D11). Low factors use their envelopes.

The scalar coefficients are bounded, including $\dot{b} = O(1)$. Differentiating K may add one scalar pair with weight $|\dot{K}|$. At high mass this weight is integrable separately. If it is tied to a high base vertex, integrate against that same variable and use the pointwise bounds (D11), (D15). At a low mass v , the inverse estimates give $|\dot{K}(v)| = O(v)$. The terms with at least one stochastic factor fall into the following cases.

1. If there are two high stochastic factors, Cauchy–Schwarz gives $D_{H,1}^2$. If there is one high stochastic factor and a high deterministic factor, the bound is $D_{H,1}x^{2+b_H}(Z^2 + Z_g)$.
2. If every high factor is deterministic, extract the bound for one of them and apply (D16) to the low factors. A sole low factor at v has the additional scalar coefficient $O(v)$ required there: immediately after its split, its two ends lie in groups with odd parity, as in the proof of Proposition 8.11. The square term cannot have this configuration. An added dotted scalar pair preserves the parity argument if it splits later than v ; if it splits earlier or at v , its own weight is $O(v)$.
3. If a stochastic factor is the only high factor, it is the factor at mass at least V_2 , so it costs $D_{H,2}$. With one low factor, the preceding parity argument gives $\int_{v \leq V_1} v e(v) dv \lesssim_\diamond V_1 Y^2$. With two or more low factors, integrate one using $\int e(v) dv \lesssim_\diamond Y^2$, and use (D11) on another to gain $x^{\delta_*/2}$. Higher fixed moments and Hölder give the same bound when these factors share a vertex.

Here is the resulting power count, before γ -averaging and up to the permitted losses. In the last two rows use $Y^2 = x^2\lambda^{-s_*}Z^2$ and $\lambda = x^{p_*}$.

selected factors	bound
two stochastic high	$x^{6-2C_H a_1} Z^4$
stochastic and deterministic high	$x^{5+b_H-C_H a_1} (Z^4 + Z_g Z^2)$
only deterministic high	$x^{5+b_H-C p_*} (Z^4 + Z_g Z^2)$
sole stochastic high, one low	$x^{5+a_1-C_H a_2-p_* s_*} Z^4$
sole stochastic high, two or more low	$x^{5+\delta_*/2-C_H a_2-p_* s_*} Z^4$.

The hierarchy in (D15) makes every exponent strictly larger than five: it gives $b_H \gg C_H a_1$, $a_1 \gg C_H a_2$, and permits p_* to be chosen smaller than all remaining gaps. The factors Z^4 and $Z_g Z^2$ have bounded γ -expectation, the latter by Cauchy–Schwarz. Thus every displayed cost has the required averaged saving. Products with no stochastic factor cancel outright.

On a direct path, terms whose maximum mass is bounded below by a fixed positive constant already satisfy (E2). The dotted weights supply $O_\diamond(\lambda)$, and $|\dot{d}| \lesssim_\diamond \lambda r^2 e$. For

a quadratic term with two distinct designated vertices, scalar parity gives the required smaller-scale ratio. If differentiation adds a scalar pair at a smaller mass v' , its weight $|\partial_{\log \lambda} K(v')| \lesssim \lambda v'$ gives that ratio instead.

All direct terms of degree $j \geq 5$ also satisfy (E2) by the normalized-measure calculation above. The same holds for the remaining size terms of those degrees: now $w \lesssim V_2 \ll \lambda$ and $\prod_s dv_s \lesssim \lambda w^{5-j} \prod_s dv_s/w$. An extra dotted weight that is unbounded must be at a higher mass and on a separate variable, where its absolute integral is available. It remains to treat only low-region terms of degrees 2, 3, 4.

Proposition 9.2 (Coefficients after compression). *Fix the dyadic restrictions on the base factors and a cut $c \gg w$ within the region where the small inverse is regular. A sufficiently large fixed separation factor is enough; on size paths require also $c \lesssim \lambda$. Group base slots into blocks that share their path through c . Their split tree below c is the skeleton, and every designated pair joins distinct blocks.*

Apply the operator $\mathcal{O}$ defined above to the pressure polynomial (E1): it is $1 + \partial_\theta$, with the formal action $\partial_\theta \Delta = -\Delta/6$, on a size path, and $\partial_{\log \lambda}$, holding Δ fixed, on a direct path. Group the resulting signed allocation coefficients by this skeleton and sum their internal placements within each block. The bound below concerns these differentiated, grouped coefficients. This is a rearrangement of a signed measure. The probabilistic expectations used to evaluate the coefficients remain positive scalar laws on each fixed topology. The spin-model pair tests have been projected to c , or to a neighboring smaller cut, so they depend only on the skeleton and can be kept outside the internal sums. For $j = 2, 3, 4$, the resulting coefficient, including the skeleton allocation weight, is bounded by

$$O_\diamond(\lambda c^{5-j}) \prod_{s \text{ skeleton fork}} dv_s/c. \quad (\text{E4})$$

If $j = 2$, the designated vertices are distinct, and $c \asymp w$, the bound includes the further factor $O_\diamond(\min_s r_s/c)$. A factor d or $\dot{d}$ may first contribute λc^2 to the coefficient, leaving a deterministic factor bounded by the ordinary envelope because $r \lesssim c$. Like b and $\dot{b}$, these weights are constant during each internal block sum.

Combine contributions only when they multiply the same product of pair tests on the same skeleton. In particular, this combines a repeated pair with the square counterterm. Give such terms the same dyadic restrictions and projection cuts. The cut, shape, and mass coordinates are fixed during coefficient differentiation. Internal summation may then be done before or after that differentiation. The spin tests receive only the formal action on Δ already specified: project the differentiated polynomial, without differentiating a random test or a projection law.

Reduction to Proposition 9.2. Start with a sufficiently small fixed cut on direct paths, choosing the low-mass threshold smaller still, and with a cut of order λ on size paths. By Corollary 8.7, changing one pair to its projection at c costs $O_{p,\circ}(z_c^0 c)$. Conditional on the positive-law prefixes at that cut, a cross-block pair has exactly its projected conditional mean. Expand a product as projected factors plus changes. Terms with just one change have zero conditional mean, so every error contains at least two changes.

For this first projection, bounded coefficient densities suffice. Their prefactor is λ on direct paths and $O_\diamond(1)$, including integrated dotted weights, on size paths. The respective costs are $O_\diamond(x^6 \lambda^{-\eta})$ and $O_\diamond(x^6 \lambda^{-C})$. Now reduce the cut by factors of two until it is comparable to w , retaining a large enough separation factor. For a step from c to $c/2$, first sum the coefficients at c and apply (E4). The product error again contains at least two changes of size $O_{p,\circ}(z_c^0 c)$; every other factor, after extracting the defect weights allowed in

the proposition, is $O_{p,\diamond}(c)$. Inactive masked pairs do not change. Hence the step costs

$$O_{\diamond}(\lambda c^5(z_c^0)^2) = O_{\diamond}(x^6 \lambda^{-\eta}/c) \lesssim_{\diamond} x^5 \lambda^{s_*/2-\eta}$$

because $c \gtrsim Y \geq U$. At the final cut, (E4) is precisely (E2): the designated fork measures have the required normalization, and the remaining skeleton variables have bounded volumes in units of c . Apply (E3) to their expectation difference. All cuts are common to the two ensembles. There are only fixedly many degrees and shapes, and the dyadic bands and successive cuts cost subpower factors. Thus Proposition 9.2 implies the remaining bounds for (I), uniformly also in the site count, with a strict power saving.

9.4 Scalar compression at a fixed cut

Proof of Proposition 9.2. We first evaluate the internal signed sum in (E1) before applying $\mathcal{O}$, and then justify commuting that operator with the sum. Absolute values are taken only after the resulting cancellations. Fix a cut c in the regular inverse region with all designated pair splits strictly below it and $c \gtrsim w \geq Y \geq U$; on size paths also $c \lesssim \lambda$. Throughout coefficient differentiation, hold this cut, the skeleton shape, its mass coordinates, and all mass restrictions fixed.

The objects in the compression formula. For a degree j term, partition its $2j$ labeled endpoint slots into blocks at c . Let D be the number of blocks, ordered by their first slots. The j ordered pairs form a graph on these blocks; every edge has two distinct ends. The skeleton allocation measure is the signed per-root measure for D ordered distinct end labels in the hierarchy ending at c , divided by c^D . Its absolute weight on each of finitely many shapes is bounded by

$$C_j c^{-1} \prod_{s \text{ skeleton fork}} dv_s/c.$$

Indeed the first label has unit per-root weight, and each of the remaining $D - 1$ placements contributes scale c . Division by c^D leaves c^{-1} . After rescaling mass to $[0, 1]$, the remaining densities are bounded. Attachments at existing forks may reuse their mass coordinates, but block identities remain distinct. There are no prescribed old leaves or forks. Thus distinct free fork coordinates are unequal almost everywhere, while a fork may have more than two children.

Next let π be a partition of the j edges into cumulant parts, keeping repeated edges as separately labeled slots. For a part $P \in \pi$, let $k_{A,P}$ be the number of its endpoints in block A , and put

$$D_{\pi} = \sum_{P \in \pi} \#\{A : k_{A,P} > 0\}.$$

Thus D counts blocks in the full skeleton, whereas D_{π} counts a block again each time it is used by another part. Neither counts endpoint multiplicities.

On the fixed skeleton, let ζ_u^A , $0 \leq u \leq c$, be the scalar prefix fields under the positive scalar law. They share their Brownian driver until their skeleton separation, and thereafter use independent transitions. Each part P uses a separate copy of this law restricted to its blocks. The normalized source sum for k slots inside one block is

$$s_k(z) = c^{-1} e^{-cV(t(c),z)} \partial_z^k e^{cV(t(c),z)}.$$

With the explicit edge weights of (E1) kept outside, the coefficient multiplying the skeleton measure is

$$\sum_{\pi} (-1)^{|\pi|-1} (|\pi| - 1)! c^{D_{\pi}} \prod_{P \in \pi} \mathbf{E}_{\text{low}} \prod_{A \text{ in } P} s_{k_{A,P}}(\zeta_c^A). \quad (\text{E5})$$

The expectation $\mathbf{E}_{\text{low}}$ is probabilistic; the preceding grouping of allocation coefficients by skeleton is a signed sum.

For example, take two copies of the edge AB between cut blocks. Their original terminal labels may differ; projection makes their spin-model pair tests identical. Abbreviate $s_1(\zeta_c^A) = \mathbf{M}_A$, $s_2(\zeta_c^A) = \mathbf{P}_A$. The one-part partition uses two blocks, whereas the two-part partition uses each block twice. Their combined internal coefficient is therefore

$$c^2 \mathbf{E}_{\text{low}}(\mathbf{P}_A \mathbf{P}_B) - c^4 (\mathbf{E}_{\text{low}}(\mathbf{M}_A \mathbf{M}_B))^2.$$

The skeleton measure contributes c^{-1} , giving net powers c and c^3 . There are two orientations of the second edge for a fixed first edge, so its factor in (E1) is $2/(2^2 2!) = 1/4$. The square counterterm uses this same skeleton, with internal coefficient c^2 and weight $-b/4$. These normalizations will give the quadratic cancellation below.

The finite signed identity. First expand each cumulant into products of positive scalar moments on its indicated edge sets. Once the skeleton assignments are fixed, there is no restriction on the internal placements within a block. A part observes only its own induced genealogy, so the internal signed sums for different parts can be performed separately, even when their actual terminal leaves coincide.

This factorization can be checked directly at finitely many levels including c . In the labeled product rule of [19, Proposition 2.5], the coefficients choosing distinct children are falling factorials of incoming-to-child mass ratios, and every terminal source slot carries its terminal mass factor. Keep the induced genealogies as discrete marks and first take the successive ratios, including an auxiliary root ratio, to be sufficiently large positive integers. The coefficients then count child identities. After distinct block identities have been assigned at c , the choices for disjoint sets of slots inside a block are independent counts, with terminal slots chosen with replacement. Their counts therefore multiply. The resulting rational coefficient identity extends by polynomial identity to the real masses, keeping terminal mass fixed and dividing by the root mass before its zero limit. This argument concerns only the coefficients; it does not extend spin tests or probability laws from integer ratios.

Evaluation of the internal source sums. For one part, the positive scalar prefix law on its restricted skeleton is independent of the internal placements after c . Conditional on these prefix fields, the source product rule sums k internal slots in a block to $e^{-cV} \partial_z^k e^{cV} = c s_k$. Different blocks use independent future kernels. Thus every used block contributes one factor c , giving c^{D_π} , and the cumulant partition formula gives (E5). Signed path limits and bounded fixed-order derivatives, as supplied by PE, pass this finite-level identity to the continuous paths. Equivalently, one may fix the prefix fields and approximate only the future clocks within the blocks. There are no prescribed internal split times to preserve in that approximation.

Differentiation at a fixed skeleton. The cut and skeleton measure stay fixed. Differentiating the scalar moments adds one covariance pair with weight $\dot{K}$. By (P8), its absolute integrated weight is finite; its non-atomic placements have bounded density for fixed cut and label count. The same bound holds along affine comparison chords of the clocks, so the finite signed differentiation rule and Fubini apply to the internal sums.

To pass from chords to ordinary derivatives, fix N . The clock difference quotients are uniformly bounded in mass: for size paths this follows from the smoothing floor and clipping, and for direct paths from the localized smooth inverse perturbation. For almost every parameter they converge at every mass in the low region and at almost every other mass.

The scalar kernels converge by fixed-size path continuity. Dominated convergence therefore passes the divided covariance insertions to the dotted insertions, also for almost every fixed skeleton. A reused split is either one of these integration coordinates or a prescribed low skeleton mass, where convergence is pointwise; constant diagonals still cancel. This proves that internal summation and a.e. coefficient differentiation commute. We may choose the low-region partition separately at each parameter when estimating the result: the coefficient has already been differentiated, so no moving indicator or cut is being differentiated.

Scalar jets and common-prefix estimates. Write $\chi = V_{zz}(0,0)$, $g = V_{zzzz}(0,0)$, $k_0 = t'(0+) > 0$, and $G(u, z) = V(t(u), z)$. The notation $O_*(A)$ controls both the value and the parameter derivative: on size paths the derivative is ∂_θ , and on direct paths it is $\lambda^{-1}\partial_{\log \lambda}$. The same convention applies to fixed L^p norms when fields are coupled by common Brownian drivers in mass coordinates. Derivatives of the explicit weights b, d will be handled separately.

By (P1)–(P2), the inverse t is smooth here, with t' bounded above and below and with the required higher derivatives bounded. The positive-order spatial derivatives of G also have bounded mass and spatial derivatives of every required fixed order, allowing polynomial growth in z after parameter differentiation. To see the mass derivatives, use

$$\partial_u G = -\frac{1}{2}t'(G_{zz} + uG_z^2).$$

For direct derivatives, differentiate the scalar PDE in λ at fixed normalized time, as in the lift proof, and then compose with the regular inverse. For size derivatives, use the rescaled scalar jets, including the terminal-mass derivative already estimated in Section 3. These operations preserve evenness in the field. In particular, on size paths,

$$\partial_\theta k_0 = -3k_0/6, \quad \partial_\theta \chi = \chi/6 + O(e^{-\Omega(M)}), \quad \partial_\theta g = 3g/6 + O(e^{-\Omega(M)}). \quad (\text{E6})$$

The normalized inverse slope and initial spatial derivatives are constant under this rescaling, up to the exponentially small terminal effect. This gives (E6). Moreover, χ is uniformly bounded above and away from zero. At every designated split,

$$b = b_\circ + O(\lambda c^2), \quad b_\circ = \chi^{-2},$$

and the error has the same bound under unscaled parameter differentiation. By (D2), each d or $\dot{d}$ can likewise contribute $O_\circ(\lambda c^2)$ to the coefficient, leaving its ordinary envelope factor.

The low prefix equation is

$$d\zeta = \sqrt{t'} dB + ut' G_z(u, \zeta) du.$$

Starting from zero, this equation gives $\zeta_u = O_*(\sqrt{u})$ and $\zeta_u - \sqrt{k_0}B_u = O_*(u^{3/2})$ in every fixed moment. For a conditional evolution starting at z , the field and its parameter derivative have norms at most $C_p(|z| + \sqrt{c})(1 + |z|)^C$. Its spatial flow derivatives have bounded moments, allowing polynomial growth in z . These bounds follow by differentiating the additive-noise equation: the drift is smooth with bounded spatial derivatives, the differentiated coefficients have polynomial growth, and each odd differentiated drift vanishes at zero.

Evaluate the block observables at their fields at c , unless another time is specified. Taylor expansion, together with even/odd parity and the definition of s_k , gives

$$\begin{aligned} M_A &:= s_1(\zeta_c^A) = G_z(c, \zeta_c^A) = \chi \zeta_c^A + O_*(c^{3/2}), \\ P_A &:= s_2(\zeta_c^A) = G_{zz}(c, \zeta_c^A) + c(G_z(c, \zeta_c^A))^2 = \chi + O_*(c), \\ s_3(\zeta_c^A) &= g \zeta_c^A + O_*(c^{3/2}), \quad s_4(\zeta_c^A) = g + O_*(c). \end{aligned} \quad (\text{E7})$$

For blocks A, B splitting at $v_{AB} < c$, the martingale property of G_z reduces their product expectation to the common prefix at v_{AB} . Thus

$$\mathbf{E}_{\text{low}} \mathbf{M}_A \mathbf{M}_B = \chi^2 k_0 v_{AB} + O_*(v_{AB}^2).$$

We also need

$$\mathbf{E}_{\text{low}} \mathbf{P}_A \mathbf{P}_B = \chi^2 + \chi^3 k_0 c^2 + \frac{1}{2} g^2 k_0^2 v_{AB}^2 + O_*(c^3). \quad (\text{E8})$$

To obtain (E8), keep the terminal cut c fixed and set $H_u = G_{zz}(u, \zeta_u) + c G_z(u, \zeta_u)^2$. The PDE and Itô's formula give drift $(c - u)t' G_{zz}^2$ and diffusion coefficient $\sqrt{t'}(G_{zzz} + 2c G_z G_{zz})$. In the product of the two block processes, the two drift integrals sum to $\chi^3 k_0 c^2 + O_*(c^3)$, by (E7). Their common quadratic variation stops at v_{AB} ; its expected rate before that time is $g^2 k_0^2 u + O_*(cu)$. Its integral is the term $\frac{1}{2} g^2 k_0^2 v_{AB}^2$, with an error of order $O_*(c^3)$.

For three blocks we use

$$\mathbf{E}_{\text{low}} \mathbf{P}_A \mathbf{M}_B \mathbf{M}_C = \chi^3 k_0 v_{BC} + O_*(c^2). \quad (\text{E9})$$

When the designated splits are distinct, this estimate retains the earlier mass. Suppose $v_{AB} = h' < v_{AC}$, so that $v_{BC} = h'$. Condition at h' on the common field z . After subtracting $\chi \mathbf{M}_B \mathbf{M}_C$, the conditional expectation on the A, C side is that of $(\mathbf{P}_A - \chi) \mathbf{M}_C$. It is an odd function of z , with spatial derivative bounded by $C(c + z^2)(1 + |z|)^C$, also after parameter differentiation at fixed z . Indeed $\mathbf{P} - \chi = O(c + |\zeta_c|^2)$, its spatial derivative is $O(|\zeta_c|)$, and $\mathbf{M} = O(|\zeta_c|)$ with bounded derivative; apply the conditional flow bounds and Hölder to their product.

The B -side conditional mean is $G_z(h', z)$, of size $O_*(|z|(1 + |z|)^C)$. Substituting the common prefix field $z = O_*(\sqrt{h'})$ therefore bounds the error in (E9) by $O_*(ch')$. The other ordering is identical after exchanging B, C . Without a designated earlier split, (E7) gives the stated $O_*(c^2)$ error.

The coefficients canceled by size rescaling. For a pure degree j term with no defect factors, the formal rule $\partial_\theta \Delta = -\Delta/6$ makes the size operator on its coefficient equal to $1 - j/6 + \partial_\theta$. Since $\ell = e^{\theta/6}$,

$$\left(1 - \frac{j}{6} + \partial_\theta\right) \ell^{-(6-j)} = 0.$$

Equation (E6) and $b_o = \chi^{-2}$ identify exactly the coefficients with this scaling:

j	coefficient monomials	size scaling
2	$b_o^2 \chi^3 k_0, \quad b_o^2 g^2 k_0^2$	ℓ^{-4}
3	$b_o^3 g^2 k_0, \quad b_o^3 \chi^3$	ℓ^{-3}
4	$b_o^4 g^2$	ℓ^{-2}

The masses multiplying these monomials are held fixed. Exponentially small terminal errors will be retained until the end. On direct paths, the operator is only $\partial_{\log \lambda}$, and the O_* convention already supplies the factor λ . For a term with a defect, use the product rule and extract at least one factor λc^2 as above. In the remaining case analysis we include the net factor $c^{D\pi-1}$; the normalized skeleton fork measure in (E4) is left outside.

The low-degree cases.

1. Suppose $j = 2$ and $D = 2$. The two edges join the same blocks, so use the repeated-edge calculation following (E5). Its coefficient combines with the square counterterm because both multiply the same projected pair square. After extracting the common skeleton measure and $c^2/4$, their coefficient before differentiation is

$$b^2 \left\{ \mathbf{E}_{\text{low}} \mathbf{P}_A \mathbf{P}_B - c^2 (\mathbf{E}_{\text{low}} \mathbf{M}_A \mathbf{M}_B)^2 \right\} - b.$$

The constant term $b^2 \chi^2 - b$, and its unscaled derivatives, are $O(\lambda c^2)$ because $b_\circ \chi^2 = 1$ and $b - b_\circ = O(\lambda c^2)$. The next two terms from (E8), after replacing b by $b_\circ$, are

$$b_\circ^2 \chi^3 k_0 c^2, \quad \frac{1}{2} b_\circ^2 g^2 k_0^2 v_{AB}^2.$$

Their direct derivatives are $O(\lambda c^2)$; their size contributions cancel by the $j = 2$ row of the table. Substitution errors from $b - b_\circ$ have the required bound. The remaining $O_*(c^3)$ errors, including the smaller product-of-means term, contribute $O(c^4)$ after the net compression factor c . On size paths this is $O(\lambda c^3)$ because $c \lesssim \lambda$; on direct paths the derivative supplies λ itself. Thus (E4) holds. A defect factor supplies λc^2 directly and gives the same bound, with the allowed $O_\circ(1)$ loss.

2. Suppose $j = 2$ and $D = 3$, with edges AB, AC . For the one-part partition, (E9) gives the leading coefficient $b_\circ^2 \chi^3 k_0 v_{BC}$, with net compression factor c^2 . Its size contribution cancels by the same row of the table; its direct derivative costs $O(\lambda c^3)$, or $O(\lambda c^2 h')$ when $v_{BC} = h'$ is the earlier split. The error is $O_*(c^2)$, or $O_*(ch')$ when the designated vertices are distinct and their earlier mass is h' . Multiplication by c^2 , followed on size paths by $c \lesssim \lambda$, gives respectively $O(\lambda c^3)$ and $O(\lambda c^2 h')$. The errors from $b - b_\circ$ and any defect factors satisfy these same bounds. For the two-part partition the product of pair moments is $O_*(v_{AB} v_{AC})$, with net factor c^3 , which is smaller still. Since $h' \lesssim \min_s r_s$, the refined bound gives the extra ratio required in (E4).
3. Suppose $j = 2$ and $D = 4$. Both partitions have net factor c^3 , and their products of moments are $O_*(c^2)$. When the designated vertices are distinct with earlier mass h' , these products are also $O_*(h')$. For a product of two pair moments this follows at once from their individual bounds. For the four-spin moment, condition on the prefixes at h' . The earlier edge has separated ends, while the ends of the later edge are still together. Hence at least two of the independent future groups have odd parity. Each such group's conditional expectation vanishes at zero starting field and has a polynomially bounded spatial derivative, also after parameter differentiation. The common prefix fields have size $O_*(\sqrt{h'})$, so the product of these two odd expectations gives $O_*(h')$. On size paths the refined contribution is therefore $O(c^3 h') \leq O(\lambda c^2 h')$, exactly the required smaller-scale bound. On direct paths the derivative supplies λ . The unrefined $O_*(c^2)$ estimate gives (E4) without the ratio. Extracting defect factors preserves both conclusions.
4. Suppose $j = 3$, for which the target is $O_\circ(\lambda c^2)$. Terms with defects or with $D_\pi \geq 4$ already have enough powers of c : on a size path one extra power is bounded by λ , and on a direct path the derivative supplies λ . It remains to consider one-part partitions with two or three blocks. For $D = 2$, each block has degree three and (E7) gives $g^2 k_0 v_{AB} + O_*(c^2)$. For $D = 3$, an odd block degree gives $O_*(c)$; otherwise all three degrees are two, giving $\chi^3 + O_*(c)$. The two leading monomials after multiplication by $b_\circ^3$ are precisely those in the $j = 3$ row of the table, so their size contributions cancel. With net factors c and c^2 , respectively, the remaining errors are $O(c^3)$, which is at most $O(\lambda c^2)$. Direct derivatives give the same target without this cancellation.

5. Suppose $j = 4$, for which the target is $O_\diamond(\lambda c)$. Only size terms with no defect and $D_\pi = 2$ lack the required power before cancellation. Such a term has one part, on two blocks of degree four each. Its moment is $g^2 + O_*(c)$. Multiplication by $b_\circ^4$ gives the last row of the table, so the leading size contribution cancels. The remainder with net factor c is $O(c^2)$, at most $O(\lambda c)$. All other terms have an additional compression power, a defect factor, or the direct derivative factor λ .

The exponentially small terminal errors retain the mass factors of their corresponding main terms. Since $M = \log^3(1/x)$, they are smaller than every fixed power of x . This proves (E4). $\square$

The row expansion, the high-mass estimates, and the low-degree coefficient bound now give both increments required in Proposition 4.1. The final parameter choice in Section 10 makes their savings simultaneous and justifies the stated uniformity for moving site weights.

10 Parameter choices and completion of the proof

Recall that $x = N^{-1/6}$ and $x^{12} \leq \lambda \leq x^{p_*}$. The parameters below are fixed independently of N . We first choose the savings used to control size changes, then the final-strength exponent p_* , and only then the truncation and loss parameters for direct changes. The essential order is that the size Taylor degrees determine the polynomial loss constants before the cutoffs and p_* are selected. Direct Taylor degrees may be larger, because their estimates do not enter that size choice. Every degree, moment order, and loss exponent below is fixed before $N \rightarrow \infty$.

1. Fix $T > 1$ and the neighborhoods, tapers, and constants c_1, c_b in (P1)–(P7), leaving strict room in the admissibility and derivative bounds. The global saving c_G in (G1) is available before choosing s_* near $1/2$ or the ultimate small p_* . In its proof, the coarse clock estimate (G3) supplies a fixed saving first. Choose a_g below that saving, then d_g small enough to absorb the fixed inverse-kernel loss in the covariance estimate following (G4), and finally the PE power slack for this global comparison small relative to d_g . The endpoint condition (P9) uses the common base temperature as explained in Section 6. It requires $\lambda \rightarrow 0$, without requiring $\lambda \leq x^{a_g}$. Thus later decreases of p_* do not decrease c_G .
2. Set $k_c = 0.03$. Choose $\tau \ll \min(10^{-4}, c_G)$, put $s_* = 1/2 - \tau$, and then choose $\eta \ll \tau$ and $\tau_0 \ll \eta$, with $s_0 = s_* - \tau_0$, as in (C1)–(C2). Choose $b_s > 0$ so that

$$2(\tau + \tau_0) + b_s < \min(c_G/12, k_c/3), \quad b_s < 2(\eta - \tau_0),$$

and then $b_\ell \ll b_s$. These inequalities leave room for the local comparison errors throughout $\lambda \geq x^{12}$. Choose also $h_e \ll \min(1, c_G)$.

Both final-strength estimates in (C2) now have a fixed saving $c > 0$ for every sufficiently small p_* . Indeed, with $\lambda = x^{p_*}$, $U = x\lambda^{-s_*/2}$, and the edge seed off, the differentiated bound in its proof has powers

$$4 + c_G - p_*, \quad 5 + p_*(2\eta - 5s_*/2), \quad 6 - 2p_*.$$

All remain separated above 4 as p_* decreases. The smoothing error $O(x^{5+c'})$ in (P11), divided by λ , has exponent $5 + c' - p_* > 4$ as well. Likewise (D11) supplies a fixed $\delta_* > 0$, for example $(1/200)/50$, under the preliminary restriction $p_* < (1/200)/100$. Neither saving depends on the remaining choice of p_* .

3. Fix the size row and pressure Taylor degrees using δ_* . In the covariance equation proving (D15), choose the row degree so that the Taylor remainder before inversion, apart from the $O(m^{-1})$ contacts, is smaller than x^6 by a fixed power. This is possible by (D11) and requires no further information about p_* . For these fixed degrees the inverse has fixed polynomial losses in $(\lambda V_i^2)^{-1}$ and $1 + M$; the low sources have a fixed loss λ^{-C} . Fix constants bounding these losses before choosing the cutoffs, large enough to cover all the estimates below.

Now choose a_0 small relative to c, δ_* , take b_H to be a sufficiently small fixed fraction of a_0 , then choose a_1 small relative to a_0, b_H , and a_2 small relative to a_1 . Finally choose p_* small relative to these gaps and to h_e , including the restriction needed for the cap on Y after (D11). With $V_i = x^{a_i}$, the high-mass terms of (E1) then have the following positive margins over x^5 :

$$\begin{aligned} 1 - 2C_H a_1 &> 0, & b_H - C_H a_1 &> 0, & b_H - C p_* &> 0, \\ a_1 - C_H a_2 - p_* s_* &> 0, & \delta_*/2 - C_H a_2 - p_* s_* &> 0. \end{aligned}$$

The first two margins cover two high stochastic factors and a high stochastic factor paired with a high deterministic factor. The third covers a high deterministic factor times the low-source estimate (D16). The last two cover a sole high stochastic factor, using respectively the parity-weighted integral $V_1 Y^2$ or a further factor $x^{\delta_*/2} Y^2$. In both cases $Y^2 = x^2 \lambda^{-s_*} Z^2$. The size projection errors $x^6 \lambda^{-C} = x^{6-Cp_*}$ also have the positive margin $1 - C p_*$. No subsequent step decreases p_* .

4. Fix the number of strengthening stages in (C11). Choose ζ small relative to the power margins in Section 7; in particular, it is smaller than p_* times each required gap in a λ -exponent, including the gaps involving b_ℓ and τ_0 . Then choose the local PE power slacks small relative to ζ , and the buffer exponent ρ small enough for (C18). The remaining finite profile and moment iterations, and their subpower losses, can be chosen with these parameters fixed.

The extra optimized excess $O(x^{4-2\zeta} u \Lambda^{1/3})$ fits the budget at (C3): after division by λu , its ratio to $x^4 \lambda^{-2s_0}$ is $x^{-2\zeta} \lambda^{k_c/3-1+2s_0}$, which is power small by these choices. The final-strength switch error $O(x^{5+c'})$ from (P11) also fits after division by λu . At the smallest dyad $u \asymp U$, its ratio to U^4 is at most $x^{c'} \lambda^{5s_*/2-1}$; larger dyads improve this bound. Both exponents are positive, so the switch leaves room for the remaining losses.

5. For the direct paths, choose the coarse cutoff exponent $\kappa > 0$ small relative to the gains in Lemma 8.8. More precisely, its large- Y event costs $x^{-4\kappa}$, so choose

$$4\kappa < \min \{p_*(1 - 2s_*), (1 - \eta)/s_* - 2\}.$$

Both quantities on the right are positive by the choices of s_*, η . The gains used in the remaining direct expansion are

$$1 - 2s_* = 2\tau > 0, \quad (1 - \eta)/2 - s_* = \tau - \eta/2 > 0, \quad s_*/2 - \eta > 0.$$

A factor λ^g with any of these exponents $g > 0$ is at most $x^{p_* g}$. Choose $\delta' > 0$ smaller than κ and small enough that

$$C_0 \delta' < p_*((1 - \eta)/2 - s_*),$$

with strict room for the later losses. Thus the direct Taylor error $x^{5-C_0 \delta'} \lambda^{(1-\eta)/2-s_*}$ saves a power over x^5 . Now fix the direct pressure Taylor degree large enough also for this δ' . The initial direct projection cost satisfies $x^6 \lambda^{-\eta} \leq x^{6-12\eta}$, so its margin over x^5 is $1 - 12\eta > 0$.

Choose the fixed moment and change-of-measure orders large enough that truncation and transfer lose only small fractions of the margins just retained. Then choose the geometric loss in (D1) and (D10), and the subpower losses, still smaller. The smooth density comparisons on intervals of length $O(b_\epsilon)$ may require higher fixed orders after this geometric choice; they change constants only. Finally take the stencil order large enough for both the fixed size pressure polynomial and the site-weight smoothing in Section 4. Enlarging this order does not require enlarging the size Taylor degree.

These choices give a common positive saving in the stopped increment bounds. For a direct strength sweep, Lemma 8.8 handles $Y > x^\kappa$; on the complement apply the allocation and compression estimates (E3), (E4) to the pressure polynomial (E1). The two multiplier sweeps use the separate switch bounds of the same lemma. The size sweep uses the capped low statistics, (D15), and the size versions of these polynomial estimates. Integrating the direct strength parameter costs only $O(\log(1/x))$. The size segment has bounded length, and the multiplier bound has at most the integrable endpoint factor $\chi_e^{-1/2}$. The fixed-degree dyadic and projection-cut sums cost subpower factors. Decreasing the common saving absorbs these integration and summation costs.

We spell out the uniformity needed when the site count varies. At each fixed size-path point θ , every estimate above is applied to the particular integer m being tested, uniformly for $N/2 \leq m \leq 3N$. In particular, the averaging in (C1)–(C2) is over γ at this fixed (θ, m) . The physical clocks are common to these site counts. Their absolute dotted-clock norms are bounded up to subpower loss as explained after (P8); integrating any remaining dotted weight in mass therefore gives a common integrable envelope in θ . Thus the bound is uniform in m before integrating θ .

Let $w_m(\theta) \geq 0$ be measurable probability weights on the allowed integers. Multiplying the parameter-averaged bound at (θ, m) by $w_m(\theta)$ and summing preserves its common envelope, since $\sum_m w_m(\theta) = 1$. A measurable choice of a single $m(\theta)$ is the special case of a point mass. Fixed-size path approximation makes the pressure tests and statistics measurable, and absolute continuity supplies their a.e. path derivatives. Fubini first removes a null set of θ for each parameter-averaged estimate. At fixed N there are only finitely many allowed m , so their exceptional sets can then be united into one null set before weighting and integrating.

For each summand the change of measure is used only on the central stop $\mathcal{K}_m(s) \leq 3x^\epsilon$. The comparisons with finitely many added rows in (A6)–(A9) have bounded covariance changes and estimate their remainders with moments of the original m -site system. They therefore require this same stop, without a new stopping condition for the added-row systems. Fix the common saving before taking the small Laplace exponent ϵ in Section 4; this exponent is distinct from the geometric slack. The density bounds on the stop hold for every fixed finite order, so they cover all transfer orders chosen above, for every sufficiently small fixed $\epsilon > 0$ and each fixed test coefficient $q > 0$ in $s = qx^{1+\epsilon}$.

For one such test, the same parameter bounds cover the finitely many segments in the comparison from N to any $N' \in [N, 2N]$. Summing their stopped bounds and then choosing γ gives the single route used in Section 4. Different tests may choose different routes: their bare endpoint laws are independent of γ . Thus the later interpolation at finitely many Laplace tests requires no parameter choice uniform over an infinite test family.

This proves Proposition 4.1, including its version with a measurably varying site count or site weights. Section 4 then proves Theorem 1.2 and proves Proposition 1.3. Removing the independent Gaussian completion and dividing by the exact finite-size standard deviation, as done there, proves Theorem 1.1 for every fixed $\beta > 1$.

11 Proofs of the fluctuation and disorder-chaos corollaries

The two consequences in the introduction now follow from the main variance asymptotic and existing comparison inequalities. We keep the finite-size normalizations explicit.

11.1 The SK–Curie–Weiss comparison

Proof of Corollary 1.4. Abbreviate $F_n^{\text{CW}} = F_n^{\text{CW}}(\beta, \gamma)$ and couple it with $F_n(\beta)$ using the same disorder. For $\Delta_n = F_n^{\text{CW}} - F_n(\beta)$, [13, Proposition 1.5] gives $\Delta_n \geq 0$ and, for every $u \geq 0$ and n ,

$$\mathbb{P}(\Delta_n \geq u) \leq C_\gamma e^{-u}, \quad C_\gamma = (1 - \gamma)^{-1/2}.$$

Integrating the tail yields the uniform bounds

$$\begin{aligned} \mathbb{E}\Delta_n^2 &= 2 \int_0^\infty u \mathbb{P}(\Delta_n > u) du \leq 2C_\gamma, \\ \|\Delta_n - \mathbb{E}\Delta_n\|_2 &\leq K_\gamma := \sqrt{2C_\gamma}. \end{aligned}$$

In particular the perturbed mean and variance are finite. Let $t_n = \sqrt{\text{Var } F_n^{\text{CW}}}$. The reverse triangle inequality in L^2 gives

$$|t_n - s_n(\beta)| \leq \|\Delta_n - \mathbb{E}\Delta_n\|_2 \leq K_\gamma.$$

Since $s_n(\beta) \sim \sqrt{a_2(\beta)}n^{1/6} \rightarrow \infty$, we have $t_n/s_n(\beta) \rightarrow 1$, which proves the variance asymptotic and makes t_n positive for all sufficiently large n . For those n , using $\|X_n(\beta)\|_2 = 1$,

$$\left\| \frac{F_n^{\text{CW}} - \mathbb{E}F_n^{\text{CW}}}{t_n} - X_n(\beta) \right\|_2 \leq \frac{K_\gamma}{t_n} + \left| \frac{s_n(\beta)}{t_n} - 1 \right| \leq \frac{2K_\gamma}{t_n} \rightarrow 0.$$

Theorem 1.1 and Slutsky's theorem now give the asserted law. $\square$

11.2 Disorder chaos in the zero-field model

Proof of Corollary 1.5. For the Gaussian field in (1.1),

$$\rho(\sigma, \tau) := \text{Cov}(H_n(\sigma), H_n(\tau)) = \frac{1}{n} \sum_{i < j} (\sigma_i \tau_i)(\sigma_j \tau_j) = \frac{nR^2 - 1}{2}.$$

Writing a single subscript for one Gibbs average, set

$$q(t) = \mathbb{E}\langle \rho \rangle_{g, g^t} = \frac{1}{n} \sum_{i < j} \mathbb{E}[\langle \sigma_i \sigma_j \rangle_g \langle \sigma_i \sigma_j \rangle_{g^t}].$$

These Gibbs means are smooth with all derivatives bounded. Applying [7, Lemma 3.3] to them at its two-sided time $t/2$ shows that q is nonnegative and nonincreasing. The variance identity [7, Theorem 3.8], applied to F_n/β , gives

$$\frac{\text{Var } F_n(\beta)}{\beta^2} = \int_0^\infty e^{-s} q(s) ds \geq (1 - e^{-t})q(t).$$

Since $q(t) = (n\mathbb{E}\langle R^2 \rangle_{g, g^t} - 1)/2$, this proves (1.7).

For replacement, apply [7, Theorem 3.14] directly to the M independent coordinates, regarding $f = F_n(\beta)/(\beta\sqrt{n})$ as a function of g . For $\partial_{ij} = \partial/\partial g_{ij}$, throughout $\mathbb{R}^M$,

$$\partial_{ij} f = n^{-1} \langle \sigma_i \sigma_j \rangle_g, \quad |\partial_{ij} f| \leq n^{-1}, \quad |\partial_{ij}^2 f| \leq \beta n^{-3/2}.$$

The Gaussian third absolute difference moment is bounded, so the theorem's remainder is $O(M\beta n^{-5/2}) = O_\beta(n^{-1/2})$, uniformly in k . Its left side and variance term give

$$\begin{aligned} \frac{1}{2} \left(\mathbb{E} \langle R^2 \rangle_{g, g^A} - \frac{1}{n} \right) &= \mathbb{E} \sum_{i < j} \partial_{ij} f(g) \partial_{ij} f(g^A) \\ &\leq \frac{M+1}{k+1} \frac{\text{Var } F_n(\beta)}{\beta^2 n} + O_\beta(n^{-1/2}). \end{aligned}$$

This proves (1.8). Finally, the variance asymptotic in the introduction gives $\text{Var } F_n(\beta) = O_\beta(n^{1/3})$. Use $1 - e^{-t} \geq (1 - e^{-1}) \min\{t, 1\}$ for the first scale and $(M+1)/(k+1) \leq M/k$ for $k \geq 1$ for the second. The stated sequence conditions make the resulting bounds tend to zero. $\square$

A Inputs from the fluctuation-scale companion

We collect the inputs from [19], with their conventions and their uses in this paper. The moment and comparison estimates are separate inputs from the exponent theorem. The target temperature is fixed in $(1, \infty)$; the auxiliary common bases $T_m \rightarrow T > 1$ are treated below. All replica tests have a fixed finite number of labels, and all moment, Taylor and recursion orders are fixed before the system size tends to infinity.

A.1 Covariance and mass normalization

The Hamiltonian in (1.1), multiplied by β , has covariance $mTQ^2/2 - T/2$. Adding the independent spin-constant Gaussian of variance $T/2$ in (1.3) gives $mTQ^2/2$, exactly the covariance of [19]. The probability spin prior subtracts $m \log 2$. Consequently the centered completed variable in the companion is our $F_m^c - \mathbf{E} F_m^c$.

Here is the conversion from our terminal mass a to terminal mass one. For $u = v/a$, put

$$\tilde{b}(u) = a^2 b(au), \quad \tilde{h}(u) = a^2 h(au), \quad \tilde{F} = aF.$$

The terminal expression becomes $\log \int e^{aH}$; at an intermediate mass $v = au$, multiplying the continuation by a changes its power mean to the one of exponent u . Thus, including self corrections,

$$f_m^{\text{PE}}(\tilde{b}, \tilde{h}) = a f_m(b, h), \quad \tilde{X}_u = X_{au}, \quad \tilde{C}_u = C_{au}, \quad H_{au} = a\tilde{H}_u, \quad W_{au} = a\tilde{W}_u.$$

In particular $vC_v \preceq H_v \preceq aC_v$. The projection denominator is unchanged:

$$(v/a) \sqrt{m(a^2 \Delta h)} = v \sqrt{m \Delta h}.$$

The same change of variables passes the deterministic-weight pinned inequality to dv with precisely the right side in (A1). At a bare size endpoint $a = \ell$, $b = T/\ell^2$, the total continuation is F_m^c/ℓ , so $x/\ell = m^{-1/6}$ when $m = Ne^\theta$.

A.2 Finite-tree calculus and its probability laws

The following list specifies both the source statement and the additional conditions retained when it is used.

1. [19, Lemmas 2.1–2.3, equation (1)] give the source Hessian, pressure variation, clock inequality, and single-spin pinned moments in Lemma 2.1. The pinned estimate requires zero deterministic external field, deterministic weights, and the full single-path law. If other branches are present, their normalized kernels are first marginalized; remaining random factors are treated afterwards by unconditional Hölder. The polynomial form (A2) is stated and proved in [19, Appendix, “Source and pressure calculus; the scope of pinned moments”]. Section 2 recalls its proof in our normalization: for fixed terminal spin the gradient of $P_v(L_v)$ is $P'_v(L_v)$ times the original gradient. The same block-Hessian inequality and Brascamp–Lieb argument apply, and gauge symmetry removes the pin without changing the centering. The coefficients of P_v are deterministic under the positive path law. This estimate is used in Lemma 8.6.
2. [19, Lemma 3.5, equation (13)] supplies (A3). The anchor is determined by the prefix and branches exterior to the future being projected. Conditional on those data, that future must retain its normalized positive kernel. The traversed field length and its positive lower mass are bounded below by inverse powers of size. In every application here, the total clocks are at most a fixed power of $\log m$, as stipulated in Section 2; hence the coefficient ratios in the finite projection iteration are polynomial in the size. The lemma itself does not require the optimized constraints of PE. The layer estimates in Sections 6–8 use this version with unit-normalized anchors and then integrate the anchor magnitude. Their Hessian blocking is the calculation in [19, proof of Proposition 3.4, preceding equation (15)].
3. [19, Proposition 2.5, Lemmas 2.6, 2.8 and 2.9, equation (2)] give the real-power allocation rule, row sums, bounded split densities, and fixed-order Taylor estimates. Each independent Gaussian increment covariance remains positive semidefinite throughout each interpolation. Spin tests and their scalar centers are held fixed during these interpolations; a varying test requires its own product-rule derivatives. These are the inputs to Lemmas 2.2–2.5, the coefficient algebra in Proposition 5.1, and the compression identity (E5). A row collapses only after its summand is independent of the new label’s detailed attachment. Allocation order is changed before absolute values are taken. A reused fork retains its atom; a new fork has a continuous mass variable.
4. [19, Corollary 2.10] bounds fixed-order source derivatives. The terminal-datum derivatives in Lemma 3.3 are treated by its diffusion representation with one bounded smooth insertion; the source bounds control the remaining fixed derivative slots. The scalar time identities are those of [19, Lemma 5.3], converted from time K/T to variance time K . They are also used for the single-coordinate freezing estimate in (D11).
5. [19, Proposition 2.14] is a limit at each fixed system size, with fixed finite observed topology. Terminal clocks and every prescribed cut or revelation side are retained. No rate uniform in system size is inferred. At a simultaneous jump we choose the joint linear filling

$$(b_-, h_-) + t(\Delta b, \Delta h), \quad 0 \leq t \leq 1.$$

If an intermediate filled revelation is observed, approximations preserve this filling and its specified observations. Integrated clock identities refer to that chosen filling; only endpoint laws are independent of it. Nonnegative contributions of filled jumps can be discarded in trace upper bounds. Fixed-size limits are used in Lemma 2.1, the scalar path construction, Lemma 4.2, and the finite-resolution passage in Proposition 5.1. The latter first uses the inverse at a finite resolution with its uniform absolute attachment bound; it does not require convergence of the inverse kernels themselves.

For the centered-coupling argument of [19, Lemmas 2.11–2.13 and 4.5], the conditioning sigma-field is generated by the common prefix, the unaffected exterior branches, and the fixed topology. It contains no observation of the affected future. This preserves the two separate conditional marginals. The weighted exterior anchor in [19, Lemma 4.8] averages the exterior joint law rather than conditioning on its terminal spins. The applications here retain these conditions. In particular Lemma 8.12 and Remark 8.13 explicitly construct a marginal law independent of the moving fork before using a centered integral.

A.3 Optimized comparisons and changed constraints

The all-mass calls in Sections 6 and 7 use [19, Definition 6.1 and Proposition 6.6], with data (3)–(6) there. Writing m for the site count and $\epsilon_{\text{PE}} > 0$ for its fixed power slack, the error parameter obeys

$$m^{-2/3+\epsilon_{\text{PE}}} d_{\text{PE}} \lesssim E \leq m^{-1/2} d_{\text{PE}}^2.$$

The bare paths and all changed competitors are admissible, total clocks are uniformly bounded, and a fixed positive matrix margin remains after the comparison. The upper bound for d_{PE}^2 has a fixed positive gap below the limiting T . The endpoint condition is (4) of PE on every required comparable mass band, with the common base T_m of (P9). The common-base extension is stated and proved in [19, Appendix, “A common prelimit temperature”]. Section 6 verifies its use here and recalls the substitutions $J_0 = T_m \chi^2 - 1$ in PE (26)–(27) and time K/T_m in PE Lemmas 5.4–5.5. This base stays fixed through each finite parent/child tree. The macroscopic limit is at the fixed $T > 1$, so [19, Lemma 5.8 and Proposition 5.9] apply at that limit. No uniform assertion near the critical temperature is required.

For clarity, the further source inputs to the modified local optimization are as follows.

1. The geometry of [19, Lemma 3.3, equations (9)–(10)] is rederived for the prescribed density box in Lemma 7.2. The first variation has no state-dependent integrating factor in this box. The original PE density cap is not assumed for its minimizers.
2. Under the stated outer-regularity hypotheses, Lemmas 7.4 and 7.5 establish the replacement raw and projection estimates (C8)–(C9) and second moments (C10), including the enhanced floor. Under the additional strong-removal hypotheses, Lemma 7.6 gives separate mass and radius estimates and one good calibration cut for all fixed moment orders. For cap/contact units these proofs use [19, Lemmas 3.11–3.12, equations (20)–(21)]: whole-cap trace bounds, controlled mean oscillation, deterministic weights, and comparable masses. Broad caps are sliced with their additional trace cost retained. These same inputs are used in Lemmas 7.11, 7.12 and 7.13. The enhanced floor is treated separately by radius slides; small physical floor mass is never used to discard radius intervals.
3. The weak-removal argument uses the child-data proof of [19, Lemma 6.2], the movement inequality of [19, Lemma 6.3], and feasible-barrier arguments of [19, Lemma 6.4]. The child has the ordinary PE constraints. The parent movements are proved feasible for the new density box before using movement. Child success gives agreement and then outer regularity for the child, as in [19, Lemma 6.5]; only then are [19, Theorem 5.1 and Proposition 5.9] used.
4. Proposition 7.7, used in the strong-removal argument, reuses the proof of [19, Proposition 4.1], rather than applying its original optimization hypotheses to the new density box. The preceding analytic estimates replace PE (12)–(13), (16)–(19). Fixed-degree tests remain those of [19, Definition 4.2]; bounded row changes, observed-block truncation, boundary covariance, test-family closure, and projected replacements are [19,

Lemmas 4.3–4.8]. The finite-depth cancellation and calibration are [19, Lemmas 4.9–4.10, equations (23)–(27)]. The division is by $\chi K_* \asymp W$, not by a stability gap. One common good cut is used for the fixed finite recursion and its required moment orders; these orders are fixed before the size limit. The resulting moving-target estimate is stated explicitly in (7.1); (7.2) gives its two integrated forms, with the mass and radius normalizations kept separate. These are the versions of PE (22) used here. The profile proof is then repeated using [19, Lemmas 5.3–5.6] and PE (28)–(29). Identification of its slope uses [19, Lemma 5.8 and Proposition 5.9] and the macroscopic transfer in Proposition 7.3.

The coarse scalar trial used for (G3) comes from the bounds in the proof of [19, Lemma 5.7], with the explicit floor, cap, and cutoff choices given in Section 6. Its proof is used quantitatively; the limiting variational statement alone would not provide the needed power saving. The scalar control representation used for coercivity and pair-kernel positivity is proved in [19, proof of Lemma 5.8] by completing the square in Itô’s formula.

A.4 The exponent and the one-dimensional input

The fixed-temperature exponent theorem [19, Theorem 1.1] is used twice in Section 4: to initialize the stopped small-positive-Laplace comparison, and to rule out zero limiting variance after a power rate of variance convergence has been obtained. The first use combines the exponent with [19, Lemmas 7.6–7.7]; along the nonbare physical paths the appropriate exponential augmentation is instead proved in Lemma 4.2 from the telescoped pinned density. That proof uses the full joint positive law and then gauge symmetry. The one-dimensional tail, positive-tail-mass and small-Laplace bounds are exactly [19, Lemma 7.7].

The exponent supplies neither a variance prefactor nor a limiting law. Those conclusions use the summable increment estimate of this paper. Finally $\text{Var } F_m^c = \text{Var } F_m + T/2$; the common Gaussian therefore disappears after division by $m^{1/6}$. The theorem is stated for log partition functions. For the physical total free energy $-F_m/\beta$, the variance constant is divided by β^2 and the standardized limiting law is reflected.

B Guide to shared notation

This table collects notation used across several sections. Local scale ratios, optimizer profiles, and rescaled discrepancy functions are defined in the arguments where they occur.

Notation	Meaning and defining location
F_n^c, Y_n, a_k	Completed free energy, scaled centered free energy, and limiting moments; (1.3) and Proposition 1.3.
N, m, n, x	Dyadic reference size N , integer size indices m, n , and $x = N^{-1/6}$. Section 4 temporarily also uses $n = Ne^\theta$ as a real mollification center.
θ, ℓ, a	Size coordinate, $\ell = e^{\theta/6}$, and terminal mass: $a = \ell$ on direct paths and $a = 4$ on smoothed paths; Section 3.
$v, t = K(v), \xi$	Replica mass, scalar variance time, and normalized scalar time $\xi = \ell^2 t$; Sections 2–3.
b, h, K, q	Matrix and field clocks, scalar reference clock, and response $q(v) = \Gamma(K(v))$; Section 2.
h_*, d, J	Reference field $h_* = K - bq$, field defect $d = h - h_*$, and row mismatch $J = b\Delta + d$; Section 2.
Q, Q^*, Δ, δ	Overlap, tilt-masked overlap, and error $\Delta = Q^* - q$. Section 9 also uses $\delta = Q^* - S_v$, so $\Delta = \delta + (S_v - q)$; (A4).

Notation	Meaning and defining location
X_v, C_v, H_v	Posterior spin mean, conditional covariance, and source Hessian; Section 2.
S_v, B_v, D_v, L_v, W_v	Deterministic overlap statistics and single-path fluctuation observables; Section 2 and (A1).
$\psi_m(s), \mathcal{K}_m(s)$	Difference of pressure tests and centered log Laplace transform $\mathcal{K}_m(s) = sm\psi_m(s)$; (A4).
λ, U, γ	Regularization strength, low scale, and deterministic path parameters; (P1) and (P3).
$Z, Z_g, e, Y = UZ$	Parameter-dependent majorants, deterministic error envelope, and derived scale; (D1) and Propositions 8.1 and 8.11.
$\nu, \mathbf{E}_v, \mathbf{E}_\gamma$	Positive-tree expectation, conditioning on the revealed prefix, and averaging over path parameters; Sections 2 and 4.
$(\cdot)_\#$	Signed allocation per root mass; Section 2.
$\lesssim_\circ, \lesssim_\diamond$	Subpower losses and the stated arbitrarily small geometric power losses; Sections 2 and 8.

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